1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

D8 MLB

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
			2012-07-03

LAST_MODIFIED=Tue Jul 3 15:06:35 2012

DRAWING
TITLE=K22
ABBREV=DRAWING
LAST_MODIFIED=Tue Jul 3 15:06:35 2012

DRAWING TITLE: SCH,D8,MLB

Apple Inc.

DRAWING NUMBER: 051-9504 SIZE: D

REVISION: 5.3.0

BRANCH

DRAWING

8 7 6 5 4 3 2 1

D

C

B

A

System Block diagram can be found on Kismet

PATH: Kismet > K70/72 > Block Diagrams > K70 Block Diagram

D8_MLB

01/05/2012

PAGE TITLE

System Block Diagram

Apple Inc.

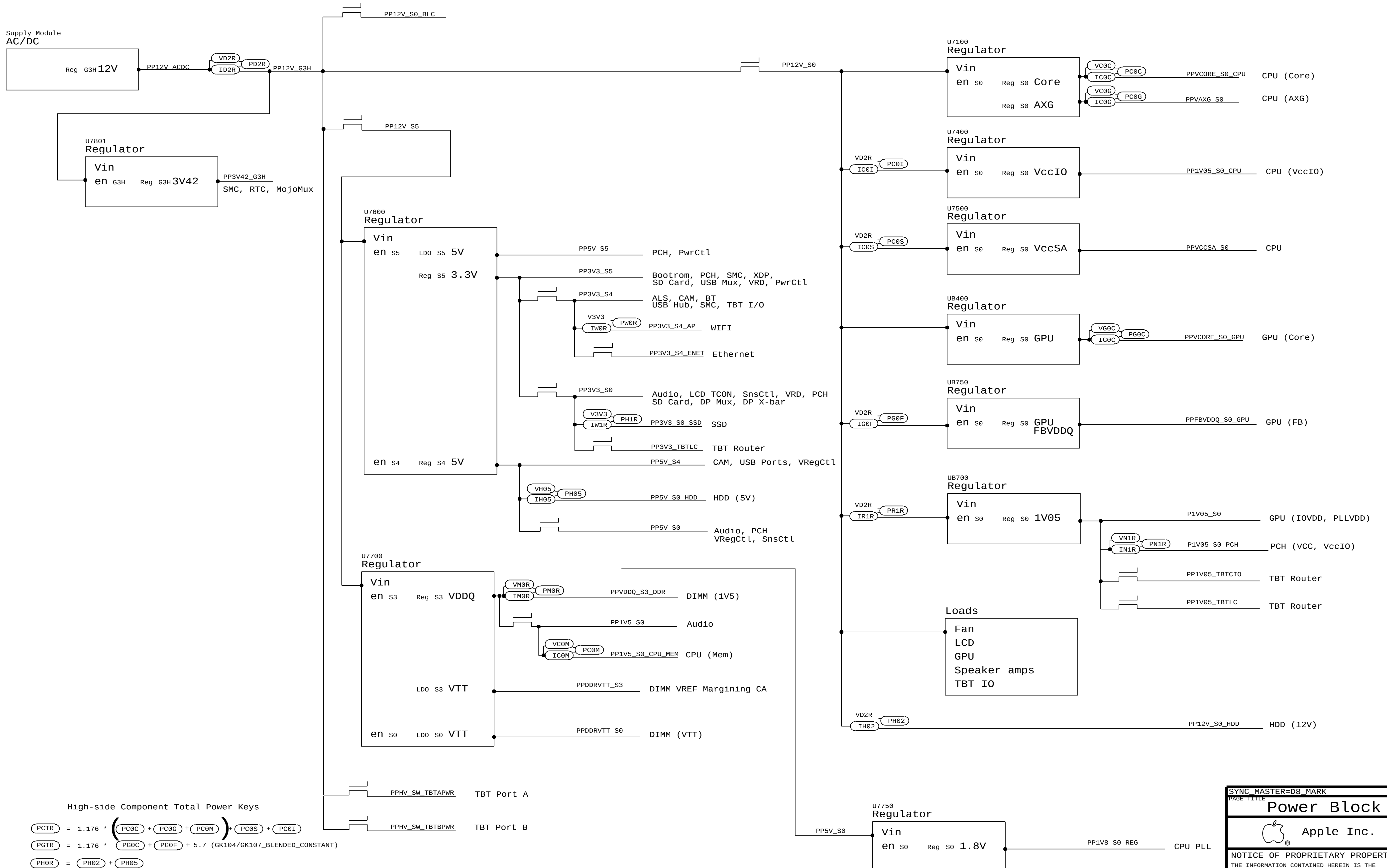
DRAWING NUMBER 051-9504

SIZE D

REVISION 5.3.0

BRANCH

8 7 6 5 4 3 2 1



High-side Component Total Power Keys

PCTR = 1.176 * (PC0C + PC0G + PC0M) + PC0S + PC0I

PGTR = 1.176 * PG0C + PG0F + 5.7 (GK104/GK107_BLENDED_CONSTANT)

PH0R = PH02 + PH05

SYNC_MASTER=D8_MARK SYNC_DATE=04/23/2012

PAGE TITLE

Power Block Diagram

Apple Inc.

DRAWING NUMBER	SIZE
051-9504	D
REVISION 5.3.0	
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-3816	PCBA,MLB,2.9G,4C,GK107,SAM,D8	D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F2FR
639-3952	PCBA,MLB,3.1G,4C,GK107,SAM,D8	D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F49T
639-4092	PCBA,MLB,2.9G,4C,GK107,HYN.D8	D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F653
639-4093	PCBA,MLB,3.1G,4C,GK107,HYN.D8	D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F654
085-4433	PCBA,MLB,DEV,D8	DEVELOPMENT,D8_DEVEL

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7122	1	MLB LABEL,48.0X4.8	EEEE_DHNM	CRITICAL	EEEE:DHNM
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F2FR	CRITICAL	EEEE:F2FR
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F49V	CRITICAL	EEEE:F49V
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F49T	CRITICAL	EEEE:F49T
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F653	CRITICAL	EEEE:F653
825-7122	1	MLB LABEL,48.0X4.8	EEEE_F654	CRITICAL	EEEE:F654

BOM Groups

BOM GROUP	BOM OPTIONS
D8_COMMON	COMMON,ALTERNATE,D8_COMMON1,D8_PROGPARTS
D8_COMMON1	XDP,RSMRST:GATE,SPEAKERID,VREF:CPU,TBTHV:P12V,PRE_BOOST:Y,FBA,FBB
D8_PROGPARTS	SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG,BLCMCU:PROG
D8_DEVEL	XDP_CONN,LPCPLUS,VREFMRGN:EXT,BKLT_PWM, DEVEL_AUDIO,TEMPSNSDEV

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4355	1	IVB,SR0TA,PRQ,N1,2.9,65W,4+1,1.1,6M,LGA	CPU	CRITICAL	CPU:4C_2P9GHZ
337S4372	1	IVB,SR0T9,PRQ,N0,3.1,65W,4+1,1.1,6M,LGA	CPU	CRITICAL	CPU:4C_3P1GHZ

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4277	1	IC,PANTHER POINT,C1,SLJC7,PRQ,BD82Z77	U1800	CRITICAL	
338S1113	1	IC,TBT,CR-4C,B1,PRQ,288 FCBGA,12X12MM	U3600	CRITICAL	
337S4280	1	IC, GPU, NV GK107-GE-PS-A2	U8000	CRITICAL	GPU:107EGE
343S0616	1	IC,BCM57766A1,ENET&SD,8X8	U3900	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	
341S3640	1	IC,EEPROM,CR,V11.1 (B1),D8	U3690	CRITICAL	TBTROM:PROG	
335S0865	1	IC,EEPROM,SERIAL,8KB,MLP8	U3690	CRITICAL	TBTROM:BLANK	
341S3641	1	IC,PROGRMD,EFI ROM,PIB,D7/D8	U5110	CRITICAL	BOOTROM:PROG	ALTERNATE:335S0812
335S0807	1	IC,64 MBIT SPI SERIAL FLASH	U5110	CRITICAL	BOOTROM:BLANK	
341S3394	1	IC,PROGRMD,SMC,A3,V2.2A32,D8	U4900	CRITICAL	SMC:PROG	
338S1098	1	IC,SMC,LX4FS1AH5BBCIGA3	U4900	CRITICAL	SMC:BLANK	
341S3646	1	IC,CAMERA FLASH,V7226,D7/D8	U4202	CRITICAL	CAMROM:PROG	
335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK	
341S3645	1	IC,ENET 1MBIT, SPI,ROM, V1.13 D8	U3990	CRITICAL	CIVROM:PROG	ALTERNATE:335S0854
335S0862	1	IC,SERIAL FLASH,2MBIT, 2.7V, REF F	U3990	CRITICAL	CIVROM:BLANK	
341S3627	1	IC,BLC,MCU, PRPOGRAMMED, V0203, D8	U9700	CRITICAL	BLCMCU:PROG	
337S3978	1	IC,BLC MCU LPC2132FBD64/01, LQFP64	U9700	CRITICAL	BLCMCU:BLANK	

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0073	1	SOCKET,MOLEX,LGA1155,CPU-LF	U1000	CRITICAL	

CPU SOCKET ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
511S0071	511S0073		ALL	TYCO SOCKET
511S0072	511S0073		ALL	FOXCONN SOCKET

D8 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-9504	1	SCH,MLB,D8	SCH1	CRITICAL	D8
820-3298	1	PCBF,MLB,D8	PCB1	CRITICAL	D8

D8 ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB diodes
157S0084	157S0058		ALL	Enet Magnetics
341S3644	341S3645		U3990	CIVROM
376S0975	376S1081		ALL	P/NCH DUAL FET
152S1668	152S1591		ALL	33UH BLC INDUCTOR
128S0365	128S0368		ALL	150UF CAPS BLK

VRAM MODULE PARTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
333S0619	4	IC,SGRAM,GDDR5,32MX32,1.5GHz,G-DIE,HF	U8400,U8450,U8500,U8550	CRITICAL	FB:BOTH_SAMSUNG
333S0620	4	IC,GDDR5,32MX32,1.5GHZ,VEGA 44NM,B-DIE	U8400,U8450,U8500,U8550	CRITICAL	FB:BOTH_HYNIX

SYNC_MASTER=D8_MLB SYNC_DATE=12/19/2011

PAGE TITLE

BOM Configuration

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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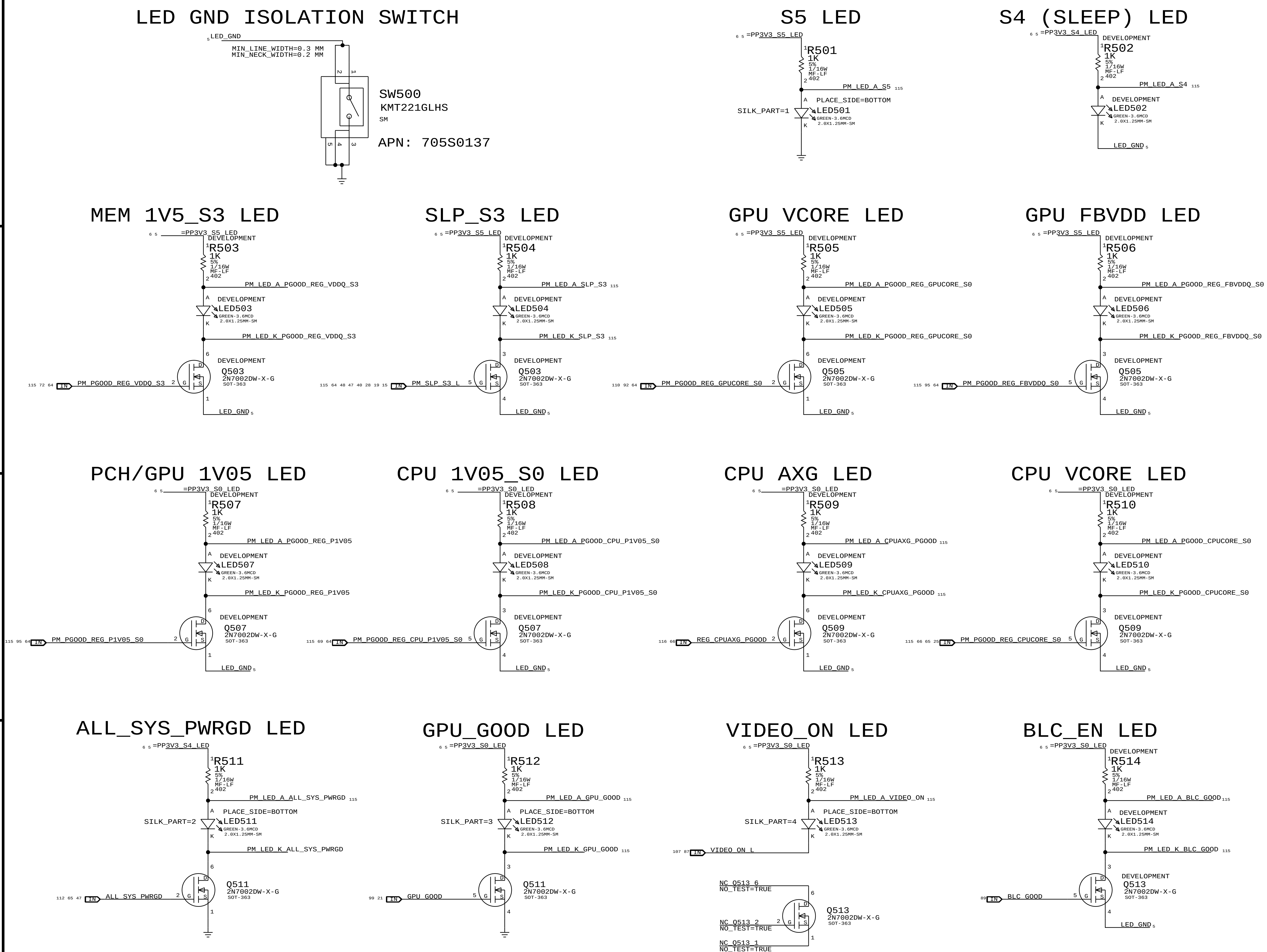

SYNC_MASTER=D8_TAVYS SYNC_DATE=06/22/2012

PAGE TITLE

DEBUG LEDS

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

BRANCH

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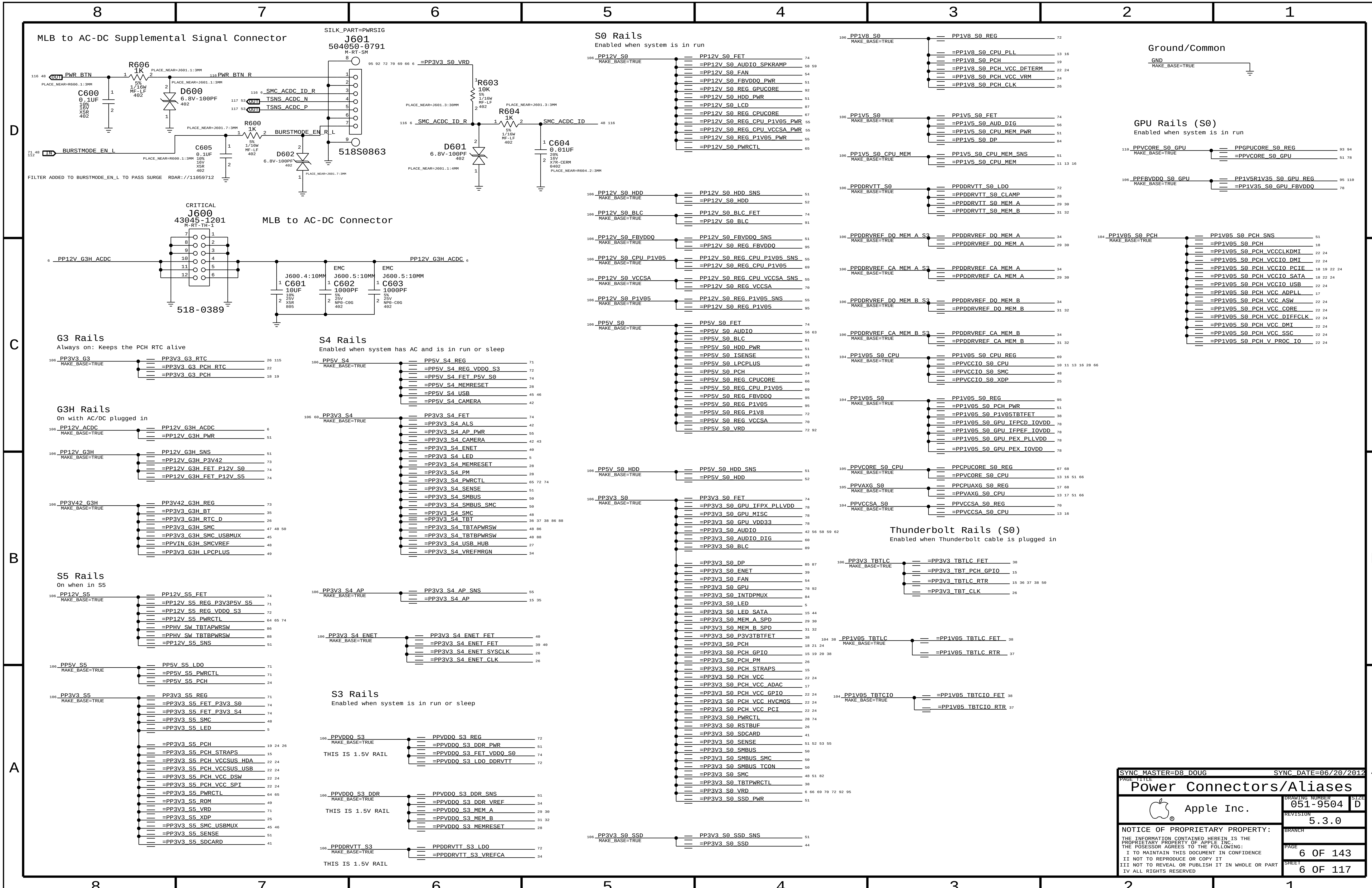
MLB to AC-DC Supplemental Signal Connector
SILK_PART=PWRSIG
J601
504050-0791
M-RT-SM
518S0863
PWR_BTN
R606 1K 5% 1/16W MF-LF 402
PWR_BTN_R
C600 0.1UF 10% 16V X5R 402
D600 6.8V-100PF 402
SMC_ACDC_ID_R
TSNS_ACDC_N
TSNS_ACDC_P
=PP3V3_S0_VRD
R603 10K 5% 1/16W MF-LF 402
R604 1K 5% 1/16W MF-LF 402
SMC_ACDC_ID_R
SMC_ACDC_ID
D601 6.8V-100PF 402
C604 0.01UF 20% 16V X7R-CERM 0402
R600 1K 5% 1/16W MF-LF 402
BURSTMODE_EN_R_L
BURSTMODE_EN_L
C605 0.1UF 10% 16V X5R 402
D602 6.8V-100PF 402
FILTER ADDED TO BURSTMODE_EN_L TO PASS SURGE RDAR://11059712
CRITICAL
J600
43045-1201
M-RT-TH-1
518-0389
MLB to AC-DC Connector
PP12V_G3H_ACDC
C601 10UF 10% 25V X5R 805
EMC
J600.4:10MM
J600.5:10MM
C602 1000PF 5% 25V NP0-C0G 402
C603 1000PF 5% 25V NP0-C0G 402
G3 Rails
Always on: Keeps the PCH RTC alive
PP3V3_G3
PP3V3_G3_RTC
=PP3V3_G3_PCH_RTC
=PP3V3_G3_PCH
G3H Rails
On with AC/DC plugged in
PP12V_ACDC
PP12V_G3H_ACDC
=PP12V_G3H_PWR
PP12V_G3H
PP12V_G3H_SNS
=PP12V_G3H_P3V42
=PP12V_G3H_FET_P12V_S0
=PP12V_G3H_FET_P12V_S5
PP3V42_G3H
PP3V42_G3H_REG
=PP3V3_G3H_BT
=PP3V3_G3H_RTC_D
=PP3V3_G3H_SMC
=PP3V3_G3H_SMC_USBMUX
=PPVIN_G3H_SMCVREF
=PP3V3_G3H_LPCPLUS
S5 Rails
On when in S5
PP12V_S5
PP12V_S5_FET
=PP12V_S5_REG_P3V3P5V_S5
=PP12V_S5_REG_VDDQ_S3
=PP12V_S5_PWRCTL
=PPHV_SW_TBTAPWRSW
=PPHV_SW_TBTBPWRSW
=PP12V_S5_SNS
PP5V_S5
PP5V_S5_LDO
=PP5V_S5_PWRCTL
=PP5V_S5_PCH
PP3V3_S5
PP3V3_S5_REG
=PP3V3_S5_FET_P3V3_S0
=PP3V3_S5_FET_P3V3_S4
=PP3V3_S5_SMC
=PP3V3_S5_LED
=PP3V3_S5_PCH
=PP3V3_S5_PCH_STRAPS
=PP3V3_S5_PCH_VCCSUS_HDA
=PP3V3_S5_PCH_VCCSUS_USB
=PP3V3_S5_PCH_VCC_DSW
=PP3V3_S5_PCH_VCC_SPI
=PP3V3_S5_PWRCTL
=PP3V3_S5_ROM
=PP3V3_S5_VRD
=PP3V3_S5_XDP
=PP3V3_S5_SMC_USBMUX
=PP3V3_S5_SENSE
=PP3V3_S5_SDCARD
S4 Rails
Enabled when system has AC and is in run or sleep
PP5V_S4
PP5V_S4_REG
=PP5V_S4_REG_VDDQ_S3
=PP5V_S4_FET_P5V_S0
=PP5V_S4_MEMRESET
=PP5V_S4_USB
=PP5V_S4_CAMERA
PP3V3_S4
PP3V3_S4_FET
=PP3V3_S4_ALS
=PP3V3_S4_AP_PWR
=PP3V3_S4_CAMERA
=PP3V3_S4_ENET
=PP3V3_S4_LED
=PP3V3_S4_MEMRESET
=PP3V3_S4_PM
=PP3V3_S4_PWRCTL
=PP3V3_S4_SENSE
=PP3V3_S4_SMBUS
=PP3V3_S4_SMBUS_SMC
=PP3V3_S4_SMC
=PP3V3_S4_TBT
=PP3V3_S4_TBTAPWRSW
=PP3V3_S4_TBTBPWRSW
=PP3V3_S4_USB_HUB
=PP3V3_S4_VREFMRGN
PP3V3_S4_AP
PP3V3_S4_AP_SNS
=PP3V3_S4_AP
PP3V3_S4_ENET
PP3V3_S4_ENET_FET
=PP3V3_S4_ENET_FET
=PP3V3_S4_ENET_SYSCLK
=PP3V3_S4_ENET_CLK
S3 Rails
Enabled when system is in run or sleep
PPVDDQ_S3
PPVDDQ_S3_REG
=PPVDDQ_S3_DDR_PWR
=PPVDDQ_S3_FET_VDDQ_S0
=PPVDDQ_S3_LDO_DDRVTT
THIS IS 1.5V RAIL
PPVDDQ_S3_DDR
PPVDDQ_S3_DDR_SNS
=PPVDDQ_S3_DDR_VREF
=PPVDDQ_S3_MEM_A
=PPVDDQ_S3_MEM_B
=PPVDDQ_S3_MEMRESET
THIS IS 1.5V RAIL
PPDDRVTT_S3
PPDDRVTT_S3_LDO
=PPDDRVTT_S3_VREFCA
THIS IS 1.5V RAIL
S0 Rails
Enabled when system is in run
PP12V_S0
PP12V_S0_FET
=PP12V_S0_AUDIO_SPKRAMP
=PP12V_S0_FAN
=PP12V_S0_FBVDDQ_PWR
=PP12V_S0_REG_GPUCORE
=PP12V_S0_HDD_PWR
=PP12V_S0_LCD
=PP12V_S0_REG_CPUCORE
=PP12V_S0_REG_CPU_P1V05_PWR
=PP12V_S0_REG_CPU_VCCSA_PWR
=PP12V_S0_REG_P1V05_PWR
=PP12V_S0_PWRCTL
PP12V_S0_HDD
PP12V_S0_HDD_SNS
=PP12V_S0_HDD
PP12V_S0_BLC
PP12V_S0_BLC_FET
=PP12V_S0_BLC
PP12V_S0_FBVDDQ
PP12V_S0_FBVDDQ_SNS
=PP12V_S0_REG_FBVDDQ
PP12V_S0_CPU_P1V05
PP12V_S0_REG_CPU_P1V05_SNS
=PP12V_S0_REG_CPU_P1V05
PP12V_S0_VCCSA
PP12V_S0_REG_CPU_VCCSA_SNS
=PP12V_S0_REG_VCCSA
PP12V_S0_P1V05
PP12V_S0_REG_P1V05_SNS
=PP12V_S0_REG_P1V05
PP5V_S0
PP5V_S0_FET
=PP5V_S0_AUDIO
=PP5V_S0_BLC
=PP5V_S0_HDD_PWR
=PP5V_S0_ISENSE
=PP5V_S0_LPCPLUS
=PP5V_S0_PCH
=PP5V_S0_REG_CPUCORE
=PP5V_S0_REG_CPU_P1V05
=PP5V_S0_REG_FBVDDQ
=PP5V_S0_REG_P1V05
=PP5V_S0_REG_P1V8
=PP5V_S0_REG_VCCSA
=PP5V_S0_VRD
PP5V_S0_HDD
PP5V_S0_HDD_SNS
=PP5V_S0_HDD
PP3V3_S0
PP3V3_S0_FET
=PP3V3_S0_GPU_IFPX_PLLVDD
=PP3V3_S0_GPU_MISC
=PP3V3_S0_GPU_VDD33
=PP3V3_S0_AUDIO
=PP3V3_S0_AUDIO_DIG
=PP3V3_S0_BLC
=PP3V3_S0_DP
=PP3V3_S0_ENET
=PP3V3_S0_FAN
=PP3V3_S0_GPU
=PP3V3_S0_INTDPMUX
=PP3V3_S0_LED
=PP3V3_S0_LED_SATA
=PP3V3_S0_MEM_A_SPD
=PP3V3_S0_MEM_B_SPD
=PP3V3_S0_P3V3TBTFET
=PP3V3_S0_PCH
=PP3V3_S0_PCH_GPIO
=PP3V3_S0_PCH_PM
=PP3V3_S0_PCH_STRAPS
=PP3V3_S0_PCH_VCC
=PP3V3_S0_PCH_VCC_ADAC
=PP3V3_S0_PCH_VCC_GPIO
=PP3V3_S0_PCH_VCC_HVCMOS
=PP3V3_S0_PCH_VCC_PCI
=PP3V3_S0_PWRCTL
=PP3V3_S0_RSTBUF
=PP3V3_S0_SDCARD
=PP3V3_S0_SENSE
=PP3V3_S0_SMBUS
=PP3V3_S0_SMBUS_SMC
=PP3V3_S0_SMBUS_TCON
=PP3V3_S0_SMC
=PP3V3_S0_TBTPWRCTL
=PP3V3_S0_VRD
=PP3V3_S0_SSD_PWR
PP3V3_S0_SSD
PP3V3_S0_SSD_SNS
=PP3V3_S0_SSD
PP1V8_S0
PP1V8_S0_REG
=PP1V8_S0_CPU_PLL
=PP1V8_S0_PCH
=PP1V8_S0_PCH_VCC_DFTERM
=PP1V8_S0_PCH_VCC_VRM
=PP1V8_S0_PCH_CLK
PP1V5_S0
PP1V5_S0_FET
=PP1V5_S0_AUD_DIG
=PP1V5_S0_CPU_MEM_PWR
=PP1V5_S0_DP
PP1V5_S0_CPU_MEM
PP1V5_S0_CPU_MEM_SNS
=PP1V5_S0_CPU_MEM
PPDDRVTT_S0
PPDDRVTT_S0_LDO
=PPDDRVTT_S0_CLAMP
=PPDDRVTT_S0_MEM_A
=PPDDRVTT_S0_MEM_B
PPDDRVREF_DQ_MEM_A_S3
PPDDRVREF_DQ_MEM_A
=PPDDRVREF_DQ_MEM_A
PPDDRVREF_CA_MEM_A_S3
PPDDRVREF_CA_MEM_A
=PPDDRVREF_CA_MEM_A
PPDDRVREF_DQ_MEM_B_S3
PPDDRVREF_DQ_MEM_B
=PPDDRVREF_DQ_MEM_B
PPDDRVREF_CA_MEM_B_S3
PPDDRVREF_CA_MEM_B
=PPDDRVREF_CA_MEM_B
PP1V05_S0_CPU
PP1V05_S0_CPU_REG
=PPVCCIO_S0_CPU
=PPVCCIO_S0_SMC
=PPVCCIO_S0_XDP
PP1V05_S0
PP1V05_S0_REG
=PP1V05_S0_PCH_PWR
=PP1V05_S0_P1V05TBTFET
=PP1V05_S0_GPU_IFPCD_IOVDD
=PP1V05_S0_GPU_IFPEF_IOVDD
=PP1V05_S0_GPU_PEX_PLLVDD
=PP1V05_S0_GPU_PEX_IOVDD
PPVCORE_S0_CPU
PPCPUCORE_S0_REG
=PPVCORE_S0_CPU
PPVAXG_S0
PPCPUAXG_S0_REG
=PPVAXG_S0_CPU
PPVCCSA_S0
PPVCCSA_S0_REG
=PPVCCSA_S0_CPU
Thunderbolt Rails (S0)
Enabled when Thunderbolt cable is plugged in
PP3V3_TBTLC
=PP3V3_TBTLC_FET
=PP3V3_TBT_PCH_GPIO
=PP3V3_TBTLC_RTR
=PP3V3_TBT_CLK
PP1V05_TBTLC
=PP1V05_TBTLC_FET
=PP1V05_TBTLC_RTR
PP1V05_TBTCIO
=PP1V05_TBTCIO_FET
=PP1V05_TBTCIO_RTR
Ground/Common
GND
MAKE_BASE=TRUE
GPU Rails (S0)
Enabled when system is in run
PPVCORE_S0_GPU
PPGPUCORE_S0_REG
=PPVCORE_S0_GPU
PPFBVDDQ_S0_GPU
PP1V5R1V35_S0_GPU_REG
=PP1V35_S0_GPU_FBVDDQ
PP1V05_S0_PCH
PP1V05_S0_PCH_SNS
=PP1V05_S0_PCH
=PP1V05_S0_PCH_VCCCLKDMI
=PP1V05_S0_PCH_VCCIO_DMI
=PP1V05_S0_PCH_VCCIO_PCIE
=PP1V05_S0_PCH_VCCIO_SATA
=PP1V05_S0_PCH_VCCIO_USB
=PP1V05_S0_PCH_VCC_ADPLL
=PP1V05_S0_PCH_VCC_ASW
=PP1V05_S0_PCH_VCC_CORE
=PP1V05_S0_PCH_VCC_DIFFCLK
=PP1V05_S0_PCH_VCC_DMI
=PP1V05_S0_PCH_VCC_SSC
=PP1V05_S0_PCH_V_PROC_IO
SYNC_MASTER=D8_DOUG
SYNC_DATE=06/20/2012
PAGE TITLE
Power Connectors/Aliases
Apple Inc.
DRAWING NUMBER
051-9504
SIZE
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5.3.0
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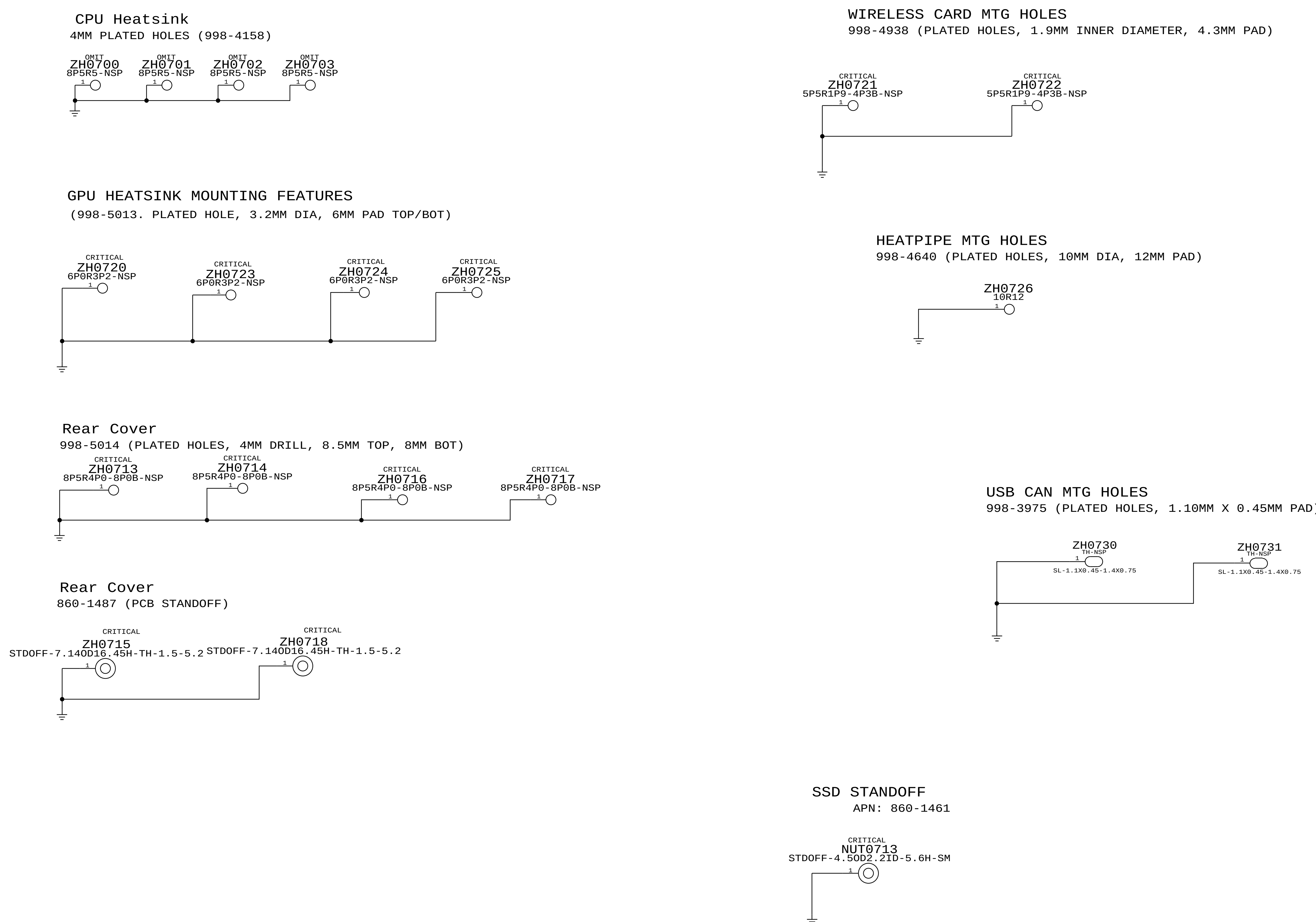

SYNC_MASTER=D8_MLB SYNC_DATE=03/25/2012

PAGE TITLE

Holes/PD parts

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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CPU Reserved

Page	Net	Alias	Properties
10	TP_CPU_RSVD<16..1>	NC_CPU_RSVD<16..1>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_RSVD<46..19>	NC_CPU_RSVD<46..19>	MAKE_BASE=TRUE NO_TEST=TRUE
10	CPU_CFG<15..12>	TP_CPU_CFG<15..12>	MAKE_BASE=TRUE

CPU Memory

Page	Net	Alias	Properties
12	TP_MEM_A_DQ_CB<7..0>	NC_MEM_A_DQ_CB<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_A_DQS_N<8>	NC_MEM_A_DQSNX<8>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_A_DQS_P<8>	NC_MEM_A_DQSPX<8>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_B_DQ_CB<7..0>	NC_MEM_B_DQ_CB<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_B_DQS_N<8>	NC_MEM_B_DQSNX<8>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_MEM_B_DQS_P<8>	NC_MEM_B_DQSPX<8>	MAKE_BASE=TRUE NO_TEST=TRUE

PCH PCIe

Page	Net	Alias	Properties
18	TP_PCIE1_D2RN	NC_PCIE1_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE1_D2RP	NC_PCIE1_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE1_R2D_CN	NC_PCIE1_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE1_R2D_CP	NC_PCIE1_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE2_D2RN	NC_PCIE2_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE2_D2RP	NC_PCIE2_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE2_R2D_CN	NC_PCIE2_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE2_R2D_CP	NC_PCIE2_R2D_PNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	DMI_MIDBUS_CLK100M_N	NC_DMI_MIDBUS_CLK100NX	MAKE_BASE=TRUE NO_TEST=TRUE
18	DMI_MIDBUS_CLK100M_P	NC_DMI_MIDBUS_CLK100PX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE0N	NC_PCIE_CLK100M_PE0NX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE0P	NC_PCIE_CLK100M_PE0PX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE4N	NC_PCIE_CLK100M_PE4NX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE4P	NC_PCIE_CLK100M_PE4PX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE5N	NC_PCIE_CLK100M_PE5NX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCIE_CLK100M_PE5P	NC_PCIE_CLK100M_PE5PX	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE6N	NC_PCIE_CLK100M_PE6NX	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE6P	NC_PCIE_CLK100M_PE6PX	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE7N	NC_PCIE_CLK100M_PE7NX	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCIE_CLK100M_PE7P	NC_PCIE_CLK100M_PE7PX	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_PE_TX_N<3..0>	NC_PE_TNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_PE_TX_P<3..0>	NC_PE_TPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_PE_RX_N<3..0>	NC_PE_RNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_PE_RX_P<3..0>	NC_PE_RPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE

PCH USB

Page	Net	Alias	Properties
20	USB_PCH_4_N	NC_USB_PCH_4NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_4_P	NC_USB_PCH_4PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_5_N	NC_USB_PCH_5NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_5_P	NC_USB_PCH_5PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_6_N	NC_USB_PCH_6NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_6_P	NC_USB_PCH_6PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_11_N	NC_USB_PCH_11NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_11_P	NC_USB_PCH_11PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_12_N	NC_USB_PCH_12NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_12_P	NC_USB_PCH_12PX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_13_N	NC_USB_PCH_13NX	MAKE_BASE=TRUE NO_TEST=TRUE
20	USB_PCH_13_P	NC_USB_PCH_13PX	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Clocks

Page	Net	Alias	Properties
18	TP_PCH_CLKOUT_DPN	NC_PCH_CLKOUT_DPNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CLKOUT_DPP	NC_PCH_CLKOUT_DPPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	PCH_CLK25M_XTALOUT	NC_PCH_CLK25M_XTALOUT	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO64_CLKOUTFLEX0	NC_PCH_GPIO64_CLKOUTFLEX0	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO65_CLKOUTFLEX1	NC_PCH_GPIO65_CLKOUTFLEX1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO66_CLKOUTFLEX2	NC_PCH_GPIO66_CLKOUTFLEX2	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_GPIO67_CLKOUTFLEX3	NC_PCH_GPIO67_CLKOUTFLEX3	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Unused Display

Page	Net	Alias	Properties
19	TP_CRT_IG_RED	NC_CRT_IG_RED	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_GREEN	NC_CRT_IG_GREEN	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_BLUE	NC_CRT_IG_BLUE	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_HSYNC	NC_CRT_IG_HSYNC	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_VSYNC	NC_CRT_IG_VSYNC	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_DDC_CLK	NC_CRT_IG_DDC_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_CRT_IG_DDC_DATA	NC_CRT_IG_DDC_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_MLN<3..0>	NC_DP_IG_B_MLNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_MLP<3..0>	NC_DP_IG_B_MLPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_AUX_N	NC_DP_IG_B_AUXNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_AUX_P	NC_DP_IG_B_AUXPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_HPD	NC_DP_IG_B_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_DDC_CLK	NC_DP_IG_B_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_B_DDC_DATA	NC_DP_IG_B_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_MLN<3..0>	NC_DP_IG_C_MLNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_MLP<3..0>	NC_DP_IG_C_MLPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_AUX_N	NC_DP_IG_C_AUXNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_AUX_P	NC_DP_IG_C_AUXPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_HPD	NC_DP_IG_C_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_CTRL_CLK	NC_DP_IG_C_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_C_CTRL_DATA	NC_DP_IG_C_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_MLN<3..0>	NC_DP_IG_D_MLNX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_MLP<3..0>	NC_DP_IG_D_MLPX<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_AUXN	NC_DP_IG_D_AUXNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_AUXP	NC_DP_IG_D_AUXPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_HPD	NC_DP_IG_D_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_CTRL_CLK	NC_DP_IG_D_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
19	DP_IG_D_CTRL_DATA	NC_DP_IG_D_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_TVCLKINN	NC_SDVO_TVCLKINNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_TVCLKINP	NC_SDVO_TVCLKINPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_STALLN	NC_SDVO_STALLNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_STALLP	NC_SDVO_STALLPX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_INTN	NC_SDVO_INTNX	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_SDVO_INTP	NC_SDVO_INTPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_L_BKLTCTL	NC_PCH_L_BKLTCTL	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_L_BKLTEN	NC_PCH_L_BKLTEN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_L_VDD_EN	NC_PCH_L_VDD_EN	MAKE_BASE=TRUE NO_TEST=TRUE

PCH SATA

Page	Net	Alias	Properties
18	TP_SATA_C_R2D_CN	NC_SATA_C_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_C_R2D_CP	NC_SATA_C_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_C_D2RN	NC_SATA_C_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_C_D2RP	NC_SATA_C_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_R2D_CN	NC_SATA_D_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_R2D_CP	NC_SATA_D_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_D2RN	NC_SATA_D_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_D_D2RP	NC_SATA_D_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_R2D_CN	NC_SATA_E_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_R2D_CP	NC_SATA_E_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_D2RN	NC_SATA_E_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_E_D2RP	NC_SATA_E_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_R2D_CN	NC_SATA_F_R2D_CNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_R2D_CP	NC_SATA_F_R2D_CPX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_D2RN	NC_SATA_F_D2RNX	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SATA_F_D2RP	NC_SATA_F_D2RPX	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Reserved

Page	Net	Alias	Properties
19	TP_PCH_RESERVE_0	NC_PCH_RESERVE_0	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_1	NC_PCH_RESERVE_1	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_2	NC_PCH_RESERVE_2	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_3	NC_PCH_RESERVE_3	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_4	NC_PCH_RESERVE_4	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_5	NC_PCH_RESERVE_5	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_6	NC_PCH_RESERVE_6	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_7	NC_PCH_RESERVE_7	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_8	NC_PCH_RESERVE_8	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_9	NC_PCH_RESERVE_9	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_10	NC_PCH_RESERVE_10	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_11	NC_PCH_RESERVE_11	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_12	NC_PCH_RESERVE_12	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_13	NC_PCH_RESERVE_13	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_14	NC_PCH_RESERVE_14	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_15	NC_PCH_RESERVE_15	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_16	NC_PCH_RESERVE_16	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_17	NC_PCH_RESERVE_17	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_18	NC_PCH_RESERVE_18	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_19	NC_PCH_RESERVE_19	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_20	NC_PCH_RESERVE_20	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_21	NC_PCH_RESERVE_21	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_22	NC_PCH_RESERVE_22	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_23	NC_PCH_RESERVE_23	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_24	NC_PCH_RESERVE_24	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_25	NC_PCH_RESERVE_25	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_26	NC_PCH_RESERVE_26	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_27	NC_PCH_RESERVE_27	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_RESERVE_28	NC_PCH_RESERVE_28	MAKE_BASE=TRUE NO_TEST=TRUE

PCH and CPU FDI

Page	Net	Alias	Properties
10	TP_CPU_FDI_TX_N<7..0>	NC_CPU_FDI_TNX<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_FDI_TX_P<7..0>	NC_CPU_FDI_TPX<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_RX_N<7..0>	NC_PCH_FDI_RNX<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_RX_P<7..0>	NC_PCH_FDI_RPX<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_FDI_FSYNC<1..0>	NC_CPU_FDI_FSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_FDI_LSYNC<1..0>	NC_CPU_FDI_LSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
10	TP_CPU_FDI_INT	NC_CPU_FDI_INT	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_FSYNC<1..0>	NC_PCH_FDI_FSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_LSYNC<1..0>	NC_PCH_FDI_LSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	PCH_FDI_INT	NC_PCH_FDI_INT	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Test Points

Page	Net	Alias	Properties
21	TP_PCH_TP1	NC_PCH_TP1	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP2	NC_PCH_TP2	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP3	NC_PCH_TP3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP4	NC_PCH_TP4	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP5	NC_PCH_TP5	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP6	NC_PCH_TP6	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP7	NC_PCH_TP7	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP8	NC_PCH_TP8	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP9	NC_PCH_TP9	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP10	NC_PCH_TP10	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP11	NC_PCH_TP11	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP12	NC_PCH_TP12	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP13	NC_PCH_TP13	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP14	NC_PCH_TP14	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP15	NC_PCH_TP15	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP16	NC_PCH_TP16	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP17	NC_PCH_TP17	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP18	NC_PCH_TP18	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP19	NC_PCH_TP19	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_TP20	NC_PCH_TP20	MAKE_BASE=TRUE NO_TEST=TRUE

PCH PCI

Page	Net	Alias	Properties
20	TP_PCI_AD<31..0>	NC_PCI_AD<31..0>	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_C_BE_L<3..0>	NC_PCI_C_BE_L<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_PAR	NC_PCI_PAR	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_RESET_L	NC_PCI_RESET_L	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCH_PCI_GNT0_L	NC_PCI_GNT0_L	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCH_INIT3V3_L	NC_PCH_INIT3V3_L	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_LPC_DREQ0_L	NC_LPC_DREQ0_L	MAKE_BASE=TRUE NO_TEST=TRUE

PCH Miscellaneous

Page	Net	Alias	Properties
18	TP_HDA_SDIN1	NC_HDA_SDIN1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_HDA_SDIN2	NC_HDA_SDIN2	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_HDA_SDIN3	NC_HDA_SDIN3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM0	NC_PCH_PWM0	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM1	NC_PCH_PWM1	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM2	NC_PCH_PWM2	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_PWM3	NC_PCH_PWM3	MAKE_BASE=TRUE NO_TEST=TRUE
21	TP_PCH_SST	NC_PCH_SST	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CL_CLK1	NC_PCH_CL_CLK1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CL_DATA1	NC_PCH_CL_DATA1	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_CL_RST1	NC_PCH_CL_RST1	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_CLK33M_OUT2	NC_PCI_CLK33M_OUT2	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCI_CLK33M_OUT3	NC_PCI_CLK33M_OUT3	MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=D8_MLB_ULTIMATE SYNC_DATE=04/02/2012

PAGE TITLE

Unused Signal Aliases

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ALIASES (BLANK)

SYNC_MASTER=K70_MLB SYNC_DATE=08/23/2011

PAGE TITLE

Signal Aliases

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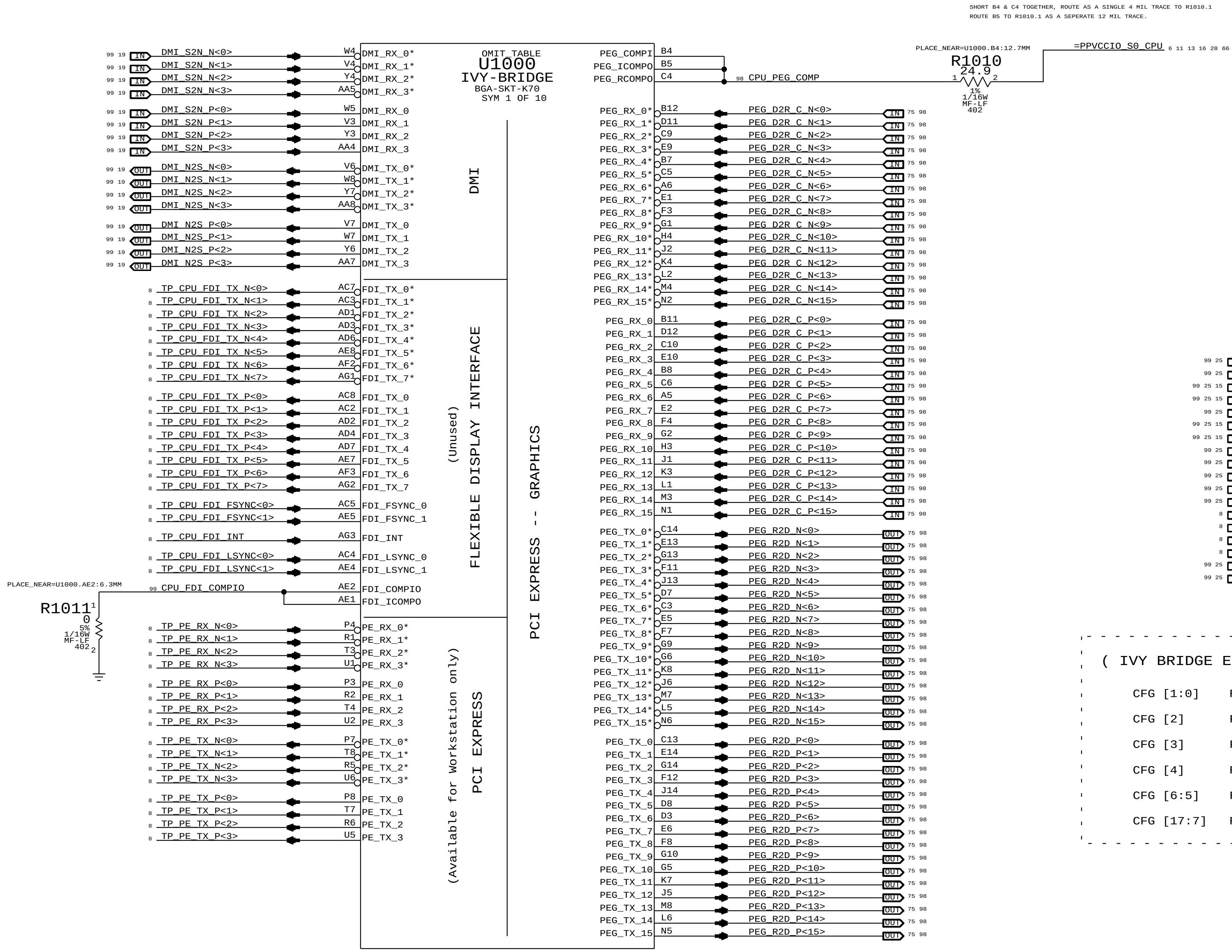

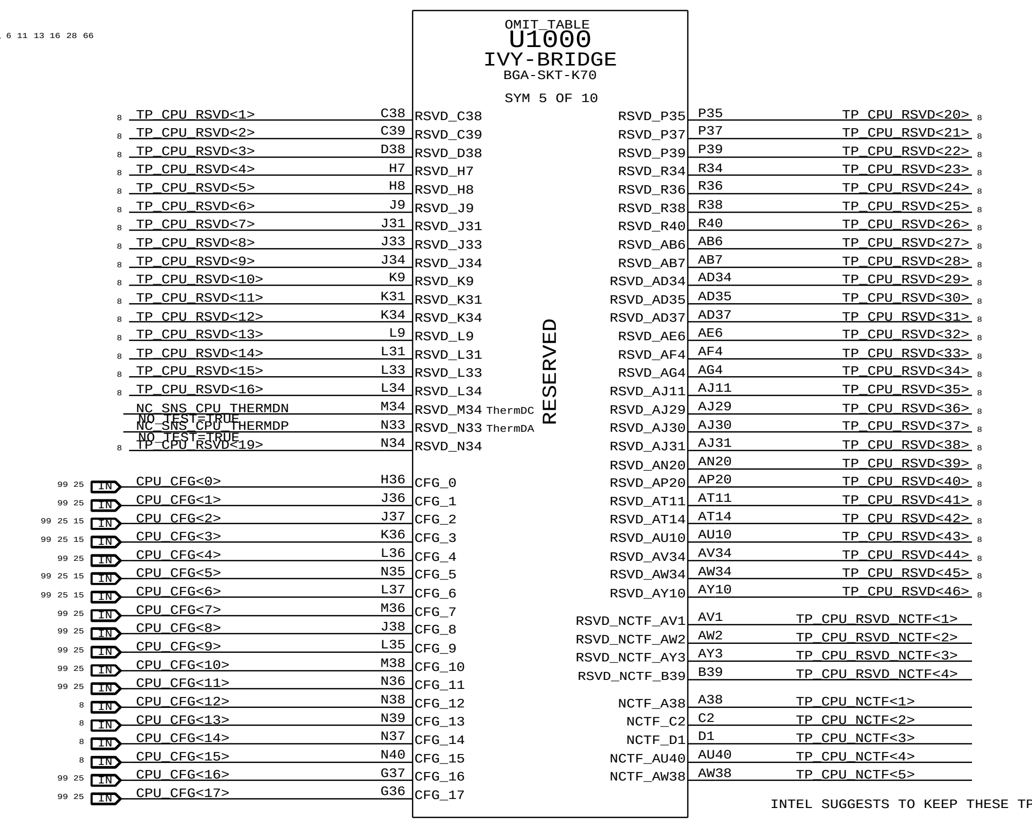

(IVY BRIDGE EDS #473717 TABLE 6-5)

CFG [1:0]	RESERVED CONFIGURATION LANE				
CFG [2]	PCIE STATIC X16 LANE REVERSAL	1 = NORMAL OPERATION	0 = LANES REVERSED		
CFG [3]	PCIE STATIC X4 LANE REVERSAL	1 = NORMAL OPERATION	0 = LANES REVERSED		
CFG [4]	RESERVED CONFIGURATION LANE				
CFG [6:5]	PCIE BIFURCATION	11 = 1 X16 (DEFAULT)	10 = 2 X8	01 = RSVD	00 = 1 X8, 2 X4
CFG [17:7]	RESERVED CONFIGURATION LANE				

SYNC_MASTER=D8_ROSITA SYNC_DATE=03/23/2012

PAGE TITLE
CPU DMI/PEG/FDI/RSVD

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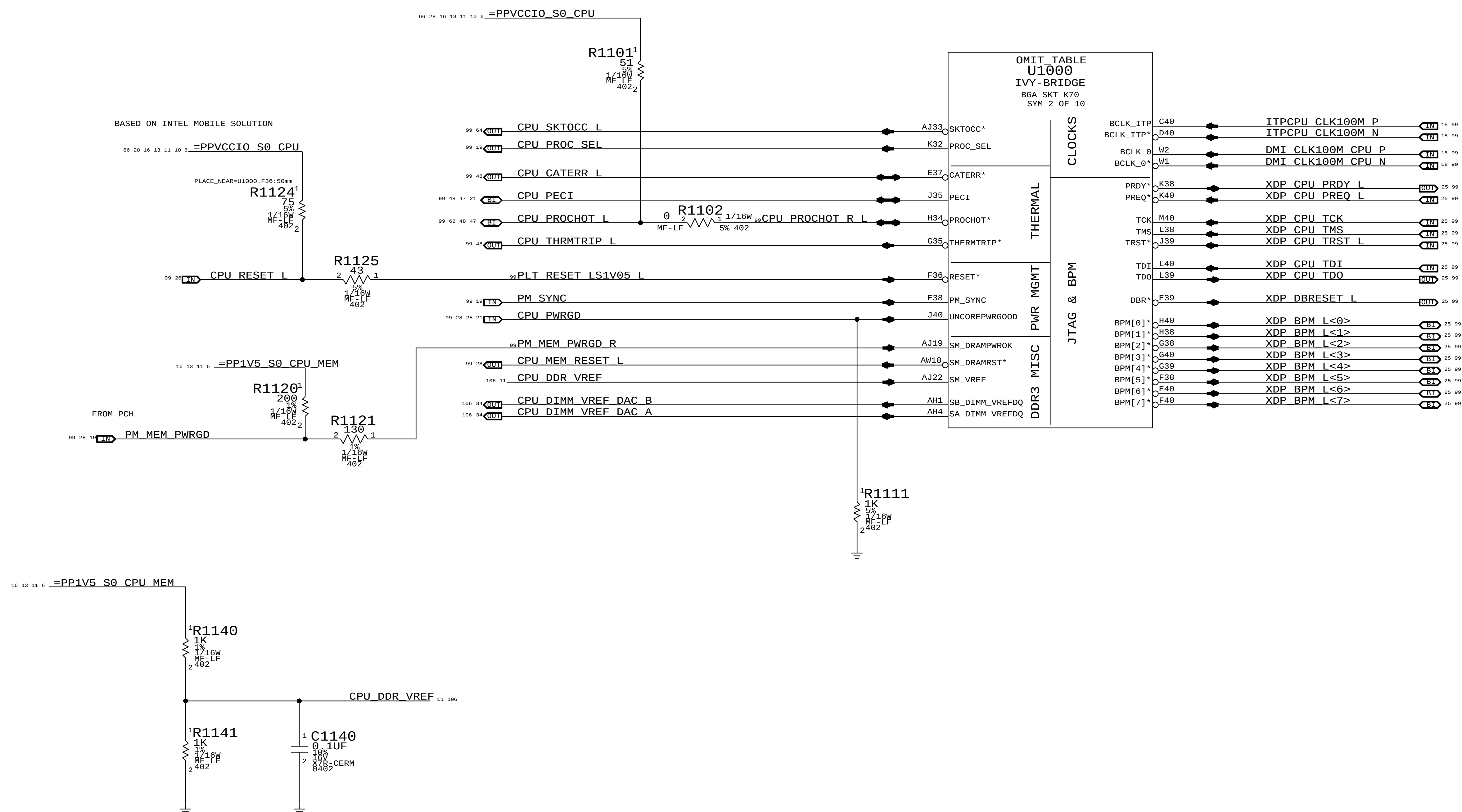

SYNC_MASTER=D7_MLB SYNC_DATE=03/15/2012

PAGE TITLE

CPU CLOCK/MISC/JTAG

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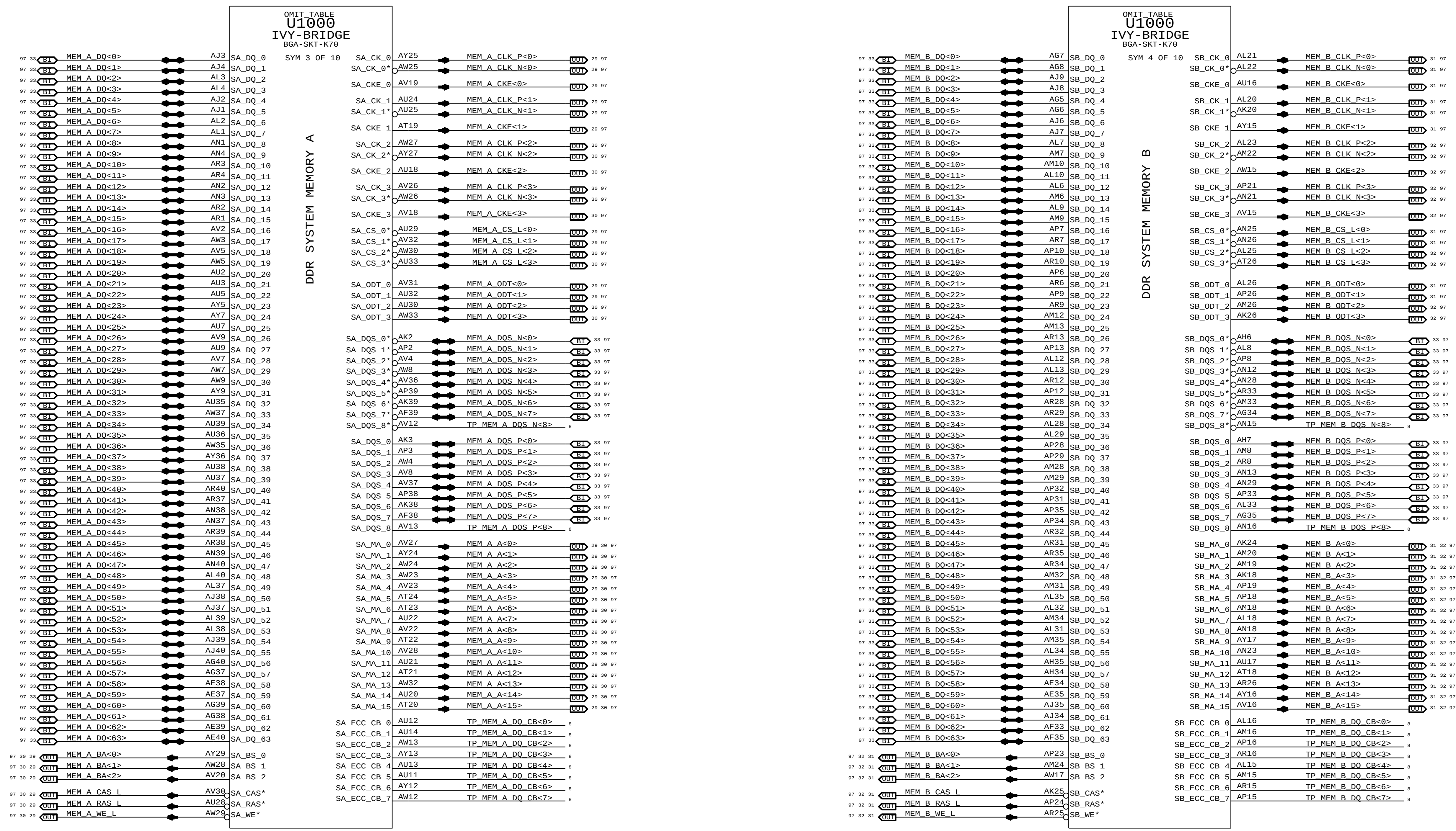

SYNC_MASTER=D7_MLB SYNC_DATE=03/15/2012

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CPU DDR3 INTERFACES

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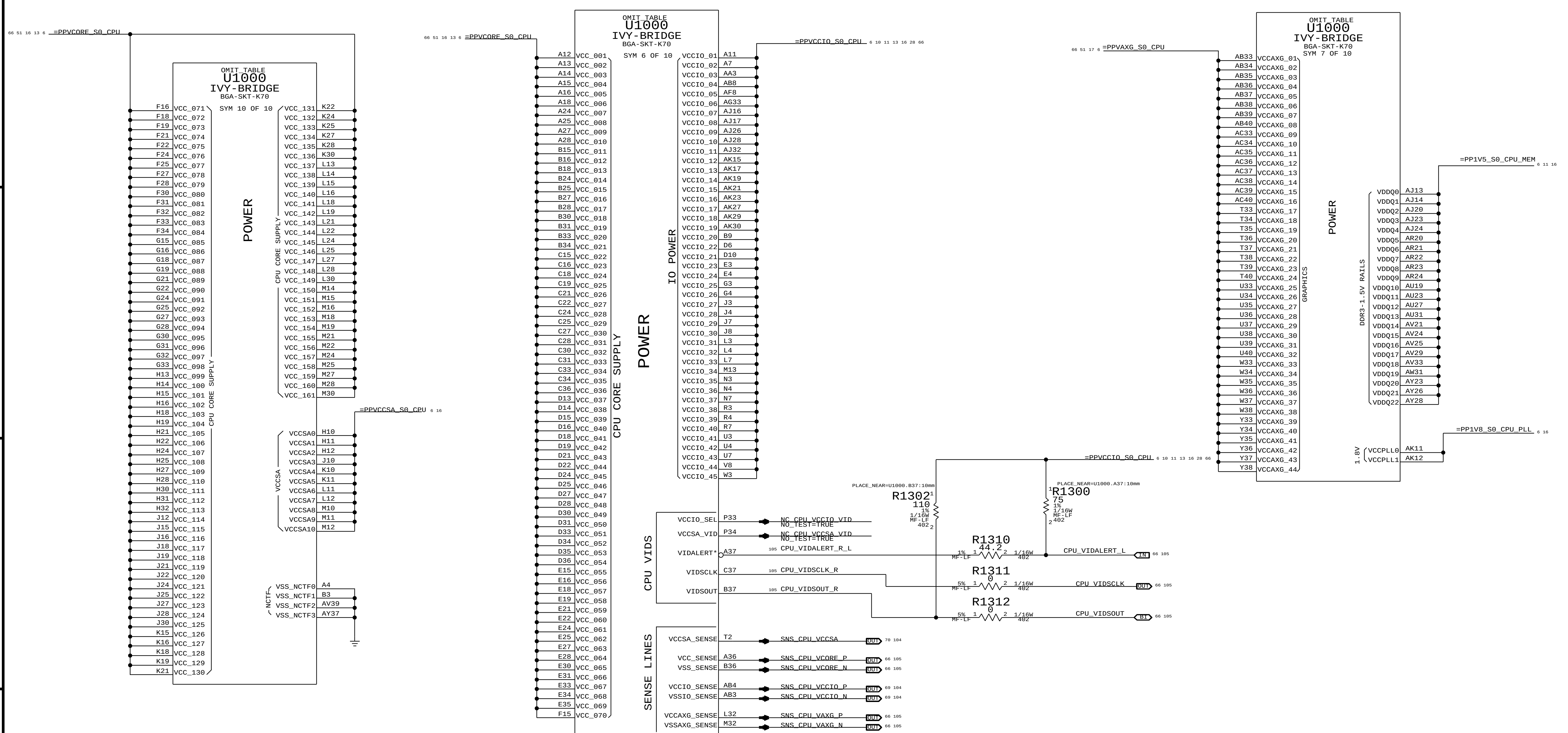

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PAGE TITLE

CPU POWER

DRAWING NUMBER	051-9504
SIZE	D
REVISION	5.3.0
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PAGE	13 OF 143
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BGA-SKT-K70
SYM 8 OF 10

VSS

A17	VSS_001	VSS_091	AM27
A23	VSS_002	VSS_092	AM3
A26	VSS_003	VSS_093	AM30
A29	VSS_004	VSS_094	AM36
A35	VSS_005	VSS_095	AM37
AA33	VSS_006	VSS_096	AM38
AA34	VSS_007	VSS_097	AM39
AA35	VSS_008	VSS_098	AM4
AA36	VSS_009	VSS_099	AM40
AA37	VSS_010	VSS_100	AM5
AA38	VSS_011	VSS_101	AN10
AA6	VSS_012	VSS_102	AN11
AB5	VSS_013	VSS_103	AN14
AC1	VSS_014	VSS_104	AN17
AC6	VSS_015	VSS_105	AN19
AD33	VSS_016	VSS_106	AN22
AD36	VSS_017	VSS_107	AN24
AD38	VSS_018	VSS_108	AN27
AD39	VSS_019	VSS_109	AN30
AD40	VSS_020	VSS_110	AN31
AD5	VSS_021	VSS_111	AN32
AD8	VSS_022	VSS_112	AN33
AE3	VSS_023	VSS_113	AN34
AE33	VSS_024	VSS_114	AN35
AE36	VSS_025	VSS_115	AN36
AF1	VSS_026	VSS_116	AN5
AF34	VSS_027	VSS_117	AN6
AF36	VSS_028	VSS_118	AN7
AF37	VSS_029	VSS_119	AN8
AF40	VSS_030	VSS_120	AN9
AF5	VSS_031	VSS_121	AP1
AF6	VSS_032	VSS_122	AP11
AF7	VSS_033	VSS_123	AP14
AG36	VSS_034	VSS_124	AP17
AH2	VSS_035	VSS_125	AP22
AH3	VSS_036	VSS_126	AP25
AH33	VSS_037	VSS_127	AP27
AH36	VSS_038	VSS_128	AP30
AH37	VSS_039	VSS_129	AP36
AH38	VSS_040	VSS_130	AP37
AH39	VSS_041	VSS_131	AP4
AH40	VSS_042	VSS_132	AP40
AH5	VSS_043	VSS_133	AP5
AH8	VSS_044	VSS_134	AR11
AJ12	VSS_045	VSS_135	AR14
AJ15	VSS_046	VSS_136	AR17
AJ18	VSS_047	VSS_137	AR18
AJ21	VSS_048	VSS_138	AR19
AJ25	VSS_049	VSS_139	AR27
AJ27	VSS_050	VSS_140	AR30
AJ36	VSS_051	VSS_141	AR36
AJ5	VSS_052	VSS_142	AR5
AK1	VSS_053	VSS_143	AT1
AK10	VSS_054	VSS_144	AT10
AK13	VSS_055	VSS_145	AT12
AK14	VSS_056	VSS_146	AT13
AK16	VSS_057	VSS_147	AT15
AK22	VSS_058	VSS_148	AT16
AK28	VSS_059	VSS_149	AT17
AK31	VSS_060	VSS_150	AT2
AK32	VSS_061	VSS_151	AT25
AK33	VSS_062	VSS_152	AT27
AK34	VSS_063	VSS_153	AT28
AK35	VSS_064	VSS_154	AT29
AK36	VSS_065	VSS_155	AT3
AK37	VSS_066	VSS_156	AT30
AK4	VSS_067	VSS_157	AT31
AK40	VSS_068	VSS_158	AT32
AK5	VSS_069	VSS_159	AT33
AK6	VSS_070	VSS_160	AT34
AK7	VSS_071	VSS_161	AT35
AK8	VSS_072	VSS_162	AT36
AK9	VSS_073	VSS_163	AT37
AL11	VSS_074	VSS_164	AT38
AL14	VSS_075	VSS_165	AT39
AL17	VSS_076	VSS_166	AT4
AL19	VSS_077	VSS_167	AT40
AL24	VSS_078	VSS_168	AT5
AL27	VSS_079	VSS_169	AT6
AL30	VSS_080	VSS_170	AT7
AL36	VSS_081	VSS_171	AT8
AL5	VSS_082	VSS_172	AT9
AM1	VSS_083	VSS_173	AU1
AM11	VSS_084	VSS_174	AU15
AM14	VSS_085	VSS_175	AU26
AM17	VSS_086	VSS_176	AU34
AM2	VSS_087	VSS_177	AU4
AM21	VSS_088	VSS_178	AU6
AM23	VSS_089	VSS_179	AU8
AM25	VSS_090	VSS_180	AV10

OMIT_TABLE
U1000
IVY-BRIDGE
BGA-SKT-K70
SYM 9 OF 10

VSS

AV11	VSS_181	VSS_271	G8
AV14	VSS_182	VSS_272	H1
AV17	VSS_183	VSS_273	H17
AV3	VSS_184	VSS_274	H2
AV35	VSS_185	VSS_275	H20
AV38	VSS_186	VSS_276	H23
AV6	VSS_187	VSS_277	H26
AW10	VSS_188	VSS_278	H29
AW11	VSS_189	VSS_279	H33
AW14	VSS_190	VSS_280	H35
AW16	VSS_191	VSS_281	H37
AW36	VSS_192	VSS_282	H39
AW6	VSS_193	VSS_283	H5
AY11	VSS_194	VSS_284	H6
AY14	VSS_195	VSS_285	H9
AY18	VSS_196	VSS_286	J11
AY35	VSS_197	VSS_287	J17
AY4	VSS_198	VSS_288	J20
AY6	VSS_199	VSS_289	J23
AY8	VSS_200	VSS_290	J26
B10	VSS_201	VSS_291	J29
B13	VSS_202	VSS_292	J32
B14	VSS_203	VSS_293	K1
B17	VSS_204	VSS_294	K12
B23	VSS_205	VSS_295	K13
B26	VSS_206	VSS_296	K14
B29	VSS_207	VSS_297	K17
B32	VSS_208	VSS_298	K2
B35	VSS_209	VSS_299	K20
B38	VSS_210	VSS_300	K23
B6	VSS_211	VSS_301	K26
C11	VSS_212	VSS_302	K29
C12	VSS_213	VSS_303	K33
C17	VSS_214	VSS_304	K35
C20	VSS_215	VSS_305	K37
C23	VSS_216	VSS_306	K39
C26	VSS_217	VSS_307	K5
C29	VSS_218	VSS_308	K6
C32	VSS_219	VSS_309	L10
C35	VSS_220	VSS_310	L17
C7	VSS_221	VSS_311	L20
C8	VSS_222	VSS_312	L23
D17	VSS_223	VSS_313	L26
D2	VSS_224	VSS_314	L29
D20	VSS_225	VSS_315	L8
D23	VSS_226	VSS_316	M1
D26	VSS_227	VSS_317	M17
D29	VSS_228	VSS_318	M2
D32	VSS_229	VSS_319	M20
D37	VSS_230	VSS_320	M23
D39	VSS_231	VSS_321	M26
D4	VSS_232	VSS_322	M29
D5	VSS_233	VSS_323	M33
D9	VSS_234	VSS_324	M35
E11	VSS_235	VSS_325	M37
E12	VSS_236	VSS_326	M39
E17	VSS_237	VSS_327	M5
E20	VSS_238	VSS_328	M6
E23	VSS_239	VSS_329	M9
E26	VSS_240	VSS_330	N8
E29	VSS_241	VSS_331	P1
E32	VSS_242	VSS_332	P2
E36	VSS_243	VSS_333	P36
E7	VSS_244	VSS_334	P38
E8	VSS_245	VSS_335	P40
F1	VSS_246	VSS_336	P5
F10	VSS_247	VSS_337	P6
F13	VSS_248	VSS_338	R33
F14	VSS_249	VSS_339	R35
F17	VSS_250	VSS_340	R37
F2	VSS_251	VSS_341	R39
F20	VSS_252	VSS_342	R8
F23	VSS_253	VSS_343	T1
F26	VSS_254	VSS_344	T5
F29	VSS_255	VSS_345	T6
F35	VSS_256	VSS_346	U8
F37	VSS_257	VSS_347	V1
F39	VSS_258	VSS_348	V2
F5	VSS_259	VSS_349	V33
F6	VSS_260	VSS_350	V34
F9	VSS_261	VSS_351	V35
G11	VSS_262	VSS_352	V36
G12	VSS_263	VSS_353	V37
G17	VSS_264	VSS_354	V38
G20	VSS_265	VSS_355	V39
G23	VSS_266	VSS_356	V40
G26	VSS_267	VSS_357	V5
G29	VSS_268	VSS_358	W6
G34	VSS_269	VSS_359	Y5
G7	VSS_270	VSS_360	Y8

SYNC_MASTER=D7_MLB SYNC_DATE=03/15/2012

PAGE TITLE
CPU GROUNDS

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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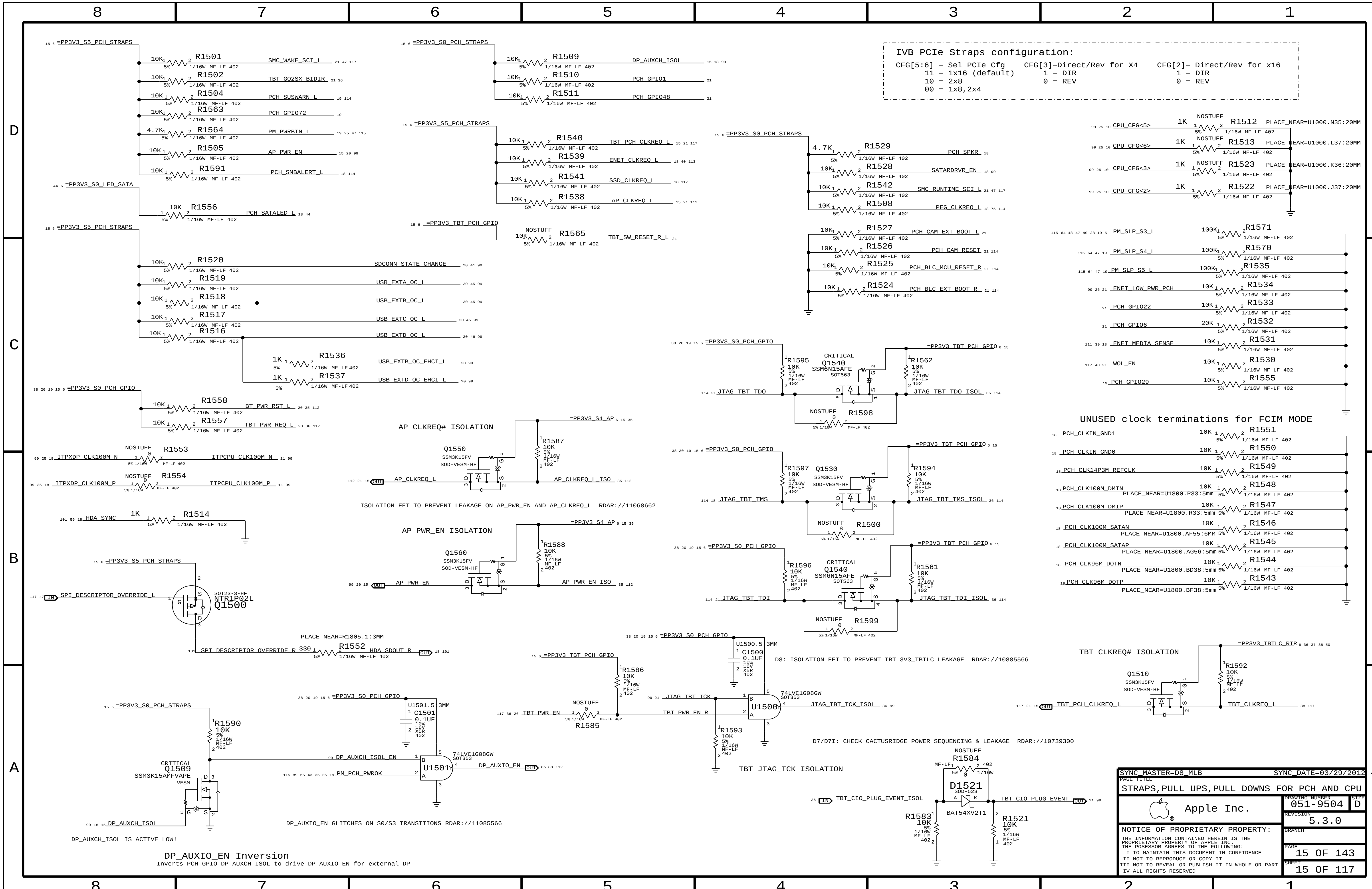
IVB PCIe Straps configuration:
CFG[5:6] = Sel PCIe Cfg
11 = 1x16 (default)
10 = 2x8
00 = 1x8,2x4
CFG[3]=Direct/Rev for X4
1 = DIR
0 = REV
CFG[2]= Direct/Rev for x16
1 = DIR
0 = REV
=PP3V3_S5_PCH_STRAPS
=PP3V3_S0_PCH_STRAPS
=PP3V3_S0_LED_SATA
=PP3V3_TBT_PCH_GPIO
=PP3V3_S0_PCH_GPIO
=PP3V3_S4_AP
=PP3V3_TBTLC_RTR
AP CLKREQ# ISOLATION
ISOLATION FET TO PREVENT LEAKAGE ON AP_PWR_EN AND AP_CLKREQ_L RDAR://11068662
AP PWR_EN ISOLATION
UNUSED clock terminations for FCIM MODE
TBT CLKREQ# ISOLATION
D8: ISOLATION FET TO PREVENT TBT 3V3_TBTLC LEAKAGE RDAR://10885566
TBT JTAG_TCK ISOLATION
D7/D7I: CHECK CACTUSRIDGE POWER SEQUENCING & LEAKAGE RDAR://10739300
DP_AUXIO_EN GLITCHES ON S0/S3 TRANSITIONS RDAR://11085566
DP_AUXCH_ISOL IS ACTIVE LOW!
DP_AUXIO_EN Inversion
Inverts PCH GPIO DP_AUXCH_ISOL to drive DP_AUXIO_EN for external DP
SYNC_MASTER=D8_MLB
SYNC_DATE=03/29/2012
PAGE TITLE
STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU
Apple Inc.
DRAWING NUMBER
051-9504
SIZE
D
REVISION
5.3.0
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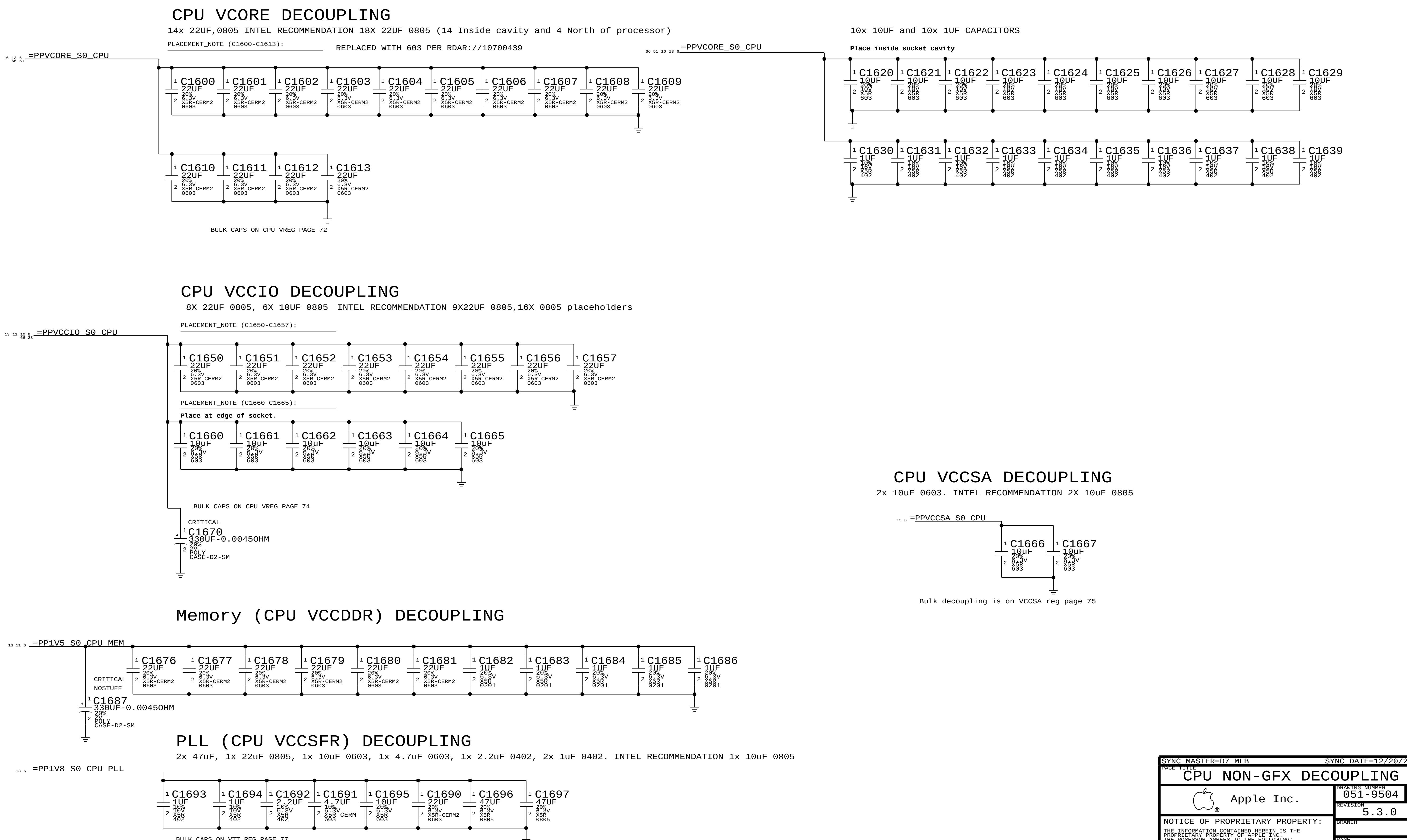

SYNC_MASTER=D7_MLB SYNC_DATE=12/20/2011

PAGE TITLE

CPU NON-GFX DECOUPLING

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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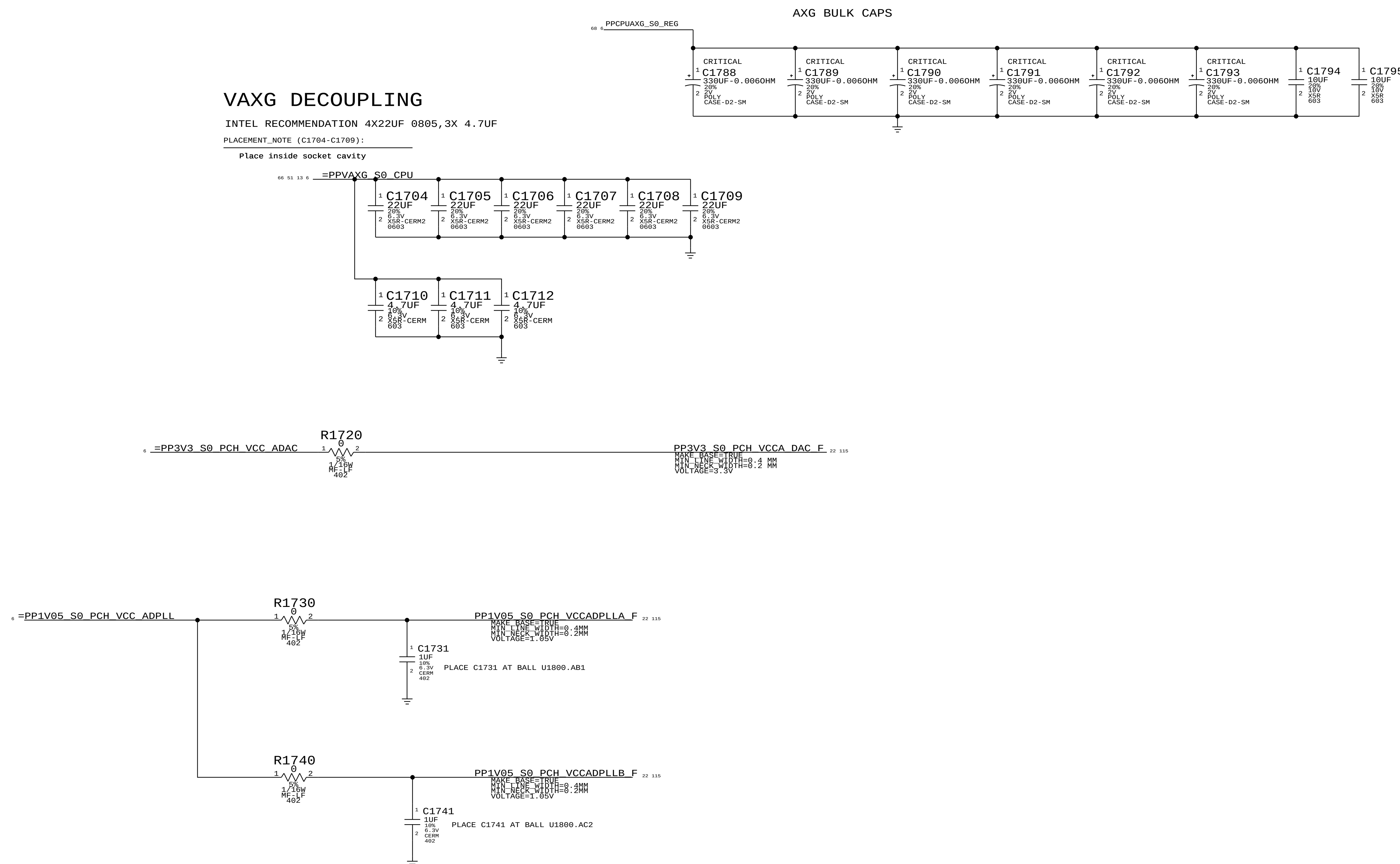

SYNC_MASTER=D8_KOSECOFF SYNC_DATE=01/26/2012

PAGE TITLE

GFX DECOUPLING & PCH PWR ALIAS

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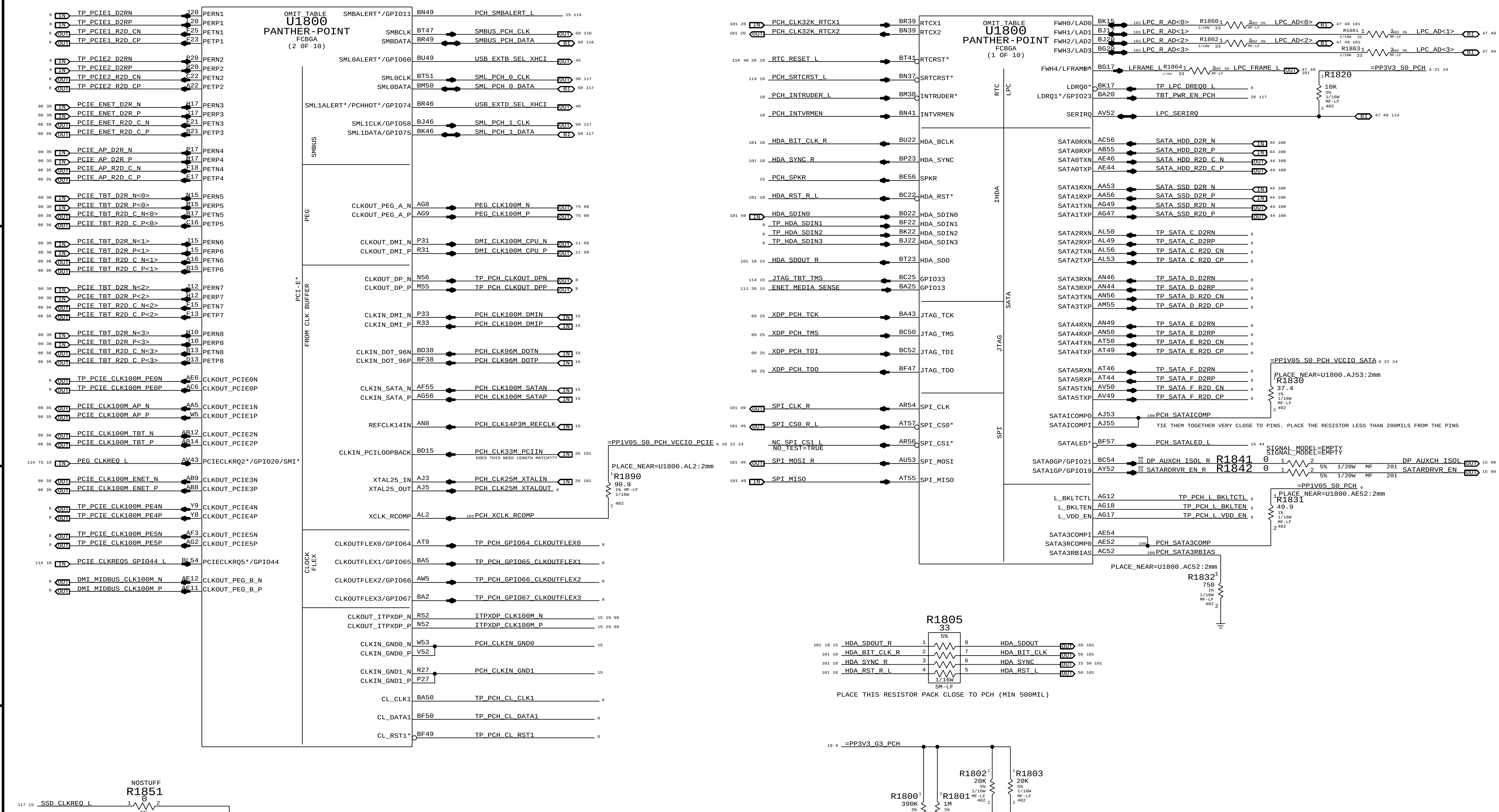

SYNC_MASTER=D8_MLB SYNC_DATE=N/A

PAGE TITLE

PCH SATA/PCIE/CLK/LPC/SPI

Apple Inc.

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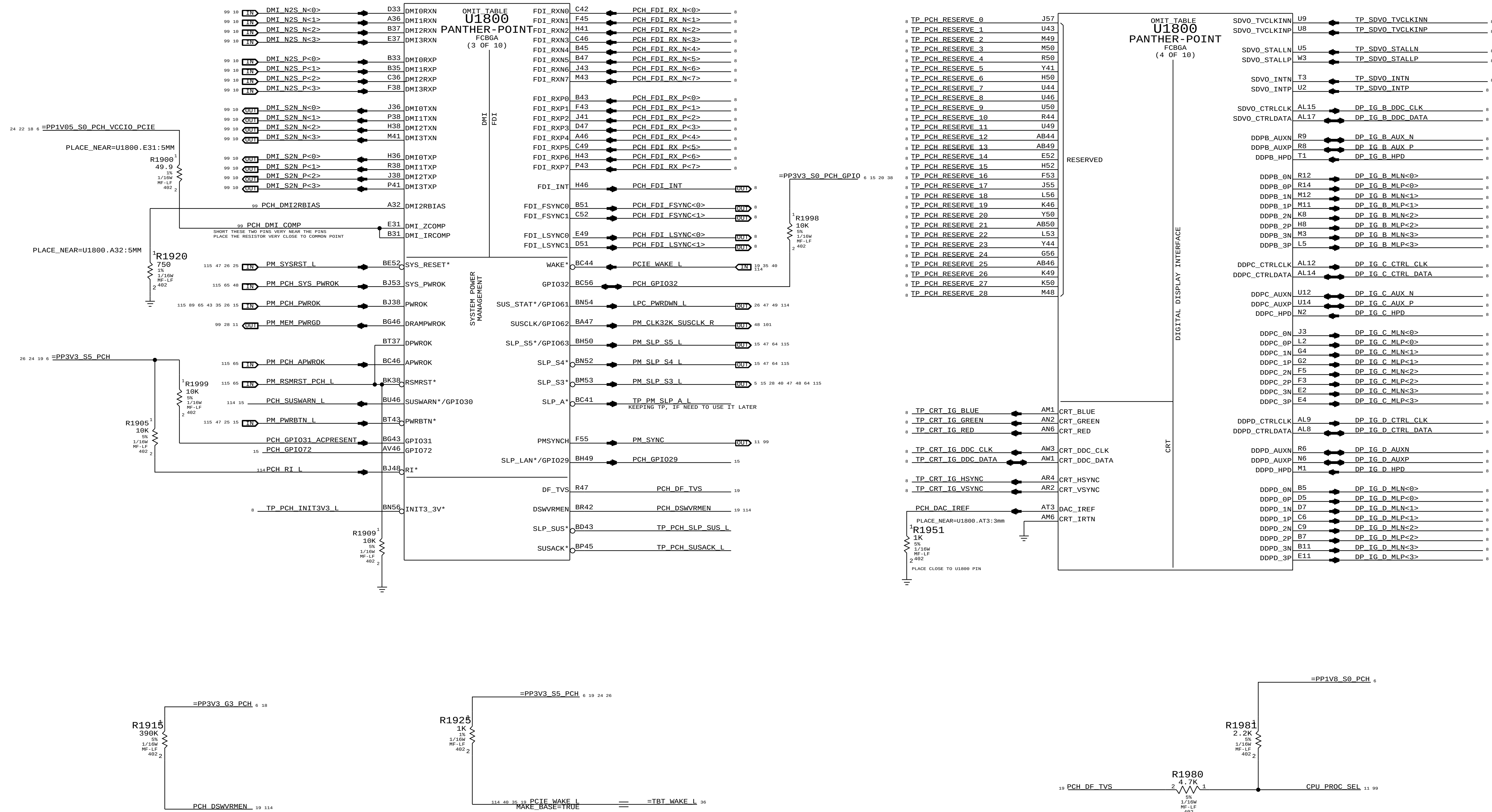

SYNC_MASTER=D7_MLB SYNC_DATE=03/15/2012

PAGE TITLE

PCH DMI/FDI/GRAPHICS

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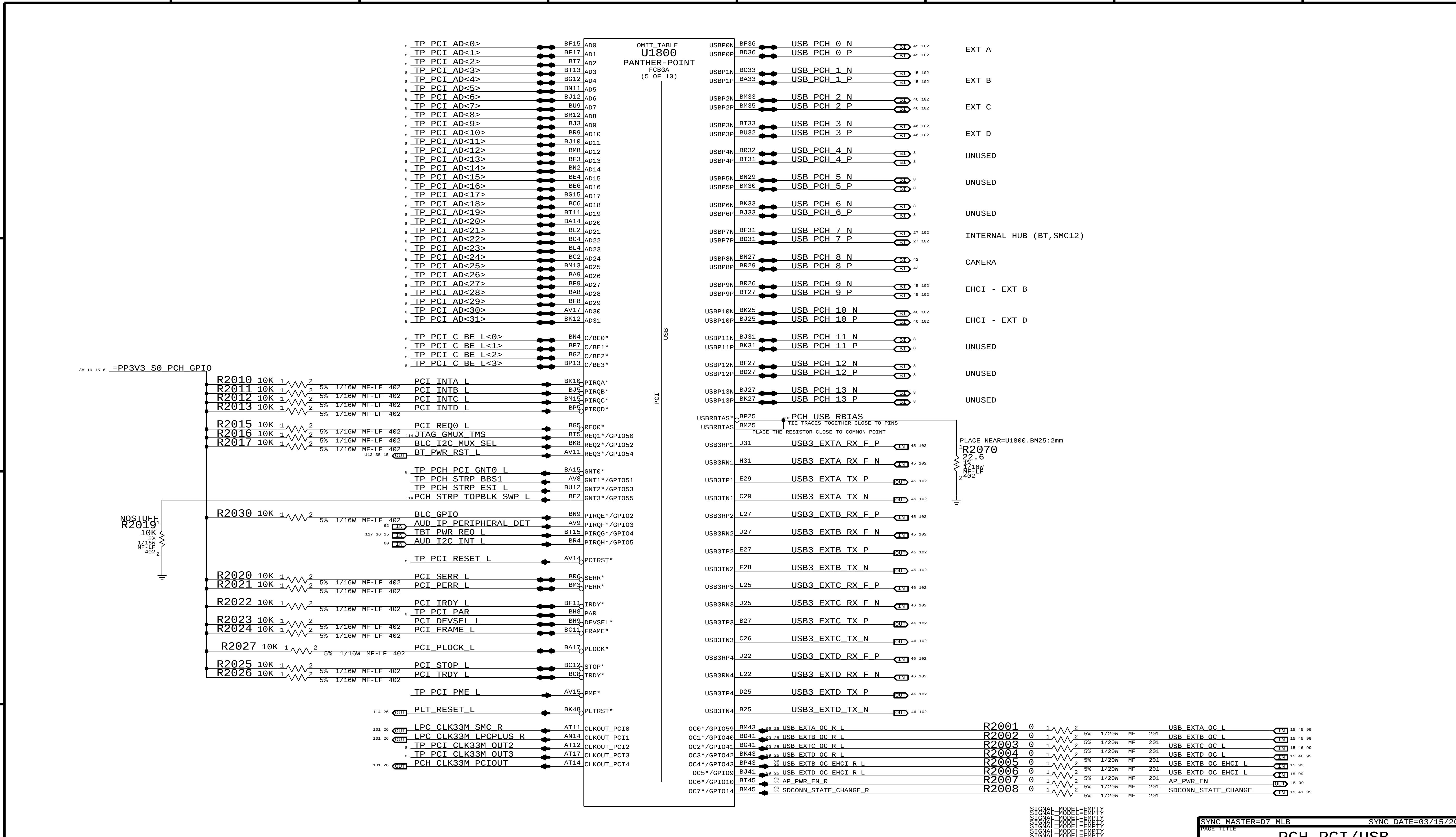

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PAGE TITLE

PCH PCI/USB

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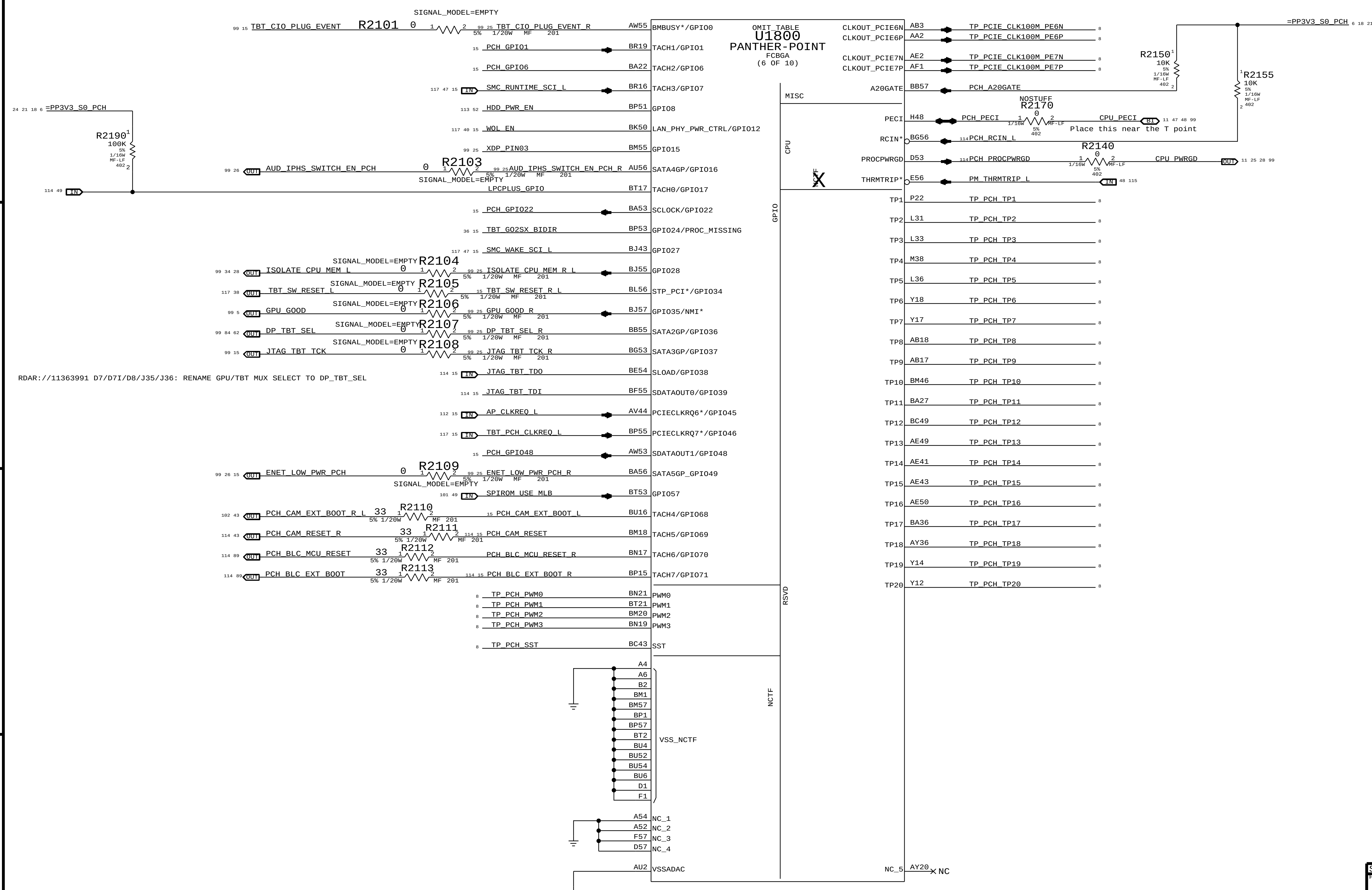

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PAGE TITLE

PCH MISC

Apple Inc.

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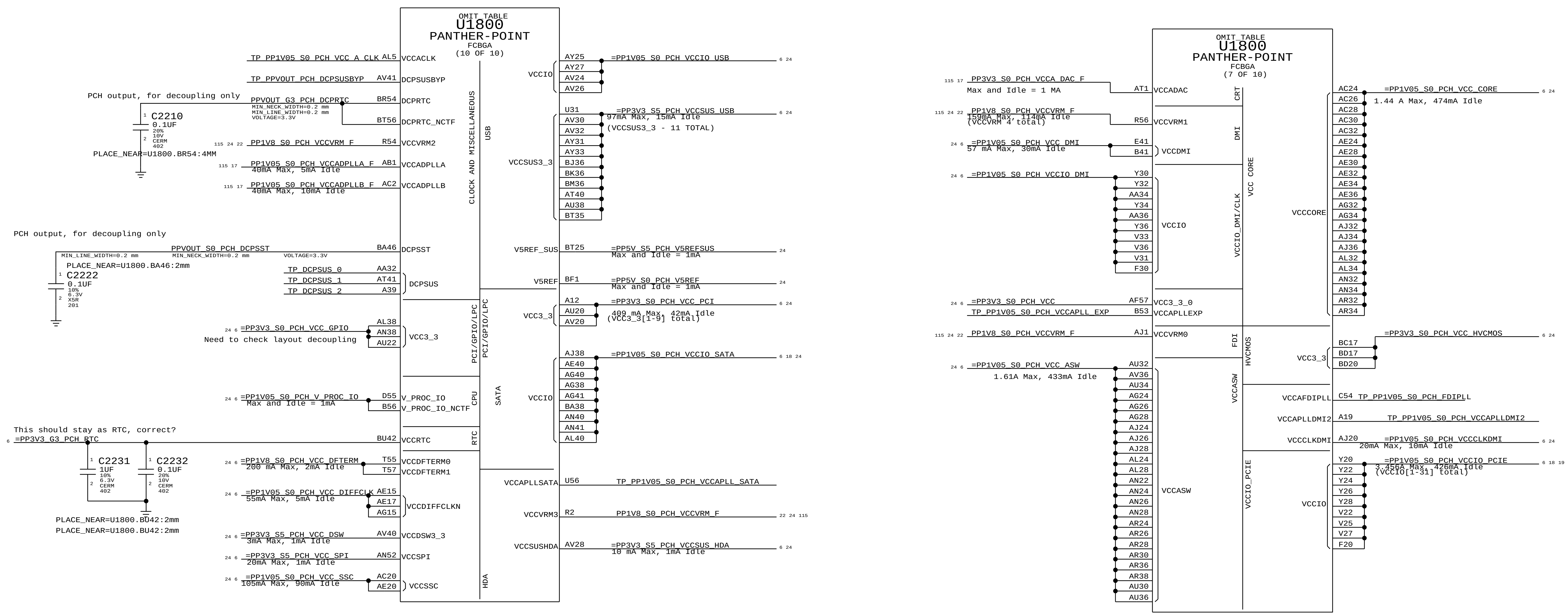

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PAGE TITLE

PCH POWER

Apple Inc.

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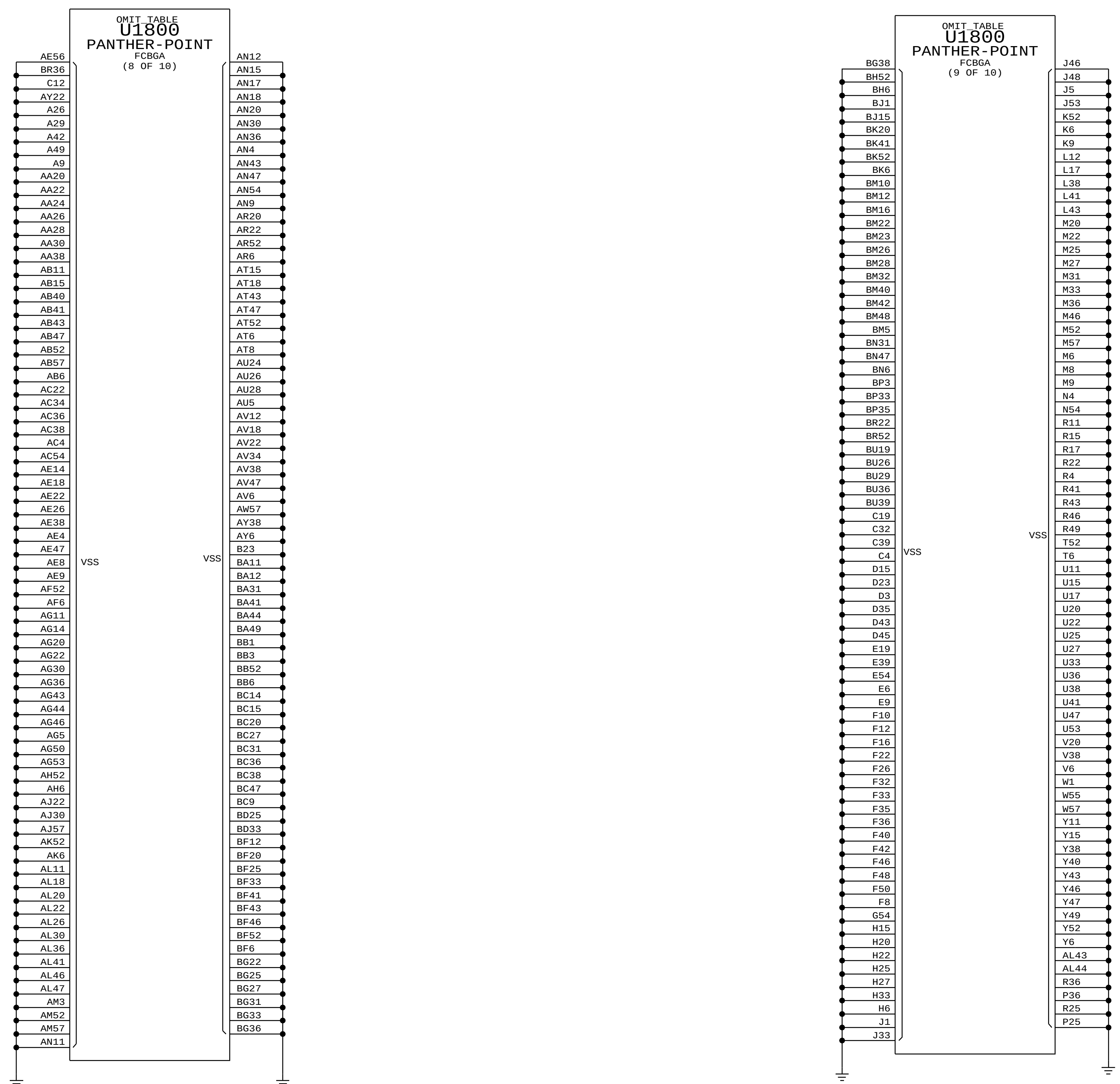

SYNC_MASTER=D7_MLB SYNC_DATE=03/15/2012

PAGE TITLE

PCH GROUNDS

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SYNC_MASTER=D8_MLB SYNC_DATE=N/A

PAGE TITLE
PCH DECOUPLING

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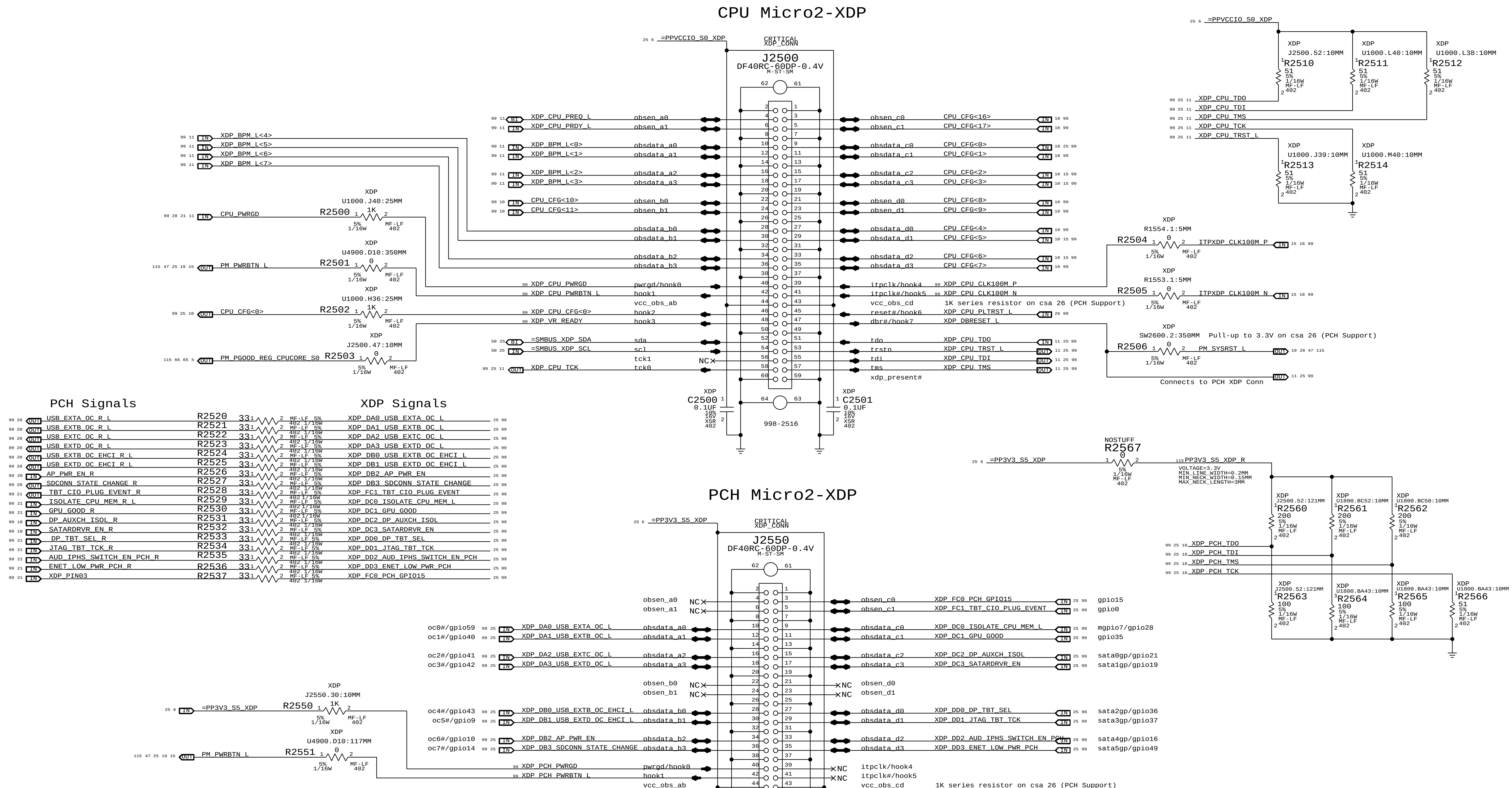

PCH/XDP Signal Isolation Notes:

- USB OC#'s not isolated, avoid USB port overcurrent events while using PCH XDP.
- Unused GPIOs 0 & 15 not isolated.
- MXM_GOOD not isolated as only LED is affected.
- For isolated GPIOs:
  - 'Output' non-XDP signals require pulls.
  - 'Output' PCH/XDP signals require pulls.

If PCH XDP not implemented, all of R2524-R2537 can be replaced with aliases. Otherwise these R's must be stuffed even in production so that PCH pins connect to appropriate non-XDP signals on PCB.

R2524-R2537 should be placed where signal path needs to split between route from PCH to J2550 and path to non-XDP signal destination.

SYNC_MASTER=D7_MLB SYNC_DATE=01/26/2012

PAGE TITLE

CPU and PCH XDP

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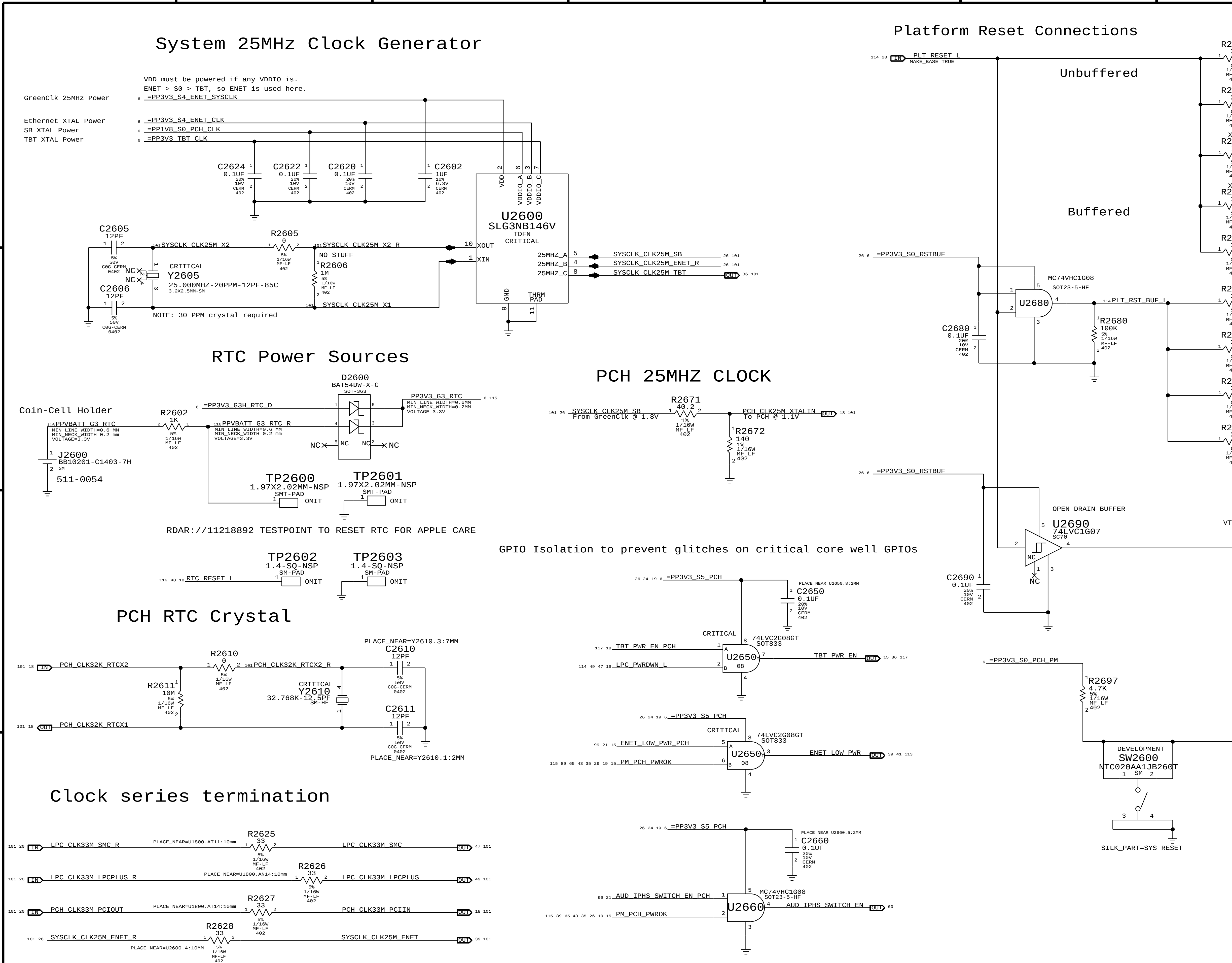

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PAGE TITLE

CHIPSET SUPPORT

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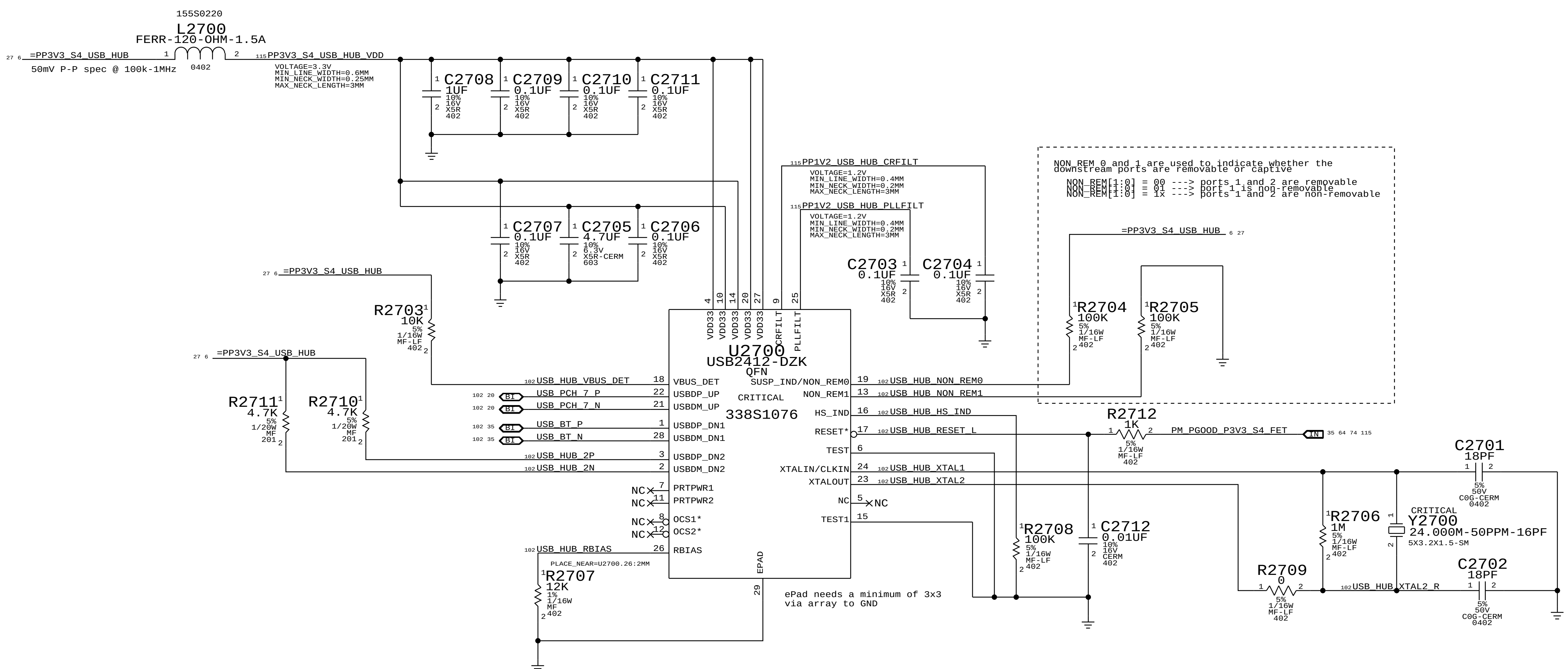

SYNC_MASTER=D8_ROSITA SYNC_DATE=03/23/2012

PAGE TITLE

USB 2.0 HUB (BT/SMC)

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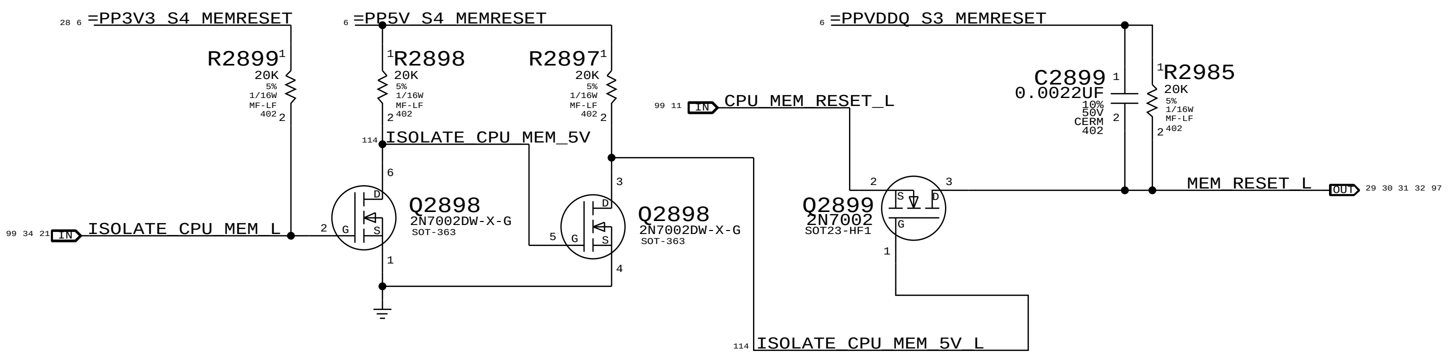
MEM_RESET_L Generator

The circuits below handle MEMVTT power during S0->S3->S0 transitions, as well
as isolating the CPU’s SM_DRAMRST# output from the SO-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behaviour of signals.
WHEN HIGH: MEM_RESET_L NOT ISOLATED.
WHEN LOW: MEM_RESET_L IS ISOLATED.
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L (Block CPU from driving MEM_RESET_L in S3)

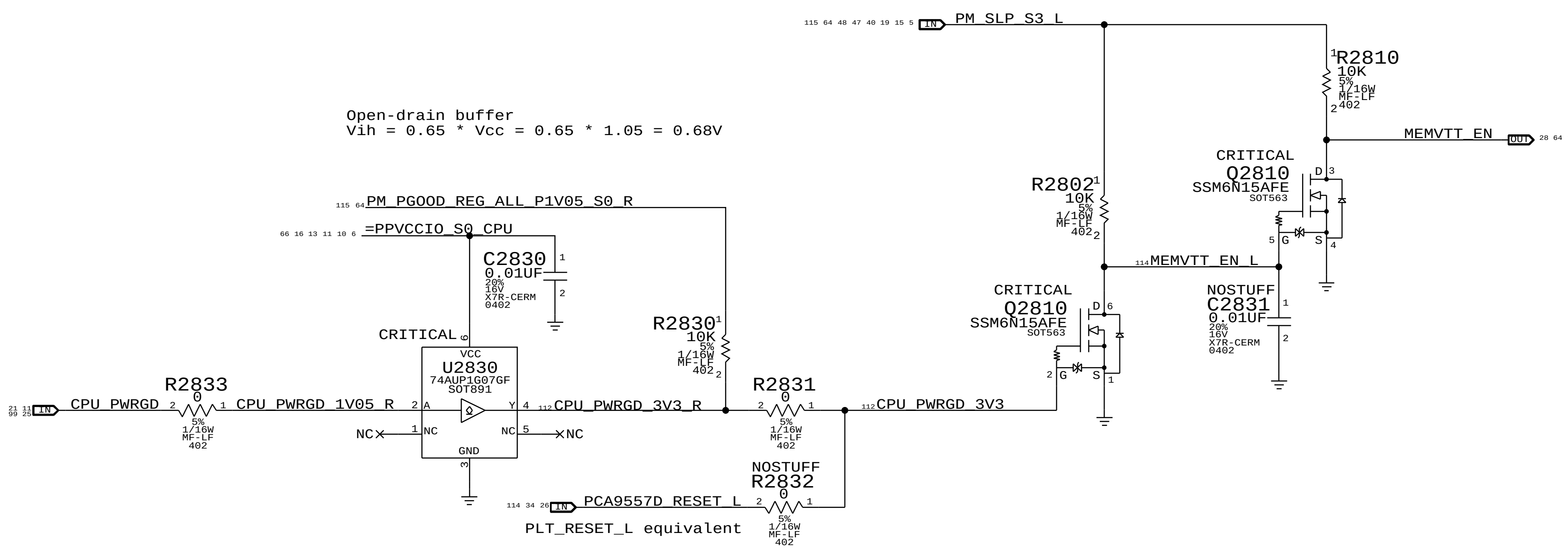


MEMVTT_EN Generator

rdar://11117167

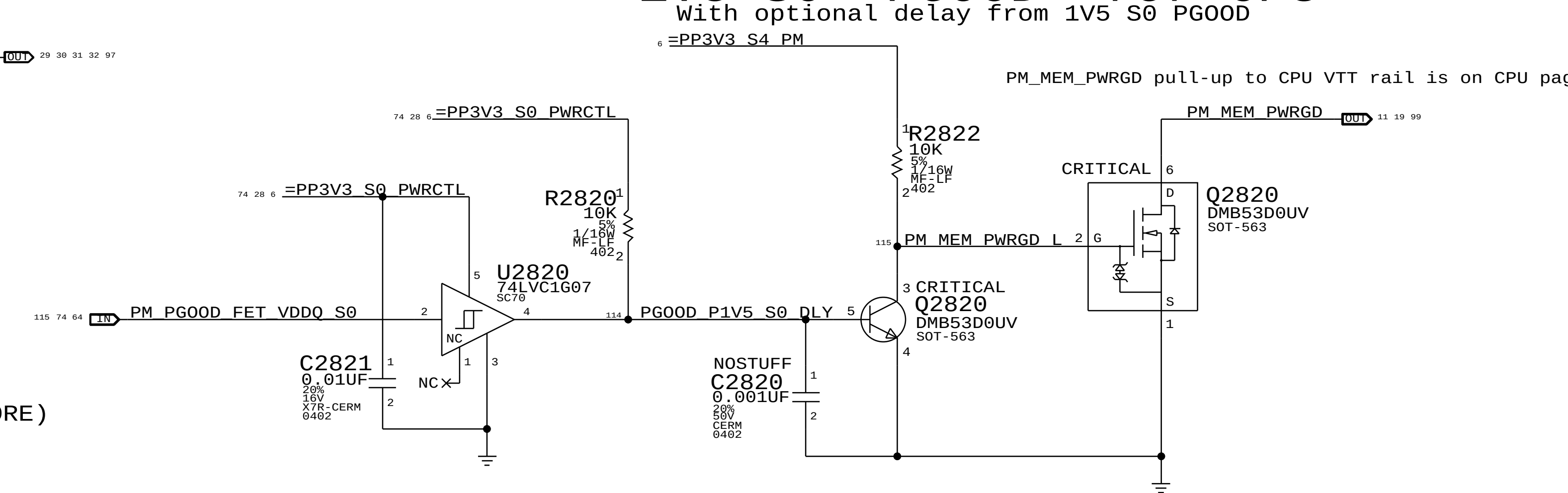
Enables MEMVTT when PCH drives CPU PWRGD.

CPU does not drive MEM_CKE until VCCORE activated but CPU 1V5 (VDDQ) leaks into it.
Clamping MEMVTT will keep the MEM_CKE low until CPU actively controls it.
MEMVTT Clamp actively holds MEMVTT rail low until MEMVTT is enabled.
MEMVTT_EN = CPU_PWRGD * PM_SLP_S3_L (VTT is enabled when PCH tells CPU to enable VCCORE)



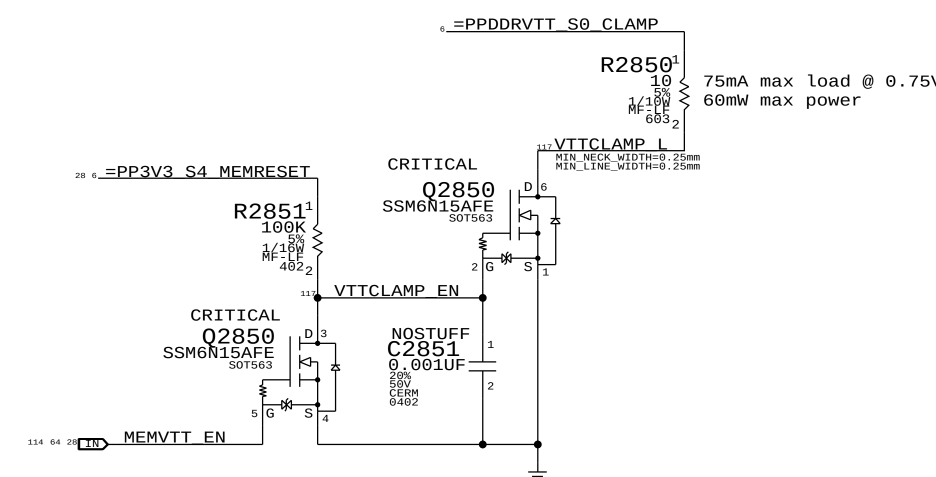
1V5 S0 "PGOOD" for CPU

With optional delay from 1V5 S0 PGOOD



MEMVTT Clamp

Ensures CKE signals are held low in S3 and in S0 before CPU PWRGD



	Step	ISOLATE_CPU_MEM_L	PM_SLP_S3_L	CPU_PWRGD	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	CPU_MEM_RESET_L	1	1
to	1	0	1	1	1	1	1
	2	0	1	1	1	1	1
S3	3	0	0	0	X	1	0
	4	0	0	0	X	1	0
to	5	0	1	1	0 (*)	1	1
	6	0	1	1	1	1	1
S0	7	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

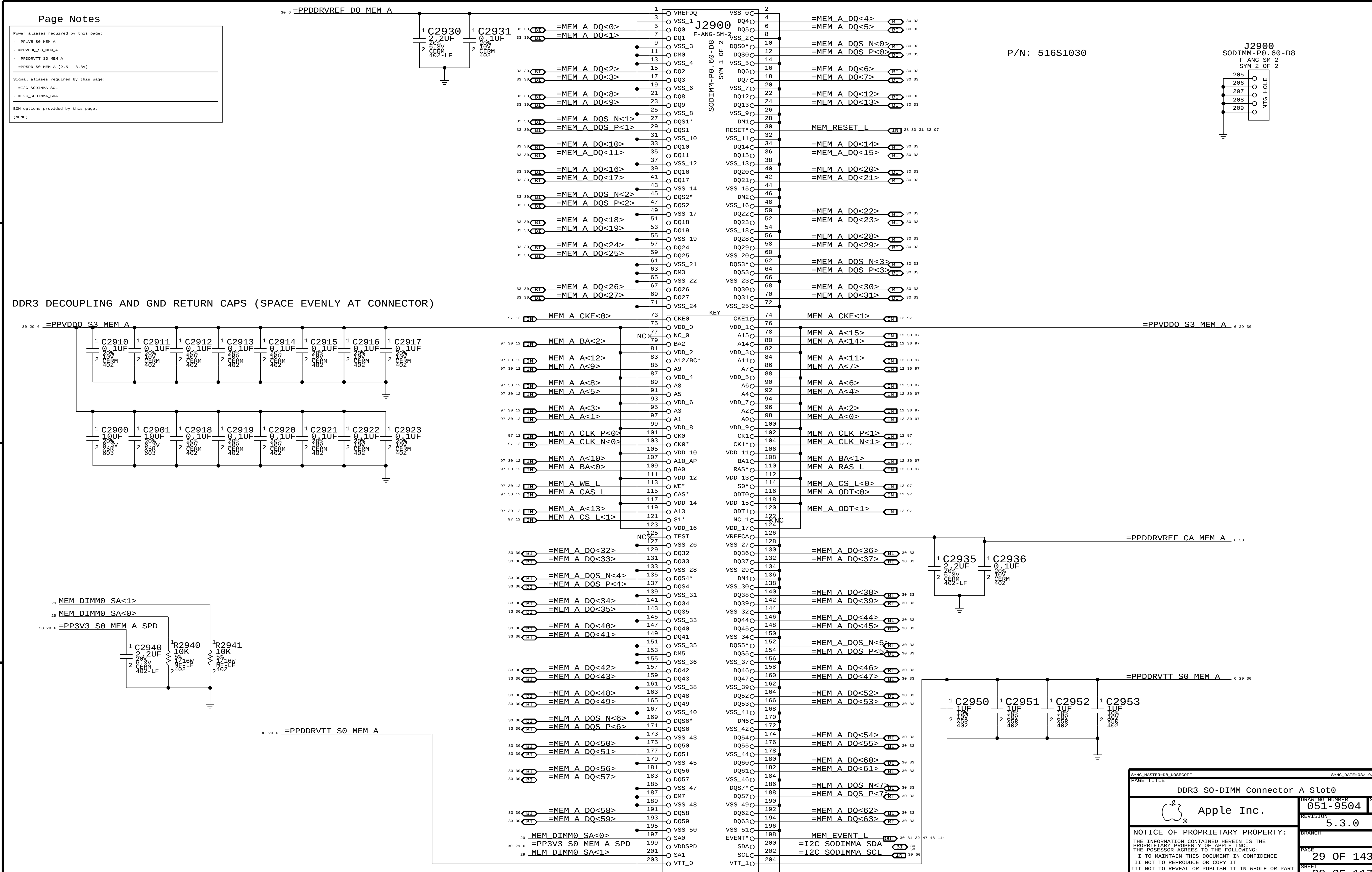
NOTE: On a S5->S0 transition, ISOLATE_CPU_MEM_L will default low.
Rails will power-up as if from S3, but MEM_RESET_L now needs to be asserted in S0. Software
must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)



SYNC_MASTER=D8_KOSECOFF SYNC_DATE=03/19/2012

PAGE TITLE

DDR3 SO-DIMM Connector A Slot0

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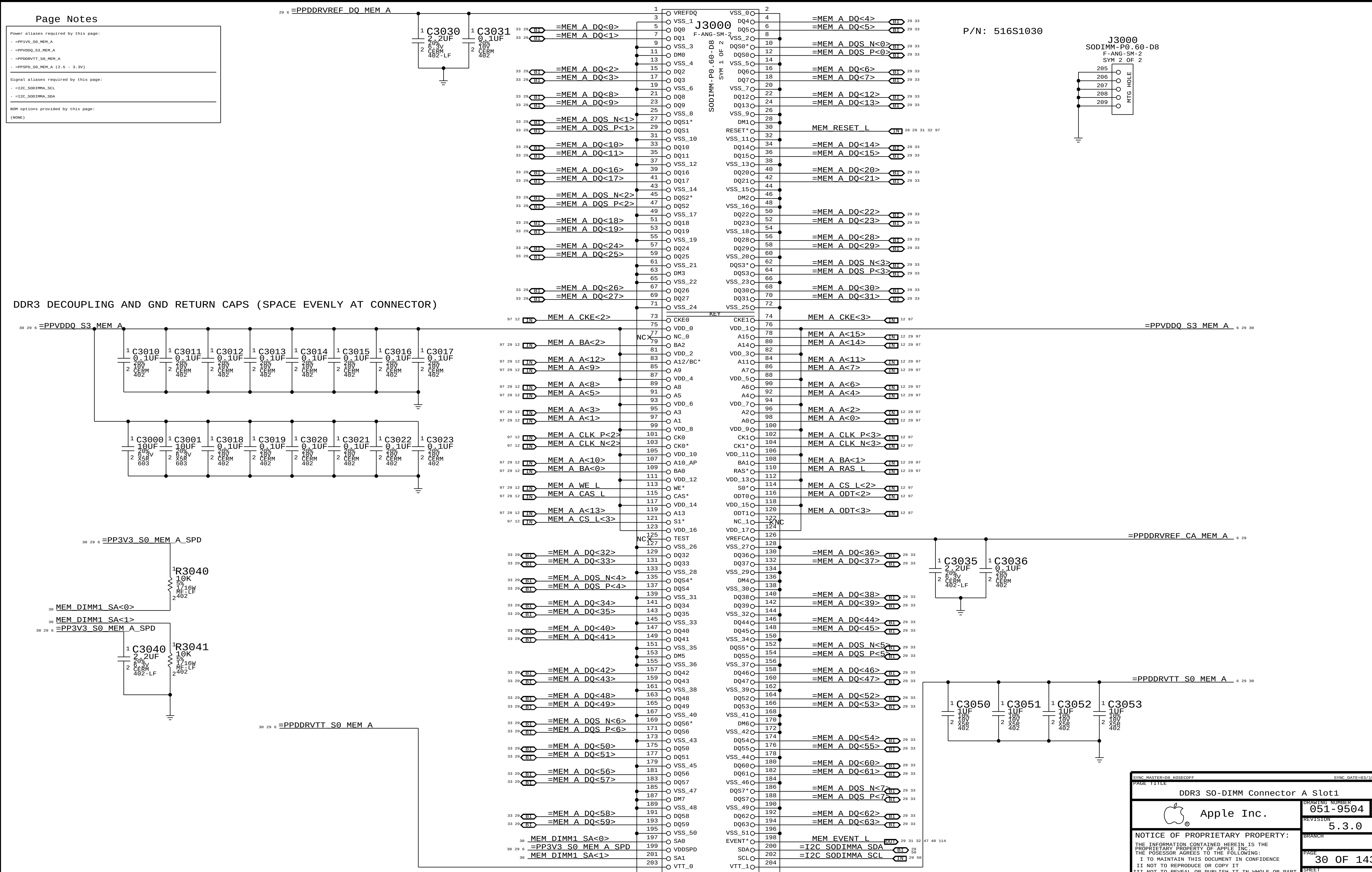
Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)



SYNC_MASTER=D8_KOSECOFF SYNC_DATE=03/19/2012

PAGE TITLE: DDR3 SO-DIMM Connector A Slot1

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Page Notes

Power aliases required by this page:

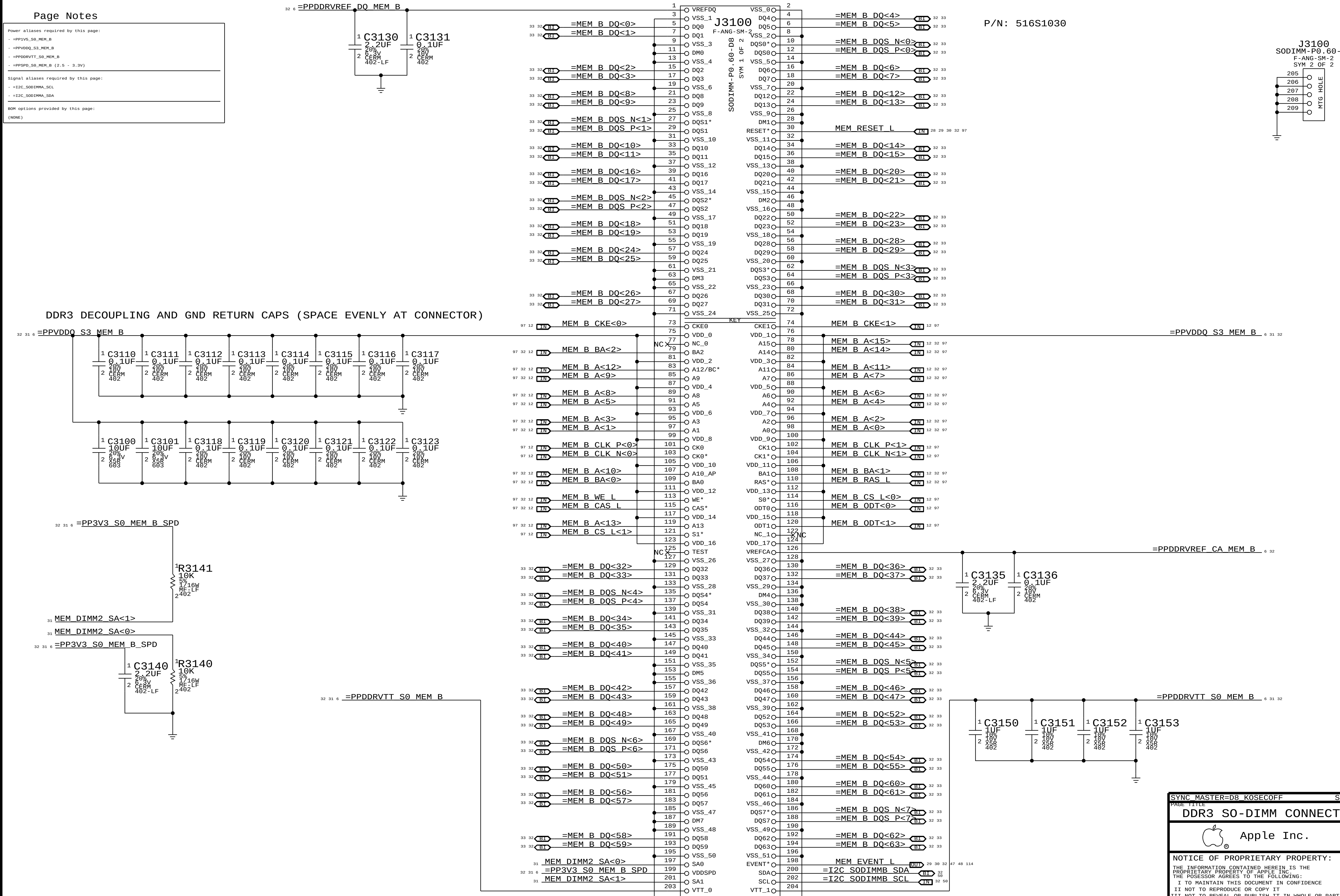
- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)



SYNC_MASTER=D8_KOSECOFF SYNC_DATE=03/19/2012

PAGE TITLE

DDR3 SO-DIMM CONNECTOR B SLOT0

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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Page Notes

Power aliases required by this page:

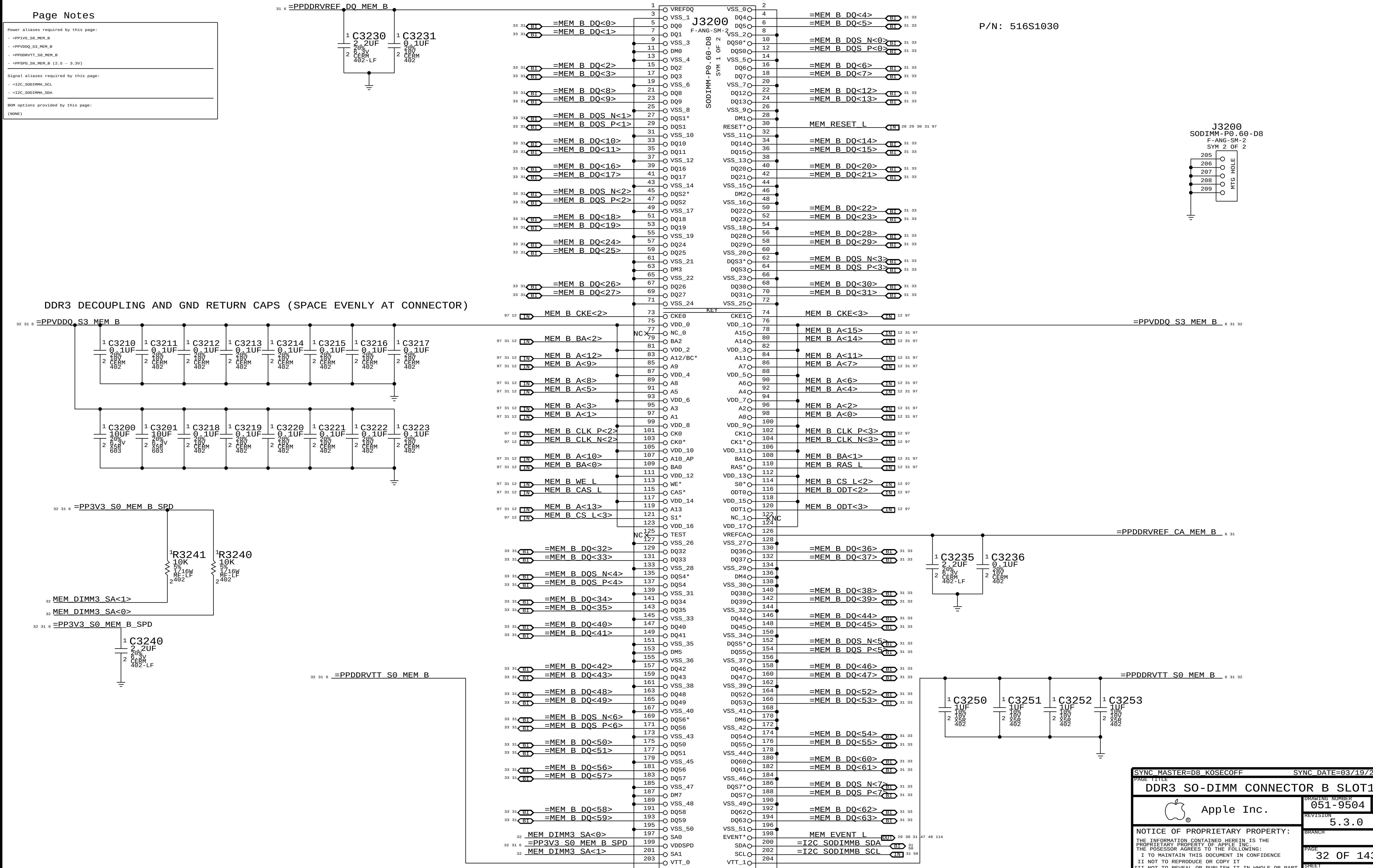
- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)



SYNC_MASTER=D8_KOSECOFF	SYNC_DATE=03/19/2012
PAGE TITLE	DDR3 SO-DIMM CONNECTOR B SLOT1
Apple Inc.	DRAWING NUMBER 051-9504 SIZE D
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NOTICE OF PROPRIETARY PROPERTY:

97 12	MEM_A_DQS_N<0>	MAKE_BASE=TRUE	=MEM_A_DQS_N<0>	29 30	97 12	MEM_B_DQS_N<0>	MAKE_BASE=TRUE	=MEM_B_DQS_N<0>	31 32
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SYNC_MASTER=D8_KOSECOFF SYNC_DATE=03/19/2012

PAGE TITLE

DDR3 ALIASES AND BITSWAPS

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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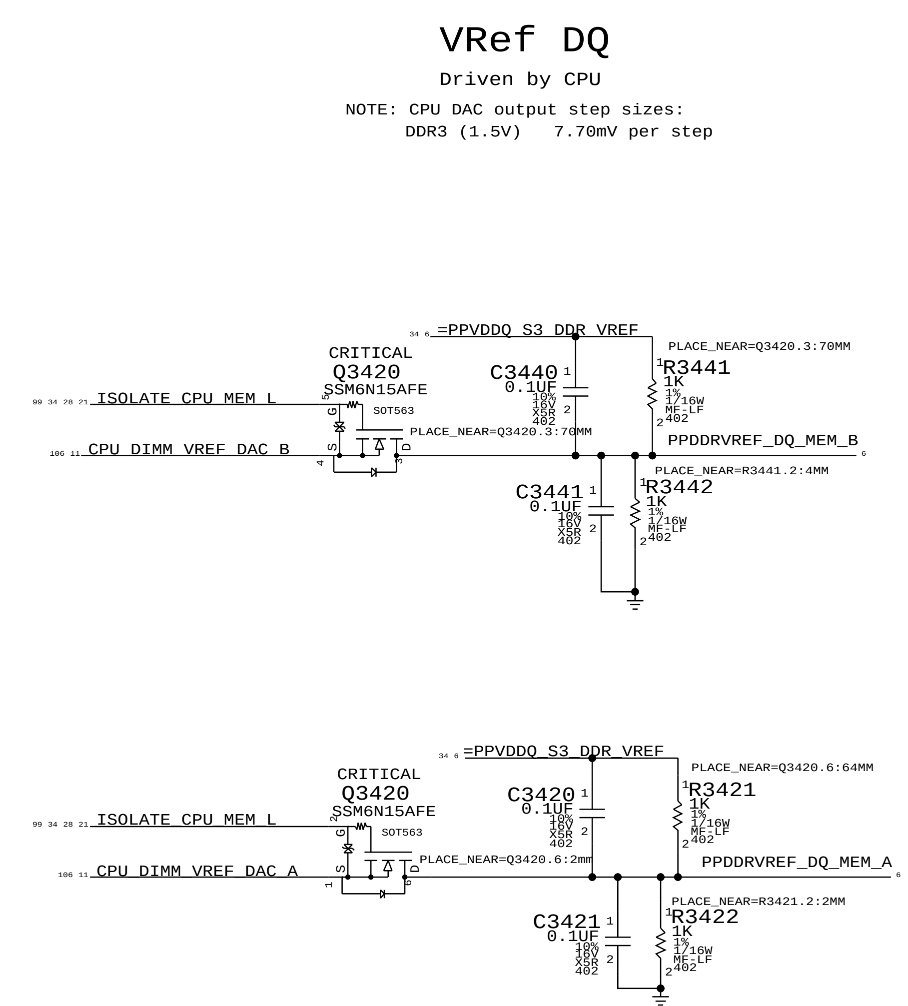

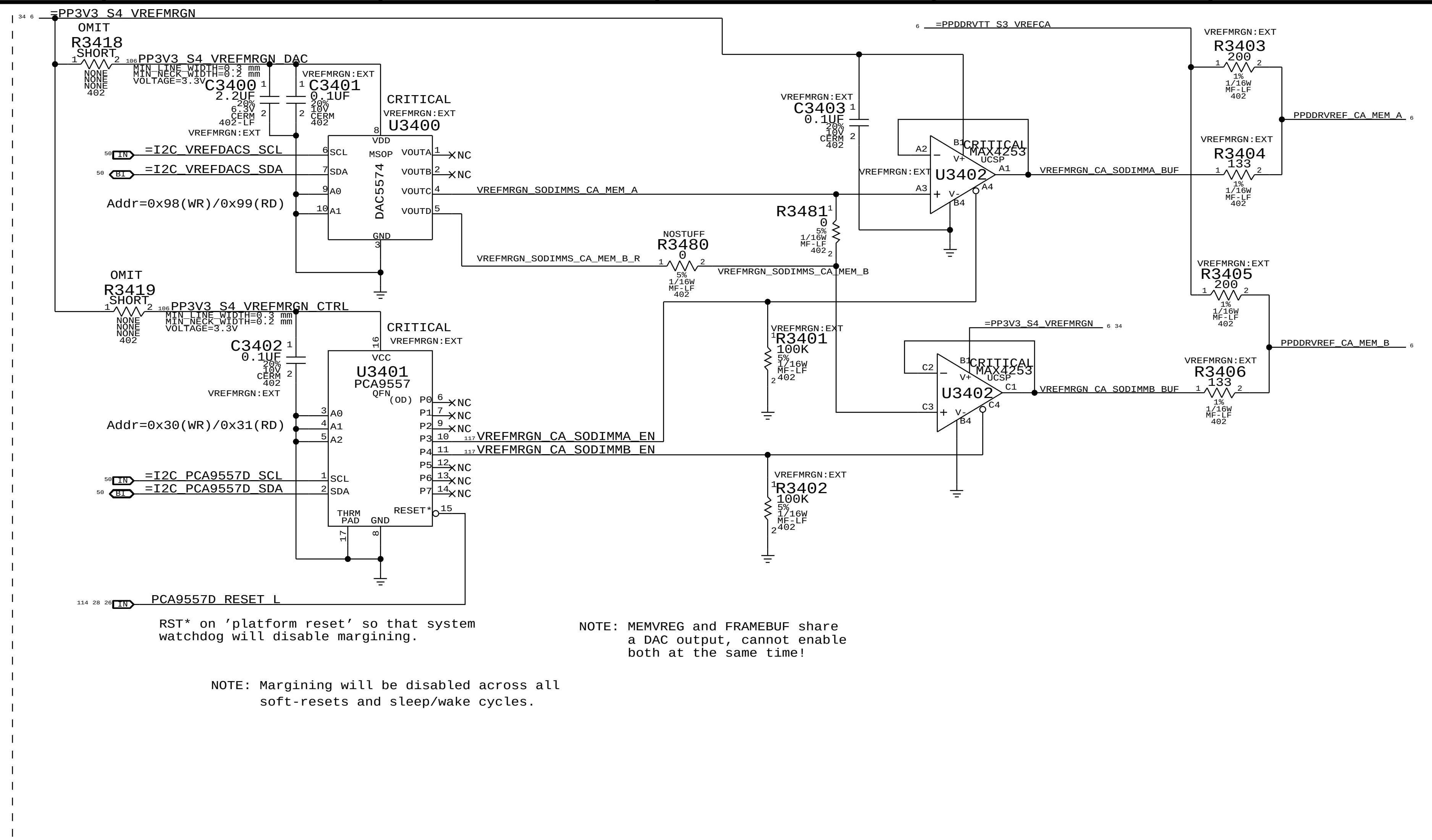

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S0004	2	RES,MTL FLM,0,5%,402,SM,LF	R3403,R3405		VREFMRGN:N

	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	C	C	D	D
PCA9557D Pin:	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)		1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)		1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)		0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)		+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output		8.59mV / step @ output	1.51mV / step @ output

SYNC_MASTER=D8_KOSECOFF SYNC_DATE=03/19/2012

PAGE TITLE
DDR3/FRAMEBUF VREF MARGINING

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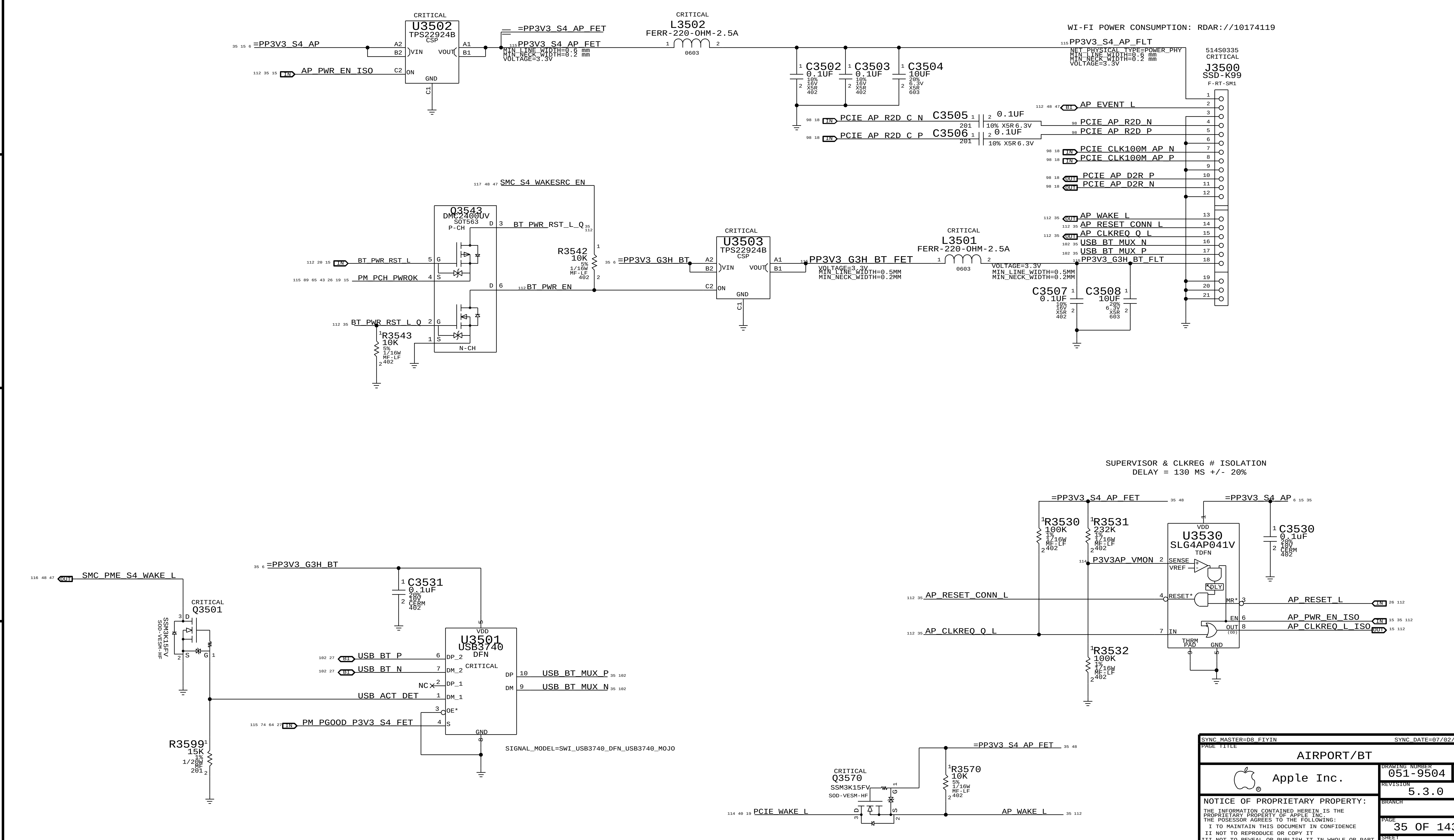
AIRPORT BLUETOOTH



SYNC_MASTER=D8_FIYIN SYNC_DATE=07/02/2012

PAGE TITLE: AIRPORT/BT

Apple Inc.

DRAWING NUMBER: 051-9504 SIZE: D

REVISION: 5.3.0

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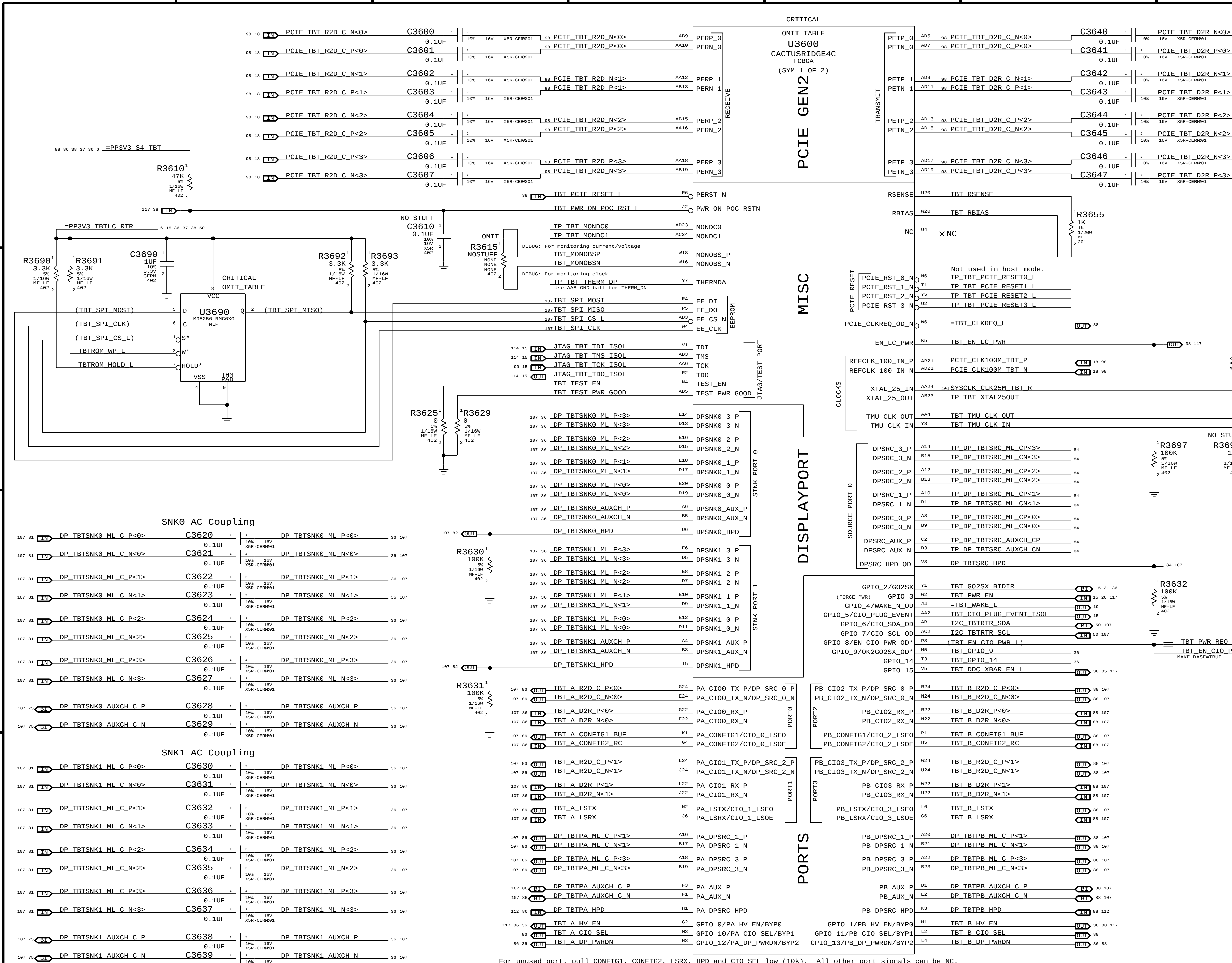

SYNC_MASTER=D8_MLB SYNC_DATE=N/A

PAGE TITLE

Thunderbolt Host (1 of 2)

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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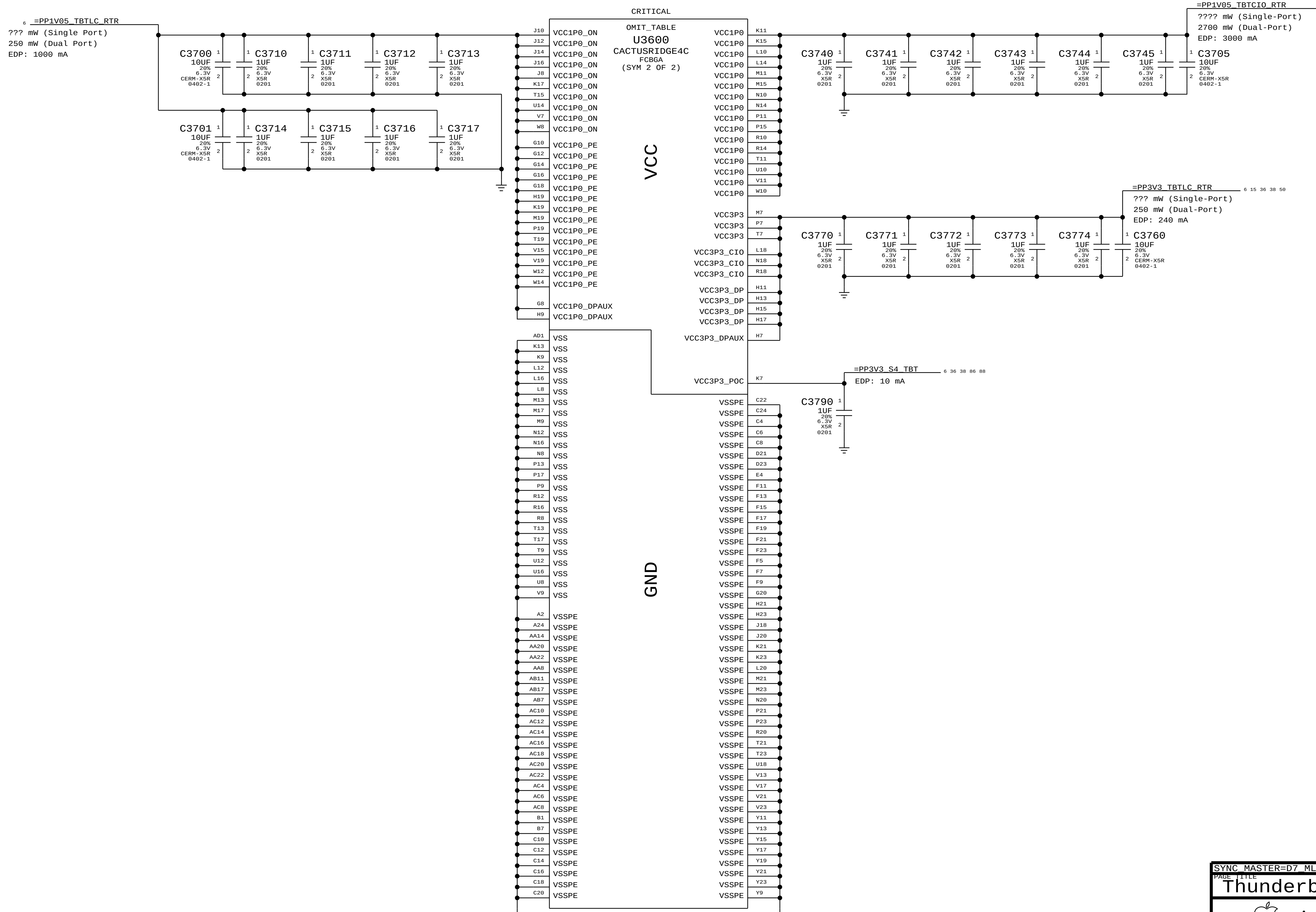

EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

SYNC_MASTER=D7_MLB SYNC_DATE=03/15/2012

PAGE TITLE: Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER: 051-9504 SIZE: D

REVISION: 5.3.0

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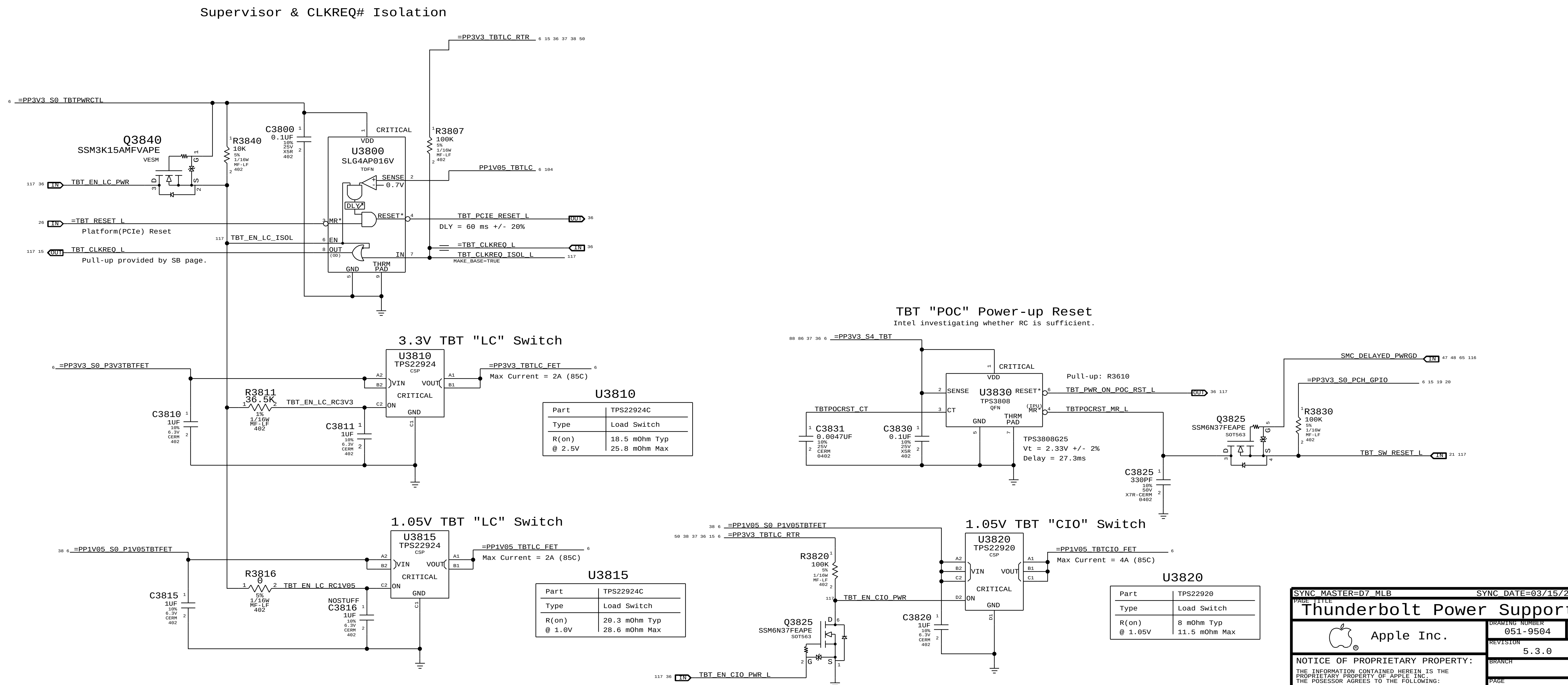
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTPWRCTL
- =PP1V05_TBT_P1V05TBTFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

Signal aliases required by this page:
- =TBT_CLKREQ_L
- =TBT_RESET_L

BOM options provided by this page:
TBTBST:Y - Stuffs 15V boost circuitry.



U3810

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

U3815

Part	TPS22924C
Type	Load Switch
R(on) @ 1.0V	20.3 mOhm Typ 28.6 mOhm Max

U3820

Part	TPS22920
Type	Load Switch
R(on) @ 1.05V	8 mOhm Typ 11.5 mOhm Max

SYNC_MASTER=D7_MLB SYNC_DATE=03/15/2012

PAGE TITLE: Thunderbolt Power Support

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DRAWING NUMBER 051-9504 SIZE D

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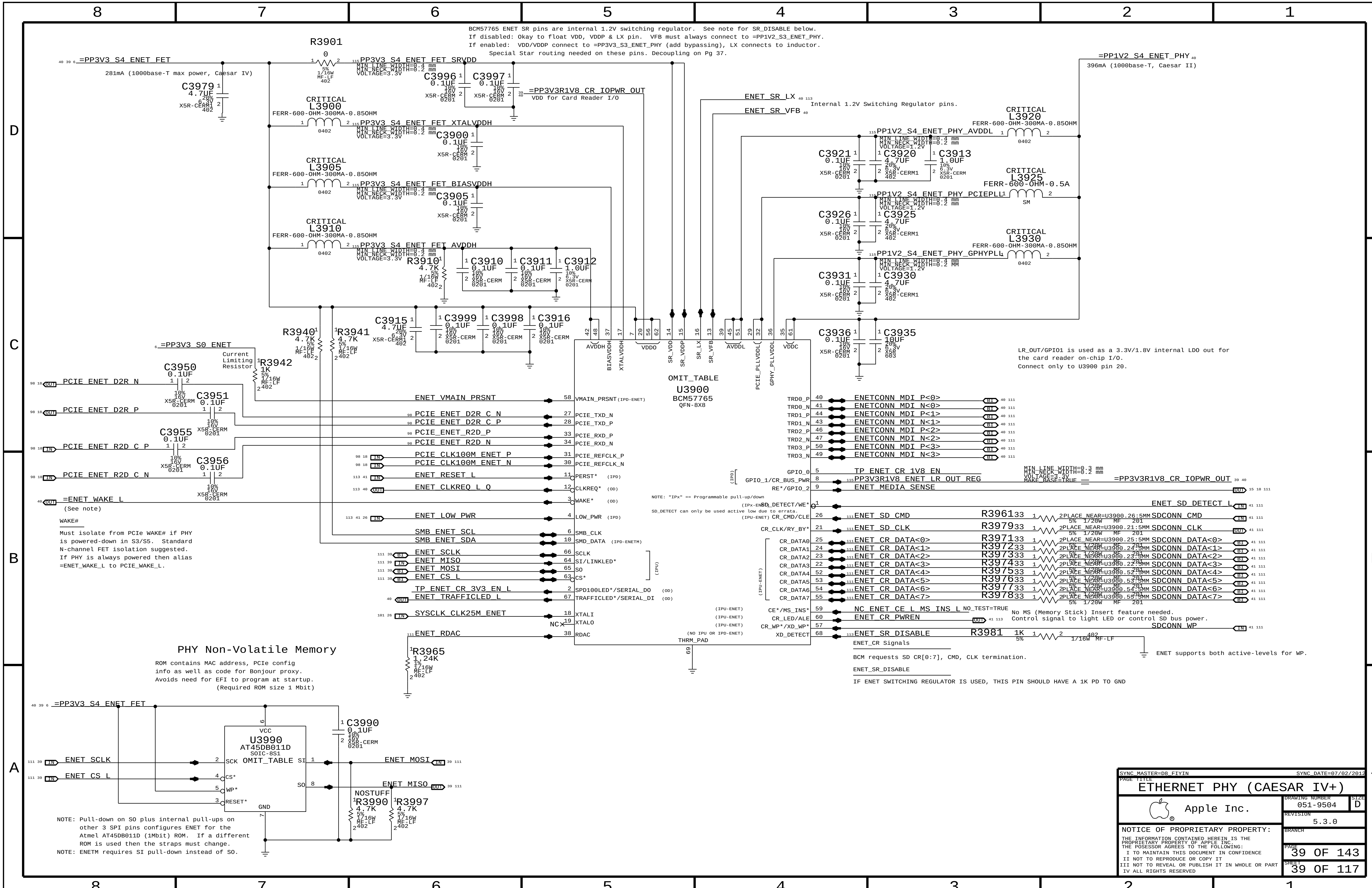

BCM57765 ENET SR pins are internal 1.2V switching regulator. See note for SR_DISABLE below.
If disabled: Okay to float VDD, VDDP & LX pin. VFB must always connect to =PP1V2_S3_ENET_PHY.
If enabled: VDD/VDDP connect to =PP3V3_S3_ENET_PHY (add bypassing), LX connects to inductor.
Special Star routing needed on these pins. Decoupling on Pg 37.
=PP3V3_S4_ENET_FET
281mA (1000base-T max power, Caesar IV)
R3901
PP3V3_S4_ENET_FET_SRVDD
=PP3V3R1V8_CR_IOPWR_OUT
VDD for Card Reader I/O
=PP1V2_S4_ENET_PHY
396mA (1000base-T, Caesar II)
ENET_SR_LX
ENET_SR_VFB
Internal 1.2V Switching Regulator pins.
CRITICAL L3900 FERR-600-OHM-300MA-0.85OHM
PP3V3_S4_ENET_FET_XTALVDDH
CRITICAL L3905 FERR-600-OHM-300MA-0.85OHM
PP3V3_S4_ENET_FET_BIASVDDH
CRITICAL L3910 FERR-600-OHM-300MA-0.85OHM
PP3V3_S4_ENET_FET_AVDDH
CRITICAL L3920 FERR-600-OHM-300MA-0.85OHM
PP1V2_S4_ENET_PHY_AVDDL
CRITICAL L3925 FERR-600-OHM-0.5A
PP1V2_S4_ENET_PHY_PCIEPLL
CRITICAL L3930 FERR-600-OHM-300MA-0.85OHM
PP1V2_S4_ENET_PHY_GPHYPLL
LR_OUT/GPIO1 is used as a 3.3V/1.8V internal LDO out for
the card reader on-chip I/O.
Connect only to U3900 pin 20.
=PP3V3_S0_ENET
Current Limiting Resistor
OMIT_TABLE
U3900
BCM57765
QFN-8X8
PCIE_ENET_D2R_N
PCIE_ENET_D2R_P
PCIE_ENET_R2D_C_P
PCIE_ENET_R2D_C_N
=ENET_WAKE_L
(See note)
WAKE#
Must isolate from PCIe WAKE# if PHY
is powered-down in S3/S5. Standard
N-channel FET isolation suggested.
If PHY is always powered then alias
=ENET_WAKE_L to PCIE_WAKE_L.
ENET_VMAIN_PRSNT
PCIE_ENET_D2R_C_N
PCIE_ENET_D2R_C_P
PCIE_ENET_R2D_P
PCIE_ENET_R2D_N
PCIE_CLK100M_ENET_P
PCIE_CLK100M_ENET_N
ENET_RESET_L
ENET_CLKREQ_L_Q
ENET_LOW_PWR
SMB_ENET_SCL
SMB_ENET_SDA
ENET_SCLK
ENET_MISO
ENET_MOSI
ENET_CS_L
TP_ENET_CR_3V3_EN_L
ENET_TRAFFICLED_L
SYSCLK_CLK25M_ENET
ENET_RDAC
ENETCONN_MDI_P<0>
ENETCONN_MDI_N<0>
ENETCONN_MDI_P<1>
ENETCONN_MDI_N<1>
ENETCONN_MDI_P<2>
ENETCONN_MDI_N<2>
ENETCONN_MDI_P<3>
ENETCONN_MDI_N<3>
TP_ENET_CR_1V8_EN
PP3V3R1V8_ENET_LR_OUT_REG
=PP3V3R1V8_CR_IOPWR_OUT
ENET_MEDIA_SENSE
NOTE: "IPx" == Programmable pull-up/down
SD_DETECT can only be used active low due to errata.
ENET_SD_DETECT_L
ENET_SD_CMD
SDCONN_CMD
ENET_SD_CLK
SDCONN_CLK
ENET_CR_DATA<0> … ENET_CR_DATA<7>
SDCONN_DATA<0> … SDCONN_DATA<7>
NC_ENET_CE_L_MS_INS_L
NO_TEST=TRUE
No MS (Memory Stick) Insert feature needed.
ENET_CR_PWREN
Control signal to light LED or control SD bus power.
SDCONN_WP
ENET_SR_DISABLE
ENET supports both active-levels for WP.
ENET_CR Signals
BCM requests SD CR[0:7], CMD, CLK termination.
ENET_SR_DISABLE
IF ENET SWITCHING REGULATOR IS USED, THIS PIN SHOULD HAVE A 1K PD TO GND
PHY Non-Volatile Memory
ROM contains MAC address, PCIe config
info as well as code for Bonjour proxy.
Avoids need for EFI to program at startup.
(Required ROM size 1 Mbit)
=PP3V3_S4_ENET_FET
U3990
AT45DB011D
SOIC-8S1
OMIT_TABLE
ENET_SCLK
ENET_CS_L
ENET_MOSI
ENET_MISO
NOSTUFF
R3990
R3997
NOTE: Pull-down on SO plus internal pull-ups on
other 3 SPI pins configures ENET for the
Atmel AT45DB011D (1Mbit) ROM. If a different
ROM is used then the straps must change.
NOTE: ENETM requires SI pull-down instead of SO.
SYNC_MASTER=D8_FIYIN
SYNC_DATE=07/02/2012
PAGE TITLE
ETHERNET PHY (CAESAR IV+)
Apple Inc.
DRAWING NUMBER 051-9504
SIZE D
REVISION 5.3.0
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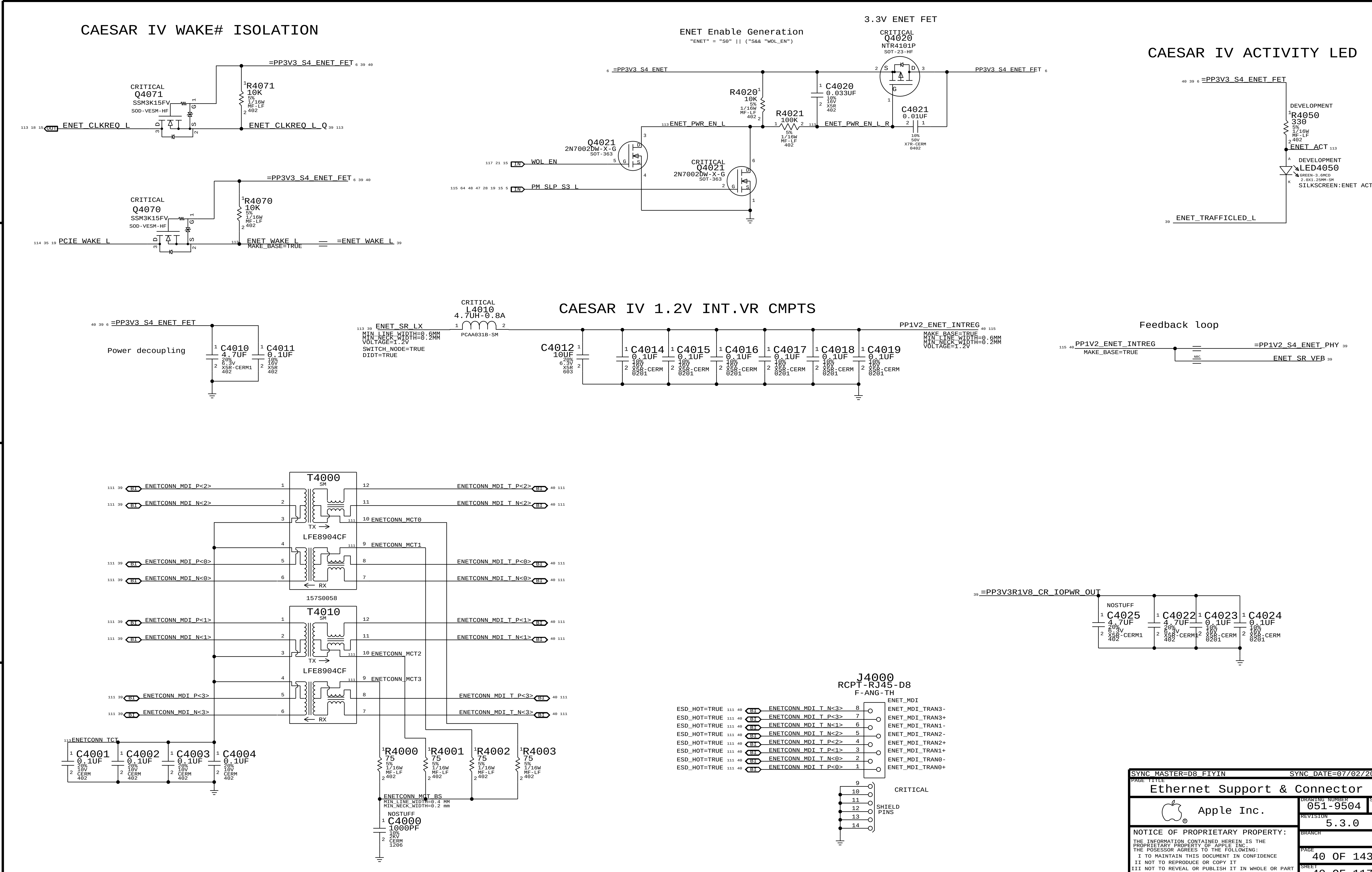
PAGE 39 OF 143
SHEET 39 OF 117



SYNC_MASTER=D8_FIYIN SYNC_DATE=07/02/2012

PAGE TITLE: Ethernet Support & Connector

Apple Inc.

DRAWING NUMBER: 051-9504 SIZE: D

REVISION: 5.3.0

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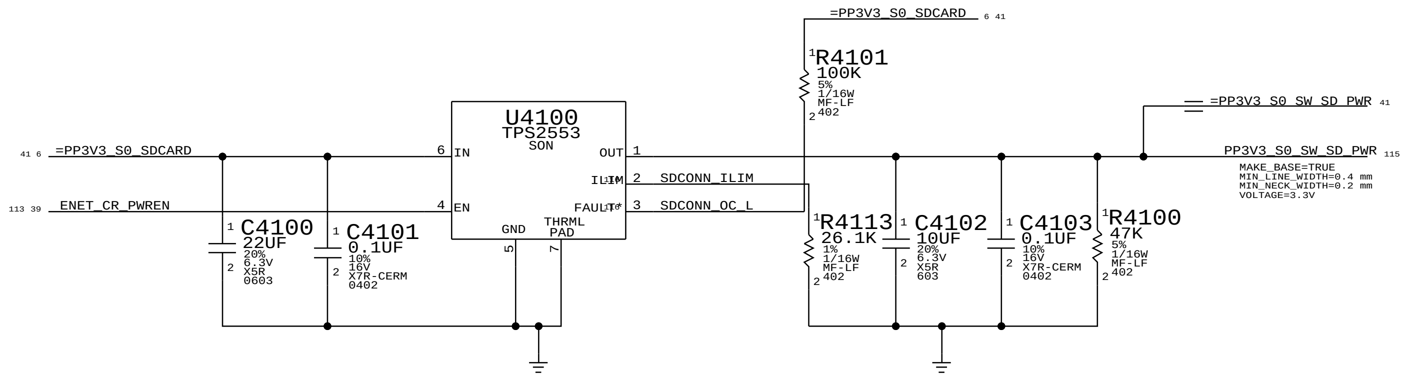
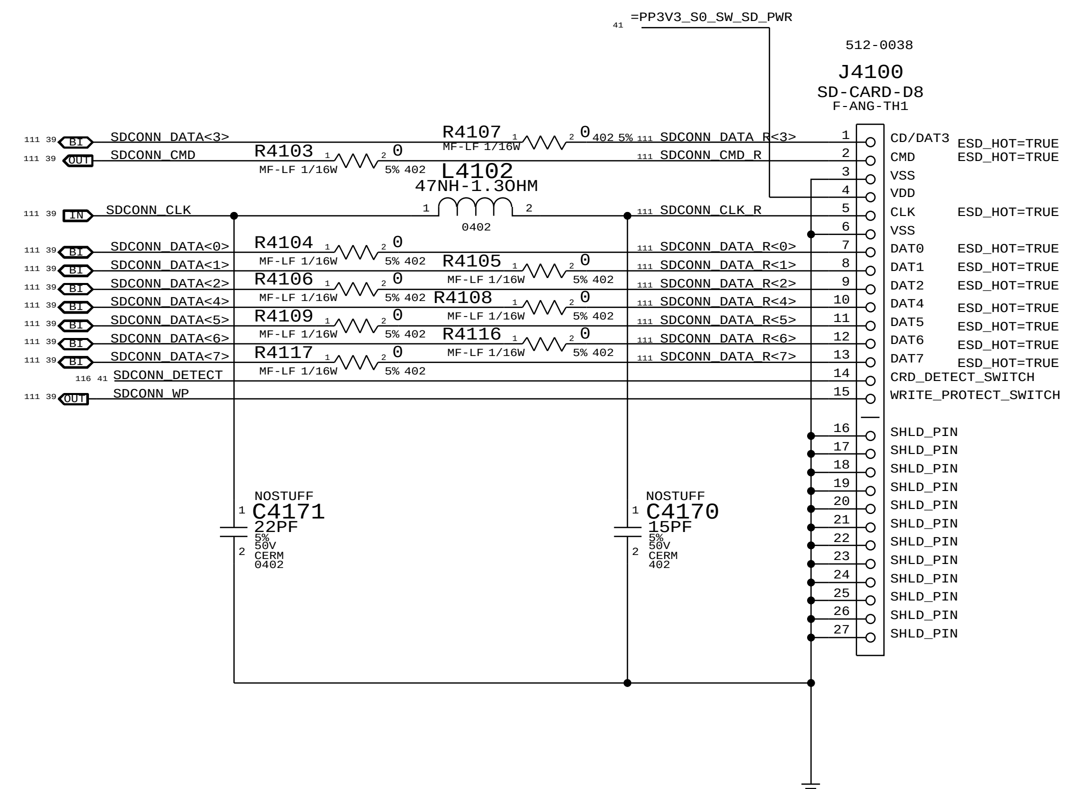
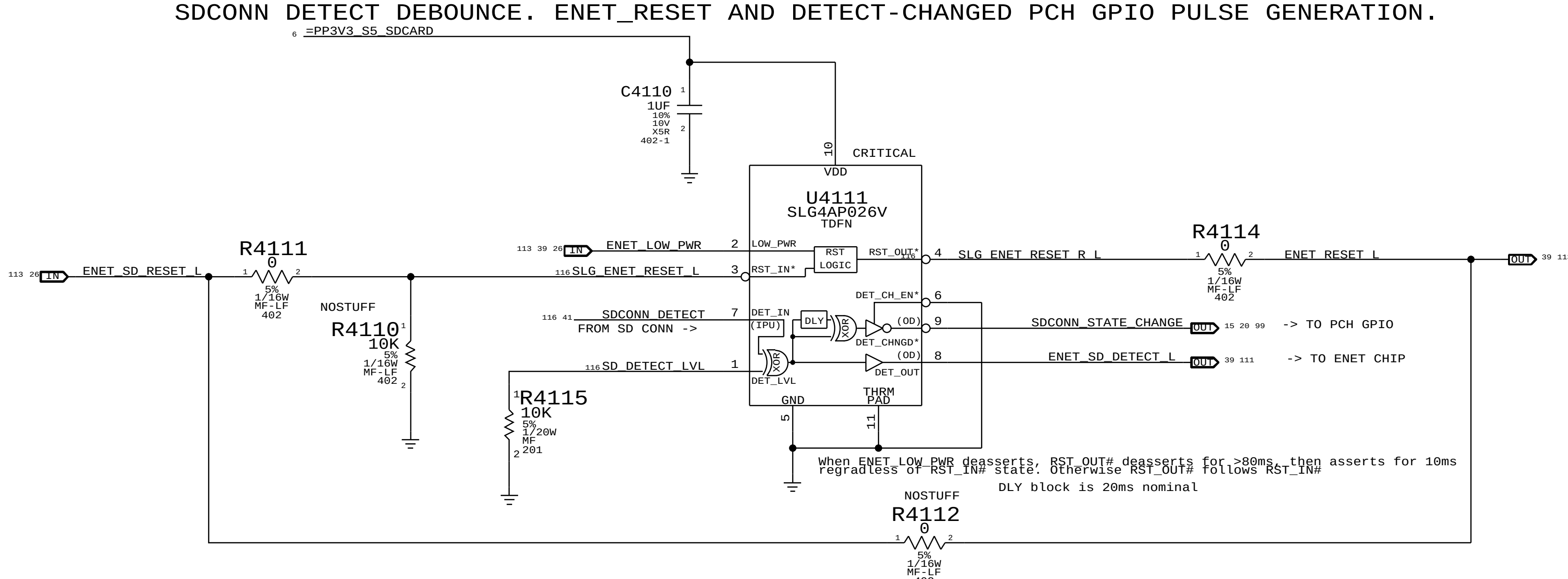
SD CARD 3.3V OVERCURRENT PROTECTION





SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.



SYNC_MASTER=D8_FIYIN SYNC_DATE=07/02/2012

PAGE TITLE

SD READER CONNECTOR

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DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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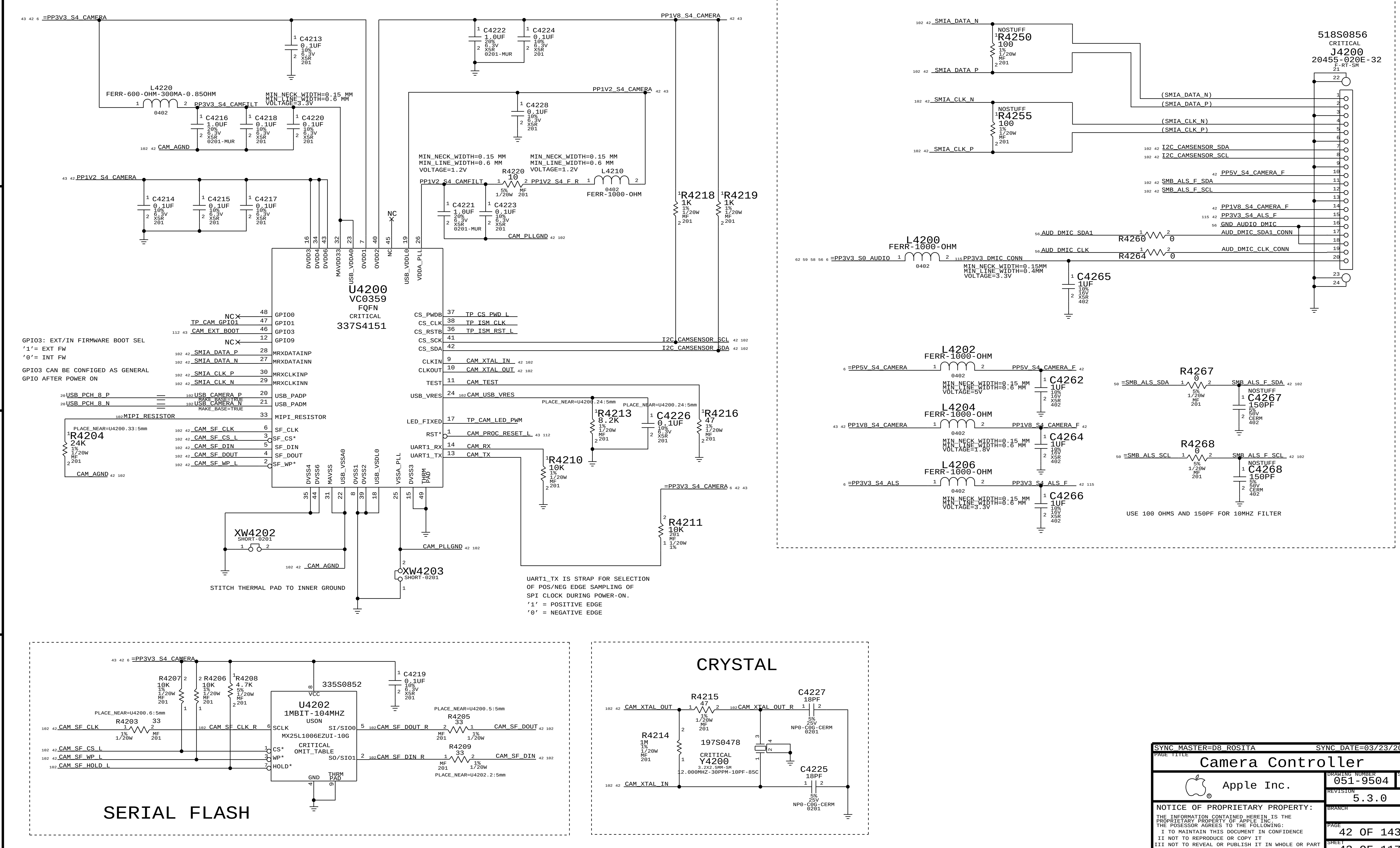
USB CAMERA CONTROLLER



SYNC_MASTER=D8_ROSITA		SYNC_DATE=03/23/2012	
PAGE TITLE: Camera Controller			
Apple Inc.	DRAWING NUMBER: 051-9504		SIZE: D
	REVISION: 5.3.0		
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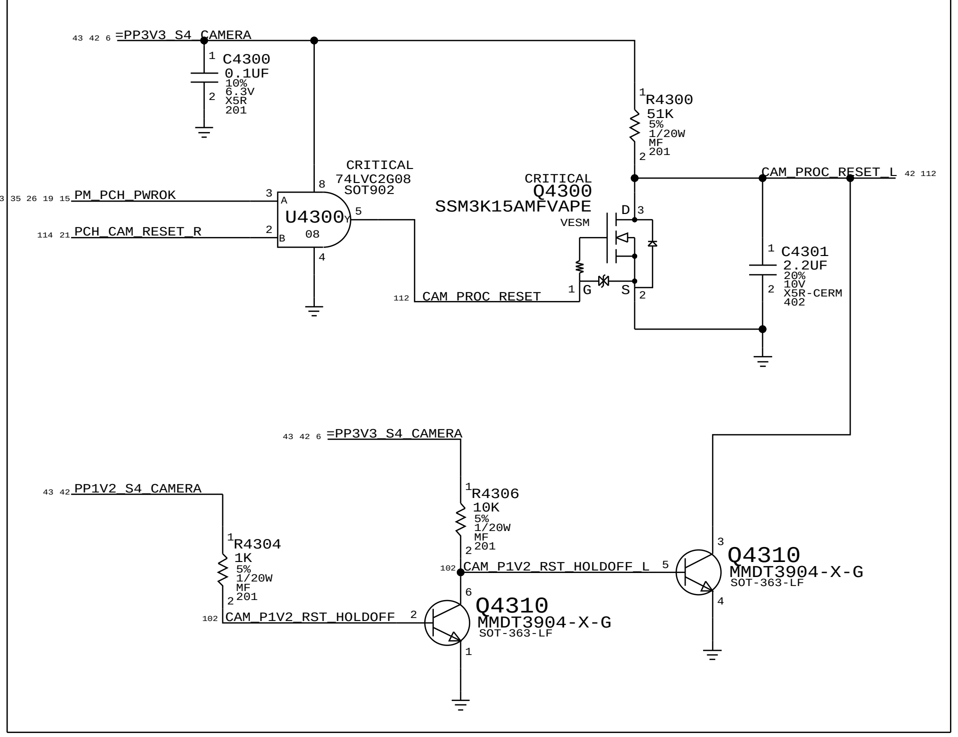
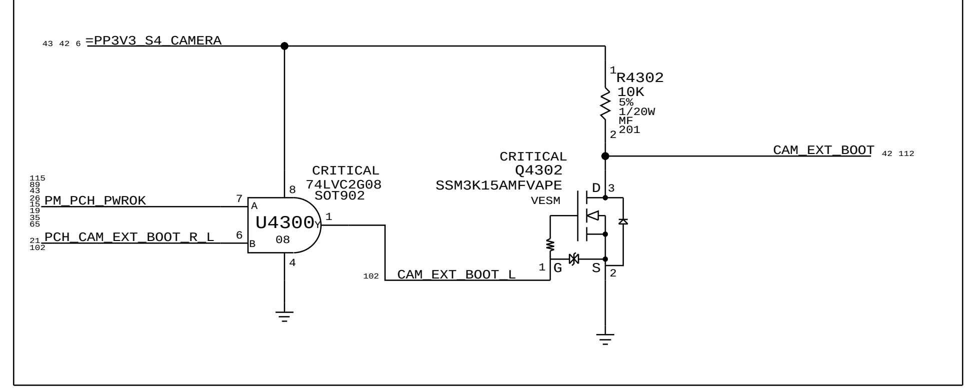
	SHEET: 42 OF 117		

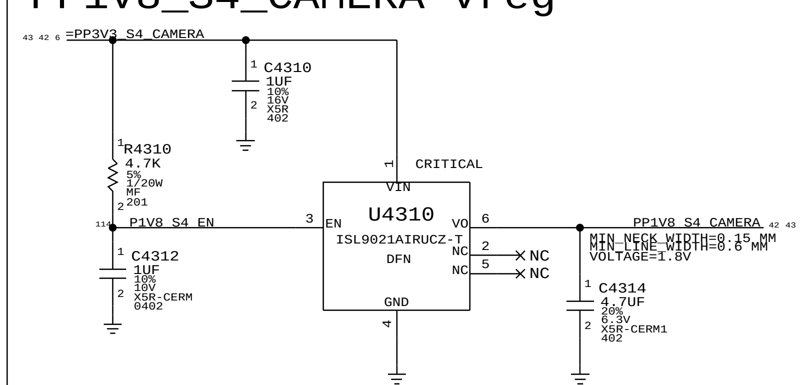
Camera Processor Reset



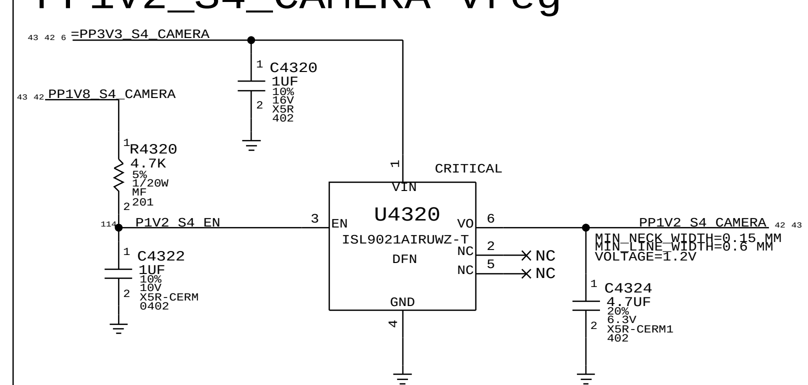
Camera Processor ExtBoot Cntl



PP1V8_S4_CAMERA Vreg



PP1V2_S4_CAMERA Vreg



SYNC_MASTER=D7_MLB SYNC_DATE=03/15/2012

PAGE TITLE
Camera Controller Support

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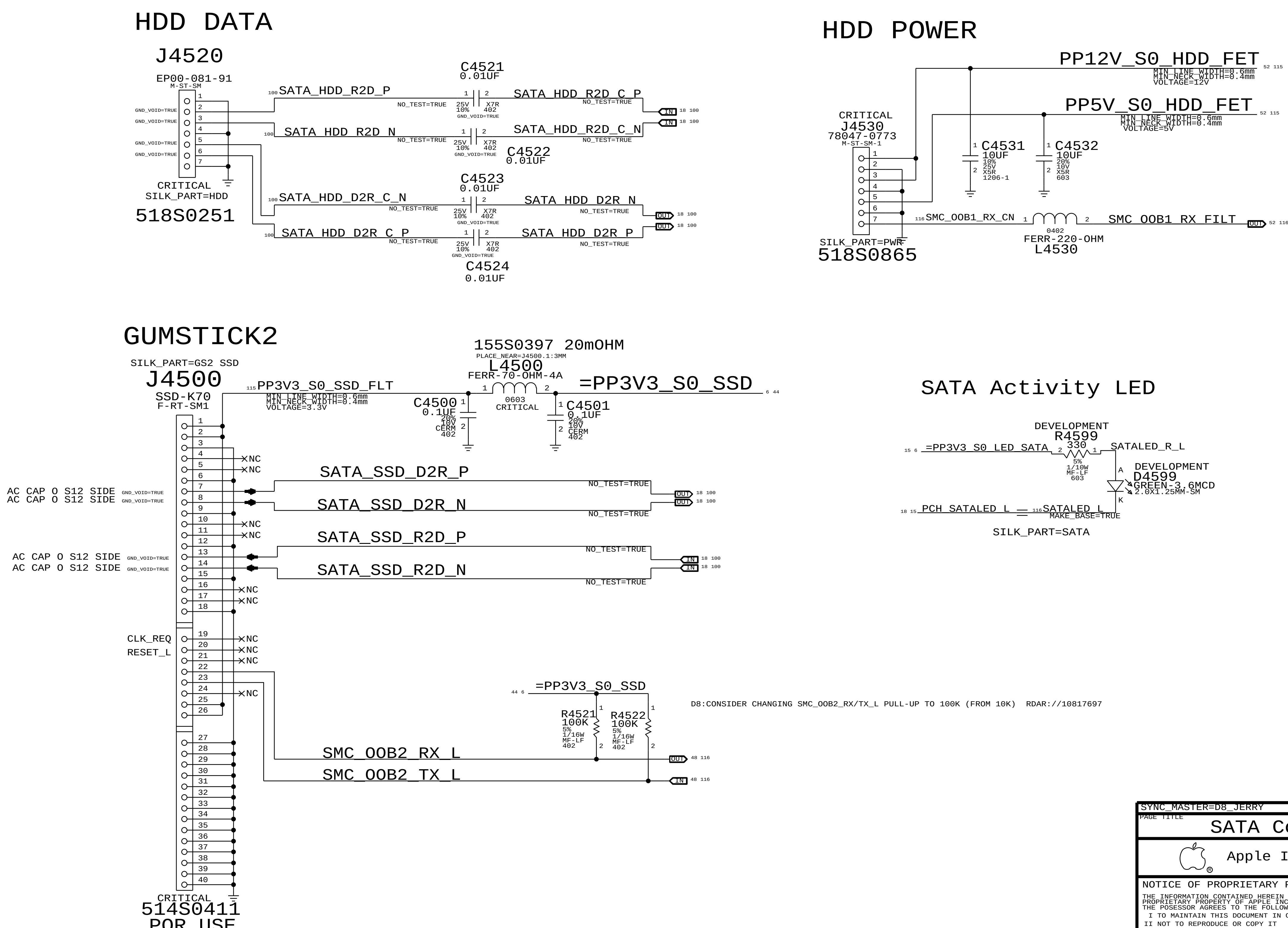

SYNC_MASTER=D8_JERRY SYNC_DATE=01/31/2012

PAGE TITLE

SATA Connectors

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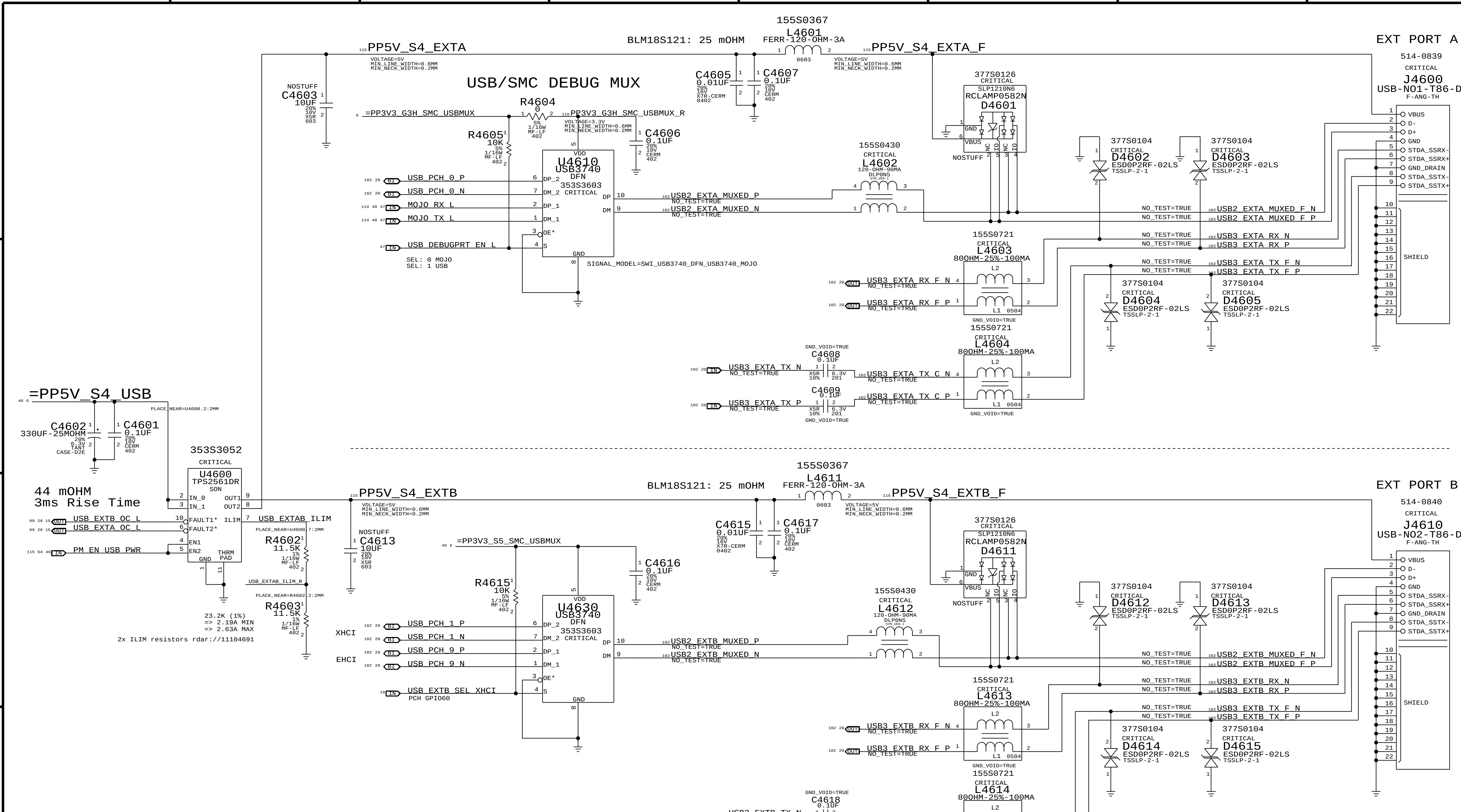

SYNC_MASTER=D8_ROSITA SYNC_DATE=03/23/2012

PAGE TITLE

EXTERNAL USB PORTS A & B

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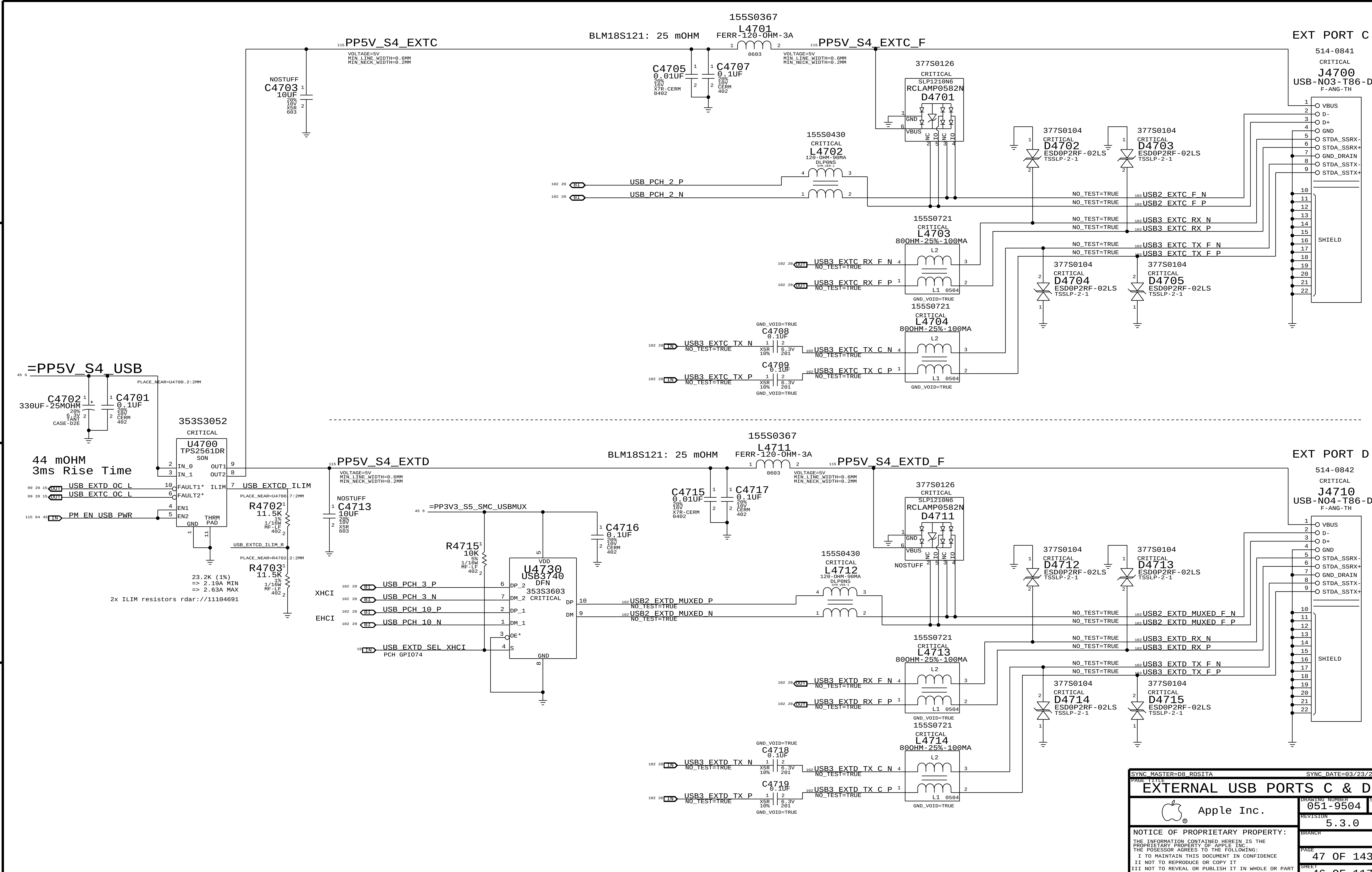

SYNC_MASTER=D8_ROSITA SYNC_DATE=03/23/2012

PAGE TITLE

EXTERNAL USB PORTS C & D

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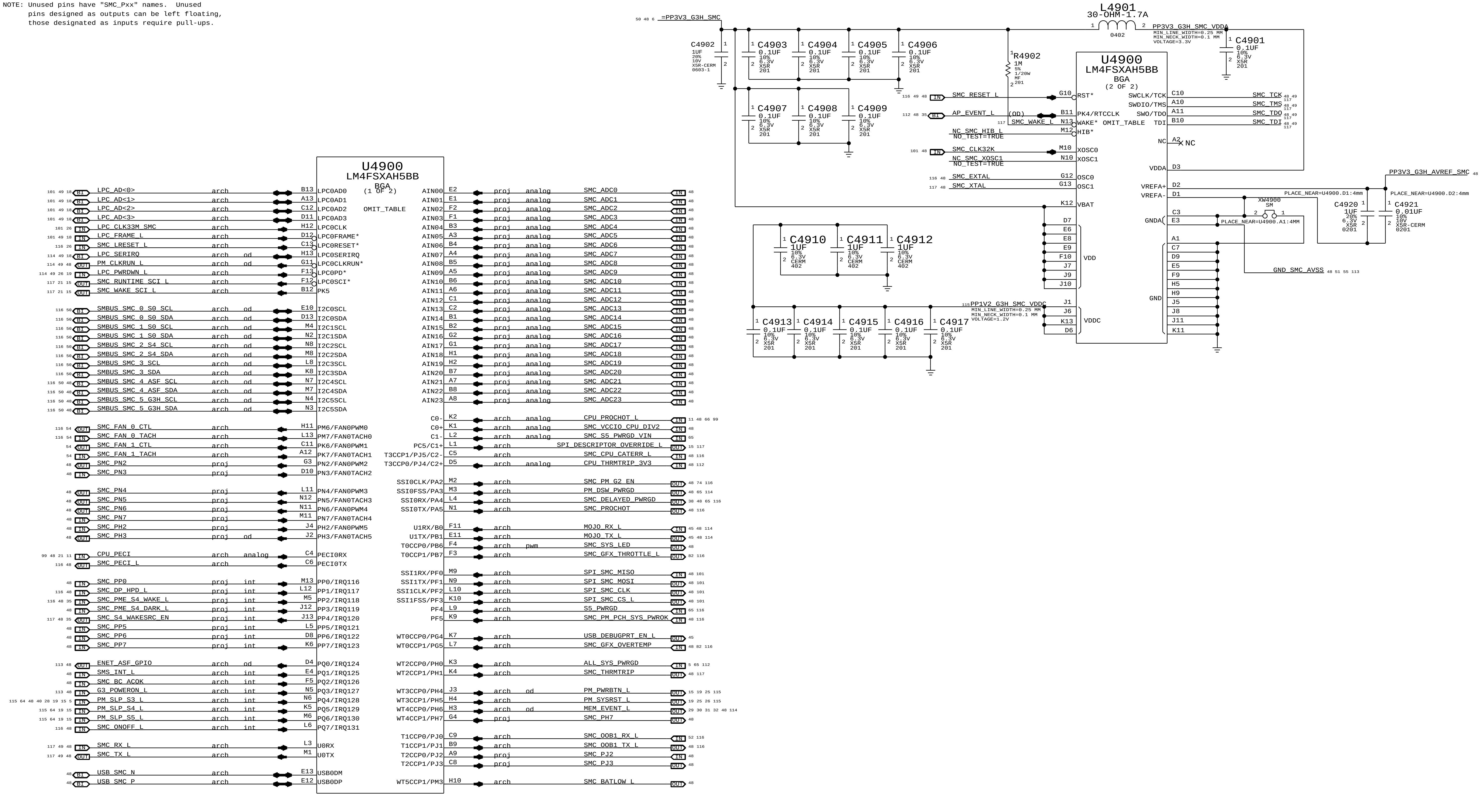
NOTE: Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designed as inputs require pull-ups.



SYNC_MASTER=D8_MARK SYNC_DATE=03/22/2012

PAGE TITLE

SMC

Apple Inc.

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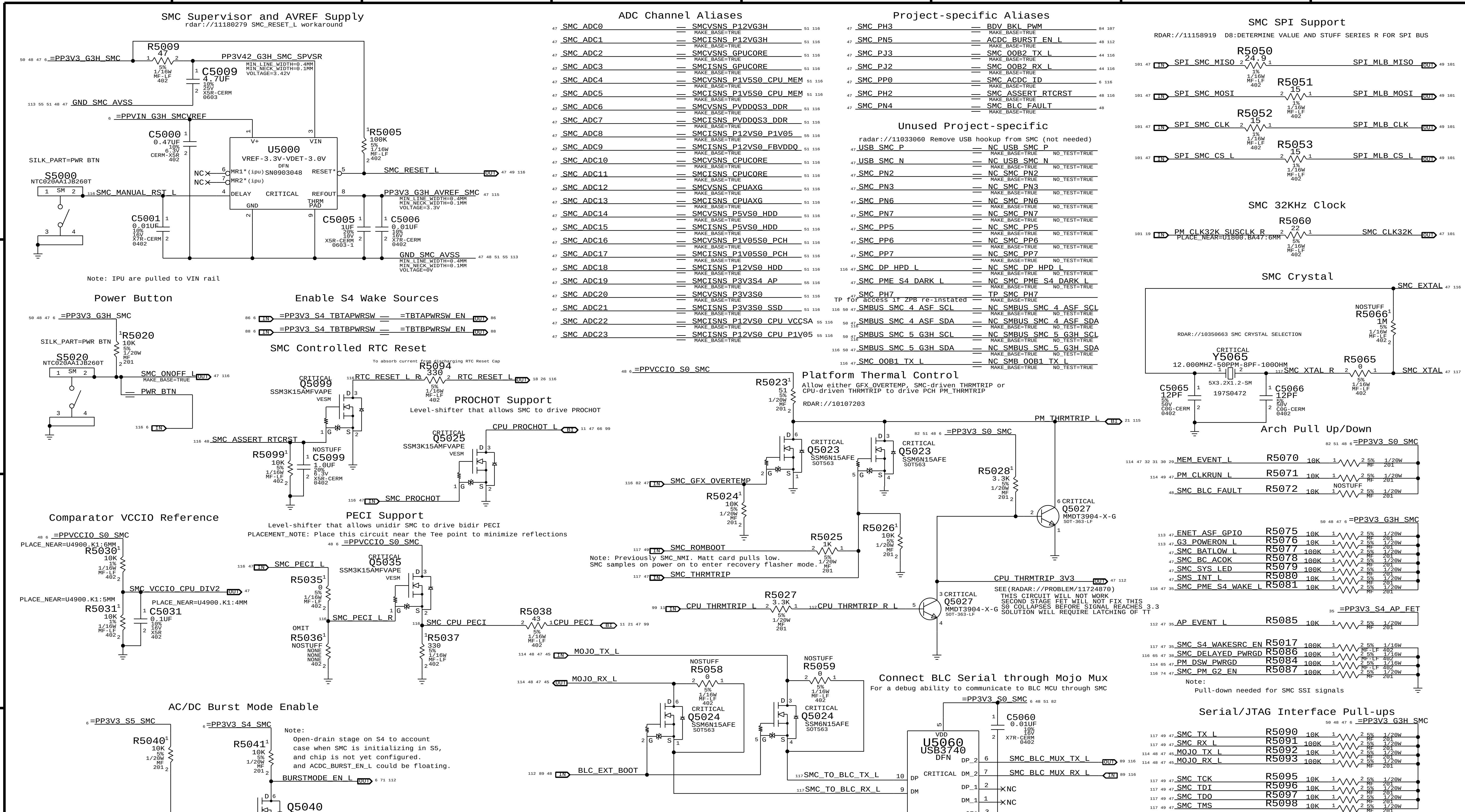

SYNC_MASTER=D8_DOUG SYNC_DATE=06/25/2012

PAGE TITLE: SMC Support

Apple Inc.

DRAWING NUMBER: 051-9504 SIZE: D

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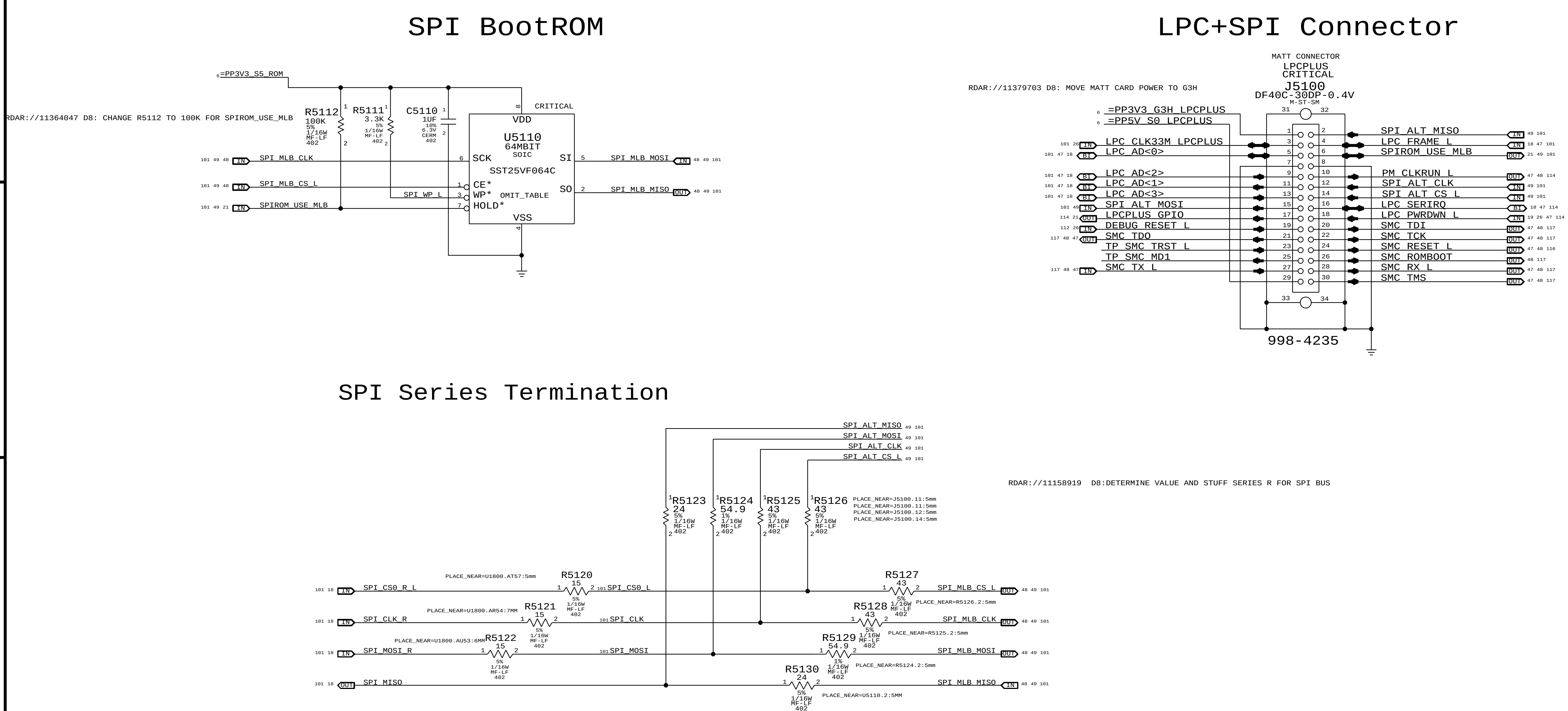

SYNC_MASTER=D8_MLB SYNC_DATE=N/A

PAGE TITLE

SPI and Debug Connector

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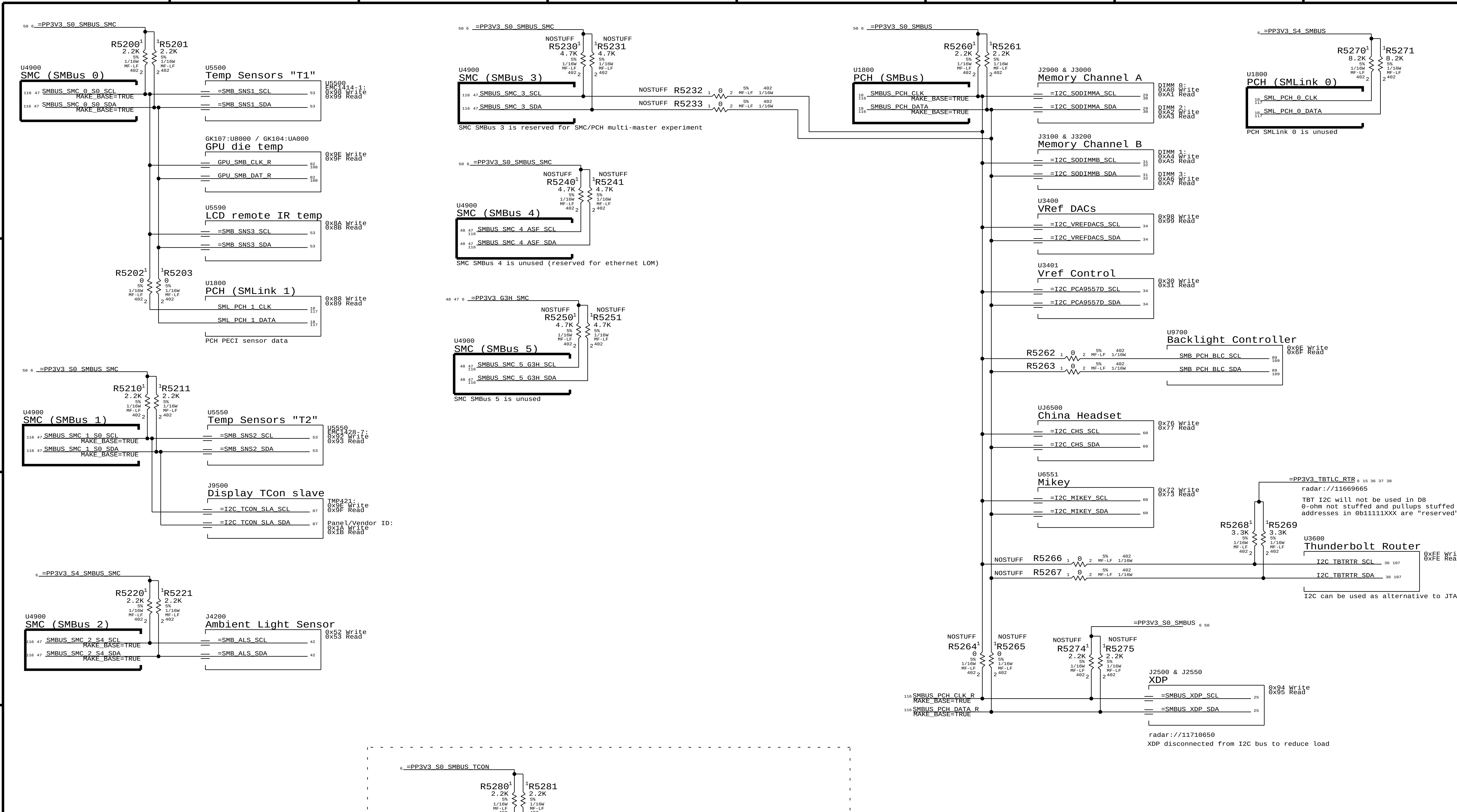

SYNC_MASTER=D8_TAVYS SYNC_DATE=06/22/2012

PAGE TITLE: SMBus Connections

Apple Inc.

DRAWING NUMBER: 051-9504 SIZE: D

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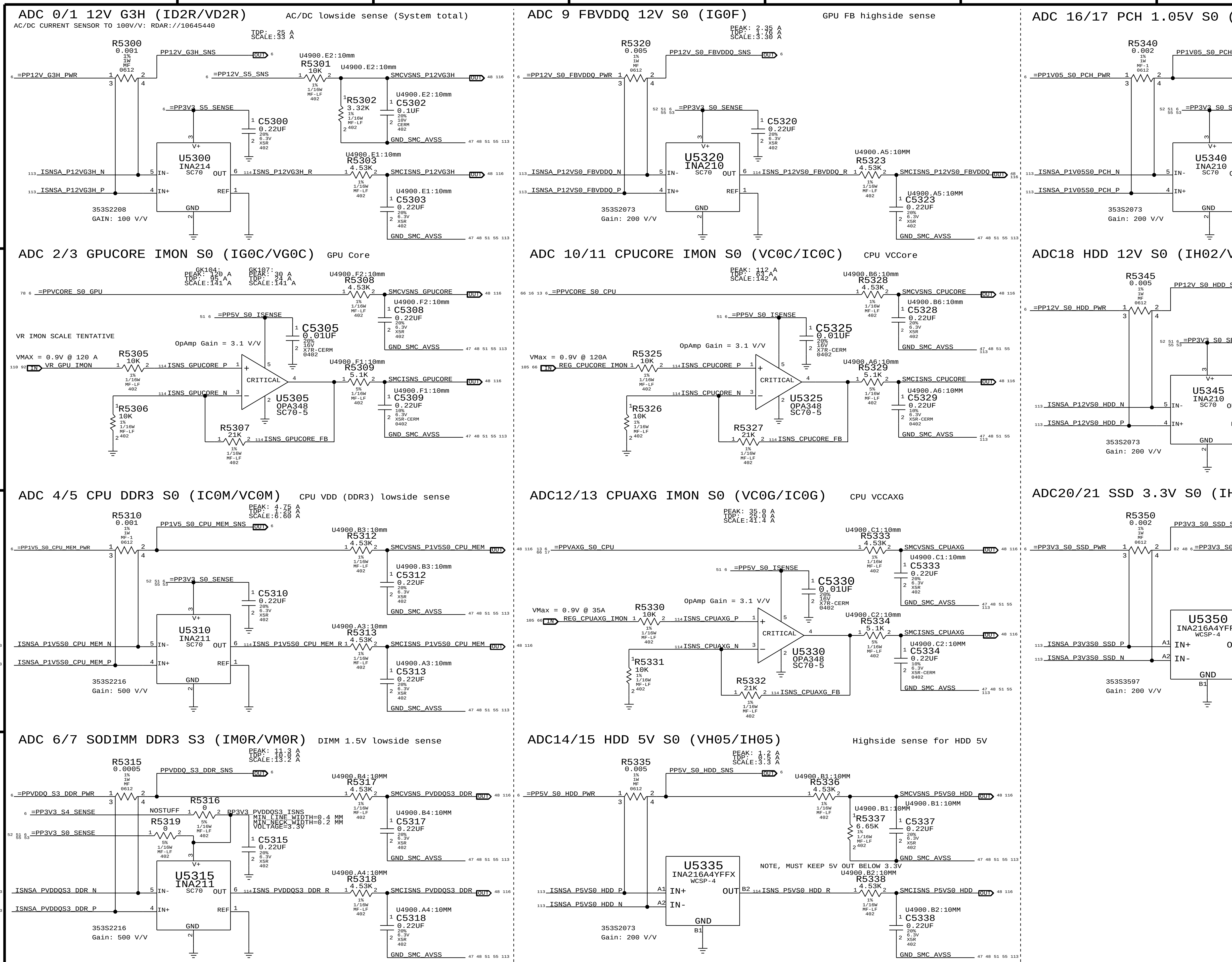
PAGE: 52 OF 143

SHEET: 50 OF 117



SYNC_MASTER=D8_DOUG SYNC_DATE=06/20/2012

PAGE TITLE

I and V Sense 1

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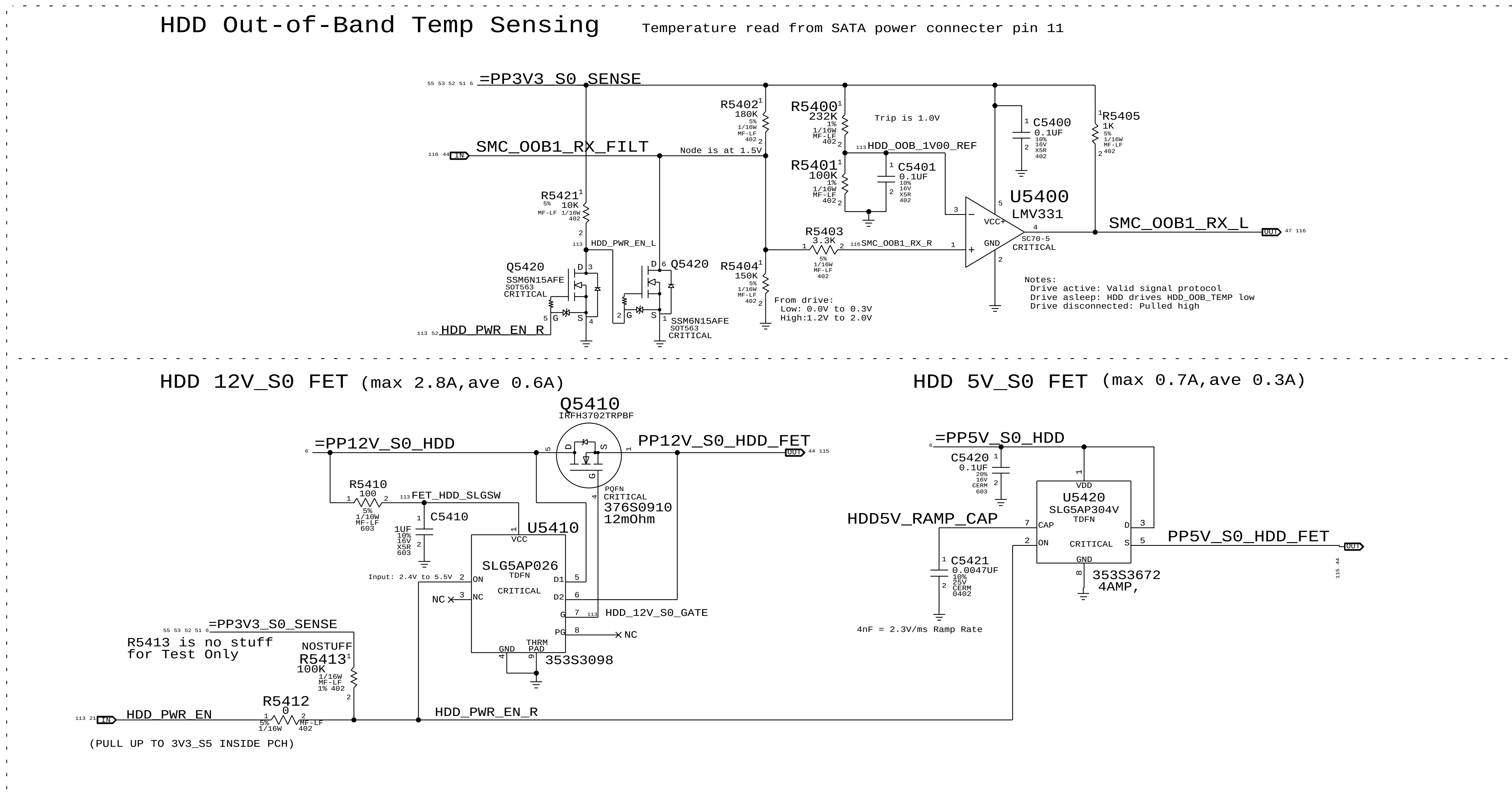

REMOVE R5413 AND SHORT R5412 AFTER HDD_PWR_EN WORKS

SYNC-MASTER=D8_JERRY SYNC-DATE=02/25/2012

PAGE TITLE: HDD/SSD Temp Sense

Apple Inc.

DRAWING NUMBER: 051-9504 SIZE: D

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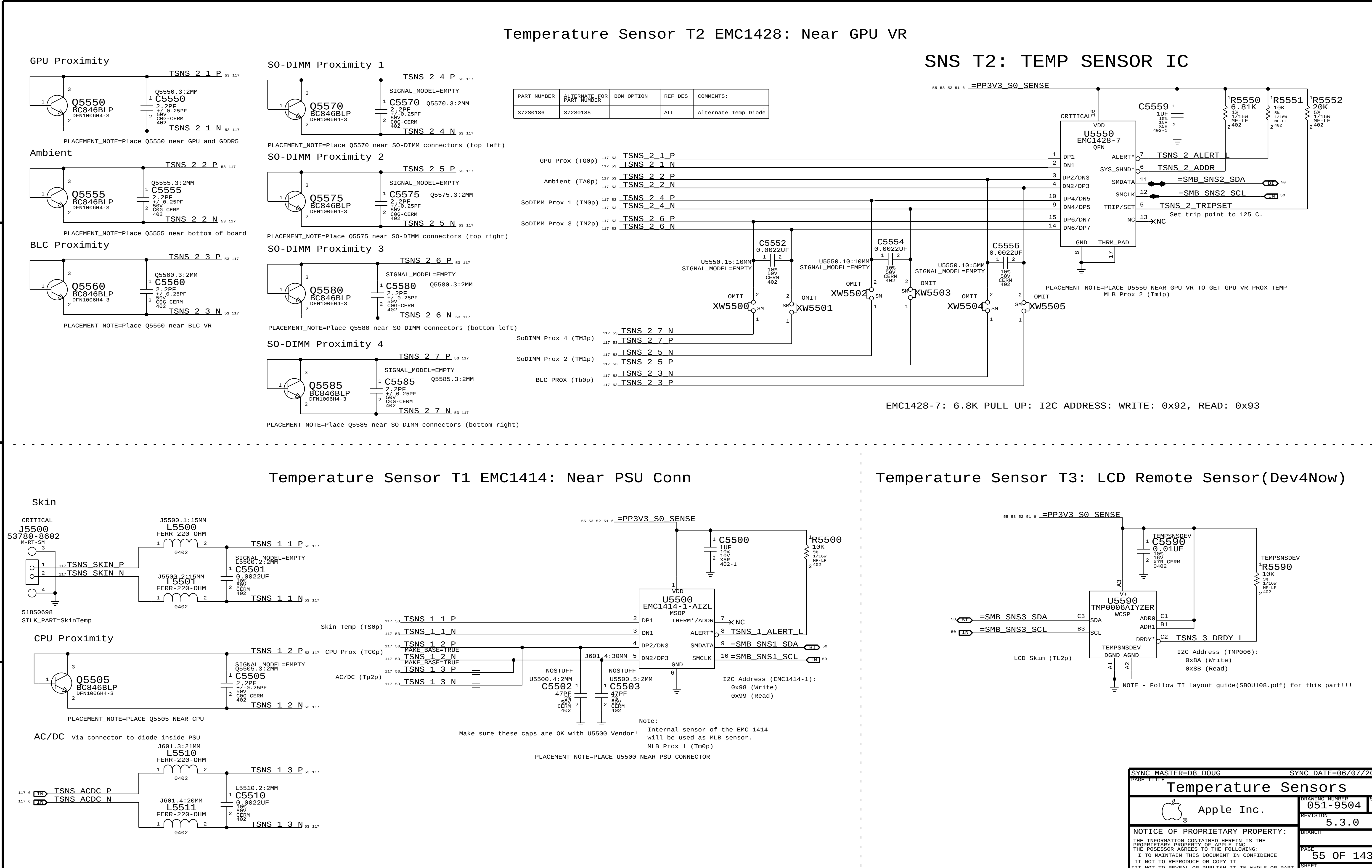

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

SYNC_MASTER=D8_DOUG SYNC_DATE=06/07/2012

PAGE TITLE

Temperature Sensors

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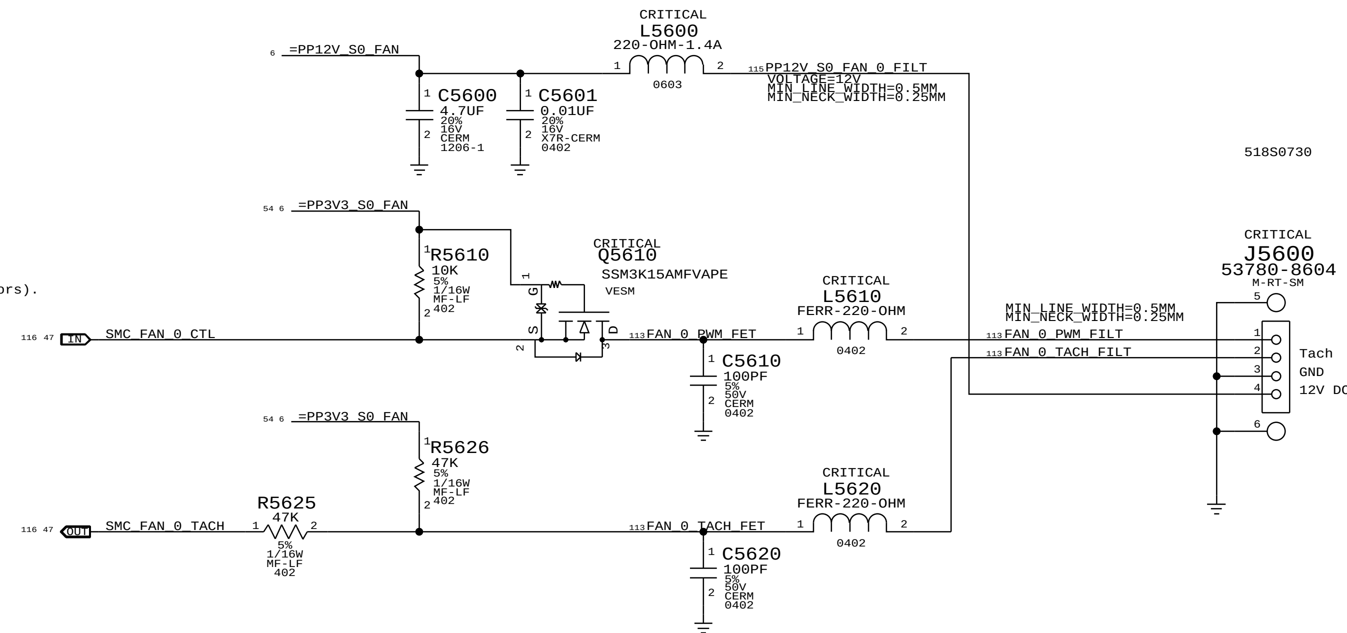
SMC Fan 0 (System)

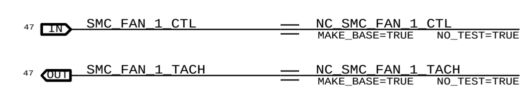
Note:
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q5610 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q5610 is at common and the SMC sinks current when Q5610 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need scematic and PCB file of fan(All Vendors).



SMC Fan 1 (Unused)



SYNC_MASTER=D7_MLB SYNC_DATE=05/24/2012

PAGE TITLE: System Fan

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DRAWING NUMBER: 051-9504 SIZE: D

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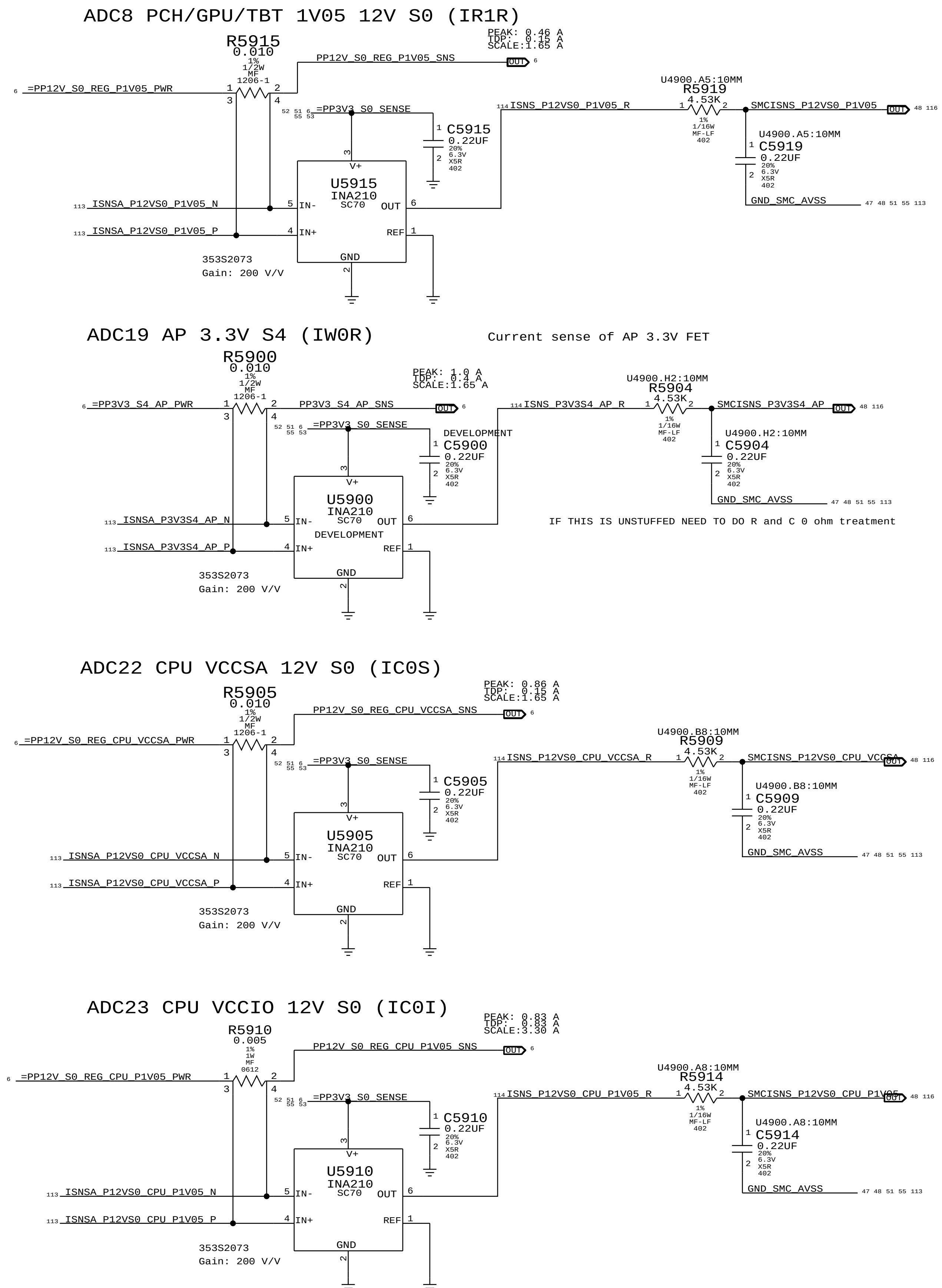

SYNC_MASTER=D8_MARK SYNC_DATE=04/23/2012

PAGE TITLE

I and V Sense 2

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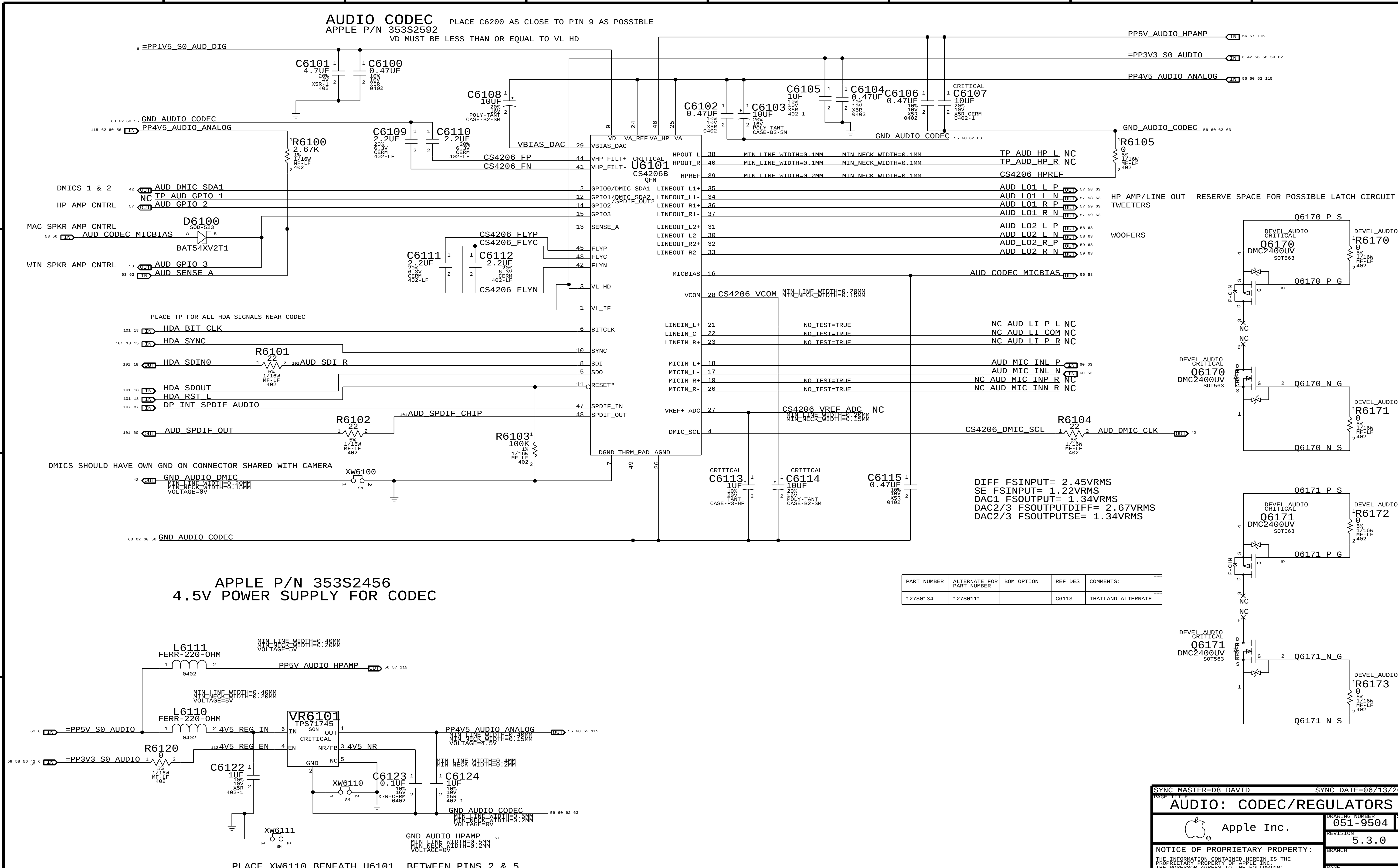

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0134	127S0111		C6113	THAILAND ALTERNATE

SYNC_MASTER=D8_DAVID SYNC_DATE=06/13/2012

PAGE TITLE

AUDIO: CODEC/REGULATORS

Apple Inc.

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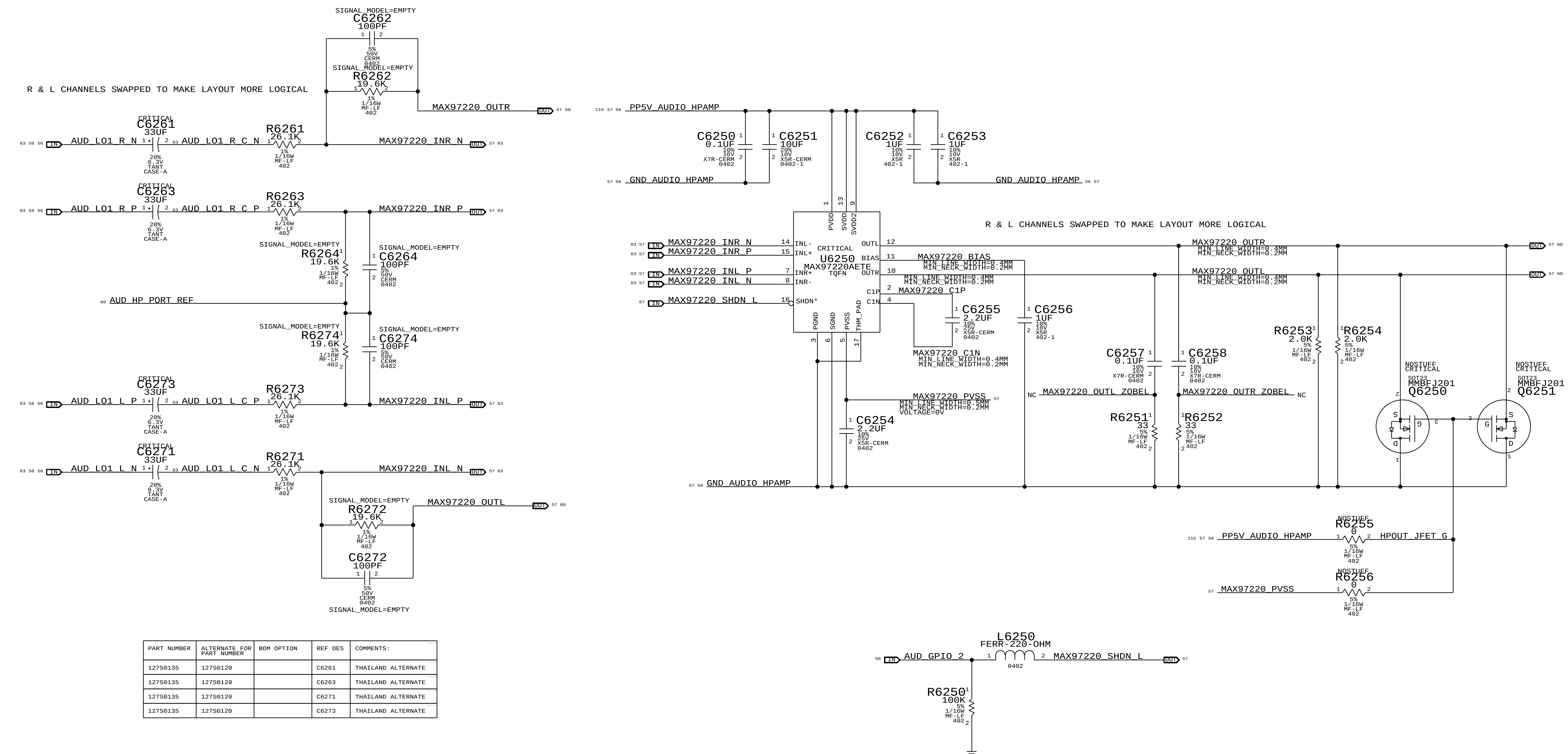

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0135	127S0120		C6261	THAILAND ALTERNATE
127S0135	127S0120		C6263	THAILAND ALTERNATE
127S0135	127S0120		C6271	THAILAND ALTERNATE
127S0135	127S0120		C6273	THAILAND ALTERNATE

SYNC_MASTER=D8_DAVID SYNC_DATE=06/13/2012

PAGE TITLE

AUDIO: HEADPHONE AMP

DRAWING NUMBER 051-9504 SIZE D

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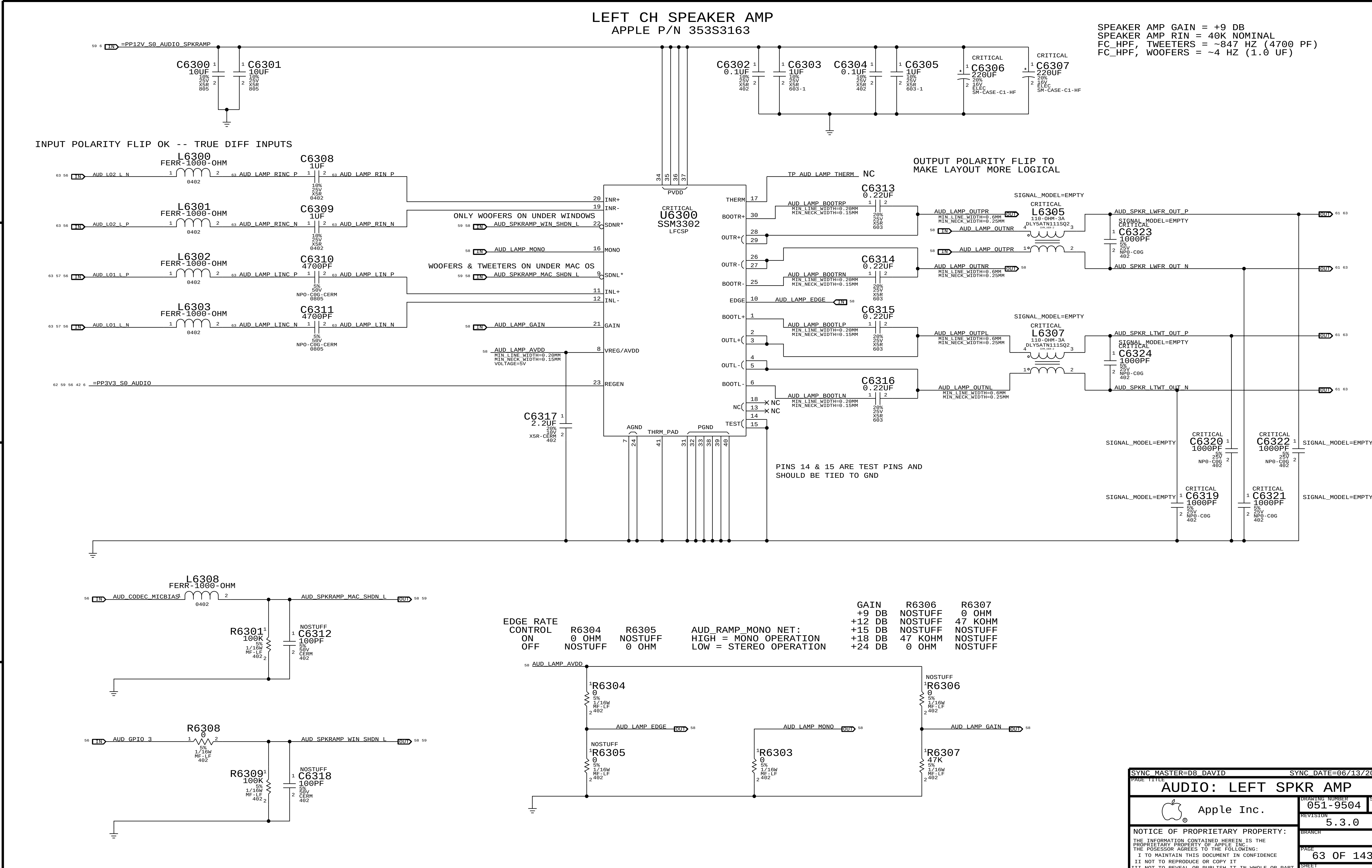

SYNC_MASTER=D8_DAVID SYNC_DATE=06/13/2012

PAGE TITLE

AUDIO: LEFT SPKR AMP

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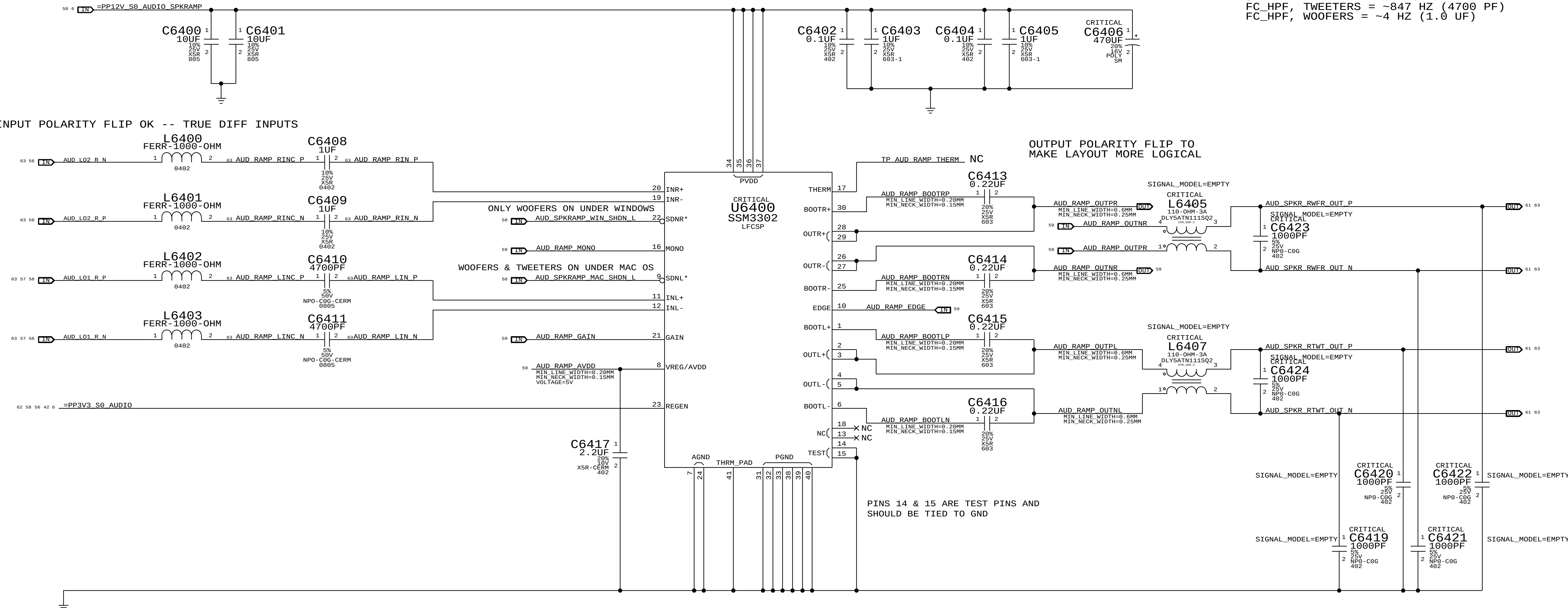
RIGHT CH SPEAKER AMP

APPLE P/N 353S3163

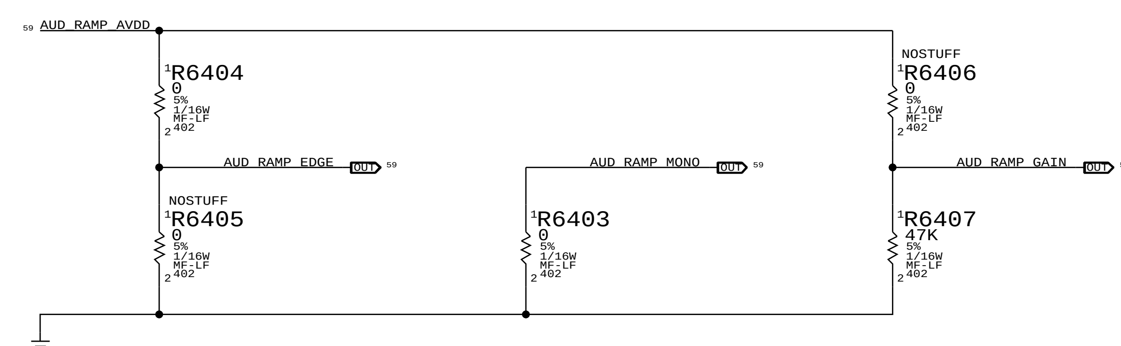
SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)



GAIN	R6406	R6407
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	47 KOHM
+15 DB	NOSTUFF	NOSTUFF
+18 DB	47 KOHM	NOSTUFF
+24 DB	0 OHM	NOSTUFF

EDGE RATE CONTROL	R6404	R6405
ON	0 OHM	NOSTUFF
OFF	NOSTUFF	0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION



SYNC_MASTER=D8_DAVID SYNC_DATE=06/13/2012

PAGE TITLE: AUDIO: RIGHT SPKR AMP

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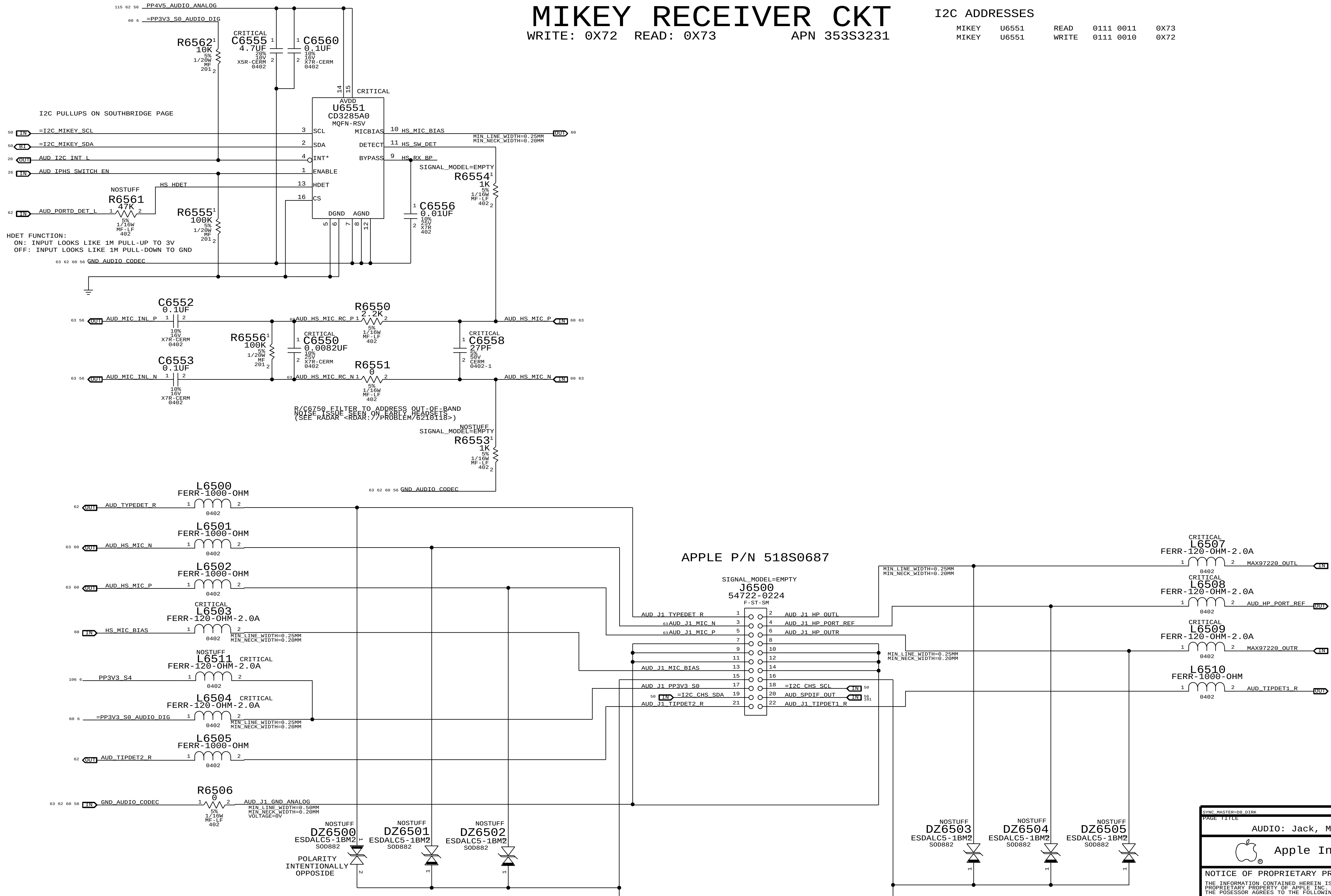
MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S3231

I2C ADDRESSES

MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72



SYNC_MASTER=D8_DIRK SYNC_DATE=06/29/2012

PAGE TITLE

AUDIO: Jack, Mikey, CHS Switch

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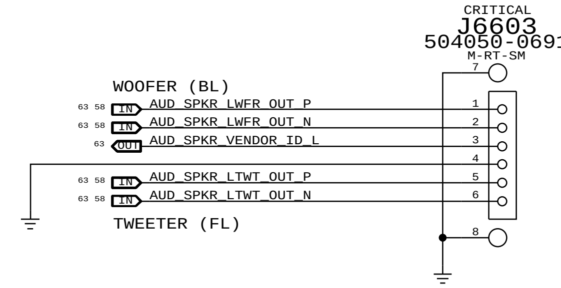
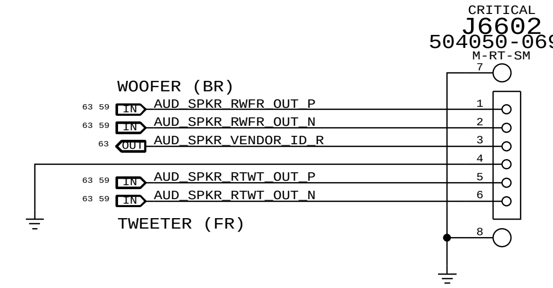
SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862





SYNC_MASTER=D8_DAVID	SYNC_DATE=06/13/2012
PAGE TITLE: Audio: Spkr/Mic Conn.	
Apple Inc.	DRAWING NUMBER: 051-9504 SIZE: D
	REVISION: 5.3.0
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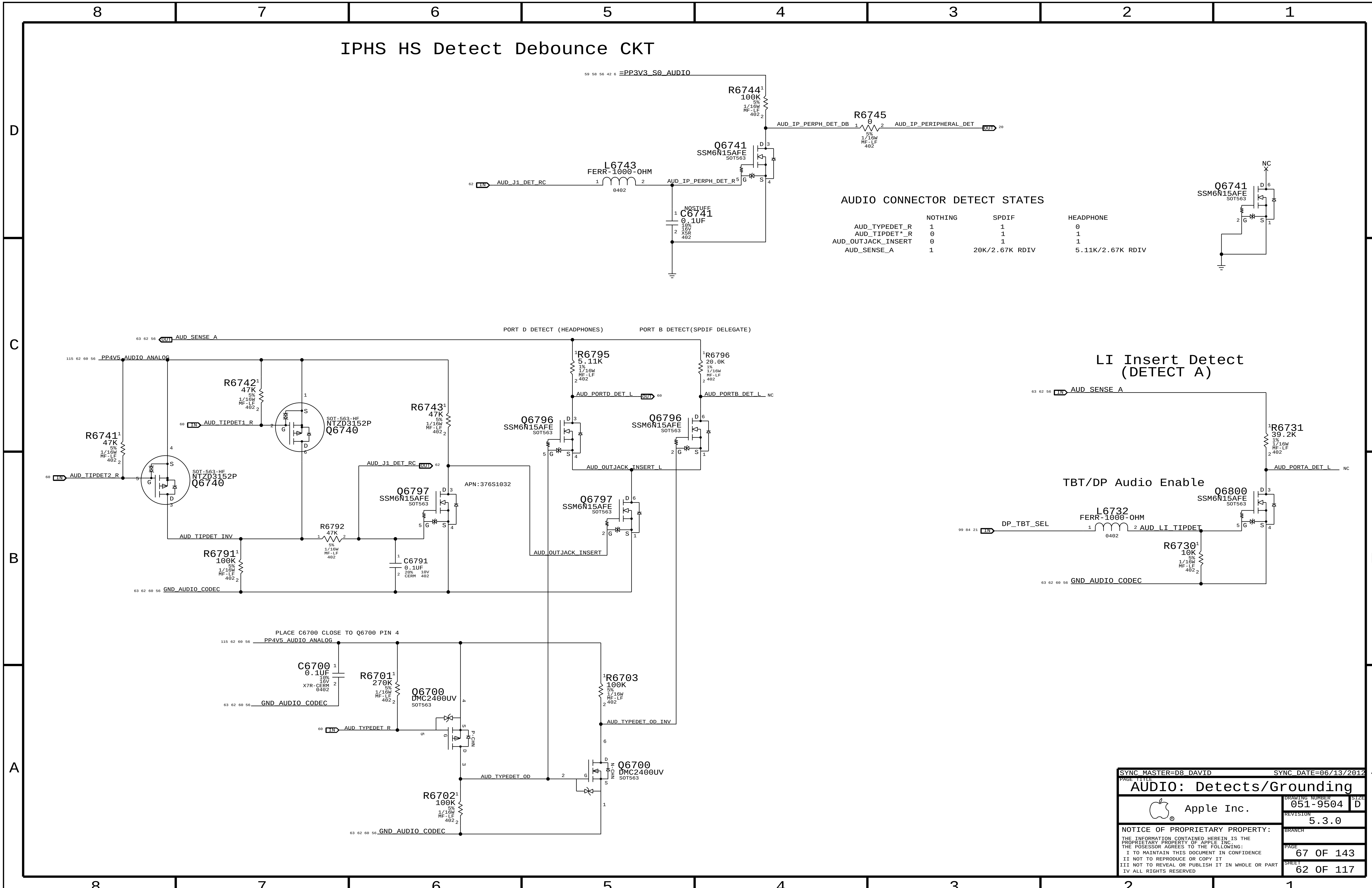
	SHEET: 61 OF 117

IPHS HS Detect Debounce CKT



AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV

SYNC_MASTER=D8_DAVID SYNC_DATE=06/13/2012

PAGE TITLE
AUDIO: Detects/Grounding

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	MICBIAS	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
SPDIF IN	0X07 (7)	0X0F (15)	N/A	0X09 (DET A)
INTERNAL MIC ARRAY	0X06 (6) 0X05 (5)	0X0E (14,LEFT & RIGHT) 0X12 (18,LEFT)	N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	PANTHER POINT GPIO 16	PANTHER POINT GPIO 5 (RCVR_INT) PANTHER POINT GPIO 3 (PERIPH DET)

OTHER DETECT

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
MULTIPLE SPKR VENDORS	N/A	N/A	N/A	0X0C (DET C)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

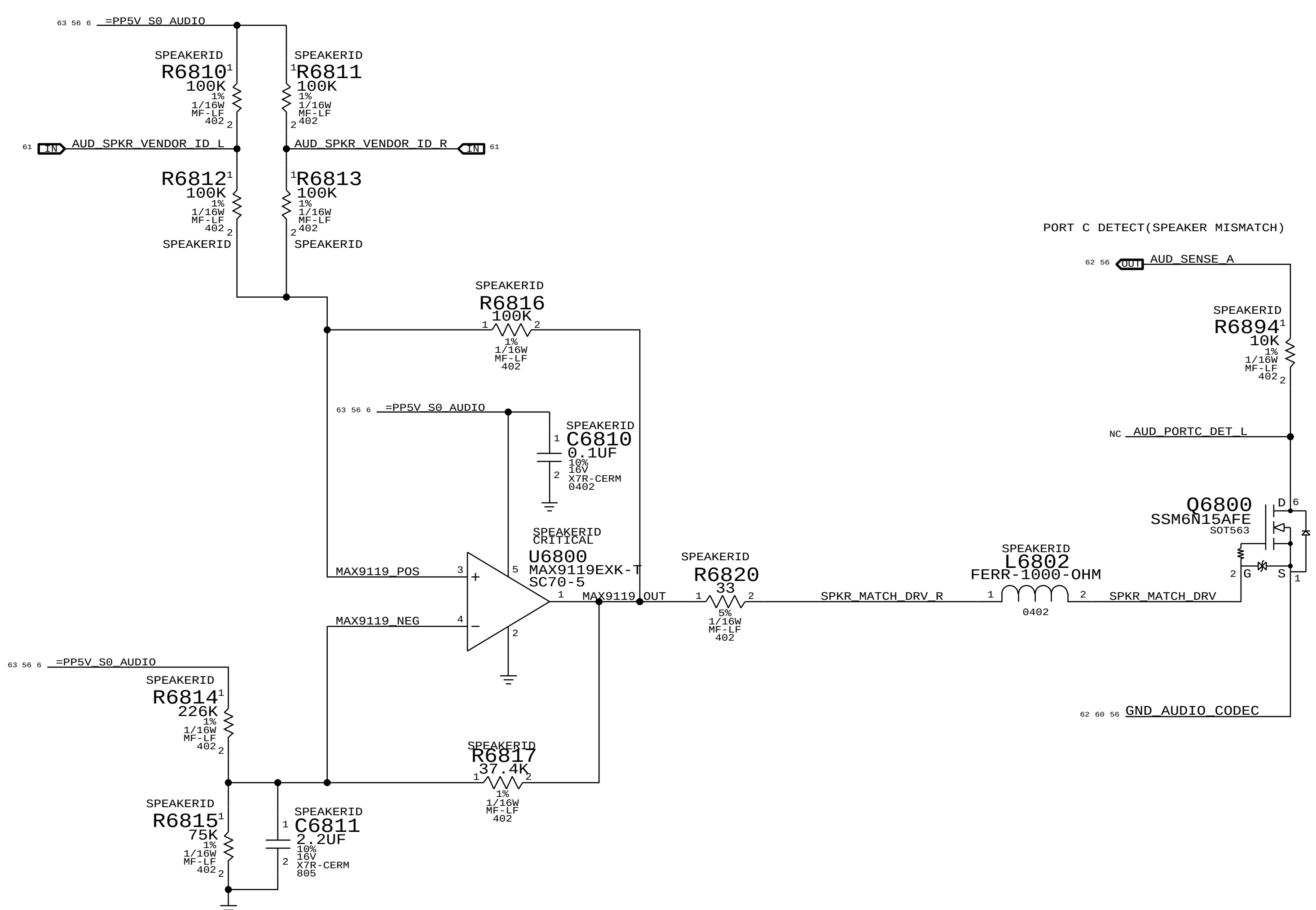
NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

NET_TYPE

	ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
I215	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_P	56 57 58
I216	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_N	56 57 58
I217	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_P	57
I218	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_N	57
I211	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_P	56 57 59
I212	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_N	56 57 59
I210	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_P	57
I209	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_N	57
I208	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_P	56 58
I206	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_N	56 58
I207	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_P	56 59
I204	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_N	56 59
I205	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_P	59
I203	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_N	59
I220	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_P	59
I219	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_N	59
I222	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_P	59
I221	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_N	59
I224	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_P	59
I223	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_N	59
I226	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_P	58
I225	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_N	58
I227	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_P	58
I229	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_N	58
I228	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_P	58
I230	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_N	58
I264	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_P	58
I263	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_N	58
I236	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_P	57
I238	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_N	57
I237	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_P	57
I239	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_N	57
I240	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_P	59 61
I241	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_N	59 61
I242	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_P	59 61
I243	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_N	59 61
I244	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_P	58 61
I246	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_N	58 61
I245	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_P	58 61
I247	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_N	58 61
I326	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_P	56 60
I327	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_N	56 60
I328	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_P	60
I329	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_RC_N	60
I254	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_P	60
I255	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HS_MIC_N	60
I324	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_P	60
I325	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_J1_MIC_N	60

CIRCUIT THEORY OF OPERATION AVAILABLE IN <RDAR://PROBLEM/9776522>



SYNC_MASTER=D8_DAVID	SYNC_DATE=06/13/2012
PAGE TITLE	AUDIO: Speaker ID
Apple Inc.	DRAWING NUMBER 051-9504 SIZE D
	REVISION 5.3.0
	BRANCH
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	SHEET 63 OF 117

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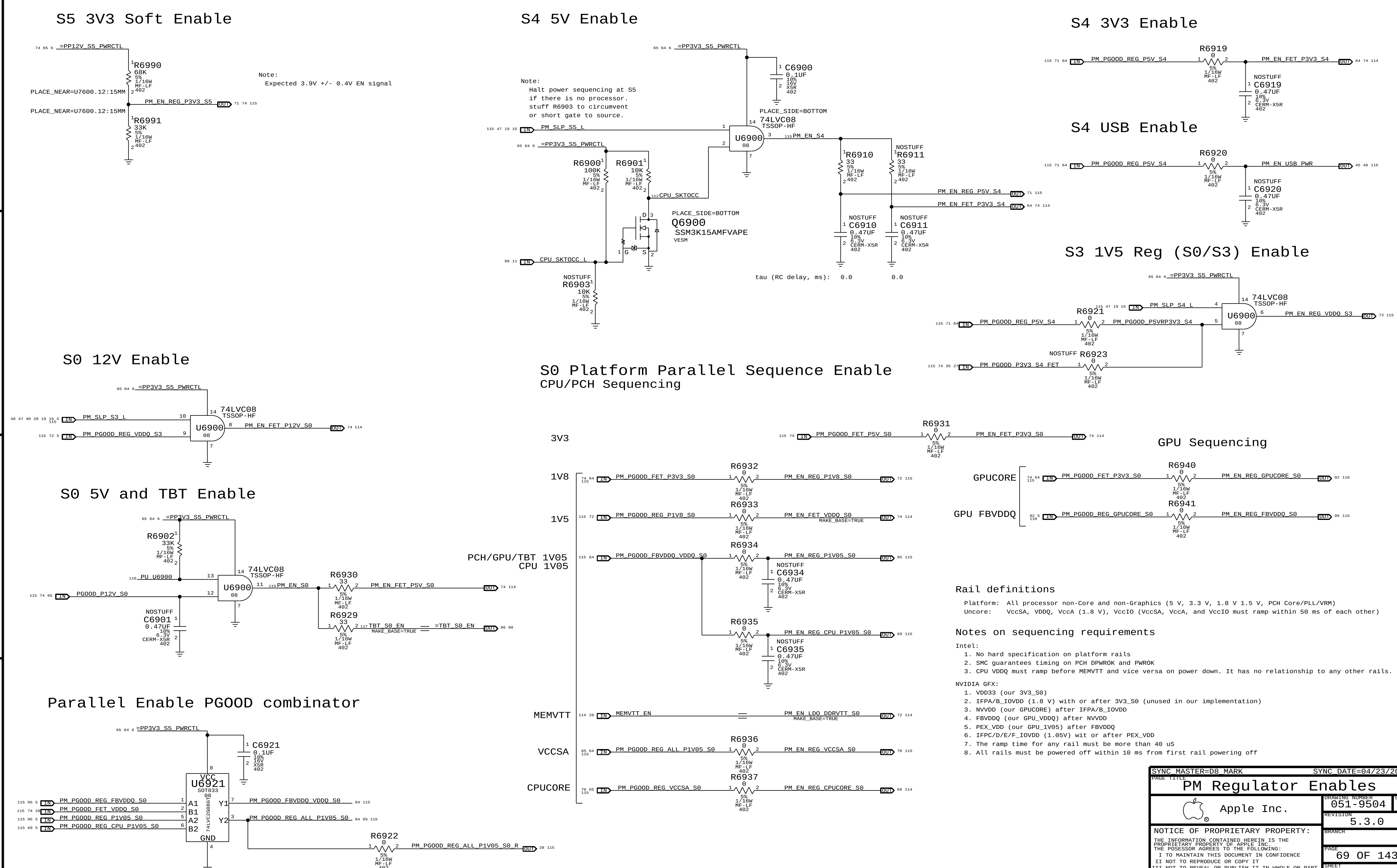

Rail definitions

Platform: All processor non-Core and non-Graphics (5 V, 3.3 V, 1.8 V 1.5 V, PCH Core/PLL/VRM)
Uncore: VccSA, VDDQ, VccA (1.8 V), VccIO (VccSA, VccA, and VccIO must ramp within 50 ms of each other)

Notes on sequencing requirements

Intel:
1. No hard specification on platform rails
2. SMC guarantees timing on PCH DPWROK and PWROK
3. CPU VDDQ must ramp before MEMVTT and vice versa on power down. It has no relationship to any other rails.

NVIDIA GFX:
1. VDD33 (our 3V3_S0)
2. IFPA/B_IOVDD (1.8 V) with or after 3V3_S0 (unused in our implementation)
3. NVVDD (our GPUCORE) after IFPA/B_IOVDD
4. FBVDDQ (our GPU_VDDQ) after NVVDD
5. PEX_VDD (our GPU_1V05) after FBVDDQ
6. IFPC/D/E/F_IOVDD (1.05V) wit or after PEX_VDD
7. The ramp time for any rail must be more than 40 uS
8. All rails must be powered off within 10 ms from first rail powering off

SYNC_MASTER=D8_MARK SYNC_DATE=04/23/2012

PAGE TITLE
PM Regulator Enables

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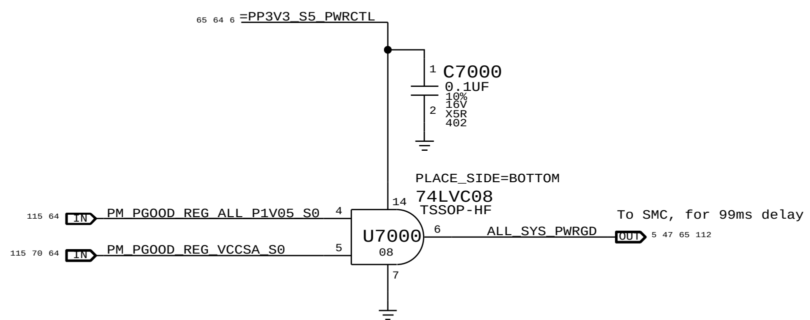
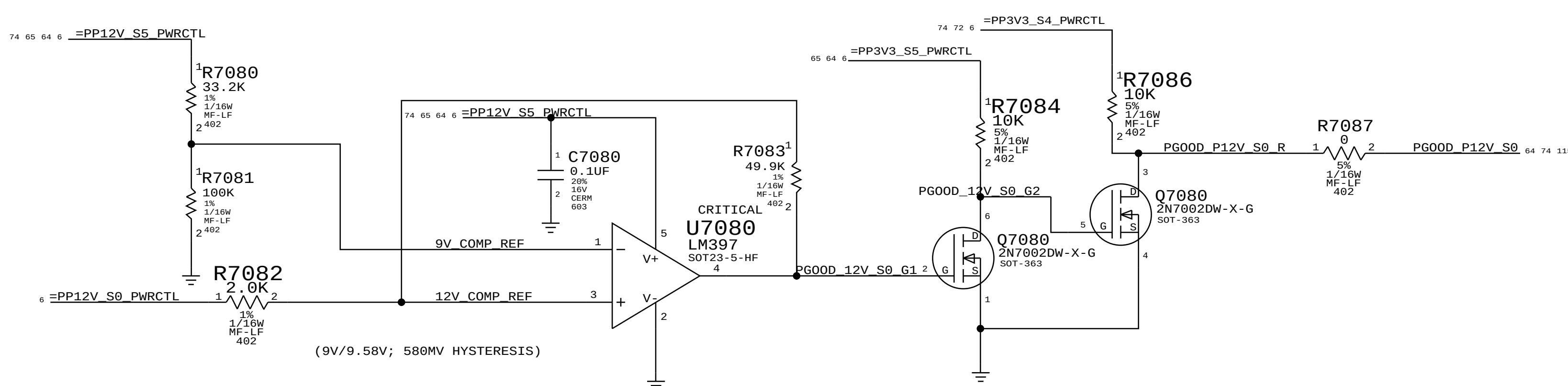
Platform Power Good Derive SMC ALL_SYS_PWRGD

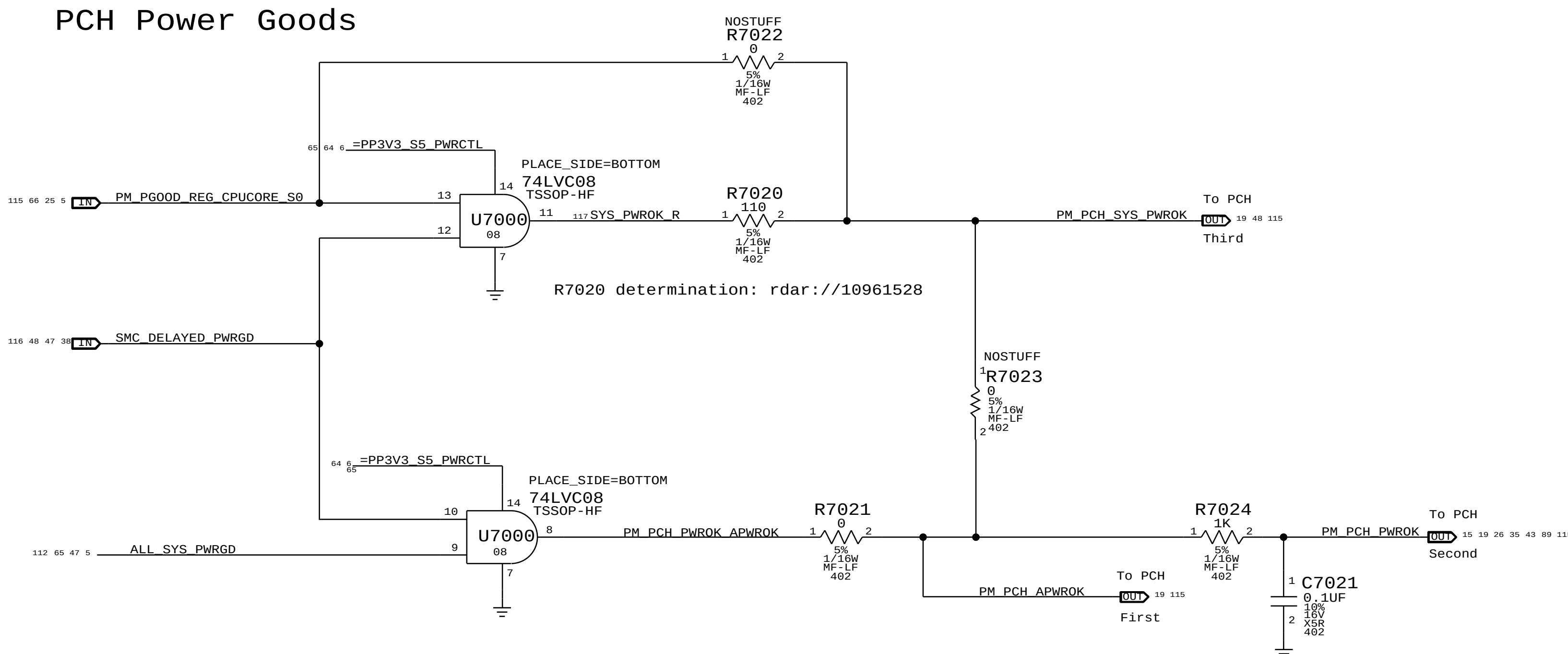
The end of the power sequence for S0 rails except CPU CORE.



PGOOD COMPARATORS FOR PP12V_S0



PCH Power Goods



radar://11043352   Need AND Gate to deassert PM_PCH_PWROK to PCH when unexpected power loss happens

Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

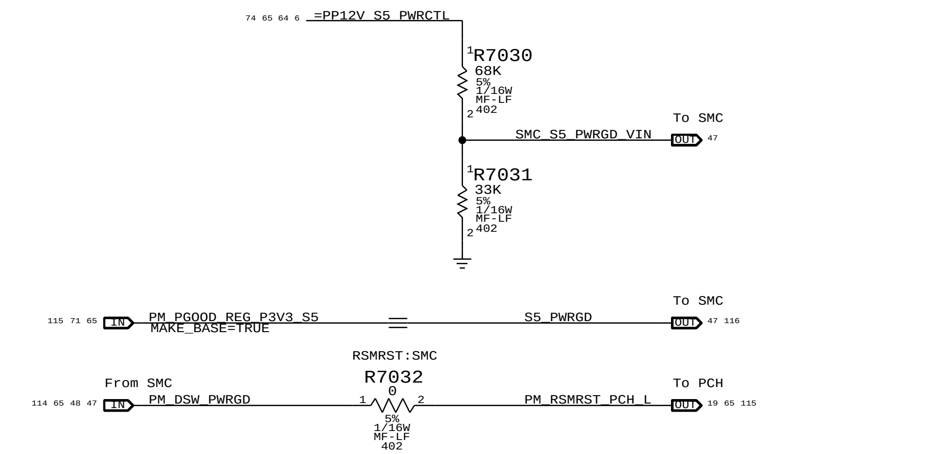
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.

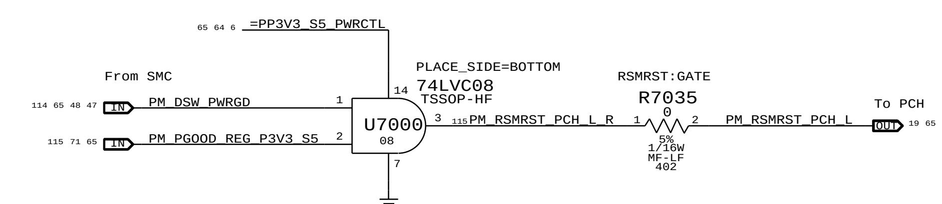
SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.

SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.



Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC_MASTER=D8_MARK   SYNC_DATE=04/23/2012

PAGE TITLE: PM Power Good

Apple Inc.

DRAWING NUMBER: 051-9504 SIZE: D

REVISION: 5.3.0

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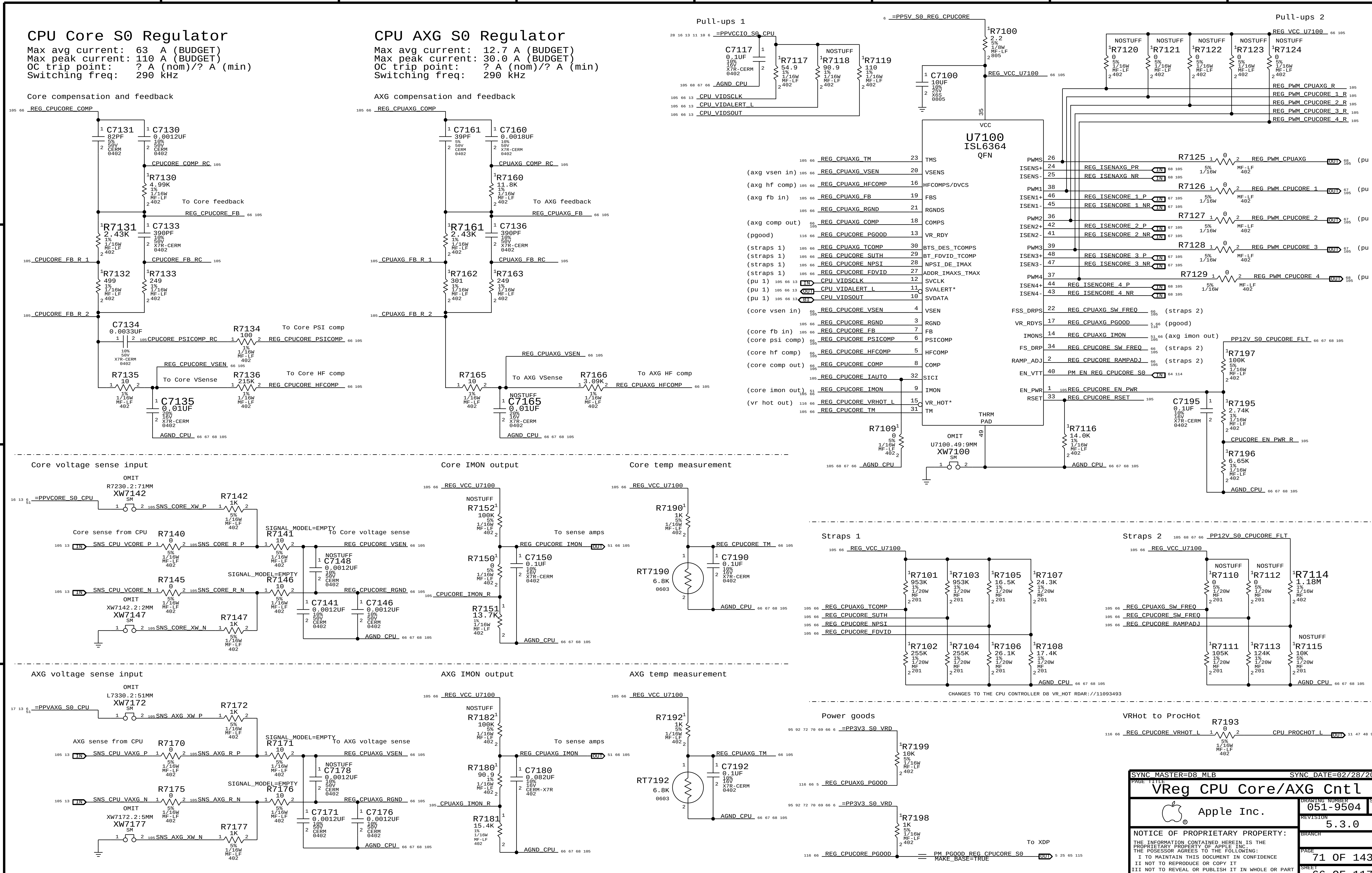
SHEET: 65 OF 117



SYNC_MASTER=D8_MLB SYNC_DATE=02/28/2012

PAGE TITLE
VReg CPU Core/AXG Cntl

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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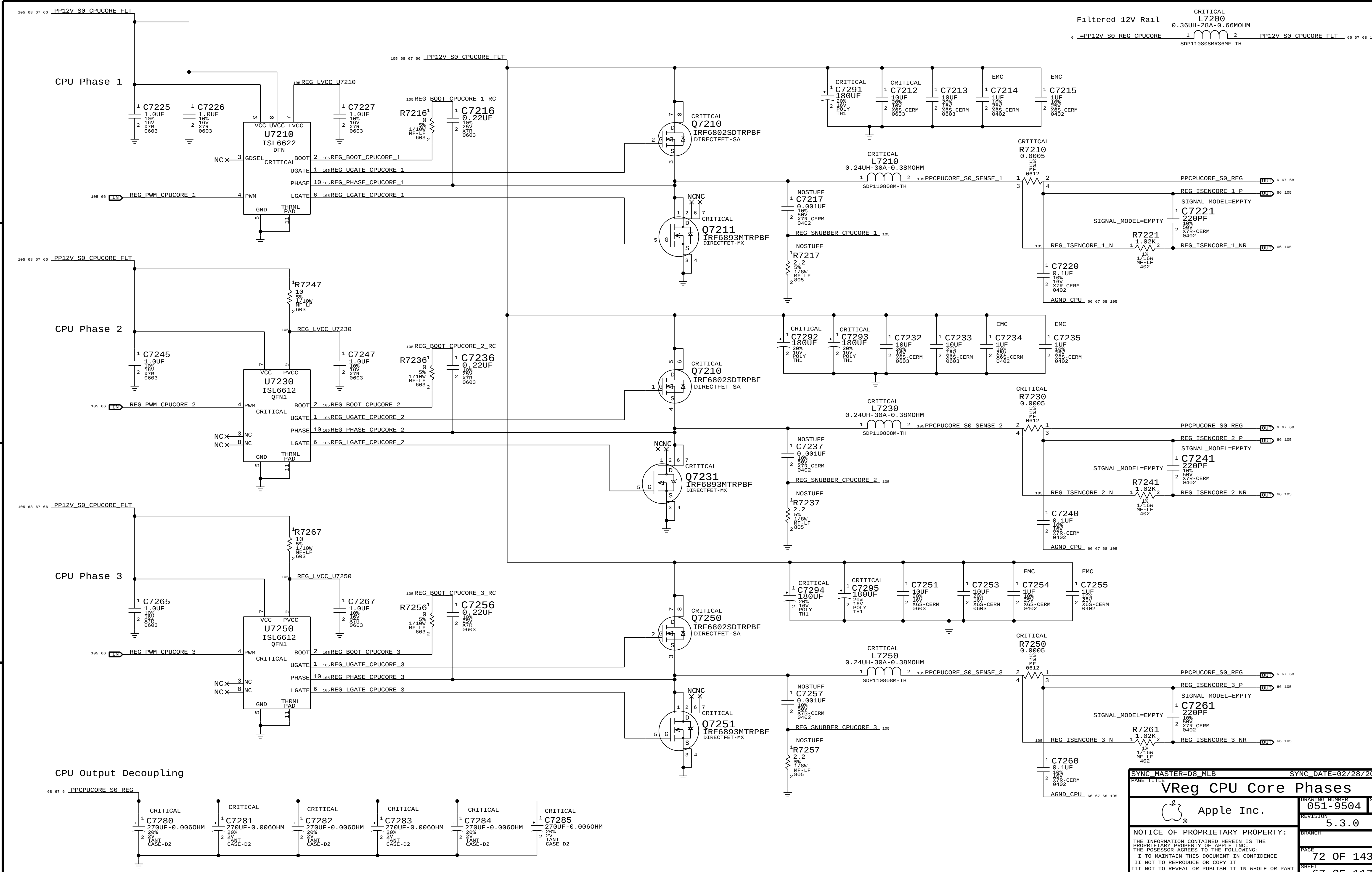

SYNC_MASTER=D8_MLB SYNC_DATE=02/28/2012

PAGE TITLE

VReg CPU Core Phases

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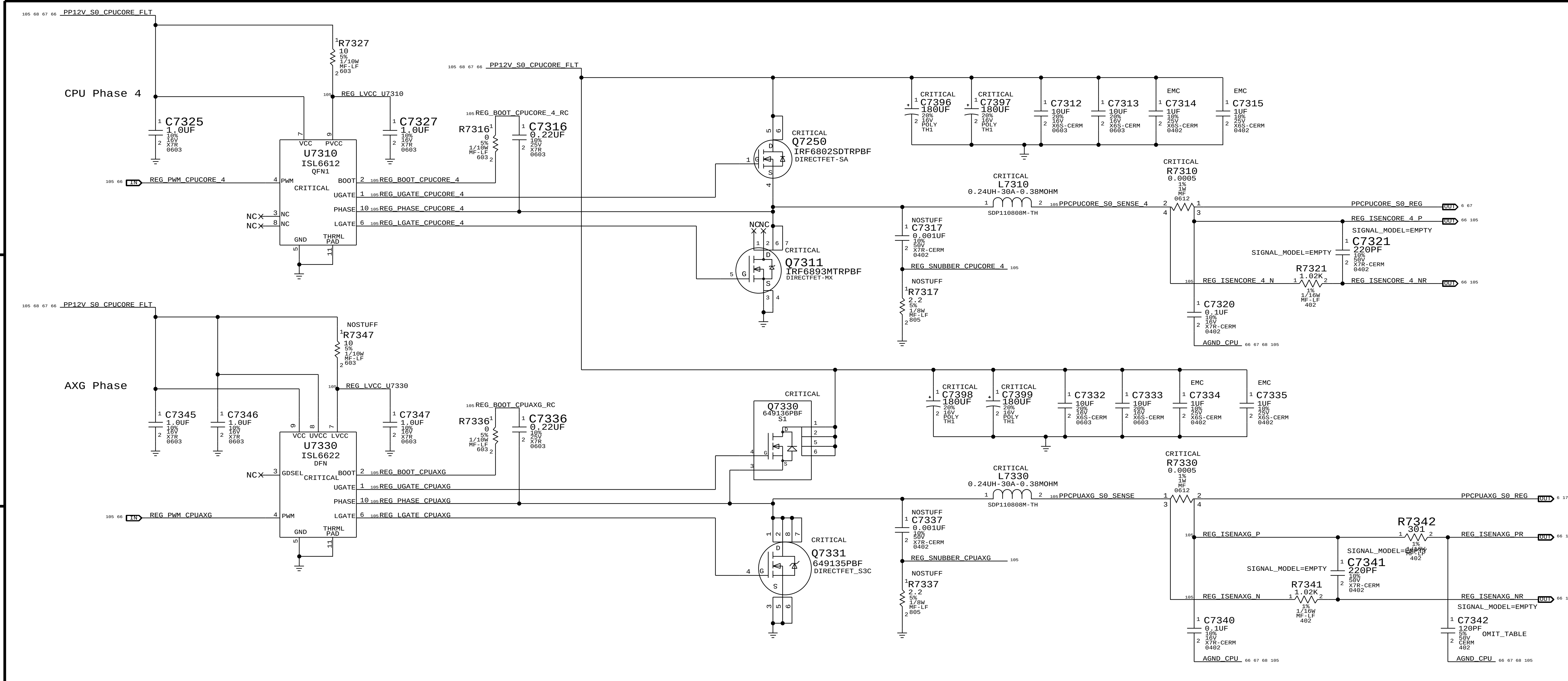

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0513	1	120PF,5%,50V,0402,SMD	C7342		

SYNC_MASTER=D8_MLB SYNC_DATE=02/28/2012

PAGE TITLE

VReg CPU AXG Phases

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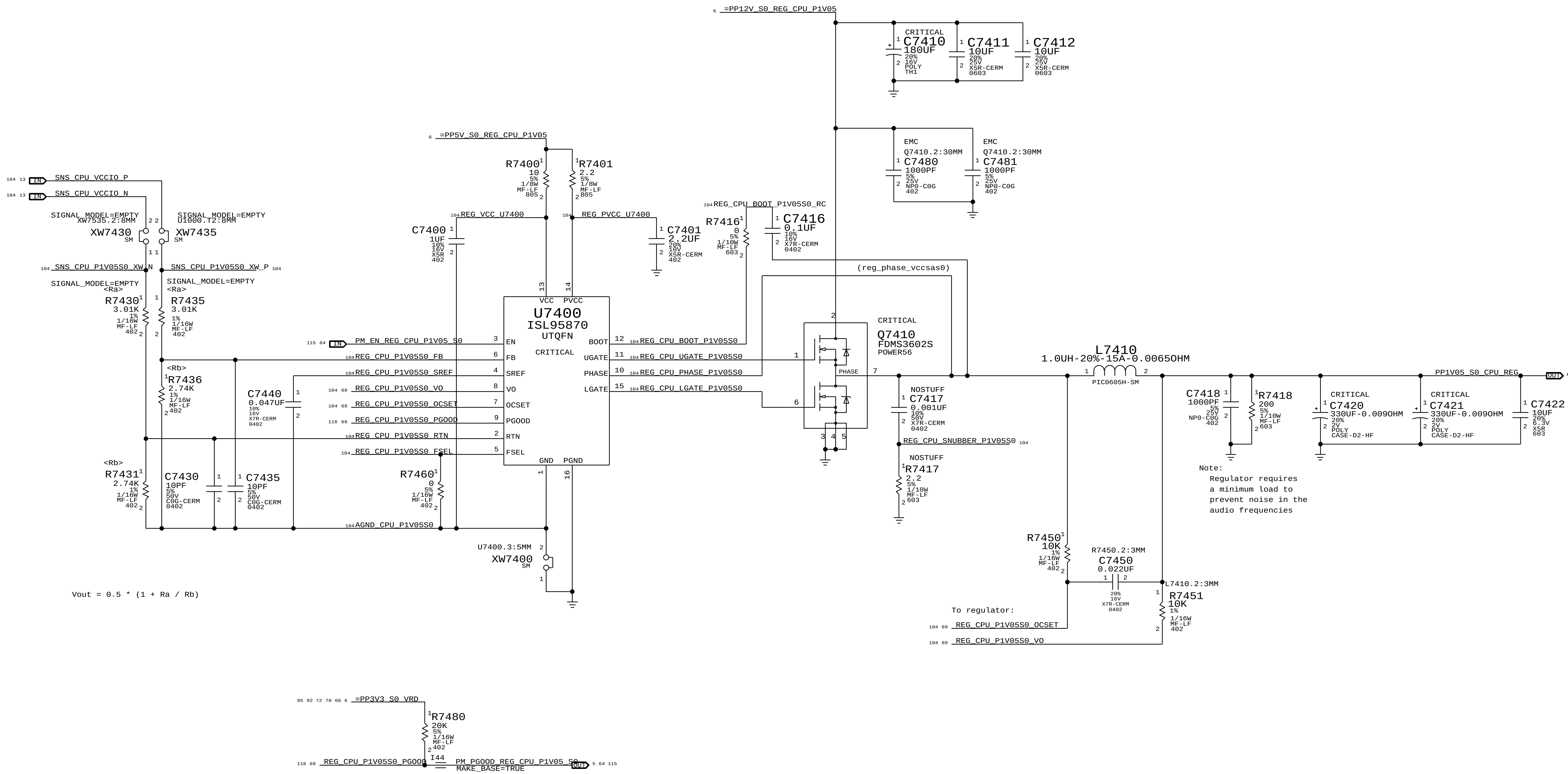
CPU VccIO (1.05V) S0 Regulator

Max avg current: 8.10 A (BUDGET)
Max peak current: 8.50 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC_MASTER=D8_KOSECOFF SYNC_DATE=02/25/2012

PAGE TITLE

VReg CPU 1.05V S0

DRAWING NUMBER 051-9504 SIZE D

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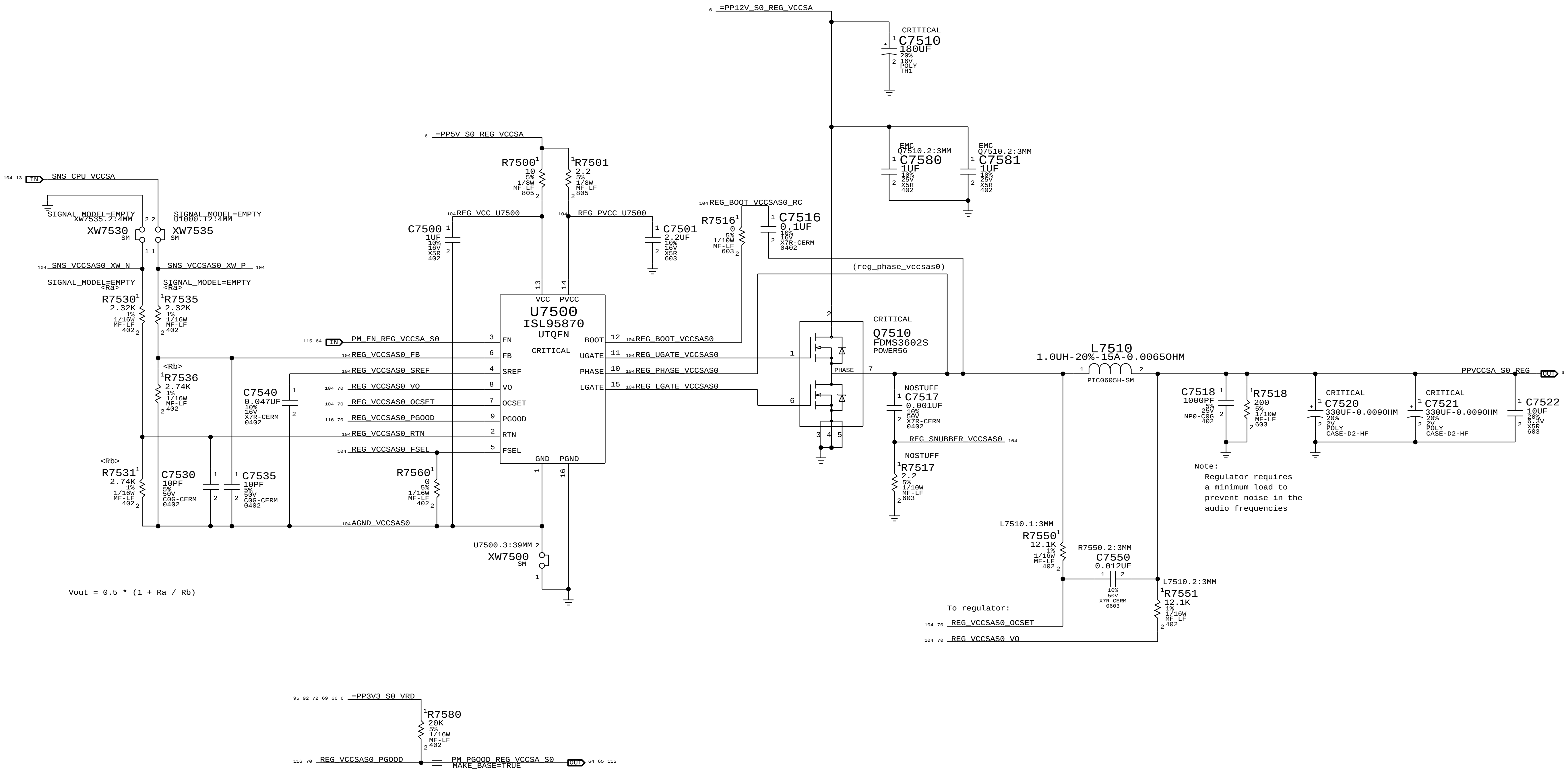
CPU VccSA (0.925V) S0 Regulator

Max avg current: 12.07 A (BUDGET)
Max peak current: 30 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC_MASTER=D8_KOSECOFF SYNC_DATE=02/25/2012

PAGE TITLE
VReg CPU VccSA S0

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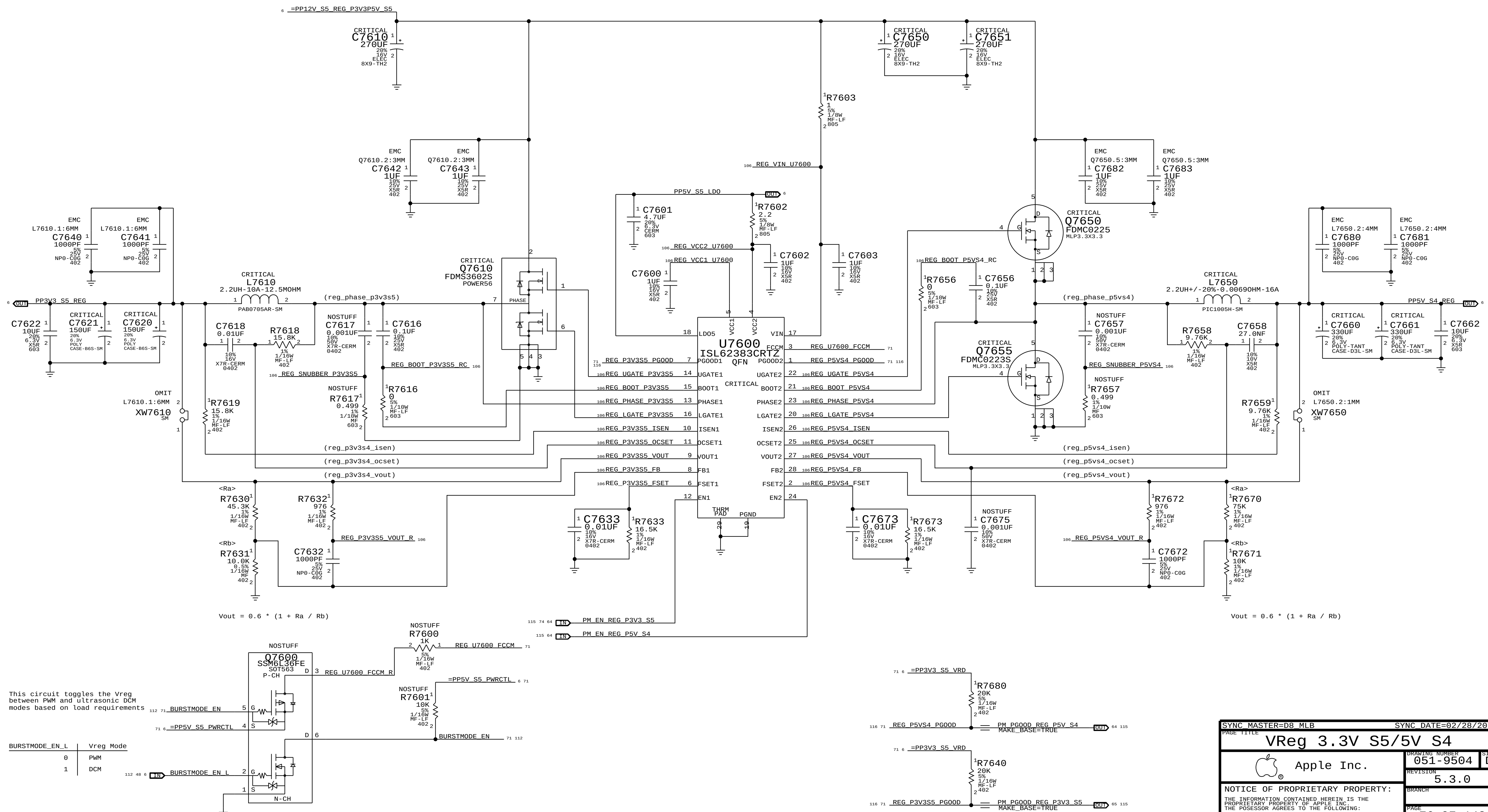
3.3V S5 Regulator

Max avg current: 6 A (design)/ 4.85 A (budget)
Max peak current: ? A (design)/ 6.6 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz

5V S4 Regulator

Max avg current: 10 A (design)/ 6.08 A (budget)
Max peak current: ? A (design)/ 6.9 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz



BURSTMODE_EN_L	Vreg Mode
0	PWM
1	DCM

SYNC_MASTER=D8_MLB SYNC_DATE=02/28/2012

PAGE TITLE: VReg 3.3V S5/5V S4

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DRAWING NUMBER: 051-9504 SIZE: D

REVISION: 5.3.0

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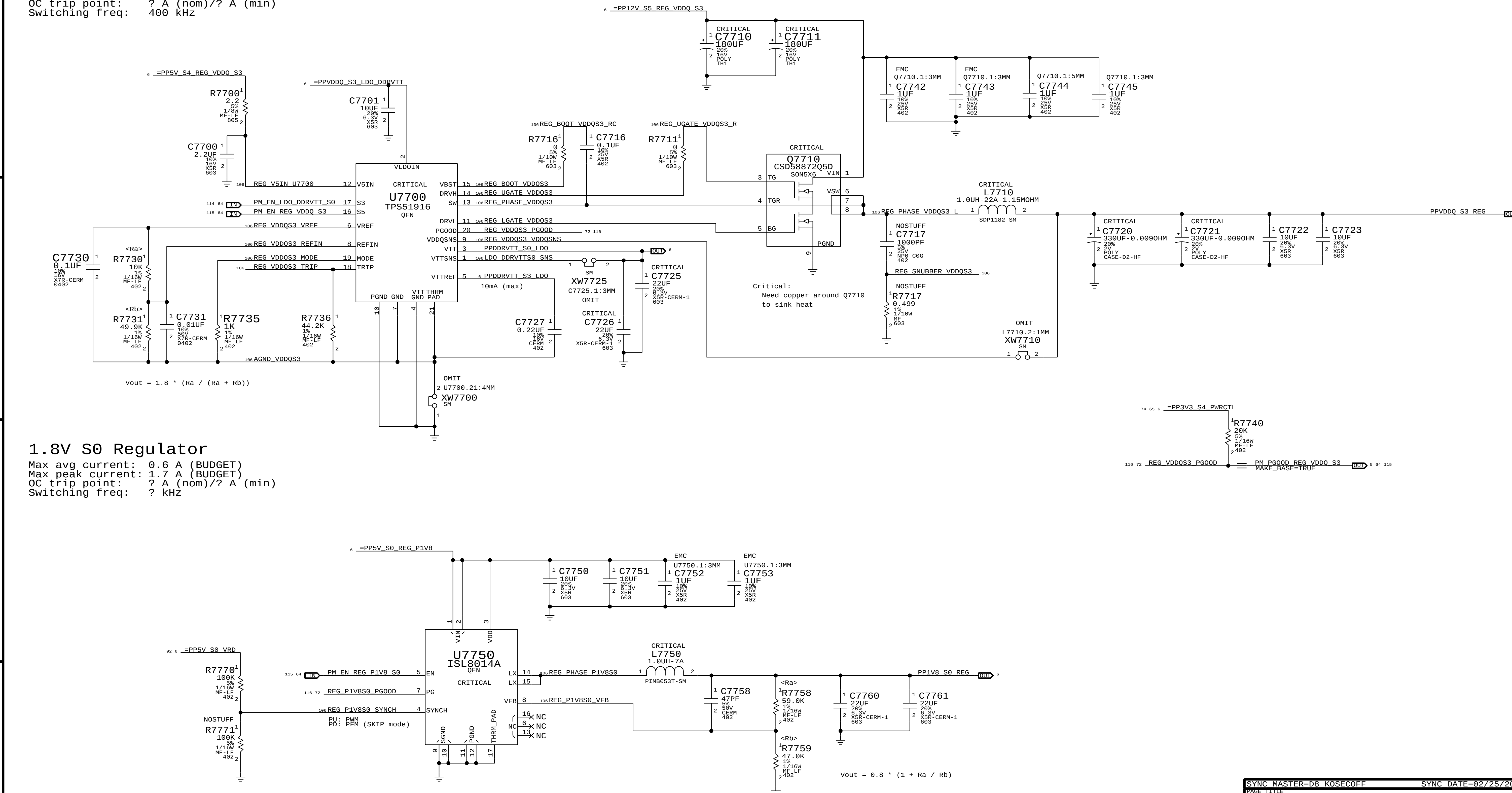
VDDQ (1.5V) S3 Regulator

Max avg current: 9.0 A (BUDGET)
Max peak current: 11.3 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 400 kHz

1.8V S0 Regulator

Max avg current: 0.6 A (BUDGET)
Max peak current: 1.7 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz



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PAGE TITLE: VReg VDDQ and 1.8V S0

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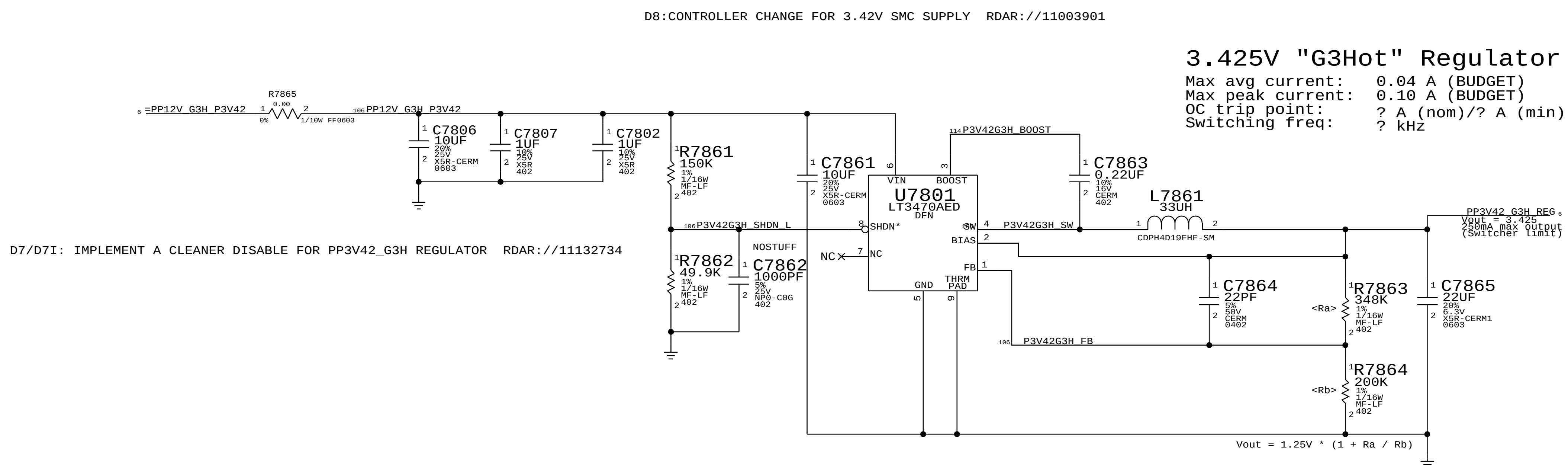

SYNC_MASTER=D8_MLB SYNC_DATE=04/11/2012

PAGE TITLE

VREG 3.42V G3HOT

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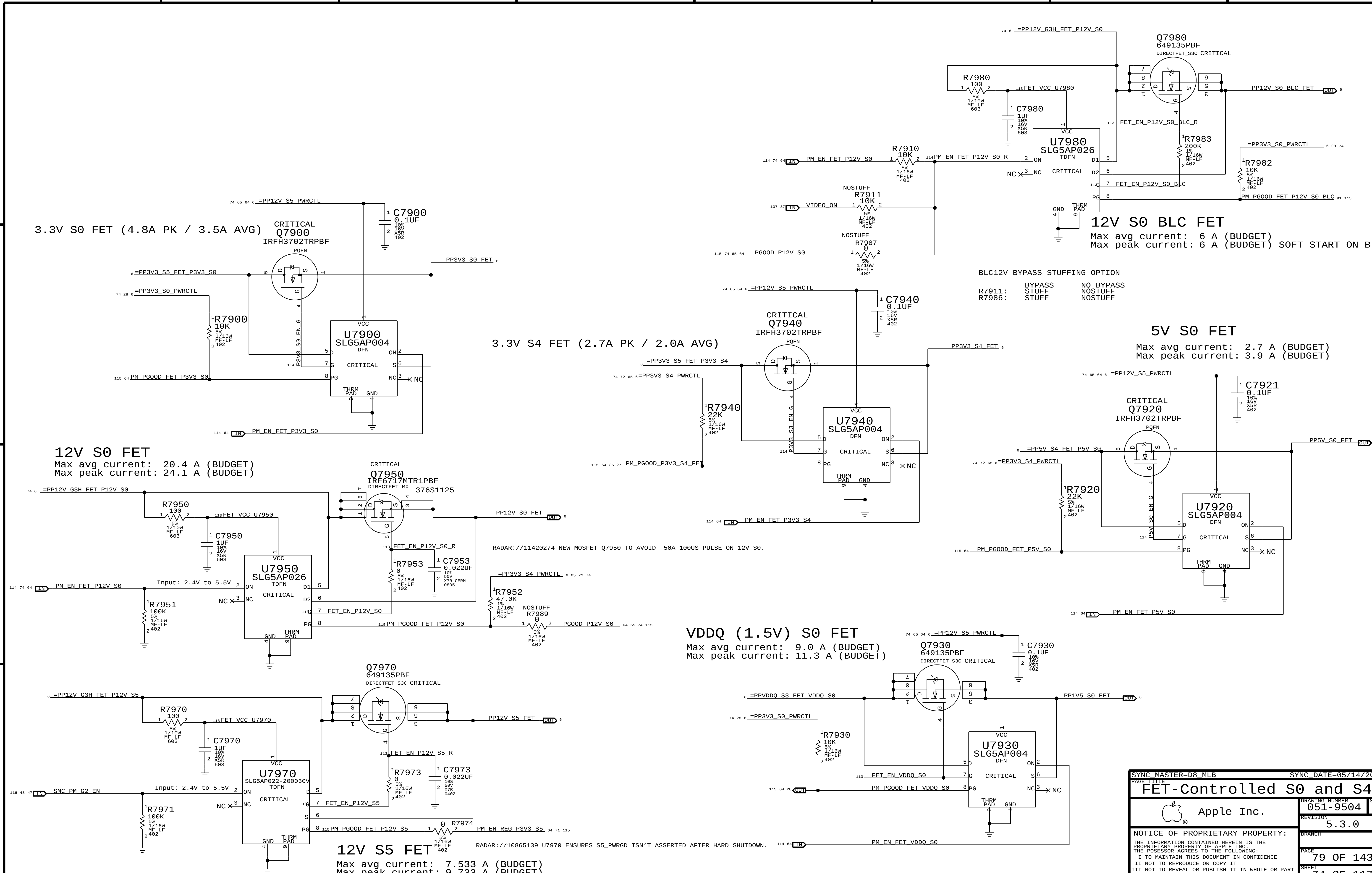

SYNC_MASTER=D8_MLB SYNC_DATE=05/14/2012

PAGE TITLE

FET-Controlled S0 and S4

Apple Inc.

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SYNC_MASTER=D8_AARON SYNC_DATE=03/13/2012

PAGE TITLE

KEPLER PCI-E

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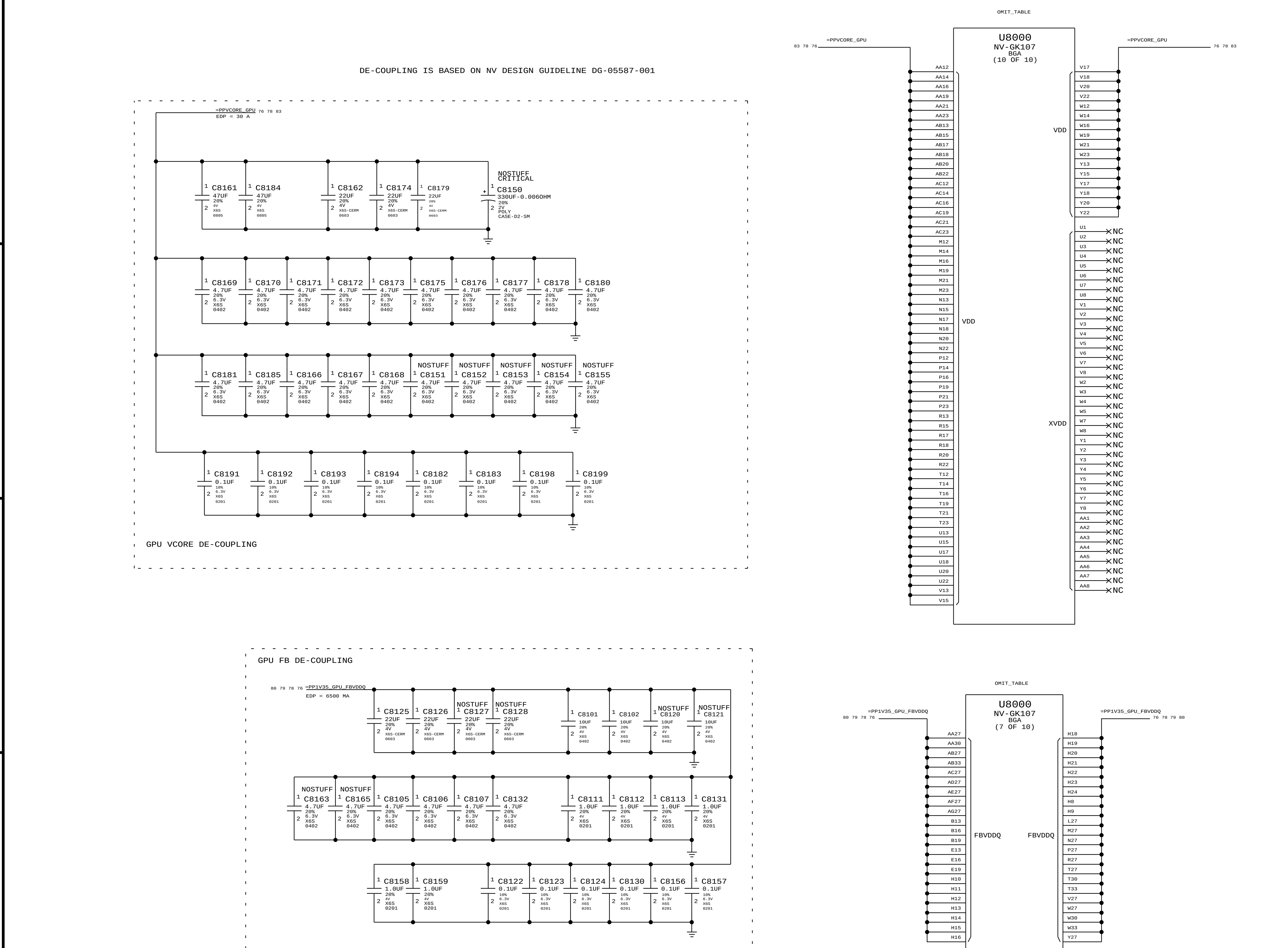

Page Notes

SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

KEPLER CORE/FB POWER

Apple Inc.

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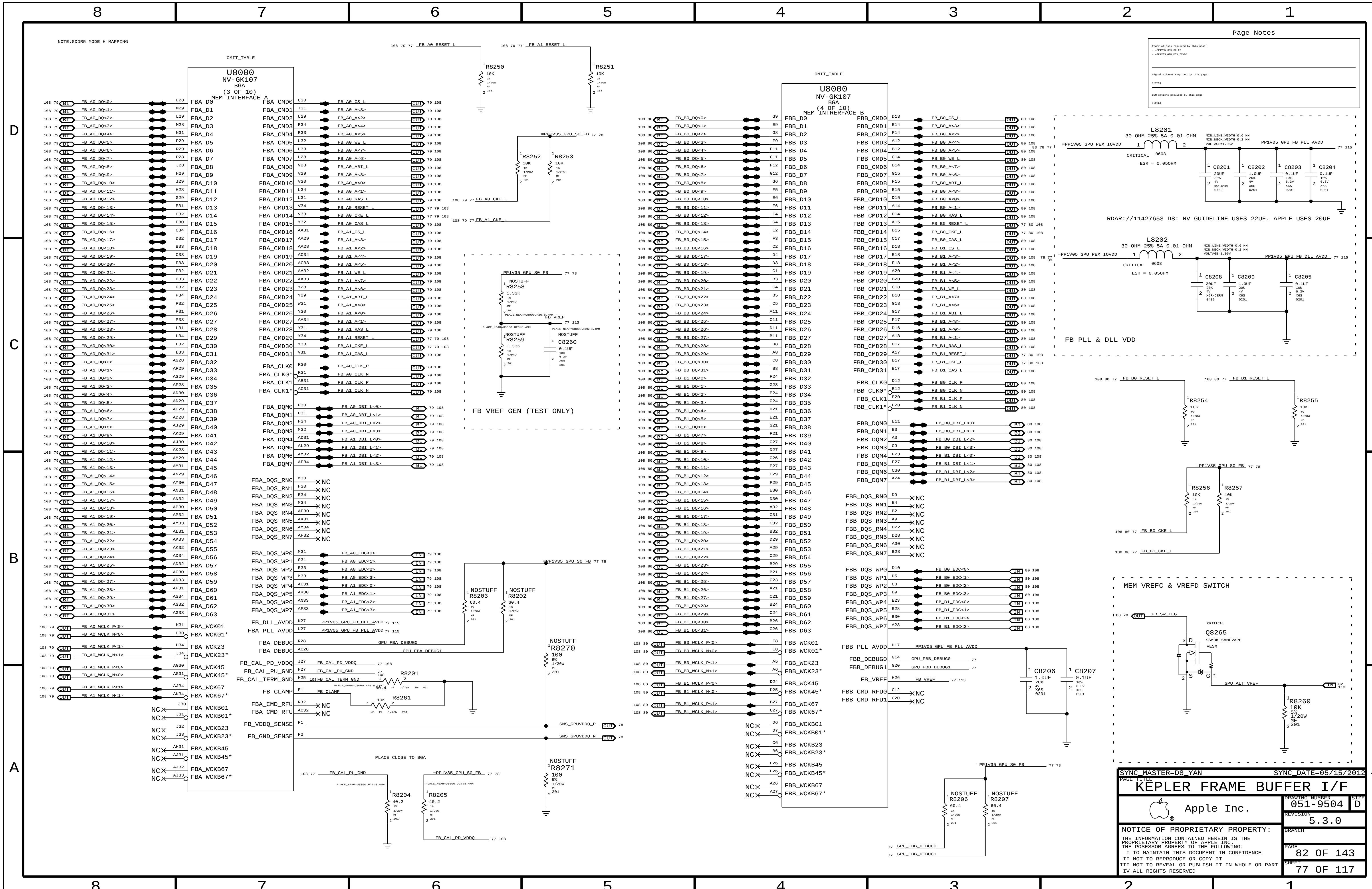

SYNC_MASTER=D8_YAN

SYNC_DATE=05/15/2012

PAGE TITLE

KEPLER FRAME BUFFER I/F

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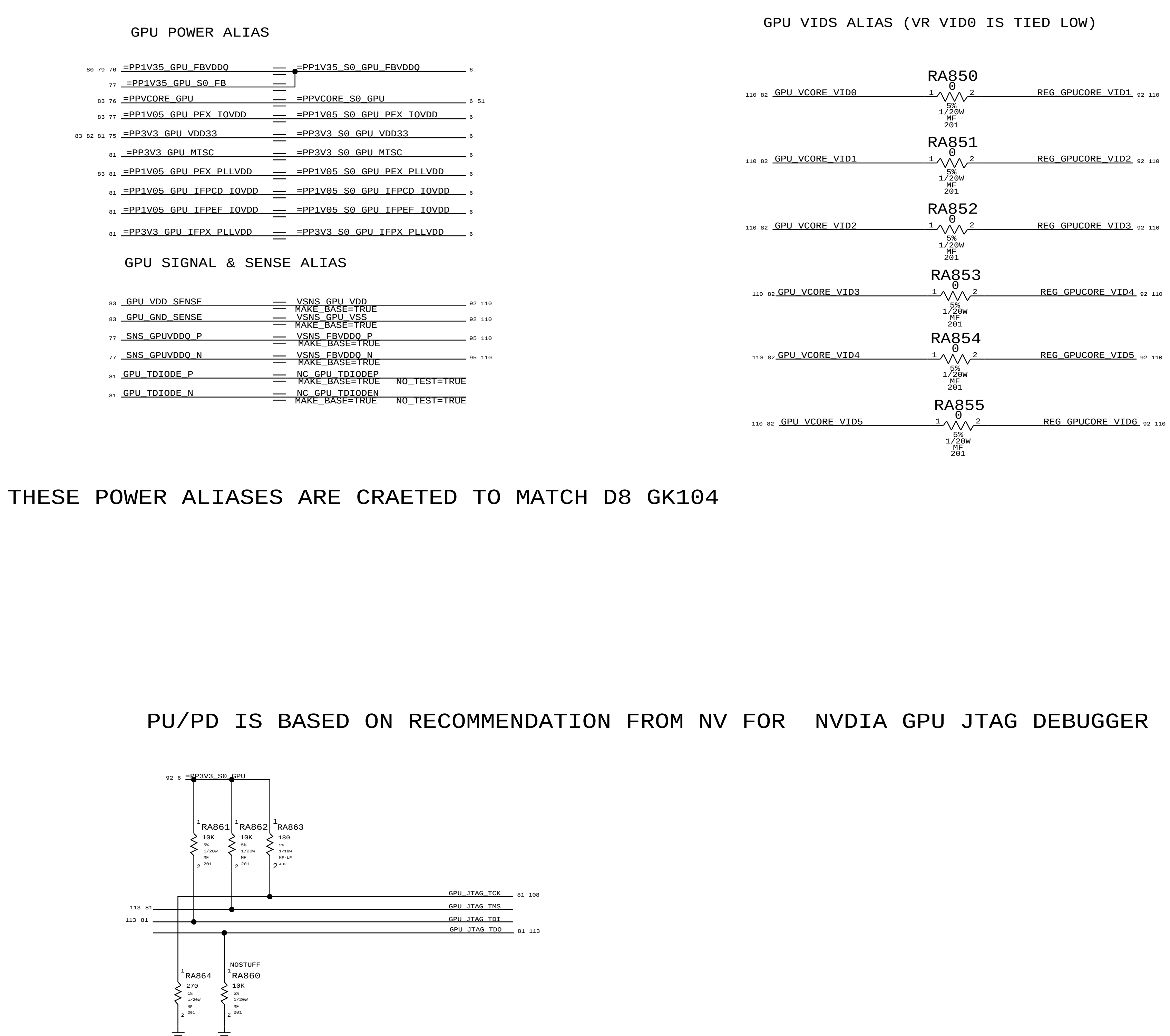

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PAGE TITLE
GPU SIGNAL & POWER ALIASES

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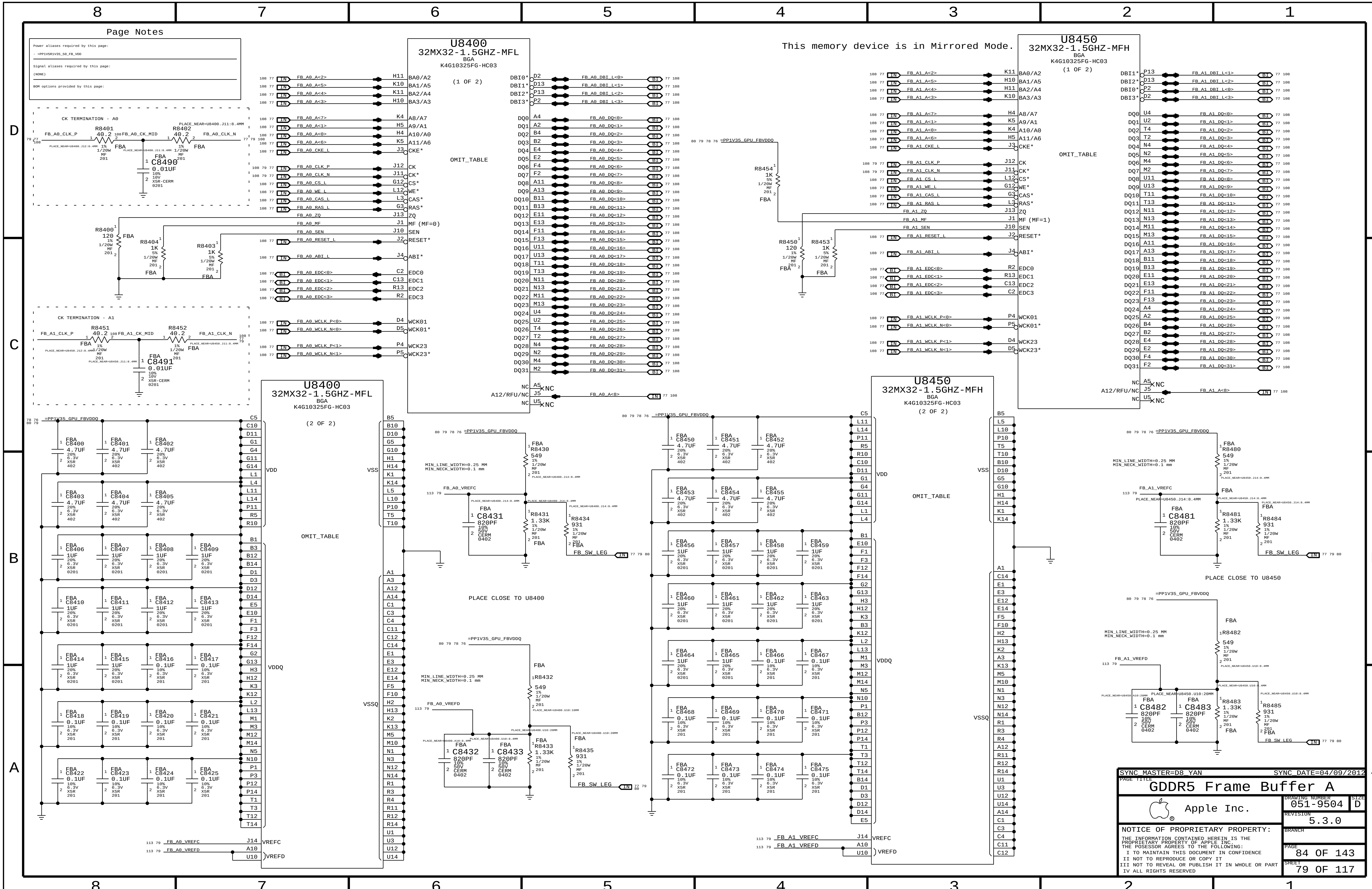

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PAGE TITLE

GDDR5 Frame Buffer A

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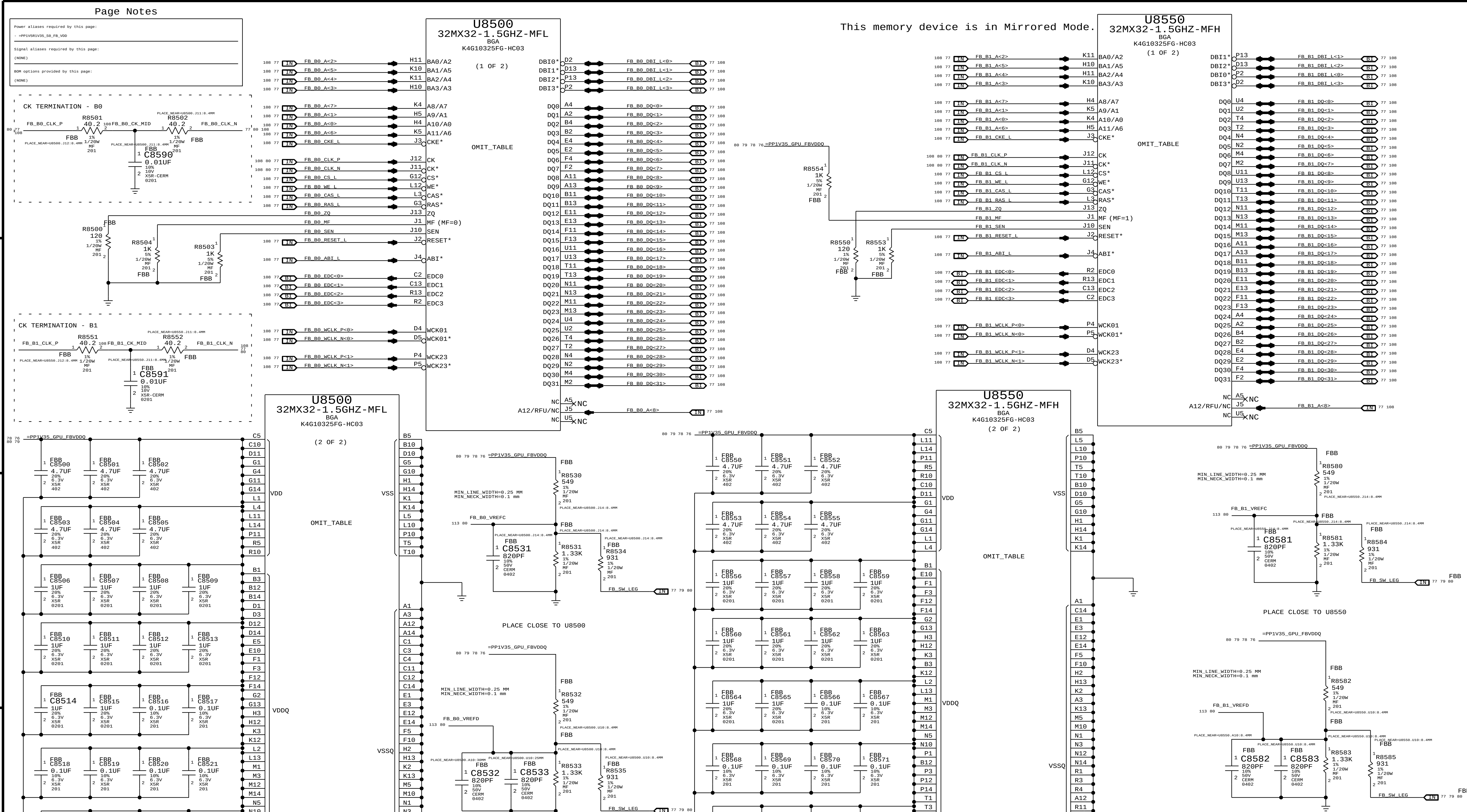

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PAGE TITLE

GDDR5 Frame Buffer B

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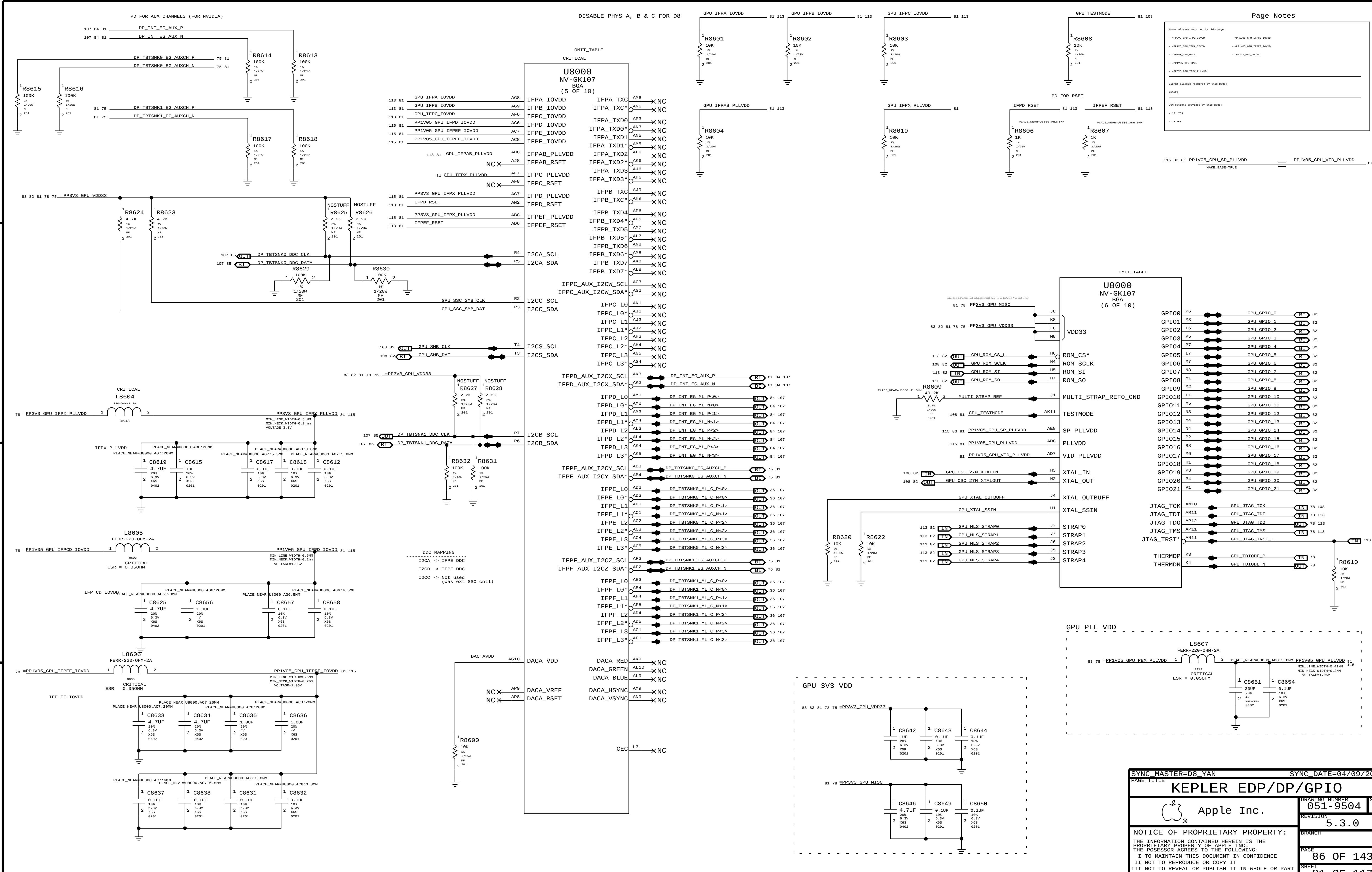

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SYNC_DATE=04/09/2012

PAGE TITLE
KEPLER EDP/DP/GPIO

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SIZE D
REVISION 5.3.0
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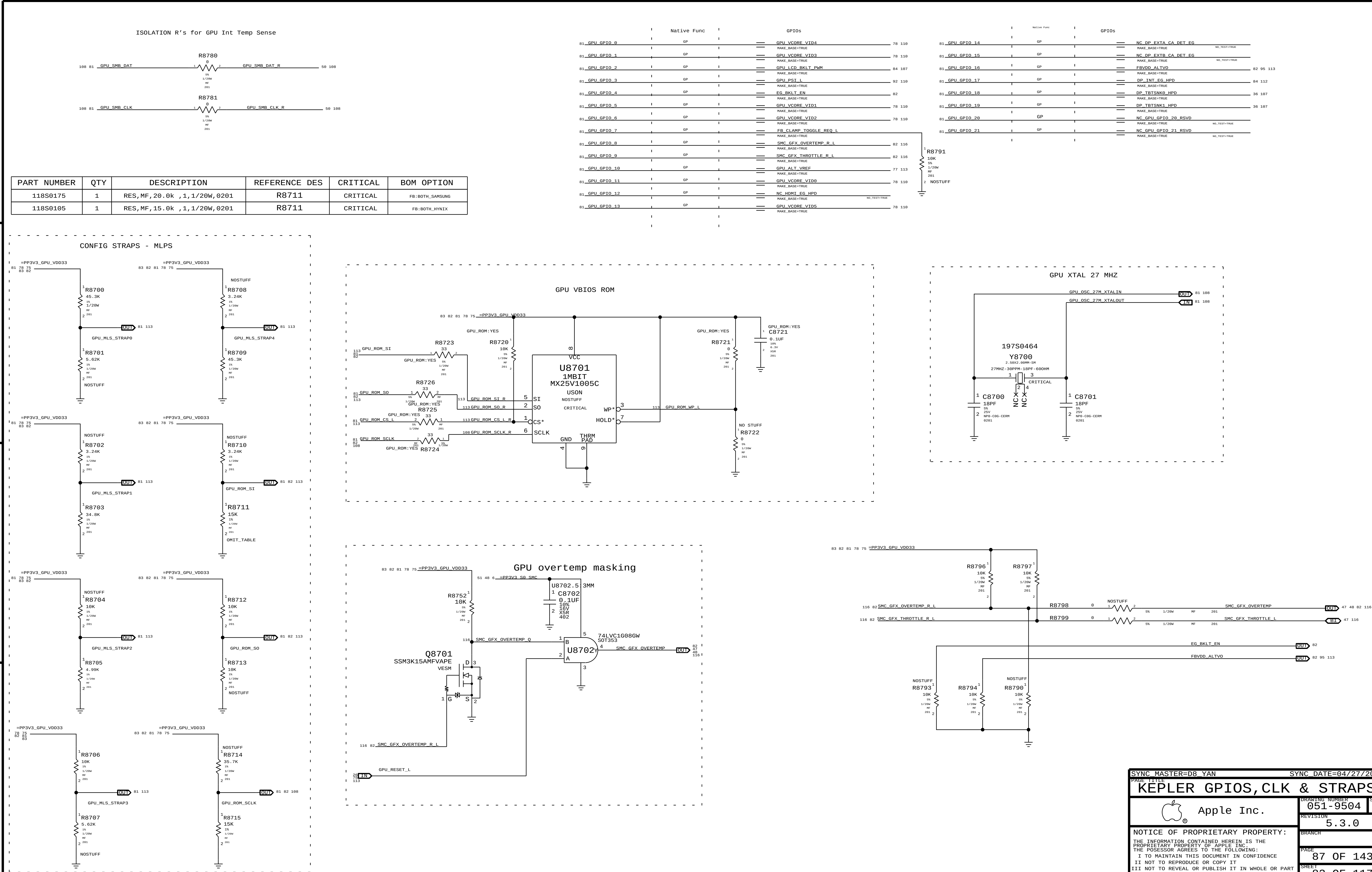

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118S0105	1	RES,MF,15.0k ,1,1/20W,0201	R8711	CRITICAL	FB:BOTH_HYNIX

SYNC_MASTER=D8_YAN SYNC_DATE=04/27/2012

PAGE TITLE

KEPLER GPIOS,CLK & STRAPS

Apple Inc.

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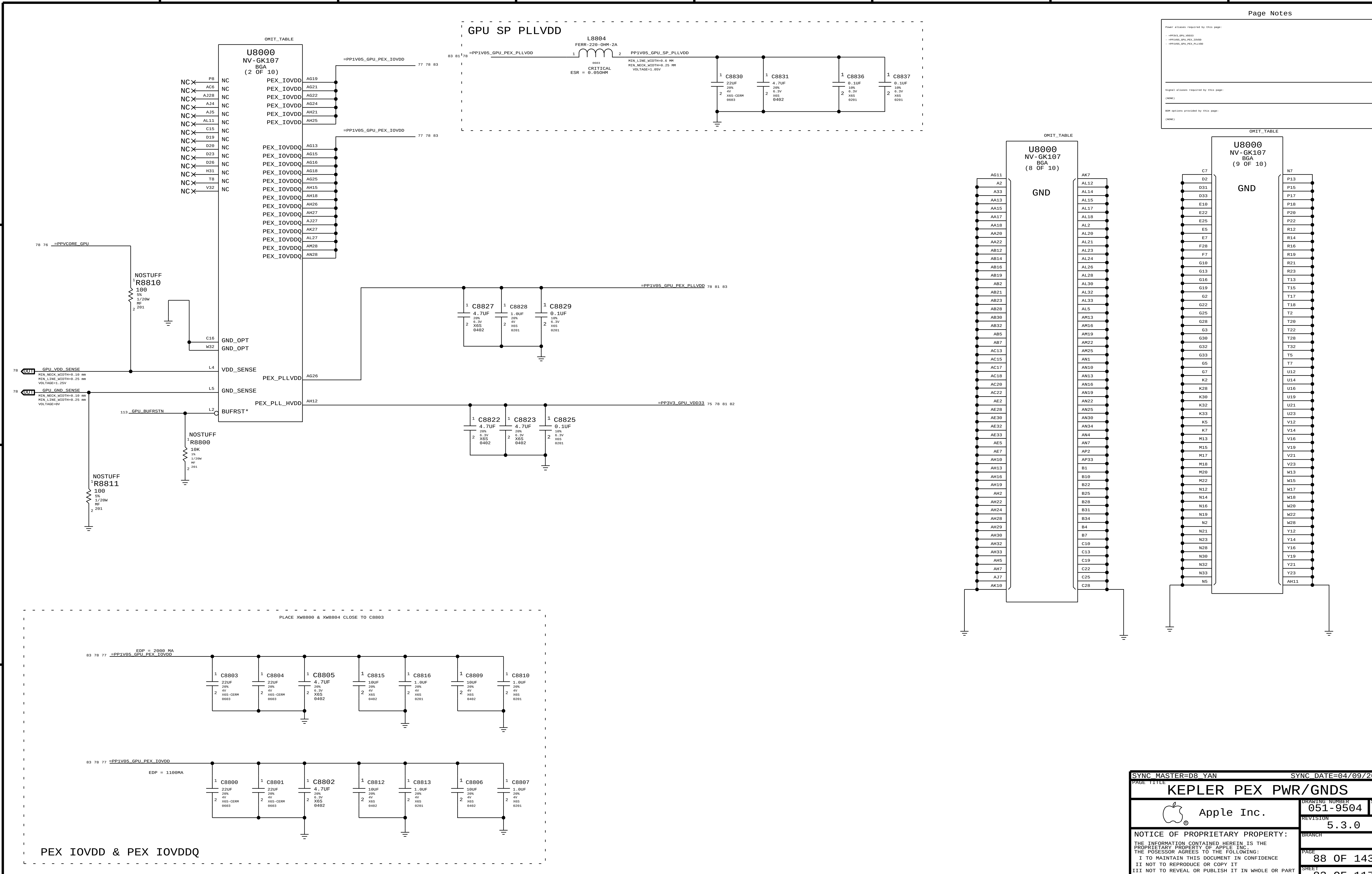

SYNC_MASTER=D8_YAN SYNC_DATE=04/09/2012

PAGE TITLE

KEPLER PEX PWR/GNDS

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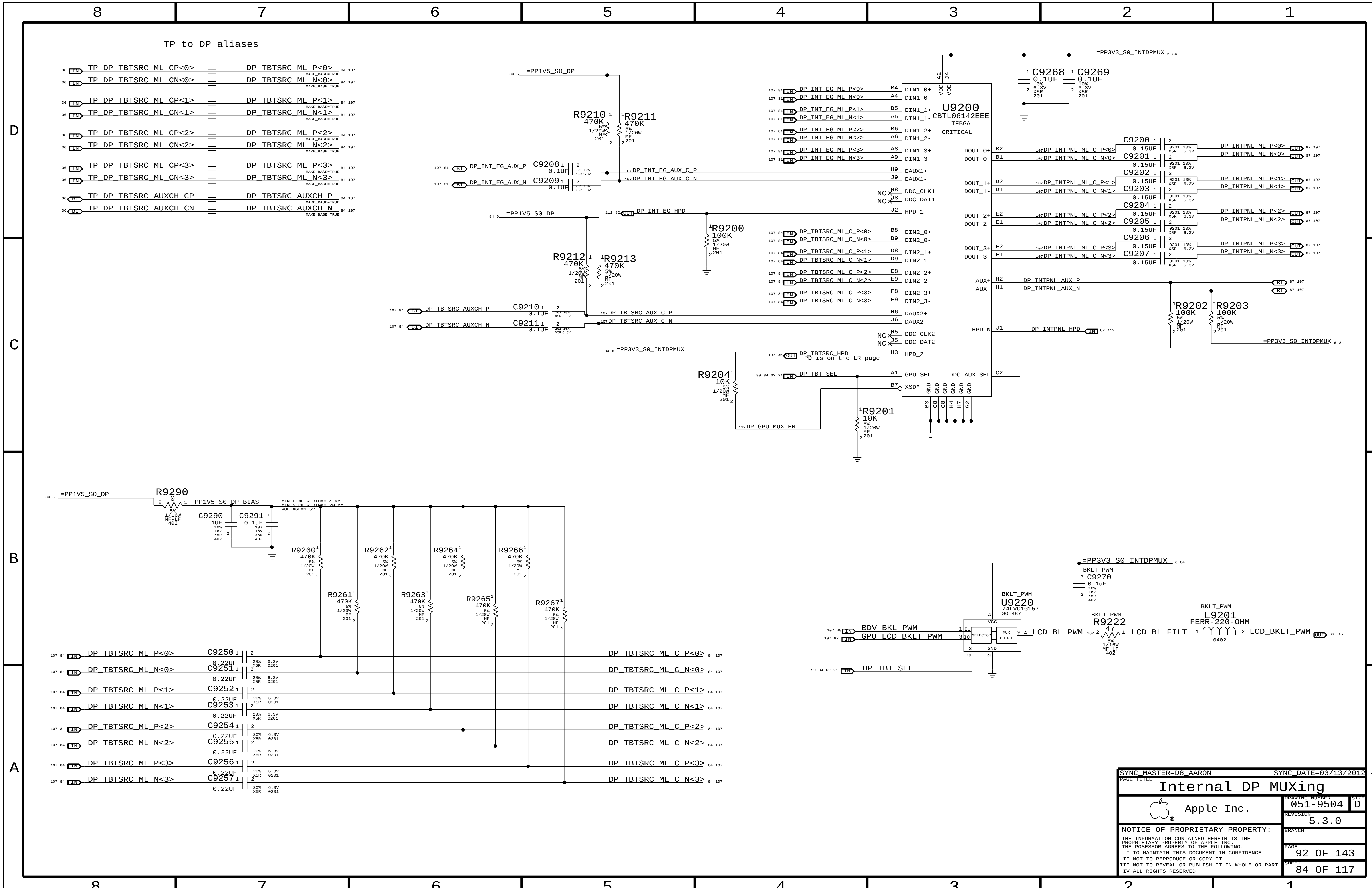

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PAGE TITLE

Internal DP MUXing

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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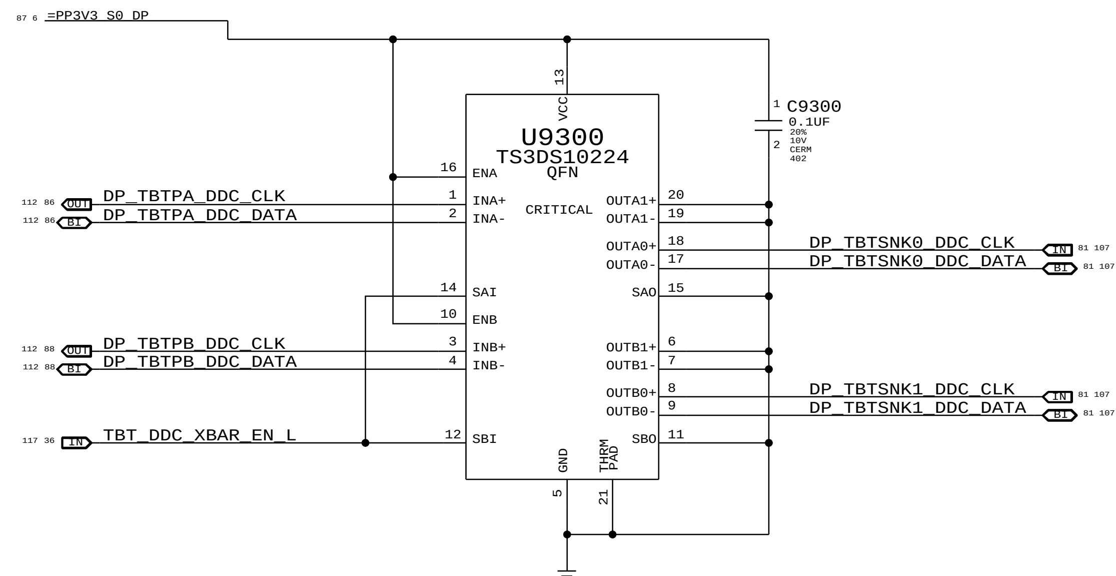
Dual-Port Host DDC Crossbar



SYNC_MASTER=D7_MLB SYNC_DATE=03/15/2012

PAGE TITLE

TBT DDC Crossbar

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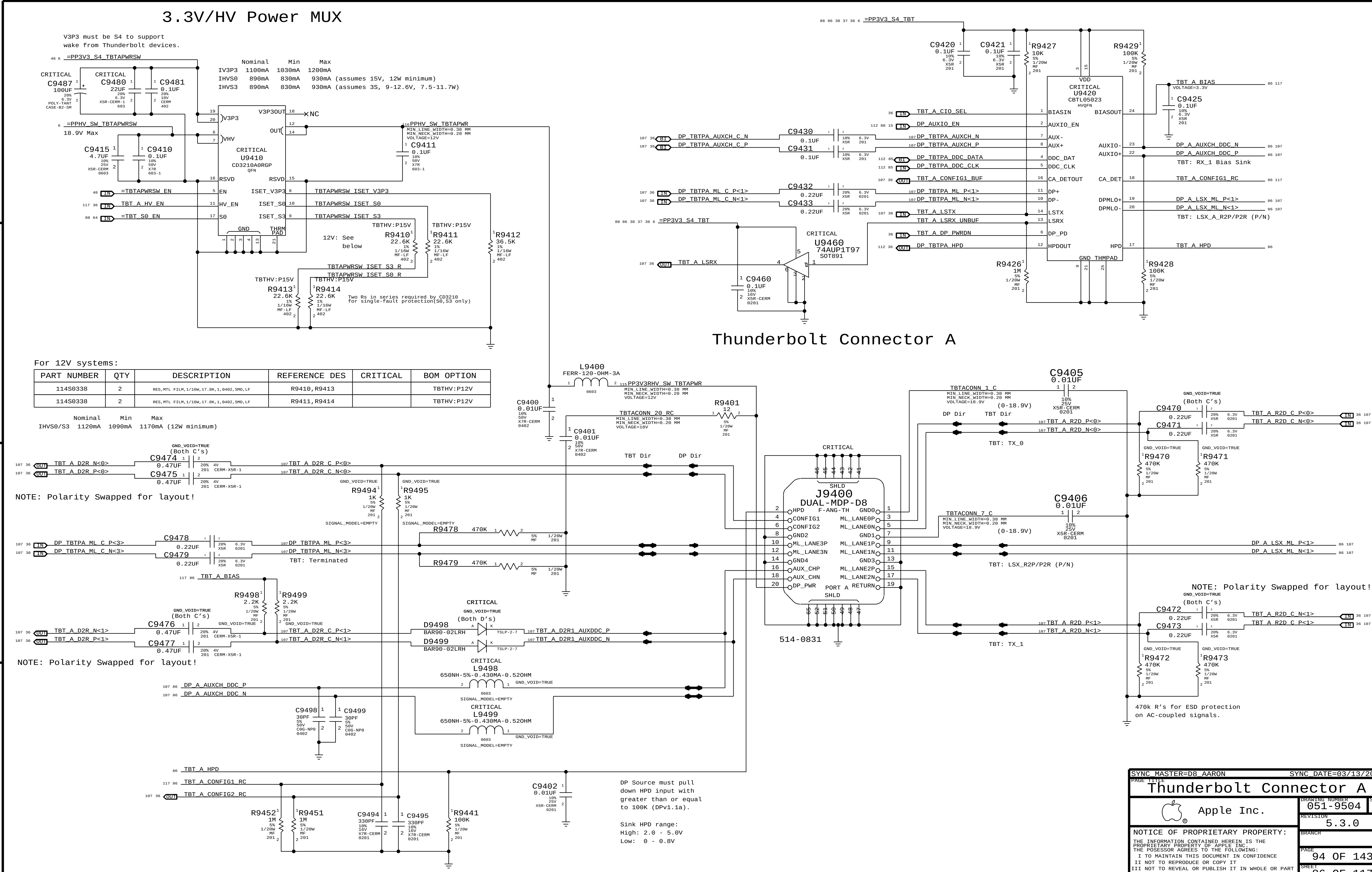

For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
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114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9411,R9414		TBTHV:P12V

SYNC_MASTER=D8_AARON SYNC_DATE=03/13/2012

PAGE TITLE: Thunderbolt Connector A

Apple Inc.

DRAWING NUMBER: 051-9504 SIZE: D

REVISION: 5.3.0

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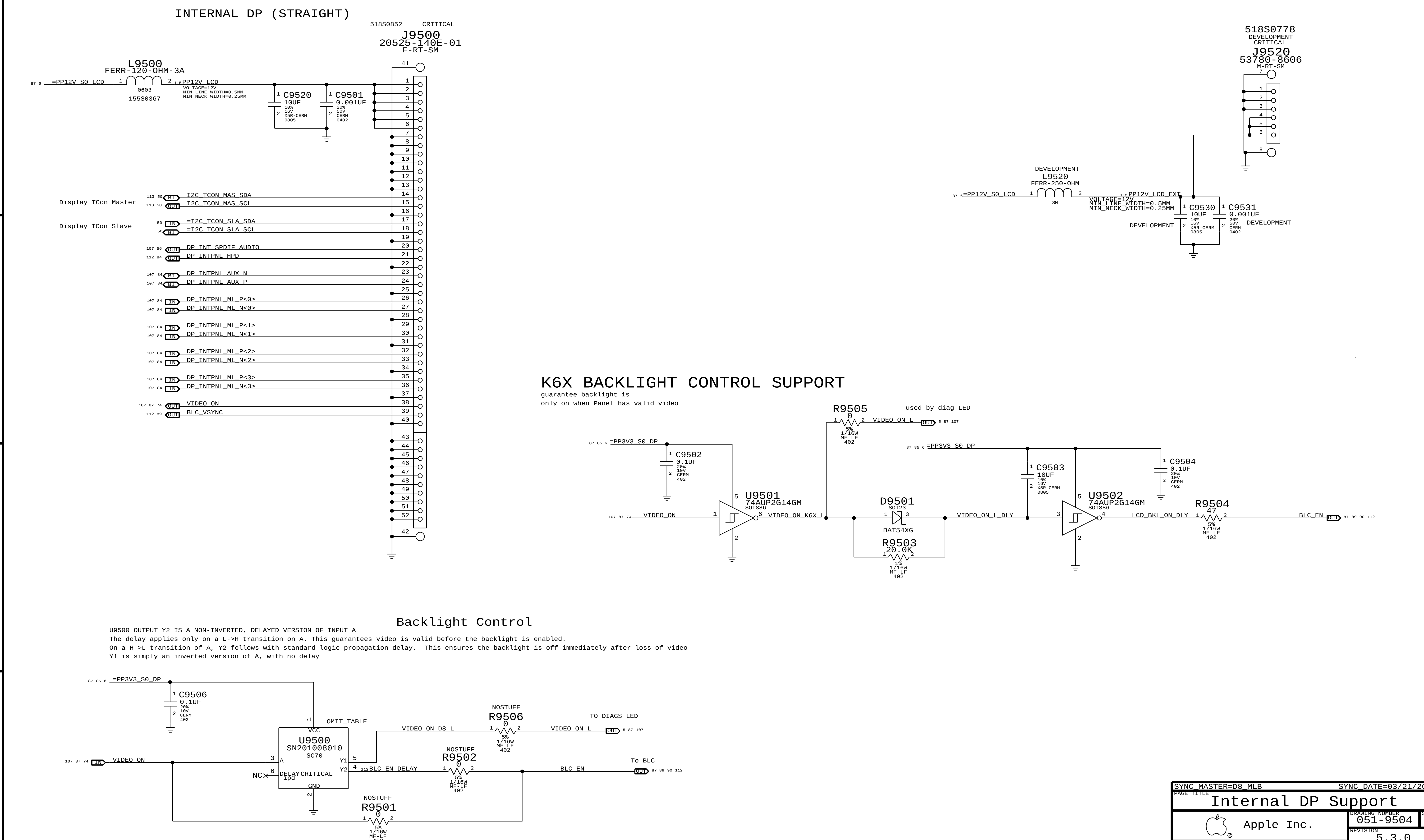

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
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SYNC_MASTER=D8_MLB SYNC_DATE=03/21/2012

PAGE TITLE

Internal DP Support

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

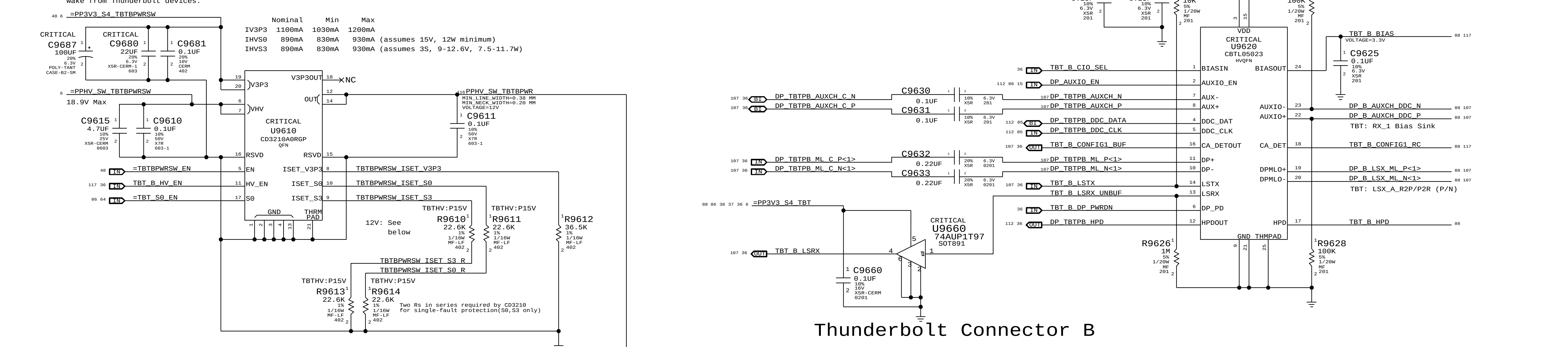
REVISION 5.3.0

BRANCH

3.3V/HV Power MUX

V3P3 must be S4 to support
wake from Thunderbolt devices.

	Nominal	Min	Max	
IV3P3	1100mA	1030mA	1200mA	
IHVS0	890mA	830mA	930mA	(assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA	(assumes 3S, 9-12.6V, 7.5-11.7W)

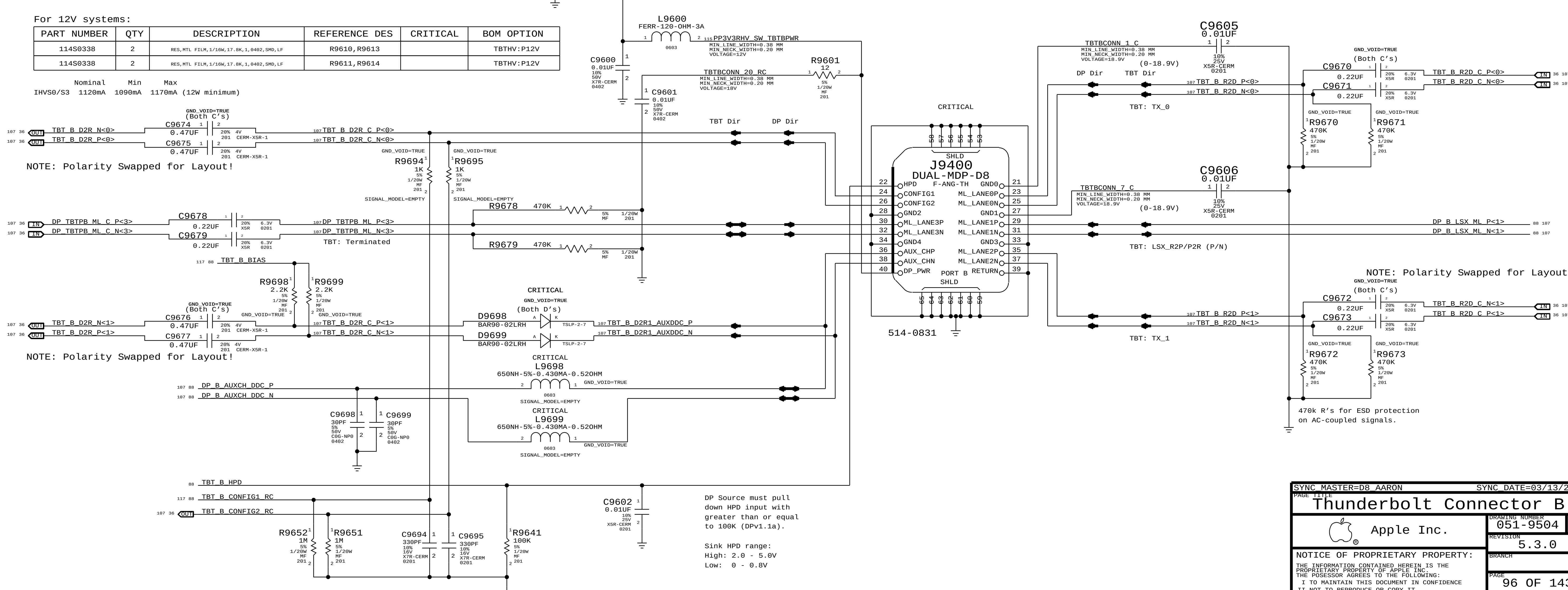


Thunderbolt Connector B

For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9610,R9613		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9611,R9614		TBTHV:P12V

	Nominal	Min	Max	
IHVS0/S3	1120mA	1090mA	1170mA	(12W minimum)



SYNC_MASTER=D8_AARON SYNC_DATE=03/13/2012

PAGE TITLE: Thunderbolt Connector B

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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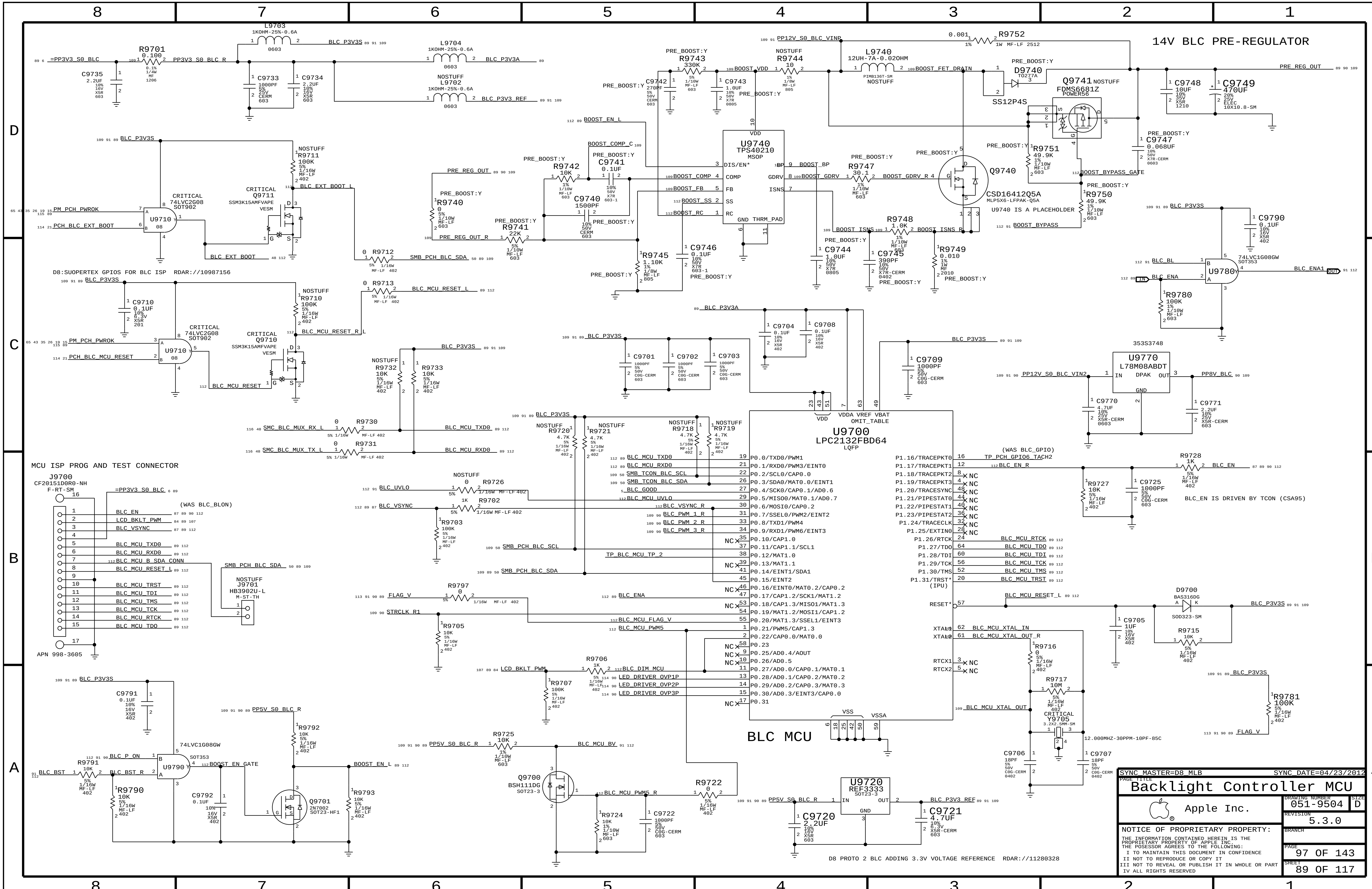

14V BLC PRE-REGULATOR
D8:SUOPERTEX GPIOS FOR BLC ISP RDAR://10987156
MCU ISP PROG AND TEST CONNECTOR
(WAS BLC_BLON)
APN 998-3605
Q9740 IS A PLACEHOLDER
U9700
LPC2132FBD64
(WAS BLC_GPIO)
BLC_EN IS DRIVEN BY TCON (CSA95)
BLC MCU
D8 PROTO 2 BLC ADDING 3.3V VOLTAGE REFERENCE RDAR://11280328
SYNC_MASTER=D8_MLB
SYNC_DATE=04/23/2012
PAGE TITLE
Backlight Controller MCU
Apple Inc.
DRAWING NUMBER
051-9504
SIZE
D
REVISION
5.3.0
BRANCH
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SHEET
89 OF 117
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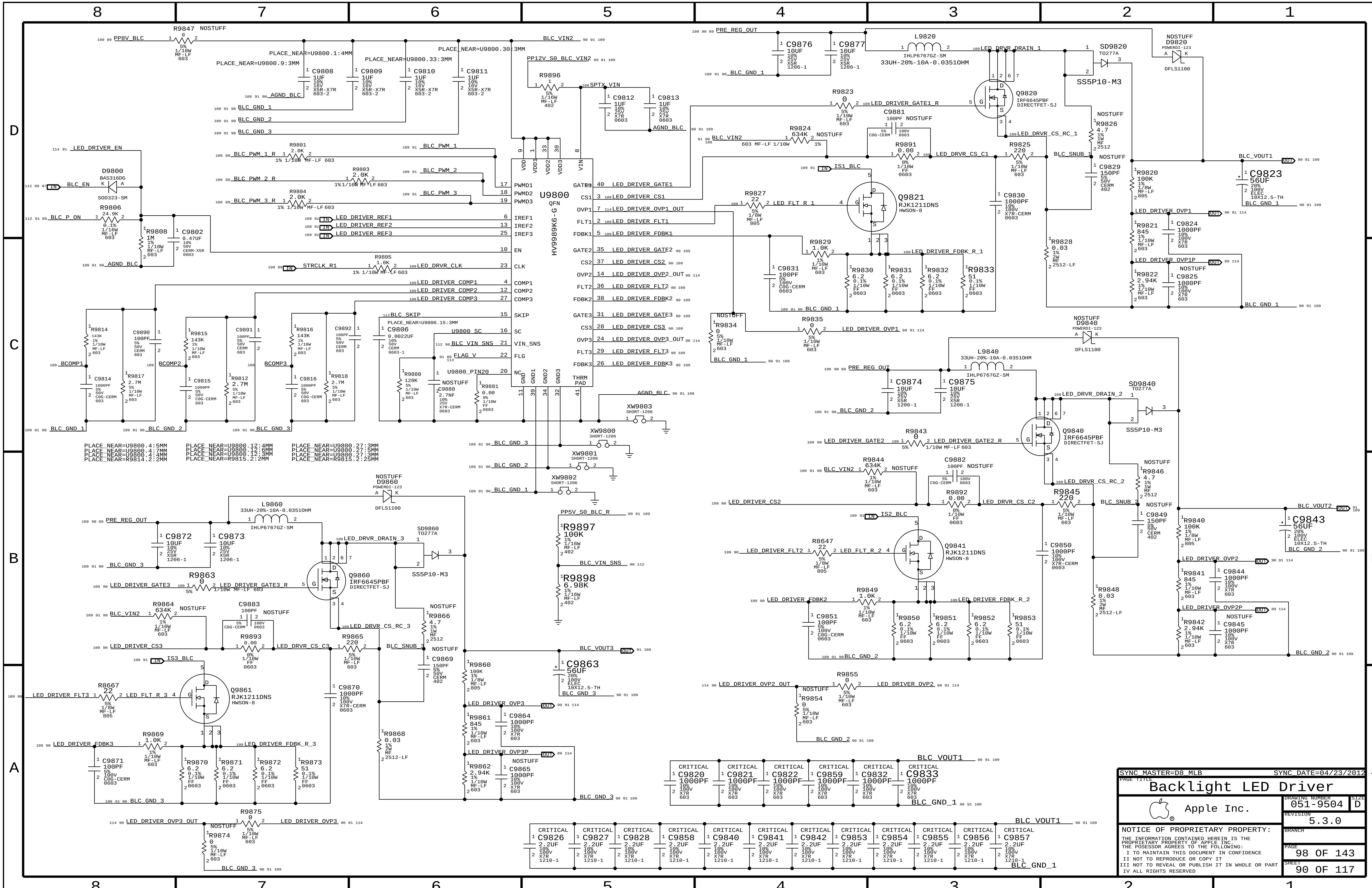

Backlight LED Driver
SYNC_MASTER=D8_MLB
SYNC_DATE=04/23/2012
Apple Inc.
DRAWING NUMBER 051-9504
SIZE D
REVISION 5.3.0
PAGE 98 OF 143
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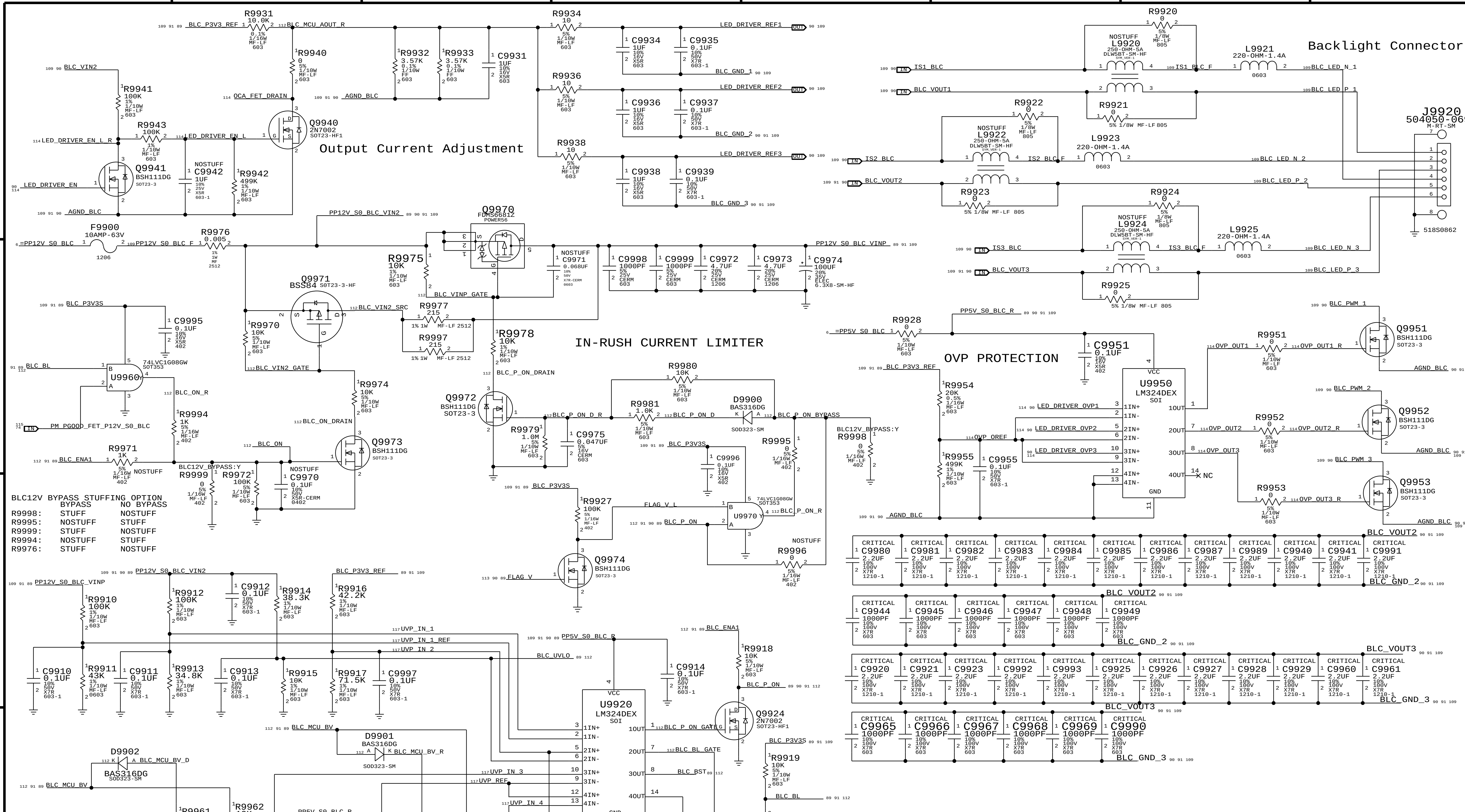
U9800
HV9989K6-G
QFN
Q9820 IRF6645PBF DIRECTFET-SJ
Q9840 IRF6645PBF DIRECTFET-SJ
Q9860 IRF6645PBF DIRECTFET-SJ
Q9821 RJK1211DNS HWSON-8
Q9841 RJK1211DNS HWSON-8
Q9861 RJK1211DNS HWSON-8
L9820 IHLP6767GZ-SM 33UH-20%-10A-0.0351OHM
L9840 33UH-20%-10A-0.0351OHM IHLP6767GZ-SM
L9860 33UH-20%-10A-0.0351OHM IHLP6767GZ-SM
SD9820 SS5P10-M3
SD9840 SS5P10-M3
SD9860 SS5P10-M3
BLC_VOUT1
BLC_VOUT2
BLC_VOUT3



SYNC_MASTER=D8_MLB SYNC_DATE=04/23/2012

PAGE TITLE

Backlight Controller

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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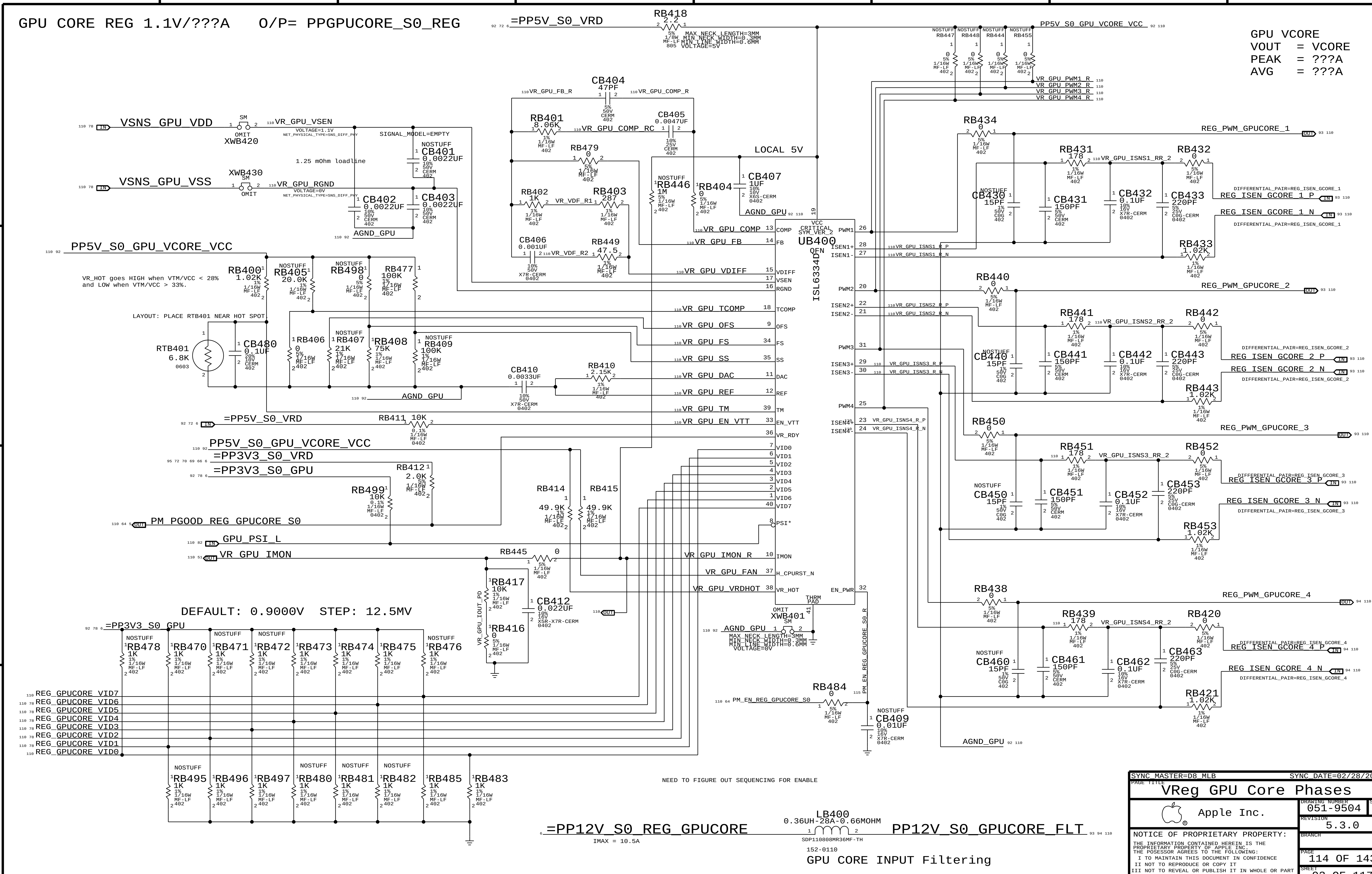

SYNC_MASTER=D8_MLB SYNC_DATE=02/28/2012

PAGE TITLE

VReg GPU Core Phases

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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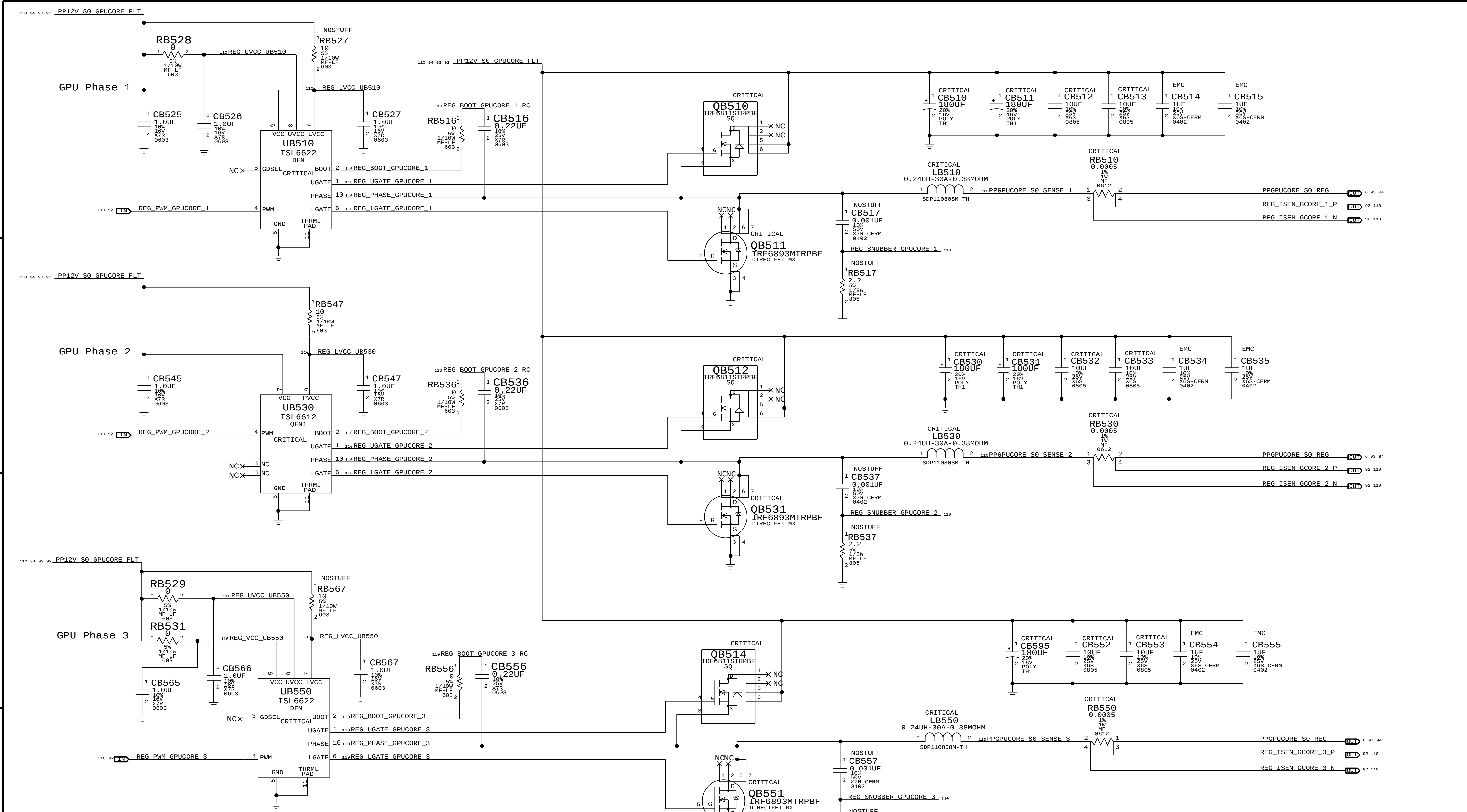
PAGE 114 OF 143

SHEET 92 OF 117



SYNC_MASTER=D8_MLB SYNC_DATE=02/25/2012

PAGE TITLE

VReg GPU Core Phases

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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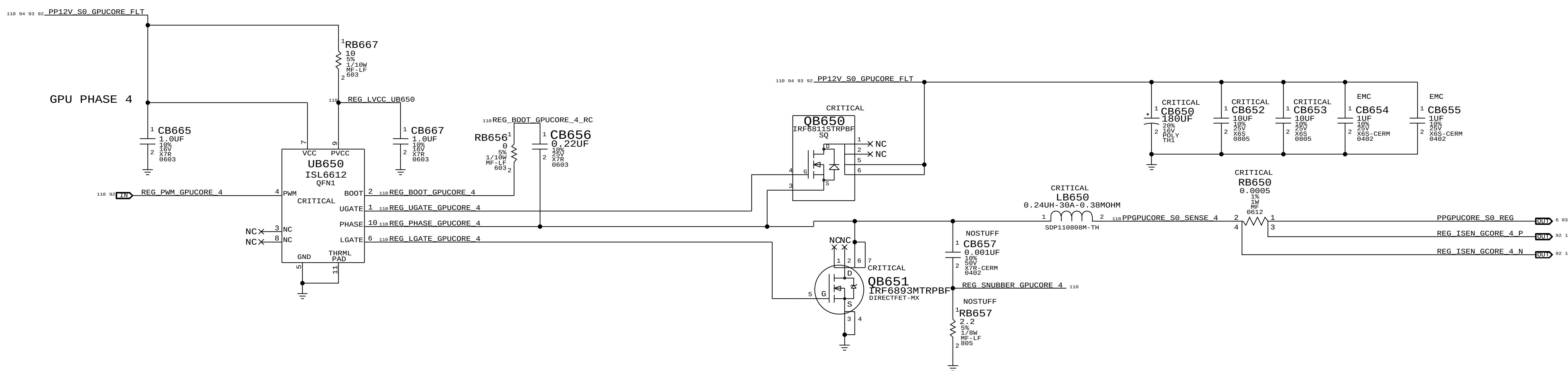

SYNC_MASTER=D8_MLB SYNC_DATE=02/06/2012

PAGE TITLE

VREG GPU CORE PHASE 4

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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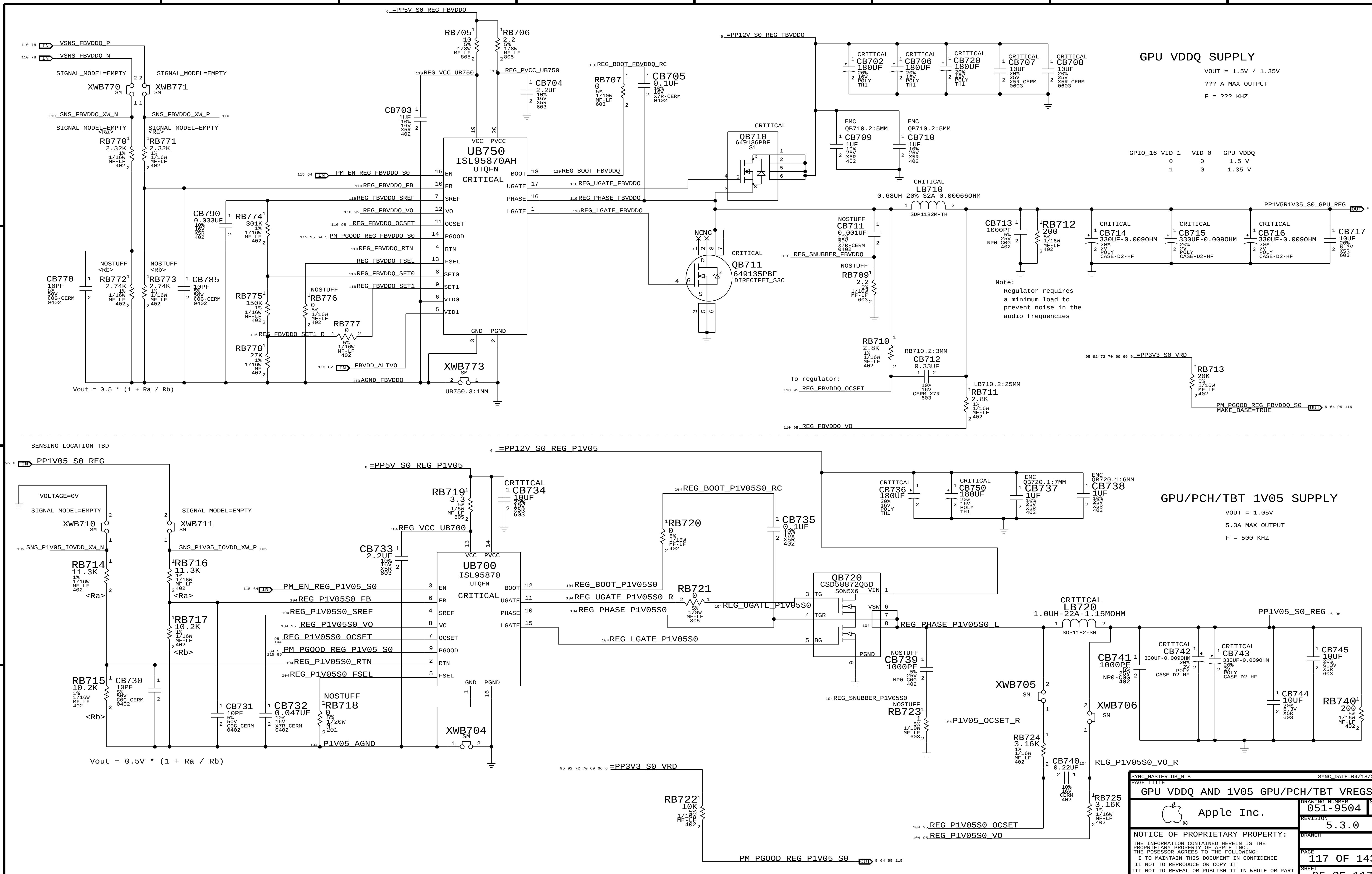

SYNC_MASTER=D8_MLB SYNC_DATE=04/18/2012

PAGE TITLE

GPU VDDQ AND 1V05 GPU/PCH/TBT VREGS

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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D

D8 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,BGA_TBT	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5,ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5,ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

C

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5,ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP,BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5,ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5,ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	ISL5,ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

B

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

A

Compensation Physical Rule Definition

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
COMP_SE	*	Y	0.305 MM	0.105 MM	3 MM	=STANDARD	=STANDARD

NOTE: line width based on 12 mil recommendation
NOTE: neck width based on 4 mil recommendation

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?
PM_ISO	*	=1:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	PM_ISO
PM	GND	*	DEFAULT

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3,ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 MM
2	Plane	1 oz
	Core	0.101 MM
3	Signal	0.5 oz
	Prepreg	0.115 MM
4	Plane	1 oz
	Core	0.076 MM
5	Signal	0.5 oz
	Prepreg	0.380 MM
6	Plane	1 oz
	Core	0.076 MM
7	Plane	1 oz
	Prepreg	0.380 MM
8	Signal	0.5 oz
	Core	0.076 MM
9	Plane	1 oz
	Prepreg	0.115 MM
10	Signal	0.5 oz
	Core	0.101 MM
11	Plane	1 oz
	Prepreg	0.071 MM
Btm	Signal	0.5 oz (Cu plated)

SYNC_MASTER=D8_KOSECOFF SYNC_DATE=03/19/2012

PAGE TITLE
D8 RULE DEFINITIONS

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_34S	BOTTOM	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S	BOTTOM	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_42S	ISL5,ISL8	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_42S_D	BOTTOM	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	0.1016 MM	0.1016 MM
DDR_42S_D	ISL5,ISL8	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF

Minimum diff spacing is 4 mil
Table 3-5, Intel Doc# 473718

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR_PHY	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	TOP,BOTTOM	=5.5X_DIELECTRIC	?
DDR_CLK_ISO	ISL3,ISL10	=4.9X_DIELECTRIC	?
DDR_CLK_ISO	*	=6.5X_DIELECTRIC	?
DDR_CTRL_ISO	TOP,BOTTOM	=4.5X_DIELECTRIC	?
DDR_CTRL_ISO	ISL3,ISL10	=4.0X_DIELECTRIC	?
DDR_CTRL_ISO	*	=5.3X_DIELECTRIC	?
DDR_CTRL2CTRL	TOP,BOTTOM	=3.0X_DIELECTRIC	?
DDR_CTRL2CTRL	ISL3,ISL10	=2.6X_DIELECTRIC	?
DDR_CTRL2CTRL	*	=3.5X_DIELECTRIC	?
DDR_CMD_ISO	TOP,BOTTOM	=4.5X_DIELECTRIC	?
DDR_CMD_ISO	ISL3,ISL10	=4.0X_DIELECTRIC	?
DDR_CMD_ISO	*	=5.3X_DIELECTRIC	?
DDR_CMD2CMD	TOP,BOTTOM	=2.3X_DIELECTRIC	?
DDR_CMD2CMD	ISL3,ISL10	=2.0X_DIELECTRIC	?
DDR_CMD2CMD	*	=2.7X_DIELECTRIC	?
DDR_DATA_ISO	TOP,BOTTOM	=4.5X_DIELECTRIC	?
DDR_DATA_ISO	ISL3,ISL10	=4.0X_DIELECTRIC	?
DDR_DATA_ISO	*	=5.3X_DIELECTRIC	?
DDR_DQ2DQ	TOP,BOTTOM	=3.2X_DIELECTRIC	900
DDR_DQ2DQ	ISL3,ISL10	=2.8X_DIELECTRIC	900
DDR_DQ2DQ	*	=3.8X_DIELECTRIC	900
DDR_DQ2DQS	TOP,BOTTOM	=3.7X_DIELECTRIC	?
DDR_DQ2DQS	ISL3,ISL10	=3.3X_DIELECTRIC	?
DDR_DQ2DQS	*	=4.4X_DIELECTRIC	?
DDR_BL2BL	TOP,BOTTOM	=4.5X_DIELECTRIC	?
DDR_BL2BL	ISL3,ISL10	=4.0X_DIELECTRIC	?
DDR_BL2BL	*	=5.3X_DIELECTRIC	?
DDR_CH2CH	TOP,BOTTOM	=9.1X_DIELECTRIC	?
DDR_CH2CH	ISL3,ISL10	=8.2X_DIELECTRIC	?
DDR_CH2CH	*	=10.9X_DIELECTRIC	?
CMD2DATA_ISO	TOP,BOTTOM	=7X_DIELECTRIC	?
CMD2DATA_ISO	ISL3,ISL10	=5X_DIELECTRIC	?
CMD2DATA_ISO	*	=5X_DIELECTRIC	?

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR_ISO	*	=4.3X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER_DDR	*	*	POWER_DDR_ISO

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD
DDR_CMD	DDR_A_DQ_BYTE*	*	CMD2DATA_ISO
DDR_CMD	DDR_A_DQS*	*	CMD2DATA_ISO
DDR_CMD	DDR_B_DQ_BYTE*	*	CMD2DATA_ISO
DDR_CMD	DDR_B_DQS*	*	CMD2DATA_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_A_DQS*	*	*	DDR_DATA_ISO	
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO	
DDR_B_DQS*	*	*	DDR_DATA_ISO	
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ	See Note (3)
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL	See Note (3)
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS	See Note (1)
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL	
DDR_A_*	DDR_B_*	*	DDR_CH2CH	See Note (2)

Note (1):
Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to contraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):
Intel suggests 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):
In order for the constraints DDR_*_DQ_BYTE* to =SAME to win out over DDR_{A,B}_DQ_BYTE* to DDR_{A,B}_DQ_BYTE* so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL.

DDR3

	Electrical Contraint Set	Physical	Spacing		
Channel A					
I178	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<1..0>	12 29
I179	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<1..0>	12 29
I255	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<3..2>	12 30
I256	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<3..2>	12 30
I180	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<1..0>	12 29
I181	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<1..0>	12 29
I182	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<1..0>	12 29
I247	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<3..2>	12 30
I248	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<3..2>	12 30
I249	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<3..2>	12 30
I183	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_A<15..0>	12 29 30
I184	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_BA<2..0>	12 29 30
I185	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_RAS_L	12 29 30
I186	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_CAS_L	12 29 30
I187	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_WE_L	12 29 30
I188	DDR_A_DQ_BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0	MEM_A_DQ<7..0>	12 33
I189	DDR_A_DQ_BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1	MEM_A_DQ<15..8>	12 33
I191	DDR_A_DQ_BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2	MEM_A_DQ<23..16>	12 33
I190	DDR_A_DQ_BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3	MEM_A_DQ<31..24>	12 33
I192	DDR_A_DQ_BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4	MEM_A_DQ<39..32>	12 33
I193	DDR_A_DQ_BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5	MEM_A_DQ<47..40>	12 33
I194	DDR_A_DQ_BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6	MEM_A_DQ<55..48>	12 33
I195	DDR_A_DQ_BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7	MEM_A_DQ<63..56>	12 33
I196	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_P<0>	12 33
I197	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_N<0>	12 33
I198	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_P<1>	12 33
I199	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_N<1>	12 33
I200	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_P<2>	12 33
I201	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_N<2>	12 33
I203	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_P<3>	12 33
I202	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_N<3>	12 33
I204	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_P<4>	12 33
I205	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_N<4>	12 33
I206	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_P<5>	12 33
I207	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_N<5>	12 33
I208	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_P<6>	12 33
I209	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_N<6>	12 33
I210	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_P<7>	12 33
I211	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_N<7>	12 33
Channel B					
I212	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<1..0>	12 31
I213	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<1..0>	12 31
I254	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<3..2>	12 32
I253	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<3..2>	12 32
I214	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<1..0>	12 31
I215	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<1..0>	12 31
I216	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<1..0>	12 31
I250	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<3..2>	12 32
I251	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<3..2>	12 32
I252	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<3..2>	12 32
I217	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_A<15..0>	12 31 32
I218	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_BA<2..0>	12 31 32
I219	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_RAS_L	12 31 32
I220	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_CAS_L	12 31 32
I221	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_WE_L	12 31 32
I222	DDR_B_DQ_BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0	MEM_B_DQ<7..0>	12 33
I223	DDR_B_DQ_BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1	MEM_B_DQ<15..8>	12 33
I225	DDR_B_DQ_BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2	MEM_B_DQ<23..16>	12 33
I224	DDR_B_DQ_BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3	MEM_B_DQ<31..24>	12 33
I226	DDR_B_DQ_BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4	MEM_B_DQ<39..32>	12 33
I227	DDR_B_DQ_BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5	MEM_B_DQ<47..40>	12 33
I228	DDR_B_DQ_BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6	MEM_B_DQ<55..48>	12 33
I229	DDR_B_DQ_BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7	MEM_B_DQ<63..56>	12 33
I230	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_P<0>	12 33
I231	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_N<0>	12 33
I232	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_P<1>	12 33
I233	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_N<1>	12 33
I234	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_P<2>	12 33
I235	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_N<2>	12 33
I237	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_P<3>	12 33
I236	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_N<3>	12 33
I238	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_P<4>	12 33
I239	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_N<4>	12 33
I240	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_P<5>	12 33
I241	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_N<5>	12 33
I242	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_P<6>	12 33
I243	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_N<6>	12 33
I244	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_P<7>	12 33
I245	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_N<7>	12 33
Reset					
I246		DDR_50S	CPU	MEM_RESET_L	28 29 30 31 32

SYNC_MASTER=D8_KOSECOFF SYNC_DATE=03/19/2012

PAGE TITLE
DDR3 Constraints

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

BRANCH

PCI EXPRESS

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	COMP_SE

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_ISO	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_ISO	*	=4X_DIELECTRIC	?
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

PCIe Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	85	16	15.75	PCIe Gen2, PCIe Gen3, DMI

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO

PCIe (PCH)

Electrical Contraint Set	Physical	Spacing	Net		Pages
x1 AirPort					
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_P	NO_TEST=TRUE	35
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_R2D_N	NO_TEST=TRUE	35
	PCIE_PHY	PCIE	PCIE_AP_R2D_C_P	NO_TEST=TRUE	18 35
	PCIE_PHY	PCIE	PCIE_AP_R2D_C_N	NO_TEST=TRUE	18 35
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_P	NO_TEST=TRUE	18 35
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE_AP_D2R_N	NO_TEST=TRUE	18 35
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_P		18 35
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_AP_N		18 35
x1 Caesar IV					
PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_P	NO_TEST=TRUE	39
PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_ENET_R2D_N	NO_TEST=TRUE	39
	PCIE_PHY	PCIE	PCIE_ENET_R2D_C_P	NO_TEST=TRUE	18 39
	PCIE_PHY	PCIE	PCIE_ENET_R2D_C_N	NO_TEST=TRUE	18 39
PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_P	NO_TEST=TRUE	18 39
PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_ENET_D2R_N	NO_TEST=TRUE	18 39
	PCIE_PHY	PCIE	PCIE_ENET_D2R_C_P	NO_TEST=TRUE	39
	PCIE_PHY	PCIE	PCIE_ENET_D2R_C_N	NO_TEST=TRUE	39
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_P		18 39
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_ENET_N		18 39

PCIE (PCH - TBT)

Electrical Contraint Set	Physical	Spacing	Net		Pages
x4 Thunderbolt					
PCIE_GEN2_R2D_PINV	PCIE_PHY	PCIE	PCIE_TBT_R2D_P<2..0>	NO_TEST=TRUE	36
PCIE_GEN2_R2D_PINV	PCIE_PHY	PCIE	PCIE_TBT_R2D_N<2..0>	NO_TEST=TRUE	36
PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_TBT_R2D_P<3>	NO_TEST=TRUE	36
PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE_TBT_R2D_N<3>	NO_TEST=TRUE	36
	PCIE_PHY	PCIE	PCIE_TBT_R2D_C_P<3..0>	NO_TEST=TRUE	18 36
	PCIE_PHY	PCIE	PCIE_TBT_R2D_C_N<3..0>	NO_TEST=TRUE	18 36
	PCIE_PHY	PCIE	PCIE_TBT_D2R_P<3..0>	NO_TEST=TRUE	18 36
	PCIE_PHY	PCIE	PCIE_TBT_D2R_N<3..0>	NO_TEST=TRUE	18 36
PCIE_GEN2_D2R_PINV	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_P<1..0>	NO_TEST=TRUE	36
PCIE_GEN2_D2R_PINV	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_N<1..0>	NO_TEST=TRUE	36
PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_P<2>	NO_TEST=TRUE	36
PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_N<2>	NO_TEST=TRUE	36
PCIE_GEN2_D2R_PINV	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_P<3>	NO_TEST=TRUE	36
PCIE_GEN2_D2R_PINV	PCIE_PHY	PCIE	PCIE_TBT_D2R_C_N<3>	NO_TEST=TRUE	36
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_P		18 36
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE_CLK100M_TBT_N		18 36

PCIe (CPU)

Electrical Contraint Set	Physical	Spacing	Net		Pages
x16 Graphics					
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<15>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<15>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<15>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<15>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<15>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<15>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<15>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<15>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<14>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<14>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<14>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<14>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<14>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<14>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<14>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<14>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<13>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<13>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<13>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<13>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<13>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<13>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<13>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<13>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<12>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<12>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<12>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<12>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<12>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<12>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<12>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<12>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<11>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<11>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<11>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<11>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<11>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<11>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<11>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<11>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<10>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<10>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<10>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<10>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<10>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<10>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<10>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<10>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<9>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<9>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<9>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<9>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<9>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<9>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<9>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<9>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<8>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<8>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<8>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<8>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<8>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<8>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<8>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<8>	NO_TEST=TRUE	10 75

PCIe (CPU)

Electrical Contraint Set	Physical	Spacing	Net		Pages
x16 Graphics					
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<7>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<7>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<7>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<7>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<7>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<7>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<7>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<7>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<6>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<6>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<6>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<6>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<6>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<6>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<6>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<6>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<5>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<5>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<5>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<5>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<5>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<5>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<5>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<5>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<4>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<4>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<4>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<4>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<4>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<4>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<4>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<4>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<3>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<3>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<3>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<3>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<3>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<3>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<3>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<3>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<2>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<2>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<2>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<2>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<2>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<2>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<2>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<2>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_P<1>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D_PINV	PCIE_PHY	PCIE	PEG_R2D_N<1>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<1>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<1>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_P<1>	NO_TEST=TRUE	75
PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG_D2R_N<1>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<1>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<1>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_P<0>	NO_TEST=TRUE	10 75
PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG_R2D_N<0>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_R2D_C_P<0>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_R2D_C_N<0>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_P<0>	NO_TEST=TRUE	75
PCIE_GEN3_D2R_PINV	PCIE_PHY	PCIE	PEG_D2R_N<0>	NO_TEST=TRUE	75
	PCIE_PHY	PCIE	PEG_D2R_C_P<0>	NO_TEST=TRUE	10 75
	PCIE_PHY	PCIE	PEG_D2R_C_N<0>	NO_TEST=TRUE	10 75
CPU PCIe Clocks					
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_P	NO_TEST=TRUE	18 75
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_N	NO_TEST=TRUE	18 75
CPU PCIe Compensation					
	COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_COMP		10

DMI

DMI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
DMI_PHY	*	DMI_85D

CPU Misc / FDI

CPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI_SE	*	Y	0.3 MM	0.25 MM	3 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CPU_PHY	*	CPU_50S
COMP_FDI_PHY	*	COMP_FDI_SE

CPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

FDI Compensation Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU	*	*	CPU_ISO
COMP_FDI	*	*	COMP_FDI_ISO

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_50S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
XDP_CLK_ISO	*	=4:1_SPACING	?

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	50	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	XDP_CLK_ISO
OBS_STROBE	*	*	XDP_CLK_ISO

DMI

Ref	Electrical Contraint Set	Physical	Spacing	Net		Pages
DMI						
I166	DMI_N2S	DMI_PHY	PCIE	DMI_N2S_P<3..0>	NO_TEST=TRUE	10 19
I165	DMI_N2S	DMI_PHY	PCIE	DMI_N2S_N<3..0>	NO_TEST=TRUE	10 19
I167	DMI_S2N	DMI_PHY	PCIE	DMI_S2N_P<3..0>	NO_TEST=TRUE	10 19
I168	DMI_S2N	DMI_PHY	PCIE	DMI_S2N_N<3..0>	NO_TEST=TRUE	10 19
I283	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_P		11 18
I284	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI_CLK100M_CPU_N		11 18
DMI Compensation						
I171		COMP_PCIE_PHY	COMP_PCIE	PCH_DMI_COMP		19
I272		COMP_PCIE_PHY	COMP_PCIE	PCH_DMI2RBIAS		19

CPU Misc.

Ref	Electrical Contraint Set	Physical	Spacing	Net	Pages
CPU Misc.					
I245		CPU_PHY	CPU	CPU_SKTOCC_L	11 64
I244		CPU_PHY	CPU	CPU_PROC_SEL	11 19
I246		CPU_PHY	CPU	CPU_CATERR_L	11 48
I251		CPU_PHY	CPU	CPU_PECI	11 21 47 48
I250		CPU_PHY	CPU	CPU_PROCHOT_L	11 47 48 66
I267		CPU_PHY	CPU	CPU_PROCHOT_R_L	11
I249		CPU_PHY	CPU	CPU_THRMTRIP_L	11 48
I248		CPU_PHY	CPU	CPU_RESET_L	11 26
I247		CPU_PHY	CPU	PLT_RESET_LS1V05_L	11
I256		CPU_PHY	CPU	PM_SYNC	11 19
I255		CPU_PHY	CPU	CPU_PWRGD	11 21 25 28
I260		CPU_PHY	CPU	PM_MEM_PWRGD	11 19 28
I254		CPU_PHY	CPU	PM_MEM_PWRGD_R	11
I253		CPU_PHY	CPU	CPU_MEM_RESET_L	11 28

FDI

Ref	Electrical Contraint Set	Physical	Spacing	Net	Pages
FDI Compensation					
I261		COMP_FDI_PHY	COMP_FDI	CPU_FDI_COMPIO	10

XDP

Ref	Electrical Contraint Set	Physical	Spacing	Net	Pages
XDP MISC.					
I174	XDP_BPM_L	XDP_PHY	XDP	XDP_BPM_L<7..0>	11 25
I175		XDP_PHY	XDP	CPU_CFG<17..16>	10 25
I257		XDP_PHY	XDP	CPU_CFG<11..0>	10 15 25
I266		XDP_PHY	XDP	XDP_DBRESET_L	11 25
I265		XDP_PHY	XDP	XDP_CPU_PWRGD	25
I264		XDP_PHY	XDP	XDP_CPU_PWRBTN_L	25
I263		XDP_PHY	XDP	XDP_CPU_CFG<0>	25
I262		XDP_PHY	XDP	XDP_VR_READY	25
I270		XDP_PHY	XDP	XDP_CPU_PLTRST_L	25 26
I269		XDP_PHY	XDP	XDP_PCH_PWRGD	25
I268		XDP_PHY	XDP	XDP_PCH_PWRBTN_L	25
I271		XDP_PHY	XDP	XDP_PCH_PLTRST_L	25 26
I176		CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_P	11 15
I177		CLK_PCIE_PHY	CLK_PCIE	ITPCPU_CLK100M_N	11 15
I178	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_P	15 18 25
I180	ITP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPXDP_CLK100M_N	15 18 25
I179		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_P	25
I181		CLK_PCIE_PHY	CLK_PCIE	XDP_CPU_CLK100M_N	25
CPU JTAG					
I182		XDP_PHY	CLK_JTAG	XDP_CPU_TCK	11 25
I183		XDP_PHY	XDP	XDP_CPU_TMS	11 25
I184		XDP_PHY	XDP	XDP_CPU_TDI	11 25
I185		XDP_PHY	XDP	XDP_CPU_TDO	11 25
I190		XDP_PHY	XDP	XDP_CPU_TRST_L	11 25
I258		XDP_PHY	XDP	XDP_CPU_PRDY_L	11 25
I259		XDP_PHY	XDP	XDP_CPU_PREQ_L	11 25
PCH JTAG					
I186		XDP_PHY	CLK_JTAG	XDP_PCH_TCK	18 25
I187		XDP_PHY	XDP	XDP_PCH_TMS	18 25
I188		XDP_PHY	XDP	XDP_PCH_TDI	18 25
I189		XDP_PHY	XDP	XDP_PCH_TDO	18 25

Chipset Test Interface

Ref	Electrical Contraint Set	Physical	Spacing	Net	Pages
OBS PCH Side					
I191	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTA_OC_R_L	20 25
I192	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTB_OC_R_L	20 25
I207	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTC_OC_R_L	20 25
I206	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTD_OC_R_L	20 25
I205	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTB_OC_EHCI_R_L	20 25
I204	XDP_PCH_OBS	XDP_PHY	XDP	USB_EXTD_OC_EHCI_R_L	20 25
I203	XDP_PCH_OBS	XDP_PHY	XDP	AP_PWR_EN_R	20 25
I202	XDP_PCH_OBS	XDP_PHY	XDP	SDCONN_STATE_CHANGE_R	20 25
I201	XDP_PCH_OBS	XDP_PHY	OBS_STROBE	TBT_CIO_PLUG_EVENT_R	21 25
I200	XDP_PCH_OBS	XDP_PHY	XDP	ISOLATE_CPU_MEM_R_L	21 25
I199	XDP_PCH_OBS	XDP_PHY	XDP	GPU_GOOD_R	21 25
I198	XDP_PCH_OBS	XDP_PHY	XDP	DP_AUXCH_ISOL_R	18 25
I197	XDP_PCH_OBS	XDP_PHY	XDP	SATARDRVR_EN_R	18 25
I196	XDP_PCH_OBS	XDP_PHY	XDP	DP_TBT_SEL_R	21 25
I195	XDP_PCH_OBS	XDP_PHY	XDP	JTAG_TBT_TCK_R	21 25
I194	XDP_PCH_OBS	XDP_PHY	XDP	AUD_IPHS_SWITCH_EN_PCH_R	21 25
I193	XDP_PCH_OBS	XDP_PHY	XDP	ENET_LOW_PWR_PCH_R	21 25
I208	XDP_PCH_OBS_UNUSED	XDP_PHY	OBS_STROBE	XDP_PIN03	21 25
OBS XDP Side					
I209		XDP_PHY	XDP	XDP_DA0_USB_EXTA_OC_L	25
I210		XDP_PHY	XDP	XDP_DA1_USB_EXTB_OC_L	25
I211		XDP_PHY	XDP	XDP_DA2_USB_EXTC_OC_L	25
I212		XDP_PHY	XDP	XDP_DA3_USB_EXTD_OC_L	25
I213		XDP_PHY	XDP	XDP_DB0_USB_EXTB_OC_EHCI_L	25
I214		XDP_PHY	XDP	XDP_DB1_USB_EXTD_OC_EHCI_L	25
I215		XDP_PHY	XDP	XDP_DB2_AP_PWR_EN	25
I216		XDP_PHY	XDP	XDP_DB3_SDCONN_STATE_CHANGE	25
I217		XDP_PHY	OBS_STROBE	XDP_FC1_TBT_CIO_PLUG_EVENT	25
I218		XDP_PHY	XDP	XDP_DC0_ISOLATE_CPU_MEM_L	25
I219		XDP_PHY	XDP	XDP_DC1_GPU_GOOD	25
I220		XDP_PHY	XDP	XDP_DC2_DP_AUXCH_ISOL	25
I221		XDP_PHY	XDP	XDP_DC3_SATARDRVR_EN	25
I222		XDP_PHY	XDP	XDP_DD0_DP_TBT_SEL	25
I223		XDP_PHY	XDP	XDP_DD1_JTAG_TBT_TCK	25
I224		XDP_PHY	XDP	XDP_DD2_AUD_IPHS_SWITCH_EN_PCH	25
I225		XDP_PHY	XDP	XDP_DD3_ENET_LOW_PWR_PCH	25
I226		XDP_PHY	OBS_STROBE	XDP_FC0_PCH_GPIO15	25
Standard usage branch off from OBS					
I227		XDP_PHY	XDP	USB_EXTA_OC_L	15 20 45
I228		XDP_PHY	XDP	USB_EXTB_OC_L	15 20 45
I230		XDP_PHY	XDP	USB_EXTC_OC_L	15 20 46
I229		XDP_PHY	XDP	USB_EXTD_OC_L	15 20 46
I232		XDP_PHY	XDP	USB_EXTB_OC_EHCI_L	15 20
I231		XDP_PHY	XDP	USB_EXTD_OC_EHCI_L	15 20
I233		XDP_PHY	XDP	AP_PWR_EN	15 20
I235		XDP_PHY	XDP	SDCONN_STATE_CHANGE	15 20 41
I234		XDP_PHY	OBS_STROBE	TBT_CIO_PLUG_EVENT	15 21
I236		XDP_PHY	XDP	ISOLATE_CPU_MEM_L	21 28 34
I237		XDP_PHY	XDP	GPU_GOOD	5 21
I238		XDP_PHY	XDP	DP_AUXCH_ISOL	15 18
I240		XDP_PHY	XDP	SATARDRVR_EN	15 18
I239		XDP_PHY	XDP	DP_TBT_SEL	21 62 84
I241		XDP_PHY	XDP	JTAG_TBT_TCK	15 21
I273		XDP_PHY	XDP	JTAG_TBT_TCK_ISOL	15 36
I243		XDP_PHY	XDP	AUD_IPHS_SWITCH_EN_PCH	21 26
I242		XDP_PHY	XDP	ENET_LOW_PWR_PCH	15 21 26
I274		XDP_PHY	XDP	DP_AUXCH_ISOL_EN	15

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PAGE TITLE

CPU MISC/DMI/FDI/XDP Constraints

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

BRANCH

SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_90D
COMP_SATA_PHY	*	SATA_50S

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=7.2X_DIELECTRIC	?
COMP_SATA_ISO	*	=5.4X_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

SATA Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
15.2.1	90	95	20	23.62	SATA Gen2, SATA Gen3

SATA Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
15-3	50	50	15	15.75	SATA Gen2, SATA Gen3

SATA

	Electrical Contraint Set	Physical	Spacing	Net	
	PCH SATA Port 0 (HDD)				
I90	SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_P	44
I92	SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_N	44
I88		SATA_PHY	SATA	SATA_HDD_R2D_C_P	18 44
I89		SATA_PHY	SATA	SATA_HDD_R2D_C_N	18 44
I94	SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_P	18 44
I95	SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_N	18 44
I91		SATA_PHY	SATA	SATA_HDD_D2R_C_P	44
I93		SATA_PHY	SATA	SATA_HDD_D2R_C_N	44
	PCH SATA Port 1 (SSD)				
I97	SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_P	18 44
I99	SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_N	18 44
I103	SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_P	18 44
I102	SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_N	18 44
	PCH SATA Compensation				
I104		COMP_SATA_PHY	COMP_SATA	PCH_SATAICOMP	18
I106		COMP_SATA_PHY	COMP_SATA	PCH_SATA3COMP	18
I105		COMP_SATA_PHY	COMP_SATA	PCH_SATA3RBIAS	18

Unused

Electrical Contraint Set	Physical	Spacing

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SATA/FDI/XDP Constraints

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DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

BRANCH

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH_ISO	*	=6.5X_DIELECTRIC	?
COMP_PCH_ISO	*	=2:1_SPACING	?
PCH_ISO	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCH	*	*	CLK_PCH_ISO
COMP_PCH	*	*	COMP_PCH_ISO
PCH	*	*	PCH_ISO

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI_ISO	*	=3.6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCI	*	*	CLK_PCI_ISO

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC_ISO	*	=1.5:1_SPACING	?
CLK_LPC_ISO	*	=3.6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LPC	*	*	LPC_ISO
CLK_LPC	*	*	CLK_LPC_ISO

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA_ISO	*	=2X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDA	*	*	HDA_ISO

Crystal

Crystal-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL_ISO	*	=4X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XTAL	*	*	XTAL_ISO

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI_ISO	*	=2X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SPI	*	*	SPI_ISO

PCI

Electrical Contraint Set	Physical	Spacing	Net	Pages
PCI Clock				
I333	CLK_PCI_50S	CLK_PCI	PCH_CLK33M_PCIIN	18 26
I334	CLK_PCI_50S	CLK_PCI	PCH_CLK33M_PCIOUT	20 26

LPC

Electrical Contraint Set	Physical	Spacing	Net	Pages
LPC				
I336	LPC_50S	LPC	LPC_AD<3..0>	18 47 49
I335	LPC_50S	LPC	LPC_R_AD<3..0>	18
I338	LPC_50S	LPC	LPC_FRAME_L	18 47 49
I337	LPC_50S	LPC	LFRAME_L	18
LPC Clocks				
I339	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS	26 49
I340	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	20 26
I341	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC	26 47
I342	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC_R	20 26

PCH Clocks

Electrical Contraint Set	Physical	Spacing	Net	Pages
PCH Reference Clock				
I385	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_SB	26
I406	CLK_PCH_50S	CLK_PCH	PCH_CLK25M_XTALIN	18 26
PCH Ref Clock Comp				
I346	PCH_50S	COMP_PCH	PCH_XCLK_RCOMP	18
PCH RTC 32K				
I347	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 26
I348	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 26
I352	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	26
SMC 32K				
I349	CLK_PCH_50S	CLK_PCH	PM_CLK32K_SUSCLK_R	19 48
I350	CLK_PCH_50S	CLK_PCH	SMC_CLK32K	47 48

25 MHz Reference Clocks

Electrical Contraint Set	Physical	Spacing	Net	Pages
25M Reference Crystal				
I379	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1	26
I380	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2	26
I381	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R	26
25M Reference Clocks				
I384	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_ENET	26 39
I386	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_ENET_R	26
I388	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_TBT	26 36
I387	CLK_PCH_50S	CLK_PCH	SYSCLK_CLK25M_TBT_R	36

HDA

Electrical Contraint Set	Physical	Spacing	Net	Pages
HDA				
I353	HDA_50S	HDA	HDA_BIT_CLK	18 56
I361	HDA_50S	HDA	HDA_BIT_CLK_R	18
I363	HDA_50S	HDA	HDA_RST_L	18 56
I362	HDA_50S	HDA	HDA_RST_R_L	18
I364	HDA_50S	HDA	HDA_SDOUT	18 56
I367	HDA_50S	HDA	HDA_SDOUT_R	15 18
I366	HDA_50S	HDA	HDA_SYNC	15 18 56
I368	HDA_50S	HDA	HDA_SYNC_R	18
I365	HDA_50S	HDA	HDA_SDIN0	18 56
I369	HDA_50S	HDA	AUD_SDI_R	56
I414	HDA_50S	HDA	SPI_DESCRIPTOR_OVERRIDE_R	15
SPDIF				
I370		HDA	AUD_SPDIF_CHIP	56
I371		HDA	AUD_SPDIF_OUT	56 60

SPI Bootrom

Electrical Contraint Set	Physical	Spacing	Net	Pages
SPI ROM				
I391	SPI_50S	SPI	SPI_CLK_R	18 49
I392	SPI_50S	SPI	SPI_CLK	49
I393	SPI_50S	SPI	SPI_ALT_CLK	49
I408	SPI_50S	SPI	SPI_SMC_CLK	47 48
I407	SPI_50S	SPI	SPI_MLB_CLK	48 49
I394	SPI_50S	SPI	SPI_CS0_R_L	18 49
I395	SPI_50S	SPI	SPI_CS0_L	49
I396	SPI_50S	SPI	SPI_ALT_CS_L	49
I409	SPI_50S	SPI	SPI_SMC_CS_L	47 48
I397	SPI_50S	SPI	SPI_MLB_CS_L	48 49
I399	SPI_50S	SPI	SPI_MOSI_R	18 49
I400	SPI_50S	SPI	SPI_MOSI	49
I401	SPI_50S	SPI	SPI_ALT_MOSI	49
I410	SPI_50S	SPI	SPI_SMC_MOSI	47 48
I411	SPI_50S	SPI	SPI_MLB_MOSI	48 49
I404	SPI_50S	SPI	SPI_MISO	18 49
I403	SPI_50S	SPI	SPI_ALT_MISO	49
I412	SPI_50S	SPI	SPI_SMC_MISO	47 48
I413	SPI_50S	SPI	SPI_MLB_MISO	48 49
I405	SPI_50S	SPI	SPIROM_USE_MLB	21 49

USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
USB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D
USB_HUB_PHY	*	USB_50S

SMSC Hub Application Note 15.17
Single-ended impedance range from 45-80 ohm is acceptable

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Comments
12.2.1	90	90	12/2.8 mils = 4.29:1	USB 2.0
13.3.1	85	85	20/2.8 mils = 7.14:1	USB 3.0
			50/2.8 mils = 17.9:1	USB 2.0/3.0

USB 2.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=4.4X_DIELECTRIC	?
USB2_ISO	TOP,BOTTOM	=4.4X_DIELECTRIC	?
USB2_CLK_ISO	*	=18X_DIELECTRIC	?
USB2_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB_HUB_ISO	*	=2:1_SPACING	?

USB 3.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_CLK_ISO	*	=18X_DIELECTRIC	?
USB3_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB3_USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	ISL10	=5X_DIELECTRIC	?

USB 2.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB2	*CLK*	*	USB2_CLK_ISO
USB2	DISPLAYPORT	*	USB2_CLK_ISO
USB2	*TBT*	*	USB2_CLK_ISO
USB2	*ENET*	*	USB2_CLK_ISO
USB2	*SD*	*	USB2_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB_HUB	*	*	USB_HUB_ISO

USB 3.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3	*	*	USB3_ISO
USB3	*CLK*	*	USB3_CLK_ISO
USB3	DISPLAYPORT	*	USB3_CLK_ISO
USB3	*TBT*	*	USB3_CLK_ISO
USB3	*ENET*	*	USB3_CLK_ISO
USB3	*SD*	*	USB3_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB3	USB3	*	USB3_USB3_ISO

CAMERA CONTROLLER

Camera Controller's SMIA Interface & MISC. Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMIA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
CAM_SE	*	Y	0.2 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D
CAM_PHY	*	CAM_SE

Camera Controller's SMIA Interface & MISC. Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?
CAM_ISO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF
CAM	*	*	CAM_ISO

USB 3.0 and USB 2.0 Trixies Muxing

	Electrical Contraint Set	Physical	Spacing	Net	Pages
External Port A (J4600)					
I320	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_P	45
I321	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTA_RX_N	45
I391		USB3_PHY	USB3	USB3_EXTA_RX_F_P	20 45
I392		USB3_PHY	USB3	USB3_EXTA_RX_F_N	20 45
I413	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_P	20 45
I414	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTA_TX_N	20 45
I415		USB3_PHY	USB3	USB3_EXTA_TX_F_P	45
I416		USB3_PHY	USB3	USB3_EXTA_TX_F_N	45
I493		USB3_PHY	USB3	USB3_EXTA_TX_C_P	45
I494		USB3_PHY	USB3	USB3_EXTA_TX_C_N	45
I417	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_PCH_0_P	20 45
I418	USB2_MUXED_MOJO_CONN	USB2_PHY	USB2	USB_PCH_0_N	20 45
I419		USB2_PHY	USB2	USB2_EXTA_MUXED_P	45
I420		USB2_PHY	USB2	USB2_EXTA_MUXED_N	45
I393		USB2_PHY	USB2	USB2_EXTA_MUXED_F_P	45
I394		USB2_PHY	USB2	USB2_EXTA_MUXED_F_N	45
External Port B (J4610)					
I421	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P	45
I422	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N	45
I423		USB3_PHY	USB3	USB3_EXTB_RX_F_P	20 45
I424		USB3_PHY	USB3	USB3_EXTB_RX_F_N	20 45
I425	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P	20 45
I427	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N	20 45
I426		USB3_PHY	USB3	USB3_EXTB_TX_F_P	45
I428		USB3_PHY	USB3	USB3_EXTB_TX_F_N	45
I495		USB3_PHY	USB3	USB3_EXTB_TX_C_P	45
I496		USB3_PHY	USB3	USB3_EXTB_TX_C_N	45
I429	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_P	20 45
I430	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_N	20 45
I436		USB2_PHY	USB2	USB_PCH_9_P	20 45
I435		USB2_PHY	USB2	USB_PCH_9_N	20 45
I432		USB2_PHY	USB2	USB2_EXTB_MUXED_P	45
I431		USB2_PHY	USB2	USB2_EXTB_MUXED_N	45
I434		USB2_PHY	USB2	USB2_EXTB_MUXED_F_P	45
I433		USB2_PHY	USB2	USB2_EXTB_MUXED_F_N	45
External Port C (J4700)					
I437	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_P	46
I439	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTC_RX_N	46
I438		USB3_PHY	USB3	USB3_EXTC_RX_F_P	20 46
I440		USB3_PHY	USB3	USB3_EXTC_RX_F_N	20 46
I441	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_P	20 46
I442	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTC_TX_N	20 46
I444		USB3_PHY	USB3	USB3_EXTC_TX_F_P	46
I443		USB3_PHY	USB3	USB3_EXTC_TX_F_N	46
I498		USB3_PHY	USB3	USB3_EXTC_TX_C_P	46
I497		USB3_PHY	USB3	USB3_EXTC_TX_C_N	46
I446	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_P	20 46
I445	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_N	20 46
I450		USB2_PHY	USB2	USB2_EXTC_F_P	46
I452		USB2_PHY	USB2	USB2_EXTC_F_N	46
External Port D (J4710)					
I453	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_P	46
I455	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_N	46
I454		USB3_PHY	USB3	USB3_EXTD_RX_F_P	20 46
I456		USB3_PHY	USB3	USB3_EXTD_RX_F_N	20 46
I457	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_P	20 46
I458	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_N	20 46
I460		USB3_PHY	USB3	USB3_EXTD_TX_F_P	46
I459		USB3_PHY	USB3	USB3_EXTD_TX_F_N	46
I499		USB3_PHY	USB3	USB3_EXTD_TX_C_P	46
I500		USB3_PHY	USB3	USB3_EXTD_TX_C_N	46
I462	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_P	20 46
I461	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_N	20 46
I463		USB2_PHY	USB2	USB_PCH_10_P	20 46
I464		USB2_PHY	USB2	USB_PCH_10_N	20 46
I465		USB2_PHY	USB2	USB2_EXTD_MUXED_P	46
I467		USB2_PHY	USB2	USB2_EXTD_MUXED_N	46
I466		USB2_PHY	USB2	USB2_EXTD_MUXED_F_P	46
I468		USB2_PHY	USB2	USB2_EXTD_MUXED_F_N	46
Camera (U4200)					
I324	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_P	42
I326	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_N	42
PCH USB Compensation					
I384		PCH_50S	COMP_PCH	PCH_USB_RBIAS	20

	Electrical Contraint Set	Physical	Spacing	Voltage	Net	Pages
Camera Controller Local Ground						
I521		GND_PHY	GND	0V	CAM_AGND	42
I522		GND_PHY	GND	0V	CAM_PLLGND	42

USB Hub

	Electrical Contraint Set	Physical	Spacing	Net	Pages
USB 2.0 Hub					
I401	USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_P	20 27
I402	USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_N	20 27
I409	USB2_HUB_BT	USB2_PHY	USB2	USB_BT_P	27 35
I410	USB2_HUB_BT	USB2_PHY	USB2	USB_BT_N	27 35
I505	USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_P	35
I506	USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_N	35
I541		USB_HUB_PHY	USB_HUB	USB_HUB_2P	27
I542		USB_HUB_PHY	USB_HUB	USB_HUB_2N	27
USB 2.0 Hub Misc.					
I516		USB_HUB_PHY	USB_HUB	USB_HUB_RESET_L	27
I520		USB_HUB_PHY	USB_HUB	USB_HUB_VBUS_DET	27
I519		USB_HUB_PHY	USB_HUB	USB_HUB_NON_REM0	27
I518		USB_HUB_PHY	USB_HUB	USB_HUB_NON_REM1	27
I517		USB_HUB_PHY	USB_HUB	USB_HUB_HS_IND	27
USB 2.0 Hub Compensation					
I471		PCH_50S	COMP_PCH	USB_HUB_RBIAS	27
USB 2.0 Hub Crystal					
I472		CLK_XTAL	XTAL	USB_HUB_XTAL1	27
I473		CLK_XTAL	XTAL	USB_HUB_XTAL2	27
I515		CLK_XTAL	XTAL	USB_HUB_XTAL2_R	27

Camera Controller

	Electrical Contraint Set	Physical	Spacing	Net	Pages
SMIA					
I501	SMIA_DATA	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_P	42
I502	SMIA_DATA	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_N	42
I528	SMIA_CLK	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_P	42
I529	SMIA_CLK	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_N	42
MISC					
I507		SPI_50S	SPI	CAM_SF_CLK	42
I508		SPI_50S	SPI	CAM_SF_CLK_R	42
I509		SPI_50S	SPI	CAM_SF_DIN	42
I510		SPI_50S	SPI	CAM_SF_DIN_R	42
I511		SPI_50S	SPI	CAM_SF_CS_L	42
I513		SPI_50S	SPI	CAM_SF_WP_L	42
I512		SPI_50S	SPI	CAM_SF_DOUT	42
I514		SPI_50S	SPI	CAM_SF_DOUT_R	42
I540		SPI_50S	SPI	CAM_SF_HOLD_L	42
I524		CAM_PHY	CAM	CAM_USB_VRES	42
I523		CAM_PHY	CAM	MIPI_RESISTOR	42
I539			PM	CAM_EXT_BOOT_L	43
I538			PM	PCH_CAM_EXT_BOOT_R_L	21 43
I537			PM	CAM_P1V2_RST_HOLDOFF	43
I536			PM	CAM_P1V2_RST_HOLDOFF_L	43
I2C					
I534		SMB_PHY	SMB	I2C_CAMSENSOR_SDA	42
I535		SMB_PHY	SMB	I2C_CAMSENSOR_SCL	42
I532		SMB_PHY	SMB	SMB_ALS_F_SDA	42
I533		SMB_PHY	SMB	SMB_ALS_F_SCL	42
Camera Controller Crystal					
I525		CLK_XTAL	XTAL	CAM_XTAL_IN	42
I526		CLK_XTAL	XTAL	CAM_XTAL_OUT	42
I527		CLK_XTAL	XTAL	CAM_XTAL_OUT_R	42

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USB/Camera Constraints

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SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
SNS_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR
SNS_PHY	*	SNS_50S

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PWR_P2MM
SENSE	GND	*	GND_P2MM

SYNC_MASTER=D8_MARK SYNC_DATE=04/23/2012

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SMBus/Sensor Constraints

Apple Inc.

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DC-DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_CTL_PHY	*	POWER_P5MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S
POWER_PHY	*	POWER_P6MM
GND_PHY	*	GND_P5MM

Constraints
Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

DC-DC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_ISO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND
VR_LGATE	*	*	SWNODE_ISO
VR_LGATE	VR_SWITCH	*	SWNODE_LG2SW

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO

Power-specific Spacing Definitions
Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?
SWNODE_LG2SW	*	=3:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

CPU VccSA

Electrical Contraint Set		Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
Input Bus								
I24		POWER_PHY	POWER	5V			REG_VCC_U7500	70
I25		POWER_PHY	POWER	5V			REG_PVCC_U7500	70
Local Ground								
I26		GND_PHY	GND	0V			AGND_VCCSAS0	70
VCCIO								
I27		POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VCCSAS0	70
I28		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VCCSAS0	70
I29		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VCCSAS0_RC	70
I31		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VCCSAS0	70
I30		VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VCCSAS0	70
I33		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_VCCSAS0	70
I32		VR_CTL_PHY	VR_CTL				REG_VCCSAS0_OCSET	70
I35		VR_CTL_PHY	VR_CTL				REG_VCCSAS0_VO	70
I34			SENSE				SNS_CPU_VCCSA	13 70
I37	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_P	70
I36	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_N	70
I38			SENSE				REG_VCCSAS0_FB	70
I39			SENSE				REG_VCCSAS0_RTN	70
I40		VR_CTL_PHY	VR_CTL				REG_VCCSAS0_SREF	70
I41		VR_CTL_PHY	VR_CTL				REG_VCCSAS0_FSEL	70
Output Bus								
I42		POWER_PHY	POWER	0.925V			PPVCCSA_S0	6

CPU VccIO PCH 1.05V S0

Electrical Contraint Set		Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
Input Bus								
I1		POWER_PHY	POWER	5V			REG_VCC_U7400	69
I2		POWER_PHY	POWER	5V			REG_PVCC_U7400	69
Local Ground								
I3		GND_PHY	GND	0V			AGND_CPU_P1V05S0	69
1.05V S0								
I5		POWER_PHY	VR_SWITCH	12V	TRUE		REG_CPU_PHASE_P1V05S0	69
I7		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_BOOT_P1V05S0	69
I6		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_BOOT_P1V05S0_RC	69
I8		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0	69
I9		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0_R	
I10		VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_CPU_LGATE_P1V05S0	69
I12		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_SNUBBER_P1V05S0	69
I11		VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_OCSET	69
I14		VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_VO	69
I13	VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE				SNS_CPU_VCCIO_P	13 69
I15	VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE				SNS_CPU_VCCIO_N	13 69
I16		SNS_DIFF_PHY	SENSE				SNS_CPU_P1V05S0_XW_P	69
I17		SNS_DIFF_PHY	SENSE				SNS_CPU_P1V05S0_XW_N	69
I19			SENSE				REG_CPU_P1V05S0_FB	69
I18			SENSE				REG_CPU_P1V05S0_RTN	69
I21		VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_SREF	69
I20		VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_FSEL	69
Output Bus								
I22		POWER_PHY	POWER	1.05V			PP1V05_S0_CPU	6

PCH/GPU/TBT 1.05V S0

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	
Input Bus							
I44	POWER_PHY	POWER	5V			REG_VCC_UB700	95
Local Ground							
I46	GND_PHY	GND	0V			P1V05_AGND	95
1.05V S0							
I47	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0	95
I48	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0_L	95
I49	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P1V05S0	95
I50	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P1V05S0_RC	95
I51	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0	95
I52	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0_R	95
I53	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_P1V05S0	95
I54	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_P1V05S0	95
I55	VR_CTL_PHY	VR_CTL				REG_P1V05S0_OCSET	95
I56	VR_CTL_PHY	VR_CTL				REG_P1V05S0_VO	95
I72	POWER_PHY	VR_SWITCH	12V			REG_P1V05S0_VO_R	95
I73	POWER_PHY	VR_SWITCH	12V			P1V05_OCSET_R	95
I61		SENSE				REG_P1V05S0_FB	95
I62		SENSE				REG_P1V05S0_RTN	95
I63	VR_CTL_PHY	VR_CTL				REG_P1V05S0_SREF	95
I64	VR_CTL_PHY	VR_CTL				REG_P1V05S0_FSEL	95
Output Bus							
I68	POWER_PHY	POWER	1.05V			PP1V05_S0_PCH	6
I66	POWER_PHY	POWER	1.05V			PP1V05_S0	6
FET Switched							
I67	POWER_PHY	POWER	1.05V			PP1V05_TBTLC	6 38
I69	POWER_PHY	POWER	1.05V			PP1V05_TBTCIO	6

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VReg Constraints

Apple Inc.

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CPU Core Phases

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		
Input Bus							
	POWER_PHY	POWER	12V			PP12V_S0_CPUCORE_FLT	66 67 68
	POWER_PHY	POWER	5V			REG_VCC_U7100	66
Local Ground							
	GND_PHY	GND	0V			AGND_CPU	66 67 68
Phase 1							
	POWER_PHY	POWER	12V			REG_LVCC_U7210	67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1	66 67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_1	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_1	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_1_RC	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_1	67
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_1	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_1	67
	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_1	67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_1_P	66 67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_1_N	67
		SENSE				REG_ISENCORE_1_NR	66 67
Phase 2							
	POWER_PHY	POWER	12V			REG_LVCC_U7230	67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2	66 67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_2	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_2	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_2_RC	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_2	67
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_2	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_2	67
	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_2	67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_2_P	66 67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_2_N	67
		SENSE				REG_ISENCORE_2_NR	66 67
Phase 3							
	POWER_PHY	POWER	12V			REG_LVCC_U7250	67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3	66 67
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_3	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_3	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_3_RC	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_3	67
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_3	67
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_3	67
	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_3	67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_3_P	66 67
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_3_N	67
		SENSE				REG_ISENCORE_3_NR	66 67
Phase 4							
	POWER_PHY	POWER	12V			REG_LVCC_U7310	68
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4	66 68
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_4	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_4	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_4_RC	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_4	68
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_4	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_4	68
	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_4	68
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_4_P	66 68
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISENCORE_4_N	68
		SENSE				REG_ISENCORE_4_NR	66 68
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CORE_XW_P	66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CORE_XW_N	66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CORE_R_P	66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CORE_R_N	66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VAXG_P	13 66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VAXG_N	13 66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VCORE_P	13 66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_CPU_VCORE_N	13 66
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_P1V05_IOVDD_XW_P	95
ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE				SNS_P1V05_IOVDD_XW_N	95

CPU AXG Phase and Core Controller

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		
AXG							
	POWER_PHY	POWER	12V			REG_LVCC_U7330	68
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG	66 68
	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG_R	66
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUAXG	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUAXG	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUAXG_RC	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUAXG	68
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUAXG	68
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUAXG	68
	POWER_PHY	POWER	1.1V			PPCPUAXG_S0_SENSE	68
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				REG_ISENAXG_P	68
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				REG_ISENAXG_N	68
		SENSE				REG_ISENAXG_PR	66 68
		SENSE				REG_ISENAXG_NR	66 68
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				SNS_AXG_R_P	66
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				SNS_AXG_R_N	66
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				SNS_AXG_XW_P	66
ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE				SNS_AXG_XW_N	66
ISL6364							
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_COMP	66
	VR_CTL_PHY	VR_CTL				CPUCORE_COMP_RC	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FB	66
	VR_CTL_PHY	VR_CTL				CPUCORE_FB_RC	66
	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_1	66
	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_2	66
	VR_CTL_PHY	VR_CTL				CPUCORE_PSICOMP_RC	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_PSICOMP	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_HFCOMP	66
		SENSE				REG_CPUCORE_VSEN	66
		SENSE				REG_CPUCORE_RGND	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IMON	51 66
	VR_CTL_PHY	VR_CTL				CPUCORE_IMON_R	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_TM	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SUTH	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_NPSI	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FDVID	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IAUTO	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SW_FREQ	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RAMPADJ	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR	66
	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R	66
	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP	66
	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB	66
	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC	66
	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1	66
	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP	66
		SENSE				REG_CPUAXG_VSEN	66
		SENSE				REG_CPUAXG_RGND	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON	51 66
	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP	66
	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ	66
	VR_VID_PHY	VR_VID				CPU_VIDSCLK	13 66
	VR_VID_PHY	VR_VID				CPU_VIDSCLK_R	13
	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	13 66
	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	13
	VR_VID_PHY	VR_VID				CPU_VIDSOUT	13 66
	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	13
Output Bus							
	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU	6
	POWER_PHY	POWER	1.1V			PPVAXG_S0	6

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CPU VReg Constraints

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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3.3V S5/5V S4

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus						
POWER_PHY	POWER	12V			REG_VIN_U7600	71
POWER_PHY	POWER	5V			REG_VCC1_U7600	71
POWER_PHY	POWER	5V			REG_VCC2_U7600	71
3.3V S5						
POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P3V3S5	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P3V3S5	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P3V3S5_RC	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P3V3S5	71
VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_P3V3S5	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P3V3S5	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_ISEN	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_OCSET	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_FSET	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT_R	71
VR_CTL_PHY	VR_CTL				REG_P3V3S5_FB	71
5V S3						
POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P5VS4	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P5VS4	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P5VS4_RC	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P5VS4	71
VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_P5VS4	71
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P5VS4	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_ISEN	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_OCSET	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_FSET	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT_R	71
VR_CTL_PHY	VR_CTL				REG_P5VS4_FB	71
Output Bus						
POWER_PHY	POWER	5V			PP5V_S5	6
POWER_PHY	POWER	5V			PP5V_S4	6
POWER_PHY	POWER	3.3V			PP3V3_S5	6
FET Switched						
POWER_PHY	POWER	5V			PP5V_S0	6
POWER_PHY	POWER	3.3V			PP3V3_S4	6 60
POWER_PHY	POWER	3.3V			PP3V3_S0	6
POWER_PHY	POWER	3.3V			PP3V3_S4_ENET	6
POWER_PHY	POWER	3.3V			PP3V3_TBTLC	6
POWER_PHY	POWER	3.3V			PP3V3_S4_AP	6
POWER_PHY	POWER	3.3V			PP3V3_S0_SSD	6

VDDQ S3 (1.5V)/VTT S0

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus						
POWER_PHY	POWER	5V			REG_V5IN_U7700	72
Local Ground						
GND_PHY	GND	0V			AGND_VDDQS3	72
VDDQ S3						
POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3	72
POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VDDQS3	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VDDQS3_RC	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3_R	72
VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VDDQS3	72
VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3	72
POWER_PHY	POWER	1.5V			PPVDDQ_S3_SENSE	
VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSNS	72
VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF	72
VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN	72
VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE	72
VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP	72
VR_CTL_PHY	VR_CTL				LDO_DDRVTTS0_SNS	72
Output Bus						
POWER_PHY	POWER	1.5V			PPVDDQ_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVTT_S0	6
FET Switched						
POWER_PHY	POWER	1.5V			PP1V5_S0	6
Sensed						
POWER_PHY	POWER	1.5V			PPVDDQ_S3_DDR	6
POWER_PHY	POWER	1.5V			PP1V5_S0_CPU_MEM	6
POWER_PHY	POWER	1.5V			PPFBVDDQ_S0_GPU	6

DDR3 Vref

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Memory Vref						
POWER_PHY	POWER	3.3V			PP3V3_S4_VREFMRGN_DAC	34
POWER_PHY	POWER	3.3V			PP3V3_S4_VREFMRGN_CTRL	34
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVREF_DQ_MEM_A_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVREF_DQ_MEM_B_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			CPU_DIMM_VREF_DAC_A	11 34
POWER_DDR_PHY	POWER_DDR	0.75V			CPU_DIMM_VREF_DAC_B	11 34
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVREF_CA_MEM_A_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVREF_CA_MEM_B_S3	6
POWER_DDR_PHY	POWER_DDR	0.75V			CPU_DDR_VREF	11
POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVTT_S3	6

1.8V S0

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
1.8V S0						
POWER_PHY	VR_SWITCH	5V	TRUE		REG_PHASE_P1V8S0	72
VR_CTL_PHY	VR_CTL				REG_P1V8S0_VFB	72
VR_CTL_PHY	VR_CTL				REG_P1V8S0_SYNCH	72
Output Bus						
POWER_PHY	POWER	1.8V			PP1V8_S0	6

HDD S0

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
HDD S0						
POWER_PHY	POWER	5V			PP5V_S0_HDD	6

12V S5

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
Input Bus						
POWER_PHY	POWER	12V			PP12V_ACDC	6
FET Switched						
POWER_PHY	POWER	12V			PP12V_S0	6
Sensed						
POWER_PHY	POWER	12V			PP12V_S5	6
POWER_PHY	POWER	12V			PP12V_G3H	6
POWER_PHY	POWER	12V			PP12V_G3H_P3V42	73
POWER_PHY	POWER	12V			PP12V_S0_FBVDDQ	6
POWER_PHY	POWER	12V			PP12V_S0_CPU_P1V05	6
POWER_PHY	POWER	12V			PP12V_S0_VCCSA	6
POWER_PHY	POWER	12V			PP12V_S0_P1V05	6
POWER_PHY	POWER	12V			PP12V_S0_HDD	6
POWER_PHY	POWER	12V			PP12V_S0_BLC	6

Ground/Common

Physical	Spacing	Voltage	DIDT	NO_TEST	Net
Common					
GND_PHY	GND	0V			GND

3V42 S0

Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Page
3V42 S0						
POWER_PHY	VR_SWITCH	5V	TRUE		P3V42G3H_SW	73
VR_CTL_PHY	VR_CTL				P3V42G3H_FB	73
VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L	73
POWER_PHY	POWER	3.3V			PP3V3_G3	6
POWER_PHY	POWER	3.3V			PP3V42_G3H	6

Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTDP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	0.075MM
TBT_GEN_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_I2C_ISO	*	=2x_DIELECTRIC	?
TBT_SPI_ISO	*	=2x_DIELECTRIC	?
TBTDP_ISO	*	=5x_DIELECTRIC	?
TBTDP_ISO	TOP,BOTTOM	=7x_DIELECTRIC	?
TBT_GEN_ISO	*	=2X_DIELECTRIC	?
BGA_TBT_AREA	*	0.075MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_TBT	BGA_TBT_AREA
TBT_I2C	*	*	TBT_I2C_ISO
TBT_SPI	*	*	TBT_SPI_ISO
TBTDP	*	*	TBTDP_ISO
TBT_GEN	*	*	TBT_GEN_ISO

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_ISO	*	=6.7X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	*	*	DP_ISO

MAX LENGTH OF DISPLAYPORT TRACES: 6 INCHES

DISPLAYPORT INTRA-PAIR MATCHING SHOULD BE 1 PS. INTER-PAIR MATCHING SHOULD BE WITHIN 150 PS.
DISPLAYPORT AUX CHANNEL INTRA-PAIR MATCHING SHOULD BE 5 PS.

TBT IC Net Properties

	Electrical Contraint Set	Physical	Spacing	Net		
I511		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	NO_TEST=TRUE	36 81
I513		DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	NO_TEST=TRUE	36 81
I512	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	NO_TEST=TRUE	36
I514	DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	NO_TEST=TRUE	36
I516		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P		36 75
I515		DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N		36 75
I517	DP_TBTSNK0_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_P		36
I518	DP_TBTSNK0_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_N		36
I519		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	NO_TEST=TRUE	36 81
I521		DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	NO_TEST=TRUE	36 81
I520	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	NO_TEST=TRUE	36
I523	DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	NO_TEST=TRUE	36
I522		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P		36 75
I524		DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N		36 75
I526	DP_TBTSNK1_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_P		36
I525	DP_TBTSNK1_AUX	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_N		36
I603	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_P<3..0>	NO_TEST=TRUE	84
I604	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_N<3..0>	NO_TEST=TRUE	84
I686	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	NO_TEST=TRUE	84
I687	DP_INTPNL_TBT_ML_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	NO_TEST=TRUE	84
I529	DP_INTPNL_TBT_AUX_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_P		84
I530	DP_INTPNL_TBT_AUX_MUX	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_N		84
I601		DP_85D	DISPLAYPORT	DP_TBTSRC_AUX_C_P		84
I602		DP_85D	DISPLAYPORT	DP_TBTSRC_AUX_C_N		84
I531		TBT_I2C_55S	TBT_I2C	I2C_TBTRTR_SCL		36 50
I532		TBT_I2C_55S	TBT_I2C	I2C_TBTRTR_SDA		36 50
I533	TBT_SPI_CLK	TBT_I2C_55S	TBT_SPI	TBT_SPI_CLK		36
I534	TBT_SPI_MOSI	TBT_I2C_55S	TBT_SPI	TBT_SPI_MOSI		36
I538	TBT_SPI_MISO	TBT_I2C_55S	TBT_SPI	TBT_SPI_MISO		36
I535	TBT_SPI_CS_L	TBT_I2C_55S	TBT_SPI	TBT_SPI_CS_L		36
I688		TBT_GEN_55S	TBT_GEN	TBT_A_CONFIG1_BUF		36 86
I689		TBT_GEN_55S	TBT_GEN	TBT_A_CONFIG2_RC		36 86
I691		TBT_GEN_55S	TBT_GEN	TBT_B_CONFIG1_BUF		36 88
I690		TBT_GEN_55S	TBT_GEN	TBT_B_CONFIG2_RC		36 88
I693		TBT_GEN_55S	TBT_GEN	TBT_A_LSTX		36 86
I692		TBT_GEN_55S	TBT_GEN	TBT_A_LSRX		36 86
I695		TBT_GEN_55S	TBT_GEN	TBT_B_LSTX		36 88
I694		TBT_GEN_55S	TBT_GEN	TBT_B_LSRX		36 88
I697		TBT_GEN_55S	TBT_GEN	DP_TBTSNK0_HPD		36 82
I696		TBT_GEN_55S	TBT_GEN	DP_TBTSNK1_HPD		36 82
I698		TBT_GEN_55S	TBT_GEN	DP_TBTSRC_HPD		36 84
I699		TBT_GEN_55S	TBT_GEN	DP_TBTSNK0_DDC_CLK		81 85
I701		TBT_GEN_55S	TBT_GEN	DP_TBTSNK0_DDC_DATA		81 85
I700		TBT_GEN_55S	TBT_GEN	DP_TBTSNK1_DDC_CLK		81 85
I702		TBT_GEN_55S	TBT_GEN	DP_TBTSNK1_DDC_DATA		81 85
I703		TBT_GEN_55S	TBT_GEN	VIDEO_ON		74 87
I704		TBT_GEN_55S	TBT_GEN	VIDEO_ON_L		5 87
I705		TBT_GEN_55S	TBT_GEN	BDV_BKL_PWM		48 84
I706		TBT_GEN_55S	TBT_GEN	GPU_LCD_BKLT_PWM		82 84
I707		TBT_GEN_55S	TBT_GEN	LCD_BL_PWM		84
I708		TBT_GEN_55S	TBT_GEN	LCD_BL_FILT		84
I709		TBT_GEN_55S	TBT_GEN	LCD_BKLT_PWM		84 89

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*: Only used on hosts supporting T29 video-in

TBT/DP Net Properties

	Electrical Contraint Set	Physical	Spacing	Net		
	Port A					
I541	TBT_A_R2D	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<0>	NO_TEST=TRUE	36 86
I542	TBT_A_R2D	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<0>	NO_TEST=TRUE	36 86
I710	TBT_A_R2D_PINV	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<1>	NO_TEST=TRUE	36 86
I711	TBT_A_R2D_PINV	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<1>	NO_TEST=TRUE	36 86
I544		TBTDP_90D	TBTDP	TBT_A_R2D_P<1..0>	NO_TEST=TRUE	86
I543		TBTDP_90D	TBTDP	TBT_A_R2D_N<1..0>	NO_TEST=TRUE	86
I595	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	NO_TEST=TRUE	36 86
I596	DP_TBTPA_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	NO_TEST=TRUE	36 86
I680	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	NO_TEST=TRUE	36 86
I681	DP_TBTPA_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	NO_TEST=TRUE	36 86
I545		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	NO_TEST=TRUE	86
I547		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	NO_TEST=TRUE	86
I678		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	NO_TEST=TRUE	86
I679		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	NO_TEST=TRUE	86
I546		DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>		86
I548		DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>		86
I549		TBTDP_90D	TBTDP	TBT_A_D2R_C_P<1..0>	NO_TEST=TRUE	86
I550		TBTDP_90D	TBTDP	TBT_A_D2R_C_N<1..0>	NO_TEST=TRUE	86
I564	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_P<1>	NO_TEST=TRUE	36 86
I563	TBT_A_D2R1	TBTDP_90D	TBTDP	TBT_A_D2R_N<1>	NO_TEST=TRUE	36 86
I675	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_P<0>	NO_TEST=TRUE	36 86
I674	TBT_A_D2R0	TBTDP_90D	TBTDP	TBT_A_D2R_N<0>	NO_TEST=TRUE	36 86
I599	DP_TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_P		36 86
I600	DP_TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_N		36 86
I565		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_P		86
I566		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_N		86
I648		DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_P		86
I649		DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_N		86
I650		TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_P	NO_TEST=TRUE	86
I651		TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_N	NO_TEST=TRUE	86
	Port B					
I652	TBT_B_R2D	TBTDP_90D	TBTDP	TBT_B_R2D_C_P<1..0>	NO_TEST=TRUE	36 88
I653	TBT_B_R2D	TBTDP_90D	TBTDP	TBT_B_R2D_C_N<1..0>	NO_TEST=TRUE	36 88
I655		TBTDP_90D	TBTDP	TBT_B_R2D_P<1..0>	NO_TEST=TRUE	88
I654		TBTDP_90D	TBTDP	TBT_B_R2D_N<1..0>	NO_TEST=TRUE	88
I657	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	NO_TEST=TRUE	36 88
I658	DP_TBTPB_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	NO_TEST=TRUE	36 88
I685	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	NO_TEST=TRUE	36 88
I684	DP_TBTPB_ML3	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	NO_TEST=TRUE	36 88
I683		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	NO_TEST=TRUE	88
I682		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	NO_TEST=TRUE	88
I656		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	NO_TEST=TRUE	88
I660		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	NO_TEST=TRUE	88
I659		DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>	NO_TEST=TRUE	88
I662		DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>	NO_TEST=TRUE	88
I661		TBTDP_90D	TBTDP	TBT_B_D2R_C_P<1..0>	NO_TEST=TRUE	88
I663		TBTDP_90D	TBTDP	TBT_B_D2R_C_N<1..0>	NO_TEST=TRUE	88
I665	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_P<1>	NO_TEST=TRUE	36 88
I664	TBT_B_D2R1	TBTDP_90D	TBTDP	TBT_B_D2R_N<1>	NO_TEST=TRUE	36 88
I676	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_P<0>	NO_TEST=TRUE	36 88
I677	TBT_B_D2R0	TBTDP_90D	TBTDP	TBT_B_D2R_N<0>	NO_TEST=TRUE	36 88
I666	DP_TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_P		36 88
I667	DP_TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_N		36 88
I669		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_P		88
I670		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_N		88
I668		DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_P		88
I672		DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_N		88
I671		TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_P	NO_TEST=TRUE	88
I673		TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_N	NO_TEST=TRUE	88

DisplayPort

	Electrical Contraint Set	Physical	Spacing	Net		
	Graphics Source					
I642	DP_INTPNL_EG_ML_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_ML_P<3..0>	NO_TEST=TRUE	81 84
I643	DP_INTPNL_EG_ML_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_ML_N<3..0>	NO_TEST=TRUE	81 84
I644	DP_INTPNL_EG_AUX_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_P	NO_TEST=TRUE	81 84
I645	DP_INTPNL_EG_AUX_MUX	DP_85D	DISPLAYPORT	DP_INT_EG_AUX_N	NO_TEST=TRUE	81 84
I646		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_C_P		84
I647		DP_85D	DISPLAYPORT	DP_INT_EG_AUX_C_N		84
	Internal Panel					
I609		DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_P<3..0>	NO_TEST=TRUE	84
I606		DP_85D	DISPLAYPORT	DP_INTPNL_ML_C_N<3..0>	NO_TEST=TRUE	84
I607	DP_INTPNL_ML_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_ML_P<3..0>	NO_TEST=TRUE	84 87
I608	DP_INTPNL_ML_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_ML_N<3..0>	NO_TEST=TRUE	84 87
I605	DP_INTPNL_AUX_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_P		84 87
I610	DP_INTPNL_AUX_CONN	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_N		84 87
	Internal DP SPDIF					
I611			HDA	DP_INT_SPDIF_AUDIO		56 87

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PAGE TITLE
TBT/DP Constraints

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

BRANCH

8 | 7 | 6 | 5 | 4 | 3 | 2 | 1

GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

	Trace-to-Trace				Isolation				
Table	Micro	Design	Strip	Design	Micro	Design	Strip	Design	Comments
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Memory address (MA). Implented 4.5 Gbps or less rules for K70.
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Address dynamic bus inversion (ADBI)
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Control (CTRL)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Clock (CLK)
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data (DQ)
5-6/5-7	7:1	7:1	7:1	7:1	7:1	7:1	7:1	7:1	Error detection pins (EDC). Using larger isolation rules,
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data dynamic bus inversion (DBI)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Forwarded clock (WCK)

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3X_DIELECTRIC	?
GDDR_ISO	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP,BOTTOM	=3X_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP,BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3X_DIELECTRIC	?
GDDR_CLK2CLK	TOP,BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP,BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP,BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP,BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP,BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP,BOTTOM	=5x_DIELECTRIC	?

Constraints (x in {A, B}, y in {0, 1})

Memory Address: MAxy[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Data: DQxy[31:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DQ	*	*	GDDR_ISO
GDDR_*_*_DQ	=SAME	*	GDDR_DQ2DQ

Address Dynamic Bus Inversion: ADBIxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Error Detection: EDCxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_EDC	*	*	GDDR_EDC_ISO
GDDR_*_*_EDC	=SAME	*	GDDR_EDC2EDC

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Data Dynamic Bus Inversion: DDBIxy[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_DBI	*	*	GDDR_ISO
GDDR_*_*_DBI	=SAME	*	GDDR_DBI2DBI

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

Forwarded Clock: WCKxy[1:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_WCK	*	*	GDDR_ISO
GDDR_*_*_WCK	=SAME	*	GDDR_WCK2WCK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

	Electrical Contraint Set	Physical	Spacing			
	Memory Address					
I642	GDDR_A0_MA	GDDR_MA_PHY	GDDR_A_0_MA	FB_A0_A<8..0>	NO_TEST=TRUE	77 79
I643	GDDR_A1_MA	GDDR_MA_PHY	GDDR_A_1_MA	FB_A1_A<8..0>	NO_TEST=TRUE	77 79
	Address Dynamic Bus Inv					
I644	GDDR_A0_ADBI	GDDR_ADBI_PHY	GDDR_A_0_ADBI	FB_A0_ABI_L	NO_TEST=TRUE	77 79
I645	GDDR_A1_ADBI	GDDR_ADBI_PHY	GDDR_A_1_ADBI	FB_A1_ABI_L	NO_TEST=TRUE	77 79
	Control					
I647	GDDR_A0_CKE	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CKE_L	NO_TEST=TRUE	77 79
I648	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CS_L	NO_TEST=TRUE	77 79
I649	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_WE_L	NO_TEST=TRUE	77 79
I650	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_CAS_L	NO_TEST=TRUE	77 79
I651	GDDR_A0_CTRL	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB_A0_RAS_L	NO_TEST=TRUE	77 79
I652	GDDR_A1_CKE	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CKE_L	NO_TEST=TRUE	77 79
I654	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CS_L	NO_TEST=TRUE	77 79
I653	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_WE_L	NO_TEST=TRUE	77 79
I655	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_CAS_L	NO_TEST=TRUE	77 79
I656	GDDR_A1_CTRL	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB_A1_RAS_L	NO_TEST=TRUE	77 79
	Clock					
I657	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_P	NO_TEST=TRUE	77 79
I658	GDDR_A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB_A0_CLK_N	NO_TEST=TRUE	77 79
I660	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_P	NO_TEST=TRUE	77 79
I659	GDDR_A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB_A1_CLK_N	NO_TEST=TRUE	77 79
	Data					
[illegible]	GDDR_A0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<7..0>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<15..8>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<23..16>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_0_DQ	FB_A0_DQ<31..24>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<7..0>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<15..8>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<23..16>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_1_DQ	FB_A1_DQ<31..24>	NO_TEST=TRUE	77 79
	Error Detection					
[illegible]	GDDR_A0_EDC0	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<0>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_EDC1	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<1>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_EDC2	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<2>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_EDC3	GDDR_EDC_PHY	GDDR_A_0_EDC	FB_A0_EDC<3>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_EDC0	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<0>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_EDC1	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<1>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_EDC2	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<2>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_EDC3	GDDR_EDC_PHY	GDDR_A_1_EDC	FB_A1_EDC<3>	NO_TEST=TRUE	77 79
	Data Dynamic Bus Inv					
[illegible]	GDDR_A0_DBI0	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<0>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_DBI1	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<1>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_DBI2	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<2>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_DBI3	GDDR_DBI_PHY	GDDR_A_0_DBI	FB_A0_DBI_L<3>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_DBI0	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<0>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_DBI1	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<1>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_DBI2	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<2>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A1_DBI3	GDDR_DBI_PHY	GDDR_A_1_DBI	FB_A1_DBI_L<3>	NO_TEST=TRUE	77 79
	Forwarded Clock					
[illegible]	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<0>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<0>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_P<1>	NO_TEST=TRUE	77 79
[illegible]	GDDR_A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB_A0_WCLK_N<1>	NO_TEST=TRUE	77 79
I689	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<0>	NO_TEST=TRUE	77 79
I690	GDDR_A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<0>	NO_TEST=TRUE	77 79
I692	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_P<1>	NO_TEST=TRUE	77 79
I691	GDDR_A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB_A1_WCLK_N<1>	NO_TEST=TRUE	77 79

GDDR5 Frame Buffer B

	Electrical Contraint Set	Physical	Spacing			
	Memory Address					
I695	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	NO_TEST=TRUE	77 80
I694	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	NO_TEST=TRUE	77 80
	Address Dynamic Bus Inv					
I696	GDDR_B0_ADBI	GDDR_ADBI_PHY	GDDR_B_0_ADBI	FB_B0_ABI_L	NO_TEST=TRUE	77 80
I697	GDDR_B1_ADBI	GDDR_ADBI_PHY	GDDR_B_1_ADBI	FB_B1_ABI_L	NO_TEST=TRUE	77 80
	Control					
I698	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	NO_TEST=TRUE	77 80
I701	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	NO_TEST=TRUE	77 80
I700	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	NO_TEST=TRUE	77 80
I702	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	NO_TEST=TRUE	77 80
I703	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	NO_TEST=TRUE	77 80
I704	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	NO_TEST=TRUE	77 80
I705	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	NO_TEST=TRUE	77 80
I706	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	NO_TEST=TRUE	77 80
I708	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	NO_TEST=TRUE	77 80
I707	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	NO_TEST=TRUE	77 80
	Clock					
I693	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	NO_TEST=TRUE	77 80
I710	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	NO_TEST=TRUE	77 80
I709	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	NO_TEST=TRUE	77 80
I711	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	NO_TEST=TRUE	77 80
	Data					
I713	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	NO_TEST=TRUE	77 80
I712	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	NO_TEST=TRUE	77 80
I714	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	NO_TEST=TRUE	77 80
I715	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	NO_TEST=TRUE	77 80
I716	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	NO_TEST=TRUE	77 80
I718	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	NO_TEST=TRUE	77 80
I717	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	NO_TEST=TRUE	77 80
I719	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	NO_TEST=TRUE	77 80
	Error Detection					
I720	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	NO_TEST=TRUE	77 80
I721	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	NO_TEST=TRUE	77 80
I723	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	NO_TEST=TRUE	77 80
I722	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	NO_TEST=TRUE	77 80
I725	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	NO_TEST=TRUE	77 80
I724	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	NO_TEST=TRUE	77 80
I726	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	NO_TEST=TRUE	77 80
I727	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	NO_TEST=TRUE	77 80
	Data Dynamic Bus Inv					
I729	GDDR_B0_DBI0	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<0>	NO_TEST=TRUE	77 80
I728	GDDR_B0_DBI1	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<1>	NO_TEST=TRUE	77 80
I730	GDDR_B0_DBI2	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<2>	NO_TEST=TRUE	77 80
I731	GDDR_B0_DBI3	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<3>	NO_TEST=TRUE	77 80
I732	GDDR_B1_DBI0	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<0>	NO_TEST=TRUE	77 80
I733	GDDR_B1_DBI1	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<1>	NO_TEST=TRUE	77 80
I734	GDDR_B1_DBI2	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<2>	NO_TEST=TRUE	77 80
I735	GDDR_B1_DBI3	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<3>	NO_TEST=TRUE	77 80
	Forwarded Clock					
I736	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	NO_TEST=TRUE	77 80
I737	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	NO_TEST=TRUE	77 80
I739	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	NO_TEST=TRUE	77 80
I738	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	NO_TEST=TRUE	77 80
I740	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	NO_TEST=TRUE	77 80
I741	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<0>	NO_TEST=TRUE	77 80
I743	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<1>	NO_TEST=TRUE	77 80
I742	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<1>	NO_TEST=TRUE	77 80

GPU

	Electrical Contraint Set	Physical	Spacing			
	Clocks					
I747		CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_O_PL		75
I748		CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_O_NG		75
I749		CLK_PCIE_PHY	CLK_PCIE	GPU_TESTMODE		81
I750		CLK_PCIE_PHY	CLK_PCIE	GPU_PEX_TERMP		75
I763		CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID	NO_TEST=TRUE	79
I764		CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID	NO_TEST=TRUE	79
I765		CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID	NO_TEST=TRUE	80
I766		CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID	NO_TEST=TRUE	80
I767		CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID	NO_TEST=TRUE	
I768		CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID	NO_TEST=TRUE	ULTIMATE
I769		CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID	NO_TEST=TRUE	ONLY
I770		CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID	NO_TEST=TRUE	
I771		CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK		78 81
I772		CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK		81 82
I773		CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R		82
I775		CLK_GPU_55S	CLK_GPU	GPU_XTALOUT		
I774		CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALOUT		81 82
I776		CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALIN		81 82
	SMB					
I564		SMB_PHY	SMB	GPU_SMB_CLK		81 82
I563		SMB_PHY	SMB	GPU_SMB_DAT		81 82
I753		SMB_PHY	SMB	GPU_SMB_CLK_R		50 82
I754		SMB_PHY	SMB	GPU_SMB_DAT_R		50 82
	PCIe Compensation					
I599		PCIE_50S	COMP_PCIE	FB_CAL_PD_VDDQ		77
I600		PCIE_50S	COMP_PCIE	FB_CAL_PU_GND		77
I762		PCIE_50S	COMP_PCIE	FB_CAL_TERM_GND		77

Frame Buffer Reset

	Electrical Contraint Set	Physical	Spacing		
	Reset				
I744	GDDR_A0_RESET	GDDR_50S	GDDR_CTRL	FB_A0_RESET_L	77 79
I755	GDDR_A1_RESET	GDDR_50S	GDDR_CTRL	FB_A1_RESET_L	77 79
I756	GDDR_B0_RESET	GDDR_50S	GDDR_CTRL	FB_B0_RESET_L	77 80
I757	GDDR_B1_RESET	GDDR_50S	GDDR_CTRL	FB_B1_RESET_L	77 80
I758	GDDR_C0_RESET	GDDR_50S	GDDR_CTRL	FB_C0_RESET_L	
I759	GDDR_C1_RESET	GDDR_50S	GDDR_CTRL	FB_C1_RESET_L	ULTIMATE
I760	GDDR_D0_RESET	GDDR_50S	GDDR_CTRL	FB_D0_RESET_L	ONLY
I761	GDDR_D1_RESET	GDDR_50S	GDDR_CTRL	FB_D1_RESET_L	

SYNC_MASTER=D8_AARON SYNC_DATE=03/13/2012

PAGE TITLE

GDDR5/GPU Constraints

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

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Backlight Controller

BLC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BLC_P6MM	*	Y	0.600 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD
BLC_P3MM	*	Y	0.300 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_BLC	*	BLC_P6MM
POWER_BLC_RET	*	BLC_P3MM
BLC_CTL_PHY	*	BLC_P3MM

BLC-specific Spacing Definitions
BLC High Voltage Output

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_HV_ISO	*	1.00MM	1000

Constraints
BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

BLC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PHASE_ISO	*	=8:1_SPACING	2000
PHASE_SW2SW	*	=1:1_SPACING	?
PHASE_SW2PWR	*	=2:1_SPACING	?
PHASE_SW2GND	*	=2:1_SPACING	?

BLC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_CTL_ISO	*	=3:1_SPACING	?

BLC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

BKLT MISCELLANEOUS

	Electrical Contraint Set	Physical	Spacing	Net	Pages
	SPI				
I564		SMB_PHY	SMB	SMB_PCH_BLC_SCL	50 89
I563		SMB_PHY	SMB	SMB_PCH_BLC_SDA	50 89
I812		SMB_PHY	SMB	SMB_TCON_BLC_SCL	50 89
I811		SMB_PHY	SMB	SMB_TCON_BLC_SDA	50 89
	12M REFERENCE CRYSTAL				
I842		CLK_XTAL	XTAL	BLC_MCU_XTAL_IN	89
I843		CLK_XTAL	XTAL	BLC_MCU_XTAL_OUT	89
I844		CLK_XTAL	XTAL	BLC_MCU_XTAL_OUT_R	89
	250K REFERENCE CLOCKS				
I845		CLK_PCH_50S	CLK_PCH	STRCLK_R1	89 90
I846		CLK_PCH_50S	CLK_PCH	LED_DRVR_CLK	90

	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
I750	POWER_PHY	POWER	12V			PP12V_S0_BLC_VIN2	89 90 91
I772	POWER_PHY	POWER	12V			PP12V_S0_BLC_VINP	89 91
I806	POWER_PHY	POWER	14V			PRE_REG_OUT	89 90
I807	POWER_PHY	POWER	3.3V			BLC_P3V3S	89 91
I867	POWER_PHY	POWER	3.3V			BLC_P3V3_REF	89 91
I809	POWER_PHY	POWER	3.3V			BLC_P3V3	
I864	POWER_PHY	POWER	3.3V			PP3V3_S0_BLC_R	89
I841	POWER_PHY	POWER	8V			SPTX_VIN	90
I810	POWER_PHY	POWER	8V			PP8V_BLC	89 90
I847	POWER_PHY	POWER	8V			BLC_VIN2	90 91
I862	POWER_PHY	POWER	14V			BOOST_FET_DRAIN	89
I863	POWER_PHY	POWER	12V			BOOST_VDD	89
I865	POWER_PHY	POWER	5V			PP5V_S0_BLC_R	89 90 91
I866	POWER_PHY	POWER	12V			PP12V_S0_BLC_F	91
Local Ground							
I751	BLC_CTL_PHY	BLC_PHASE	0V			BLC_GND_1	90 91
I804	BLC_CTL_PHY	BLC_PHASE	0V			BLC_GND_2	90 91
I805	BLC_CTL_PHY	BLC_PHASE	0V			BLC_GND_3	90 91
I816	GND_PHY	GND	0V			AGND_BLC	90 91
Backlight							
I755	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1	90
I756	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1_R	90
I798	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2	90
I797	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2_R	90
I800	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3	90
I799	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3_R	90
I795	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_1	90
I794	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_2	90
I796	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_3	90
I801	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS1	90
I803	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS2	90
I802	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS3	90
I828	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C1	90
I827	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C2	90
I826	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C3	90
I833	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_1	90
I832	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_2	90
I831	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_3	90
I830	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK1	90
I834	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK2	90
I829	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK3	90
I835	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1_R	89 90
I836	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2_R	89 90
I837	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3_R	89 90
I840	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1	90 91
I839	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2	90 91
I838	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3	90 91
I819	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF1	90 91
I818	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF2	90 91
I817	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF3	90 91
I758	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP1	90
I759	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP2	90
I781	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP3	90
I822	BLC_CTL_PHY	BLC_CTL				BCOMP1	90
I821	BLC_CTL_PHY	BLC_CTL				BCOMP2	90
I820	BLC_CTL_PHY	BLC_CTL				BCOMP3	90
I849	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT1	90
I848	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT2	90
I850	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT3	90
I851	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_1	90
I852	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_2	90
I853	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_3	90
I856	BLC_CTL_PHY	BLC_CTL				PRE_REG_OUT_R	89
I855	BLC_CTL_PHY	BLC_CTL				BOOST_FB	89
I854	BLC_CTL_PHY	BLC_CTL				BOOST_COMP	89
I857	BLC_CTL_PHY	BLC_CTL				BOOST_COMP_C	89
I860	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV	89
I861	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV_R	89
I859	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS	89
I858	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS_R	89
I752	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_1	90
I753	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_2	90
I754	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_3	90
OUTPUT BUS							
I789	POWER_BLC_RET	BLC_HV	80V			IS1_BLC_F	91
I788	POWER_BLC_RET	BLC_HV	80V			IS2_BLC_F	91
I790	POWER_BLC_RET	BLC_HV	80V			IS3_BLC_F	91
I791	POWER_BLC_RET	BLC_HV	80V			IS1_BLC	90 91
I792	POWER_BLC_RET	BLC_HV	80V			IS2_BLC	90 91
I793	POWER_BLC_RET	BLC_HV	80V			IS3_BLC	90 91
I783	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_1	91
I782	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_1	91
I784	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_2	91
I785	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_2	91
I786	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_3	91
I787	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_3	91
I825	POWER_BLC	BLC_HV	80V			BLC_VOUT1	90 91
I824	POWER_BLC	BLC_HV	80V			BLC_VOUT2	90 91
I823	POWER_BLC	BLC_HV	80V			BLC_VOUT3	90 91

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PAGE TITLE
BLC Constraints

Apple Inc.

DRAWING NUMBER 051-9504 SIZE D

REVISION 5.3.0

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GPU CORE PHASES

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER_PHY	POWER	12V			PP12V_S0_GPUCORE_FLT	92 93 94
	POWER_PHY	POWER	5V			PP5V_S0_GPU_VCORE_VCC	92
Local Ground							
	GND_PHY	GND	0V			AGND_GPU	92
Phase 1							
	POWER_PHY	POWER	12V			REG_LVCC_UB510	93
	POWER_PHY	POWER	12V			REG_UVCC_UB510	93
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_1	92 93
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM1_R	92
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_1	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_1	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_1_RC	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_1	93
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_GPUCORE_1	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_1	93
	POWER_PHY	POWER	0.9V			PPGPUCORE_S0_SENSE_1	93
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_1_P	92 93
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_1_N	92 93
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS1_R_P	92
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS1_R_N	92
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS1_RR_2	92
Phase 2							
	POWER_PHY	POWER	12V			REG_LVCC_UB530	93
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_2	92 93
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM2_R	92
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_2	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_2	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_2_RC	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_2	93
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_GPUCORE_2	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_2	93
	POWER_PHY	POWER	0.9V			PPGPUCORE_S0_SENSE_2	93
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_2_P	92 93
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_2_N	92 93
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS2_R_P	92
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS2_R_N	92
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS2_RR_2	92
Phase 3							
	POWER_PHY	POWER	12V			REG_VCC_UB550	93
	POWER_PHY	POWER	12V			REG_UVCC_UB550	93
	POWER_PHY	POWER	12V			REG_LVCC_UB550	93
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_3	92 93
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM3_R	92
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_3	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_3	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_3_RC	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_3	93
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_GPUCORE_3	93
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_3	93
	POWER_PHY	POWER	0.9V			PPGPUCORE_S0_SENSE_3	93
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_3_P	92 93
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_3_N	92 93
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS3_R_P	92
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS3_R_N	92
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS3_RR_2	92
Phase 4							
	POWER_PHY	POWER	12V			REG_LVCC_UB650	94
	VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_4	92 94
	VR_CTL_PHY	VR_CTL				VR_GPU_PWM4_R	92
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_4	94
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_4	94
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_4_RC	94
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_4	94
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_GPUCORE_4	94
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_4	94
	POWER_PHY	POWER	0.9V			PPGPUCORE_S0_SENSE_4	94
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_4_P	92 94
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				REG_ISEN_GCORE_4_N	92 94
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS4_R_P	92
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VR_GPU_ISNS4_R_N	92
ISNS_GPU_CORE		SENSE				VR_GPU_ISNS4_RR_2	92

GPU CORE CONTROLLER

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
ISL6334							
	VR_CTL_PHY	VR_CTL				VR_GPU_COMP	92
	VR_CTL_PHY	VR_CTL				VR_GPU_COMP_R	92
	VR_CTL_PHY	VR_CTL				VR_GPU_COMP_RC	92
	VR_CTL_PHY	VR_CTL				VR_GPU_FB	92
	VR_CTL_PHY	VR_CTL				VR_GPU_FB_R	92
	VR_CTL_PHY	VR_CTL				VR_GPU_VDIFF	92
	VR_CTL_PHY	VR_CTL				VR_VDF_R1	92
	VR_CTL_PHY	VR_CTL				VR_VDF_R2	92
	VR_CTL_PHY	VR_CTL				VR_GPU_TCOMP	92
	VR_CTL_PHY	VR_CTL				VR_GPU_OFS	92
	VR_CTL_PHY	VR_CTL				VR_GPU_FS	92
		VR_CTL				VR_GPU_EN_VTT	92
		VR_CTL				PM_EN_REG_GPUCORE_S0	64 92
		VR_CTL				PM_PGOOD_REG_GPUCORE_S0	5 64 92
		VR_CTL				GPU_PSI_L	82 92
	VR_CTL_PHY	VR_CTL				VR_GPU_IOUT	
	VR_CTL_PHY	VR_CTL				VR_GPU_IMON	51 92 110
	VR_CTL_PHY	VR_CTL				VR_GPU_FAN	92 110
		VR_CTL				VR_GPU_VRDHOT	92
		VR_CTL				VR_GPU_EN_PWR	
	VR_CTL_PHY	VR_CTL				VR_GPU_SS	92
	VR_CTL_PHY	VR_CTL				VR_GPU_DAC	92
	VR_CTL_PHY	VR_CTL				VR_GPU_REF	92
	VR_CTL_PHY	VR_CTL				VR_GPU_TM	92
	VR_CTL_PHY	VR_CTL				VR_GPU_IMON	51 92 110
	VR_CTL_PHY	VR_CTL				VR_GPU_FAN	92 110
	VR_CTL_PHY	VR_CTL				VR_GPU_VRHOT	
		VR_CTL				GPU_PSI_R	
		VR_CTL				GPU_PSI_L_R	
	VR_CTL_PHY	VR_CTL				VR_GPU_IMON_R	92
VSNS_GPU_VDD	SNS_DIFF_PHY	SENSE				VSNS_GPU_VDD	78 92
VSNS_GPU_VSS	SNS_DIFF_PHY	SENSE				VSNS_GPU_VSS	78 92
	SNS_DIFF_PHY	SENSE				VR_GPU_VSEN	92
	SNS_DIFF_PHY	SENSE				VR_GPU_RGND	92
GPU VIDS							
		VR_VID				REG_GPUCORE_VID7	92
		VR_VID				REG_GPUCORE_VID6	78 92
		VR_VID				REG_GPUCORE_VID5	78 92
		VR_VID				REG_GPUCORE_VID4	78 92
		VR_VID				REG_GPUCORE_VID3	78 92
		VR_VID				REG_GPUCORE_VID2	78 92
		VR_VID				REG_GPUCORE_VID1	78 92
		VR_VID				REG_GPUCORE_VID0	92
		VR_VID				GPU_VCORE_VID6	
		VR_VID				GPU_VCORE_VID5	78 82
		VR_VID				GPU_VCORE_VID4	78 82
		VR_VID				GPU_VCORE_VID3	78 82
		VR_VID				GPU_VCORE_VID2	78 82
		VR_VID				GPU_VCORE_VID1	78 82
		VR_VID				GPU_VCORE_VID0	78 82
Output Bus							
	POWER_PHY	POWER	0.9V			PPVCORE_S0_GPU	6

GPU FBVDDQ

Electrical Contraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Net	Pages
Input Bus							
	POWER_PHY	POWER	5V			REG_VCC_UB750	95
	POWER_PHY	POWER	5V			REG_PVCC_UB750	95
Local Ground							
	GND_PHY	GND	0V			AGND_FBVDDQ	95
FBVDDQ							
	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_FBVDDQ	95
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_FBVDDQ	95
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_FBVDDQ_RC	95
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_FBVDDQ	95
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_FBVDDQ_R	
	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_FBVDDQ	95
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_FBVDDQ	95
		SENSE				VSNS_FBVDDQ	
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VSNS_FBVDDQ_P	78 95
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				VSNS_FBVDDQ_N	78 95
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				SNS_FBVDDQ_XW_P	95
ISNS_GPU_CORE	SNS_DIFF_PHY	SENSE				SNS_FBVDDQ_XW_N	95
		SENSE				FBVDDQ_SENSE_R	
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_OCSET	
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_VO	95
		SENSE				REG_FBVDDQ_FB	95
		SENSE				REG_FBVDDQ_RTN	95
Output Bus							
	POWER_PHY	POWER	1.5V			PP1V5R1V35_S0_GPU_REG	6 95

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GPU VREG CONSTRAINTS

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Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	
ENET_DIFF	*	*	ENET_DIFF_ISO	
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF	
ENET_TRANS	*	*	ENET_TRANS_ISO	2 kV isolation
COMP_ENET	*	*	COMP_ENET_ISO	
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF	

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

VENI, VIDI, VICI!

	Electrical Contraint Set	Physical	Spacing	Net		
	Ethernet					
I474	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_P<3..0>		39 40
I475	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_N<3..0>		39 40
I476		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_P<3..0>		40
I477		ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_N<3..0>		40
I504			ENET_TRANS	ENETCONN_MCT3		40
I503			ENET_TRANS	ENETCONN_MCT2		40
I502			ENET_TRANS	ENETCONN_MCT1		40
I501			ENET_TRANS	ENETCONN_MCT0		40
I478		ENET_COMP_PHY	COMP_ENET	ENET_RDAC		39
	SD					
I479	SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0>	NO_TEST=TRUE	39
I480		SD_PHY	SD	SDCONN_DATA<7..0>		39 41
I499		SD_PHY	SD	SDCONN_DATA_R<7..0>	NO_TEST=TRUE	41
I481	SD_CMD	SD_PHY	SD	ENET_SD_CMD	NO_TEST=TRUE	39
I482		SD_PHY	SD	SDCONN_CMD	NO_TEST=TRUE	39 41
I500		SD_PHY	SD	SDCONN_CMD_R	NO_TEST=TRUE	41
I483	SD_CLK	SD_PHY	SD	ENET_SD_CLK	NO_TEST=TRUE	39
I484		SD_PHY	SD	SDCONN_CLK	NO_TEST=TRUE	39 41
I486		SD_PHY	SD	SDCONN_CLK_R	NO_TEST=TRUE	41
I485		SD_PHY	SD	ENET_MEDIA_SENSE		15 18 39
I488		SD_PHY	SD	ENET_SD_DETECT_L		39 41
I487		SD_PHY	SD	SDCONN_WP		39 41
	CIV SPI					
I490		SPI_50S	SPI	ENET_SCLK		39
I489		SPI_50S	SPI	ENET_MISO		39
I492		SPI_50S	SPI	ENET_MOSI		39
I491		SPI_50S	SPI	ENET_CS_L		39

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4V5_*

	Physical	Spacing	Netname	
I1		PM	4V5_REG_EN	56
I2	POWER_PHY	POWER	4V5_REG_IN	56

ACDC_*

	Spacing	Netname	
I3	PM	ACDC_BURST_EN	48
I4	PM	ACDC_BURST_EN_L	48

ALL_*

	Spacing	Netname	
I5	PM	ALL_SYS_PWRGD	5 47 65

AP_*

	Spacing	Netname	
I6	GENERIC_ISO	AP_CLKREQ_L	15 21
I7	GENERIC_ISO	AP_CLKREQ_L_ISO	15 35
I8	GENERIC_ISO	AP_CLKREQ_Q_L	35
I9	GENERIC_ISO	AP_EVENT_L	35 47 48
I10	GENERIC_ISO	AP_PWR_EN_ISO	15 35
I11	PM	AP_RESET_CONN_L	35
I12	PM	AP_RESET_L	26 35
I13	PM	AP_WAKE_L	35

BLC12V_*

	Spacing	Netname
I14	PM	BLC12V_FAULT
I15	PM	BLC12V_FAULT_L
I16	PM	BLC12V_UVLO
I17	PM	BLC12V_UVLO_OUT
I18	PM	BLC12V_UVLO_REF

BLC_*

	Physical	Spacing	Voltage	Netname	
I19		GENERIC_ISO		BLC_BL	89 91
I20		GENERIC_ISO		BLC_BL_GATE	91
I21		GENERIC_ISO		BLC_BST	89 91
I22		GENERIC_ISO		BLC_BST_R	89
I23		GENERIC_ISO		BLC_BYPASS_GATE	91
I24		GENERIC_ISO		BLC_DIM_MCU	89
I25		PM		BLC_EN	87 89 90
I26		GENERIC_ISO		BLC_ENA	89
I27		GENERIC_ISO		BLC_ENA1	89 91
I28		PM		BLC_EN_DELAY	87
I29		PM		BLC_EN_R	89

BLC_*

	Physical	Spacing	Voltage	Netname	
I30		PM		BLC_EXT_BOOT	48 89
I31		PM		BLC_EXT_BOOT_L	89
I32	BLC_CTL_PHY	BLC_CTL	5V	BLC_MCU_AOUT_R	91
I33		GENERIC_ISO		BLC_MCU_BV	89 91
I34		GENERIC_ISO		BLC_MCU_BV_D	91
I35		GENERIC_ISO		BLC_MCU_BV_R	91
I36	SMB_PHY	SMB		BLC_MCU_B_SDA_CONN	89
I37		GENERIC_ISO		BLC_MCU_FLAG_V	89
I38		GENERIC_ISO		BLC_MCU_PWM5	89
I39		GENERIC_ISO		BLC_MCU_PWM5_R	89
I40		PM		BLC_MCU_RESET	89
I41		PM		BLC_MCU_RESET_L	89
I42		PM		BLC_MCU_RESET_R_L	89
I43	XDP_PHY	CLK_JTAG		BLC_MCU_RTCK	89
I44		GENERIC_ISO		BLC_MCU_RXD0	89
I45	XDP_PHY	XDP		BLC_MCU_TCK	89
I46	XDP_PHY	XDP		BLC_MCU_TDI	89
I47	XDP_PHY	XDP		BLC_MCU_TDO	89
I48	XDP_PHY	XDP		BLC_MCU_TMS	89
I49	XDP_PHY	XDP		BLC_MCU_TRST	89
I50	XDP_PHY	XDP		BLC_MCU_TXD0	89
I51		GENERIC_ISO		BLC_MCU_UVLO	89
I52		GENERIC_ISO		BLC_ON	91
I53		GENERIC_ISO		BLC_ON_DRAIN	91
I54		GENERIC_ISO		BLC_ON_R	91
I55		GENERIC_ISO		BLC_P_ON	89 90 91
I56		GENERIC_ISO		BLC_P_ON_BYPASS	91
I57		GENERIC_ISO		BLC_P_ON_D	91
I58		GENERIC_ISO		BLC_P_ON_DRAIN	91
I59		GENERIC_ISO		BLC_P_ON_D_R	91
I60		GENERIC_ISO		BLC_P_ON_GATE	91
I61		GENERIC_ISO		BLC_P_ON_R	91
I62		GENERIC_ISO		BLC_SKIP	90
I63		GENERIC_ISO		BLC_SNUB_1	90
I64		GENERIC_ISO		BLC_SNUB_2	90
I65		GENERIC_ISO		BLC_SNUB_3	90
I66		GENERIC_ISO		BLC_UVLO	89 91
I67		GENERIC_ISO		BLC_VIN2_GATE	91
I68		GENERIC_ISO		BLC_VIN2_SRC	91
I69		GENERIC_ISO		BLC_VINP_GATE	91
I70		GENERIC_ISO		BLC_VIN_SNS	90
I71		GENERIC_ISO		BLC_VSYNC	87 89
I72		GENERIC_ISO		BLC_VSYNC_R	89

BOOST_*

	Spacing	Netname	
I73	GENERIC_ISO	BOOST_BP	89
I74	GENERIC_ISO	BOOST_BYPASS	89 91
I75	GENERIC_ISO	BOOST_BYPASS_GATE	89
I76	PM	BOOST_EN_GATE	89
I77	PM	BOOST_EN_L	89
I78	GENERIC_ISO	BOOST_RC	89
I79	GENERIC_ISO	BOOST_SS	89

BT_*

	Spacing	Netname	
I80	PM	BT_PWR_EN	35
I81	PM	BT_PWR_RST_L	15 20 35
I82	PM	BT_PWR_RST_L_Q	35

BURSTMODE_*

	Spacing	Netname	
I83	PM	BURSTMODE_EN	71
I84	PM	BURSTMODE_EN_L	6 48 71

CAM_*

	Spacing	Netname	
I85	PM	CAM_EXT_BOOT	42 43
I86	PM	CAM_PROC_RESET	43
I87	PM	CAM_PROC_RESET_L	42 43

CPU_*

	Physical	Spacing	Netname	
I88		PM	CPU_PWRGD_1V05_R	28
I89		PM	CPU_PWRGD_3V3	28
I90		PM	CPU_PWRGD_3V3_R	28
I91		GENERIC_ISO	CPU_SKTOCC	64
I92	CPU_PHY	CPU	CPU_THRMTRIP_3V3	47 48
I93	CPU_PHY	CPU	CPU_THRMTRIP_R_L	48

DEBUG_*

	Spacing	Netname	
I94	PM	DEBUG_RESET_L	26 49

DP_*

	Physical	Spacing	Netname	
I95		PM	DP_AUXIO_EN	15 86 88
I96		PM	DP_GPU_MUX_EN	84
I97		GENERIC_ISO	DP_INTPNL_HPD	84 87
I98		GENERIC_ISO	DP_INT_EG_HPD	82 84
I99	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_CLK	85 86
I100	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_DATA	85 86
I101		GENERIC_ISO	DP_TBTPA_HPD	36 86
I102	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_CLK	85 88
I103	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_DATA	85 88
I104		GENERIC_ISO	DP_TBTPB_HPD	36 88

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ENETCONN_*

	Spacing	Netname	
I1	GENERIC_ISO	ENETCONN_TCT	40

ENET_*

	Physical	Spacing	Netname	
I2		GENERIC_ISO	ENET_ACT	40
I3		GENERIC_ISO	ENET_ASF_GPIO	47 48
I4		GENERIC_ISO	ENET_CLKREQ_L	15 18 40
I5		PM	ENET_CLKREQ_L_Q	39 40
I6		PM	ENET_CR_1V8_EN	
I7		PM	ENET_CR_1V8_EN_R	
I8		PM	ENET_CR_3V3_EN_L	
I9		PM	ENET_CR_3V3_EN_L_R	
I10		PM	ENET_CR_PWREN	39 41
I11	XDP_PHY	XDP	ENET_LOW_PWR	26 39 41
I12		PM	ENET_PWR_EN_L	40
I13		PM	ENET_PWR_EN_L_R	40
I14		PM	ENET_RESET_L	39 41
I15		PM	ENET_SD_RESET_L	26 41
I16		GENERIC_ISO	ENET_SR_DISABLE	39
I17		GENERIC_ISO	ENET_SR_LX	39 40
I18		PM	ENET_WAKE_L	40

FAN_*

	Spacing	Netname	
I19	GENERIC_ISO	FAN_0_PWM_FET	54
I20	GENERIC_ISO	FAN_0_PWM_FILT	54
I21	GENERIC_ISO	FAN_0_TACH_FET	54
I22	GENERIC_ISO	FAN_0_TACH_FILT	54

FBVDD_*

	Spacing	Netname	
I23	GENERIC_ISO	FBVDD_ALTVO	82 95

FB_*

	Spacing	Netname	
I24	GENERIC_ISO	FB_A0_VREFC	79
I25	GENERIC_ISO	FB_A0_VREFD	79
I26	GENERIC_ISO	FB_A1_VREFC	79
I27	GENERIC_ISO	FB_A1_VREFD	79
I28	GENERIC_ISO	FB_B0_VREFC	80
I29	GENERIC_ISO	FB_B0_VREFD	80
I30	GENERIC_ISO	FB_B1_VREFC	80
I31	GENERIC_ISO	FB_B1_VREFD	80
I32	GENERIC_ISO	FB_VREF	77

FET_*

	Physical	Spacing	Voltage	Netname	
I33		GENERIC_ISO		FET_EN_P12V_S0	74
I34		GENERIC_ISO		FET_EN_P12V_S0_BLC	74
I35		GENERIC_ISO		FET_EN_P12V_S0_BLC_R	74
I36		GENERIC_ISO		FET_EN_P12V_S0_R	74
I37		GENERIC_ISO		FET_EN_P12V_S5	74
I38		GENERIC_ISO		FET_EN_P12V_S5_R	74
I39		GENERIC_ISO		FET_EN_VDDQ_S0	74
I40		GENERIC_ISO		FET_HDD_SLGSW	52
I41	POWER_PHY	POWER	12V	FET_VCC_U7950	74
I42	POWER_PHY	POWER	12V	FET_VCC_U7970	74
I43	POWER_PHY	POWER	12V	FET_VCC_U7980	74

FLAG_*

	Spacing	Netname	
I44	GENERIC_ISO	FLAG_V	89 90 91

G3_*

	Spacing	Netname	
I45	GENERIC_ISO	G3_POWERON_L	47 48

GND_*

	Spacing	Netname	
I46	SENSE	GND_SMC_AVSS	47 48 51 55

GPU_*

	Physical	Spacing	Netname	
I47		GENERIC_ISO	GPU_ALT_VREF	77 82
I48		GENERIC_ISO	GPU_BUFRSTN	83
I49		GENERIC_ISO	GPU_IFPAB_PLLVDD	81
I50		GENERIC_ISO	GPU_IFPA_IOVDD	81
I51		GENERIC_ISO	GPU_IFPB_IOVDD	81
I52		GENERIC_ISO	GPU_IFPC_IOVDD	81
I53	XDP_PHY	XDP	GPU_JTAG_TDI	78 81
I54	XDP_PHY	XDP	GPU_JTAG_TDO	78 81
I55	XDP_PHY	XDP	GPU_JTAG_TMS	78 81
I56	XDP_PHY	XDP	GPU_JTAG_TRST_L	81
I57		GENERIC_ISO	GPU_MLS_STRAP0	81 82
I58		GENERIC_ISO	GPU_MLS_STRAP1	81 82
I59		GENERIC_ISO	GPU_MLS_STRAP2	81 82
I60		GENERIC_ISO	GPU_MLS_STRAP3	81 82
I61		GENERIC_ISO	GPU_MLS_STRAP4	81 82
I62		GENERIC_ISO	GPU_RESET_L	26 75 82
I63	SPI_50S	SPI	GPU_ROM_CS_L	81 82
I64	SPI_50S	SPI	GPU_ROM_CS_L_R	82
I65	SPI_50S	SPI	GPU_ROM_SI	81 82
I66	SPI_50S	SPI	GPU_ROM_SI_R	82
I67	SPI_50S	SPI	GPU_ROM_SO	81 82
I68	SPI_50S	SPI	GPU_ROM_SO_R	82
I69	SPI_50S	SPI	GPU_ROM_WP_L	82

HDD_*

	Spacing	Netname	
I70	GENERIC_ISO	HDD_12V_S0_GATE	52
I71	GENERIC_ISO	HDD_OOB_1V00_REF	52
I72	PM	HDD_PWR_EN	21 52
I73	GENERIC_ISO	HDD_PWR_EN_L	52
I74	GENERIC_ISO	HDD_PWR_EN_R	52

I2C_*

	Physical	Spacing	Netname	
I75	SMB_PHY	SMB	I2C_TCON_MAS_SCL	50 87
I76	SMB_PHY	SMB	I2C_TCON_MAS_SDA	50 87

IFPD_*

	Spacing	Netname	
I77	GENERIC_ISO	IFPD_RSET	81

IFPEF_*

	Spacing	Netname	
I78	GENERIC_ISO	IFPEF_RSET	81

ISNSA_*

	Electrical	Physical	Spacing	Netname	
I79	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_N	51
I80	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_P	51
I81	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_N	55
I82	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_P	55
I83	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_N	55
I84	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_P	55
I85	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDDQ_N	51
I86	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDDQ_P	51
I87	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_N	51
I88	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_P	51
I89	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_N	55
I90	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_P	55
I91	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_N	51
I92	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_P	51
I93	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_N	51
I94	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_P	51
I95	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0_SSD_N	51
I96	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0_SSD_P	51
I97	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_N	55
I98	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_P	55
I99	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5VS0_HDD_N	51
I100	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5VS0_HDD_P	51
I101	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_N	51
I102	SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_P	51

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ISNS_*

	Physical	Spacing	Netname	
I1	SNS_PHY	SENSE	ISNS_CPUAXG_FB	51
I2	SNS_PHY	SENSE	ISNS_CPUAXG_N	51
I3	SNS_PHY	SENSE	ISNS_CPUAXG_P	51
I4	SNS_PHY	SENSE	ISNS_CPUCORE_FB	51
I5	SNS_PHY	SENSE	ISNS_CPUCORE_N	51
I6	SNS_PHY	SENSE	ISNS_CPUCORE_P	51
I7	SNS_PHY	SENSE	ISNS_GPUCORE_FB	51
I8	SNS_PHY	SENSE	ISNS_GPUCORE_N	51
I9	SNS_PHY	SENSE	ISNS_GPUCORE_P	51
I10	SNS_PHY	SENSE	ISNS_P12VG3H_R	51
I11	SNS_PHY	SENSE	ISNS_P12VS0_CPU_P1V05_R	55
I12	SNS_PHY	SENSE	ISNS_P12VS0_CPU_VCCSA_R	55
I13	SNS_PHY	SENSE	ISNS_P12VS0_FBVDDQ_R	51
I14	SNS_PHY	SENSE	ISNS_P12VS0_HDD_R	51
I15	SNS_PHY	SENSE	ISNS_P12VS0_P1V05_R	55
I16	SNS_PHY	SENSE	ISNS_P1V05S0_PCH_R	51
I17	SNS_PHY	SENSE	ISNS_P1V5S0_CPU_MEM_R	51
I18	SNS_PHY	SENSE	ISNS_P3V3S0_SSD_R	51
I19	SNS_PHY	SENSE	ISNS_P3V3S4_AP_R	55
I20	SNS_PHY	SENSE	ISNS_P5VS0_HDD_R	51
I21	SNS_PHY	SENSE	ISNS_PVDDQS3_DDR_R	51

ISOLATE_*

	Spacing	Netname	
I22	PM	ISOLATE_CPU_MEM_5V	28
I23	PM	ISOLATE_CPU_MEM_5V_L	28

JTAG_*

	Physical	Spacing	Netname	
I24	XDP_PHY	XDP	JTAG_GMUX_TMS	20
I25	XDP_PHY	XDP	JTAG_TBT_TDI	15 21
I26	XDP_PHY	XDP	JTAG_TBT_TDI_ISOL	15 36
I27	XDP_PHY	XDP	JTAG_TBT_TDO	15 21
I28	XDP_PHY	XDP	JTAG_TBT_TDO_ISOL	15 36
I29	XDP_PHY	XDP	JTAG_TBT_TMS	15 18
I30	XDP_PHY	XDP	JTAG_TBT_TMS_ISOL	15 36

LED_*

	Spacing	Netname	
I31	PM	LED_DRIVER_EN	90 91
I32	PM	LED_DRIVER_EN_L	91
I33	PM	LED_DRIVER_EN_L_R	91
I34	GENERIC_ISO	LED_DRIVER_OVP1	90 91
I35	GENERIC_ISO	LED_DRIVER_OVP1P	89 90
I36	GENERIC_ISO	LED_DRIVER_OVP1_OUT	90
I37	GENERIC_ISO	LED_DRIVER_OVP2	90 91
I38	GENERIC_ISO	LED_DRIVER_OVP2P	89 90
I39	GENERIC_ISO	LED_DRIVER_OVP2_OUT	90
I40	GENERIC_ISO	LED_DRIVER_OVP3	90 91
I41	GENERIC_ISO	LED_DRIVER_OVP3P	89 90
I42	GENERIC_ISO	LED_DRIVER_OVP3_OUT	90

LPCPLUS_*

	Spacing	Netname	
I43	GENERIC_ISO	LPCPLUS_GPIO	21 49

LPC_*

	Spacing	Netname	
I44	GENERIC_ISO	LPC_PWRDWN_L	19 26 47 49
I45	GENERIC_ISO	LPC_SERIRQ	18 47 49

MEMVTT_*

	Spacing	Netname	
I46	PM	MEMVTT_EN	28 64
I47	PM	MEMVTT_EN_L	28

MEM_*

	Spacing	Netname	
I48	GENERIC_ISO	MEM_EVENT_L	29 30 31 32 47 48

MOJO_*

	Physical	Spacing	Netname	
I49	XDP_PHY	XDP	MOJO_RX_L	45 47 48
I50	XDP_PHY	XDP	MOJO_TX_L	45 47 48

OCA_*

	Spacing	Netname	
I51	GENERIC_ISO	OCA_FET_DRAIN	91

OVP_*

	Spacing	Netname	
I52	GENERIC_ISO	OVP_OREF	91
I53	GENERIC_ISO	OVP_OUT1	91
I54	GENERIC_ISO	OVP_OUT1_R	91
I55	GENERIC_ISO	OVP_OUT2	91
I56	GENERIC_ISO	OVP_OUT2_R	91
I57	GENERIC_ISO	OVP_OUT3	91
I58	GENERIC_ISO	OVP_OUT3_R	91

P1V2_*

	Spacing	Netname	
I59	PM	P1V2_S4_EN	43

P1V8_*

	Spacing	Netname	
I60	PM	P1V8_S4_EN	43

P3V3AP_*

	Spacing	Netname	
I61	GENERIC_ISO	P3V3AP_VMON	35

P3V3_*

	Spacing	Netname	
I62	GENERIC_ISO	P3V3_S0_EN_G	74
I63	GENERIC_ISO	P3V3_S3_EN_G	74

P3V42G3H_*

	Spacing	Netname	
I64	GENERIC_ISO	P3V42G3H_BOOST	73

P5V_*

	Spacing	Netname	
I65	GENERIC_ISO	P5V_S0_EN_G	74

PCA9557D_*

	Spacing	Netname	
I66	GENERIC_ISO	PCA9557D_RESET_L	26 28 34

PCH_*

	Physical	Spacing	Netname	
I67		PM	PCH_BLC_EXT_BOOT	21 89
I68		PM	PCH_BLC_EXT_BOOT_R	15 21
I69		PM	PCH_BLC_MCU_RESET	21 89
I70		PM	PCH_BLC_MCU_RESET_R	15 21
I71		PM	PCH_CAM_RESET	15 21
I72		PM	PCH_CAM_RESET_R	21 43
I73		GENERIC_ISO	PCH_DSWVRMEN	19
I74	CPU_PHY	CPU	PCH_PECI	21
I75	CPU_PHY	CPU	PCH_PROCPWRGD	21
I76		GENERIC_ISO	PCH_RCIN_L	21
I77		GENERIC_ISO	PCH_RI_L	19
I78		GENERIC_ISO	PCH_SMBALERT_L	15 18
I79		GENERIC_ISO	PCH_SRTCRST_L	18
I80		GENERIC_ISO	PCH_STRP_TOPBLK_SWP_L	20
I81		GENERIC_ISO	PCH_SUSWARN_L	15 19

PCIE_*

	Spacing	Netname	
I82	GENERIC_ISO	PCIE_CLKREQ5_GPIO44_L	18
I83	PM	PCIE_WAKE_L	19 35 40

PEG_*

	Spacing	Netname	
I84	GENERIC_ISO	PEG_CLKREQ_L	15 18 75

PGOOD_*

	Spacing	Netname	
I85	PM	PGOOD_P1V5_S0_DLY	28

PLT_*

	Spacing	Netname	
I86	PCH	PLT_RESET_L	20 26
I87	PCH	PLT_RST_BUF_L	26

PM_*

	Spacing	Netname	
I88	PM	PM_CLKRUN_L	47 48 49
I89	PM	PM_DSW_PWRGD	47 48 65
I90	PM	PM_EN_FET_P12V_S0	64 74
I91	PM	PM_EN_FET_P12V_S0_R	74
I92	PM	PM_EN_FET_P3V3_S0	64 74
I93	PM	PM_EN_FET_P3V3_S4	64 74
I94	PM	PM_EN_FET_P5V_S0	64 74
I95	PM	PM_EN_FET_VDDQ_S0	64 74
I96	PM	PM_EN_LDO_DDRVTT_S0	64 72
I97	PM	PM_EN_REG_CPUCORE_S0	64 66

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PM_*

	Spacing	Netname	
I1	PM	PM_EN_REG_CPU_P1V05_S0	64 69
I2	PM	PM_EN_REG_FBVDDQ_S0	64 95
I3	PM	PM_EN_REG_GPUCORE_S0_R	92
I4	PM	PM_EN_REG_P1V05_S0	64 95
I5	PM	PM_EN_REG_P1V8_S0	64 72
I6	PM	PM_EN_REG_P3V3_S5	64 71 74
I7	PM	PM_EN_REG_P5V_S4	64 71
I8	PM	PM_EN_REG_VCCSA_S0	64 70
I9	PM	PM_EN_REG_VDDQ_S3	64 72
I10	PM	PM_EN_S0	64
I11	PM	PM_EN_S4	64
I12	PM	PM_EN_USB_PWR	45 46 64
I13	PM	PM_LED_A_ALL_SYS_PWRGD	5
I14	PM	PM_LED_A_BLC_GOOD	5
I15	PM	PM_LED_A_CPUAXG_PGOOD	5
I16	PM	PM_LED_A_GPU_GOOD	5
I17	PM	PM_LED_A_PGOOD_CPUCORE_S0	5
I18	PM	PM_LED_A_PGOOD_CPU_P1V05_S0	5
I19	PM	PM_LED_A_PGOOD_REG_FBVDDQ_S0	5
I20	PM	PM_LED_A_PGOOD_REG_GPUCORE_S0	5
I21	PM	PM_LED_A_PGOOD_REG_P1V05	5
I22	PM	PM_LED_A_PGOOD_REG_VDDQ_S3	5
I23	PM	PM_LED_A_S4	5
I24	PM	PM_LED_A_S5	5
I25	PM	PM_LED_A_SLP_S3	5
I26	PM	PM_LED_A_VIDEO_ON	5
I27	PM	PM_LED_K_ALL_SYS_PWRGD	5
I28	PM	PM_LED_K_BLC_GOOD	5
I29	PM	PM_LED_K_CPUAXG_PGOOD	5
I30	PM	PM_LED_K_GPU_GOOD	5
I31	PM	PM_LED_K_PGOOD_CPUCORE_S0	5
I32	PM	PM_LED_K_PGOOD_CPU_P1V05_S0	5
I33	PM	PM_LED_K_PGOOD_REG_FBVDDQ_S0	5
I34	PM	PM_LED_K_PGOOD_REG_GPUCORE_S0	5
I35	PM	PM_LED_K_PGOOD_REG_P1V05	5
I36	PM	PM_LED_K_PGOOD_REG_VDDQ_S3	5
I37	PM	PM_LED_K_SLP_S3	5
I38	PM	PM_MEM_PWRGD_L	28
I39	PM	PM_PCH_APWROK	19 65
I40	PM	PM_PCH_PWROK	15 19 26 35 43 65 89
I41	PM	PM_PCH_PWROK_APWROK	65
I42	PM	PM_PCH_SYS_PWROK	19 48 65
I43	PM	PM_PGOOD_FBVDDQ_VDDQ_S0	64
I44	PM	PM_PGOOD_FET_P12V_S0	74
I45	PM	PM_PGOOD_FET_P12V_S0_BLC	74 91
I46	PM	PM_PGOOD_FET_P12V_S5	74
I47	PM	PM_PGOOD_FET_P3V3_S0	64 74
I48	PM	PM_PGOOD_FET_P5V_S0	64 74
I49	PM	PM_PGOOD_FET_VDDQ_S0	28 64 74
I50	PM	PM_PGOOD_P3V3_S4_FET	27 35 64 74
I51	PM	PM_PGOOD_REG_ALL_P1V05_S0	64 65
I52	PM	PM_PGOOD_REG_ALL_P1V05_S0_R	28 64
I53	PM	PM_PGOOD_REG_CPUCORE_S0	5 25 65 66
I54	PM	PM_PGOOD_REG_CPU_P1V05_S0	5 64 69
I55	PM	PM_PGOOD_REG_FBVDDQ_S0	5 64 95
I56	PM	PM_PGOOD_REG_P1V05_S0	5 64 95
I57	PM	PM_PGOOD_REG_P1V8_S0	64 72
I58	PM	PM_PGOOD_REG_P3V3_S5	65 71
I59	PM	PM_PGOOD_REG_P5V_S4	64 71
I60	PM	PM_PGOOD_REG_VCCSA_S0	64 65 70
I61	PM	PM_PGOOD_REG_VDDQ_S3	5 64 72
I62	PM	PM_PWRBTN_L	15 19 25 47
I63	PM	PM_RSMRST_PCH_L	19 65
I64	PM	PM_RSMRST_PCH_L_R	65
I65	PM	PM_SLP_S3_L	5 15 19 28 40 47 48 64
I66	PM	PM_SLP_S4_L	15 19 47 64
I67	PM	PM_SLP_S5_L	15 19 47 64
I68	PCH	PM_SYSRST_L	19 25 26 47
I69	PCH	PM_THRMTRIP_L	21 48
I129	PM	PGOOD_P12V_S0_R	65
I130	PM	PGOOD_P12V_S0	64 65 74

PP12V_*

	Physical	Spacing	Voltage	Netname	
I70	POWER_PHY	POWER	12V	PP12V_LCD	87
I71	POWER_PHY	POWER	12V	PP12V_LCD_EXT	87
I72	POWER_PHY	POWER	12V	PP12V_S0_FAN_0_FILT	54
I73	POWER_PHY	POWER	12V	PP12V_S0_HDD_FET	44 52

PP1V05_*

	Physical	Spacing	Voltage	Netname	
I74	POWER_PHY	POWER	1.05V	PP1V05_GPU_FB_DLL_AVDD	77
I75	POWER_PHY	POWER	1.05V	PP1V05_GPU_FB_PLL_AVDD	77
I76	POWER_PHY	POWER	1.05V	PP1V05_GPU_IFPD_IOVDD	81
I77	POWER_PHY	POWER	1.05V	PP1V05_GPU_IFPEF_IOVDD	81
I78	POWER_PHY	POWER	1.05V	PP1V05_GPU_PLLVDD	81
I79	POWER_PHY	POWER	1.05V	PP1V05_GPU_SP_PLLVDD	81 83
I80	POWER_PHY	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLA_F	17 22
I81	POWER_PHY	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLB_F	17 22

PP1V2_*

	Physical	Spacing	Voltage	Netname	
I82	POWER_PHY	POWER	1.2V	PP1V2_ENET_INTREG	40
I83	POWER_PHY	POWER	1.2V	PP1V2_G3H_SMC_VDDC	47
I84	POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_AVDDL	39
I85	POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_GPHYPLL	39
I86	POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_PCIEPLL	39
I87	POWER_PHY	POWER	1.2V	PP1V2_USB_HUB_CRFILT	27
I88	POWER_PHY	POWER	1.2V	PP1V2_USB_HUB_PLLFILT	27

PP1V5_*

	Physical	Spacing	Voltage	Netname	
I89	POWER_PHY	POWER	1.5V	PP1V5_S0_DP_BIAS	84

PP1V8_*

	Physical	Spacing	Voltage	Netname	
I90	POWER_PHY	POWER	1.8V	PP1V8_S0_PCH_VCCVRM_F	22 24

PP3V3R1V8_*

	Physical	Spacing	Netname	
I92	POWER_PHY	POWER	PP3V3R1V8_ENET_LR_OUT_REG	39

PP3V3RHV_*

	Physical	Spacing	Voltage	Netname	
I93	POWER_PHY	POWER	12V	PP3V3RHV_SW_TBTAPWR	86
I94	POWER_PHY	POWER	12V	PP3V3RHV_SW_TBTBPWR	88

PP3V3_*

	Physical	Spacing	Voltage	Netname	
I95	POWER_PHY	POWER	3.3V	PP3V3_DMIC_CONN	42
I96	POWER_PHY	POWER	3.3V	PP3V3_G3H_AVREF_SMC	47 48
I97	POWER_PHY	POWER	3.3V	PP3V3_G3H_BT_FET	35
I98	POWER_PHY	POWER	3.3V	PP3V3_G3H_BT_FLT	35
I99	POWER_PHY	POWER	3.3V	PP3V3_G3H_SMC_USBMUX_R	45
I100	POWER_PHY	POWER	3.3V	PP3V3_G3H_SMC_VDDA	47
I101	POWER_PHY	POWER	3.3V	PP3V3_G3_RTC	6 26
I102	POWER_PHY	POWER	3.3V	PP3V3_GPU_IFPX_PLLVDD	81
I103	POWER_PHY	POWER	3.3V	PP3V3_PVDDQS3_ISNS	51
I104	POWER_PHY	POWER	3.3V	PP3V3_S0_PCH_VCCA_DAC_F	17 22
I105	POWER_PHY	POWER	3.3V	PP3V3_S0_SSD_FLT	44
I106	POWER_PHY	POWER	3.3V	PP3V3_S0_SW_SD_PWR	41
I107	POWER_PHY	POWER	3.3V	PP3V3_S4_ALS_F	42
I108	POWER_PHY	POWER	3.3V	PP3V3_S4_AP_FET	35
I109	POWER_PHY	POWER	3.3V	PP3V3_S4_AP_FLT	35
I110	POWER_PHY	POWER	3.3V	PP3V3_S4_ENET_FET_AVDDH	39
I111	POWER_PHY	POWER	3.3V	PP3V3_S4_ENET_FET_BIASVDDH	39
I112	POWER_PHY	POWER	3.3V	PP3V3_S4_ENET_FET_SRVDD	39
I113	POWER_PHY	POWER	3.3V	PP3V3_S4_ENET_FET_XTALVDDH	39
I114	POWER_PHY	POWER	3.3V	PP3V3_S4_USB_HUB_VDD	27
I115	POWER_PHY	POWER	3.3V	PP3V3_S5_XDP_R	25

PP4V5_*

	Physical	Spacing	Voltage	Netname	
I116	POWER_PHY	POWER	4.5V	PP4V5_AUDIO_ANALOG	56 60 62

PP5V_*

	Physical	Spacing	Voltage	Netname	
I117	POWER_PHY	POWER	5V	PP5V_AUDIO_HPAMP	56 57
I118	POWER_PHY	POWER	5V	PP5V_S0_HDD_FET	44 52
I119	POWER_PHY	POWER	5V	PP5V_S0_PCH_V5REF	24
I120	POWER_PHY	POWER	5V	PP5V_S4_EXTA	45
I121	POWER_PHY	POWER	5V	PP5V_S4_EXTA_F	45
I122	POWER_PHY	POWER	5V	PP5V_S4_EXTB	45
I123	POWER_PHY	POWER	5V	PP5V_S4_EXTB_F	45
I124	POWER_PHY	POWER	5V	PP5V_S4_EXTC	46
I125	POWER_PHY	POWER	5V	PP5V_S4_EXTC_F	46
I126	POWER_PHY	POWER	5V	PP5V_S4_EXTD	46
I127	POWER_PHY	POWER	5V	PP5V_S4_EXTD_F	46
I128	POWER_PHY	POWER	5V	PP5V_S5_PCH_V5REFSUS	24

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PPHV_*

	Physical	Spacing	Voltage	Netname	
I1	POWER_PHY	POWER	12V	PPHV_SW_TBTAPWR	86
I2	POWER_PHY	POWER	12V	PPHV_SW_TBTBPWR	88

PPVBATT_*

	Physical	Spacing	Voltage	Netname	
I3	POWER_PHY	POWER	3.3V	PPVBATT_G3_RTC	26
I4	POWER_PHY	POWER	3.3V	PPVBATT_G3_RTC_R	26

PU_*

	Spacing	Netname	
I5	GENERIC_ISO	PU_U6900	64

PWR_*

	Spacing	Netname	
I6	PM	PWR_BTN	6 48
I7	PM	PWR_BTN_R	6

REG_*

	Physical	Spacing	Netname	
I8		PM	REG_CPUAXG_PGOOD	5 66
I9		PM	REG_CPUCORE_PGOOD	66
I10	CPU_PHY	CPU	REG_CPUCORE_VRHOT_L	66
I11		PM	REG_CPU_P1V05S0_PGOOD	69
I12	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET0	95
I13	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET1	95
I14	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET1_R	95
I15	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SREF	95
I16		PM	REG_P1V8S0_PGOOD	72
I17		PM	REG_P3V3S5_PGOOD	71
I18		PM	REG_P5VS4_PGOOD	71
I19		PM	REG_VCCSAS0_PGOOD	70
I20		PM	REG_VDDQS3_PGOOD	72

RTC_*

	Spacing	Netname	
I21	PM	RTC_RESET_L	18 26 48
I22	PM	RTC_RESET_L_R	48

S5_*

	Spacing	Netname	
I23	PM	S5_PWRGD	47 65

SATALED_*

	Spacing	Netname	
I24	GENERIC_ISO	SATALED_L	44

SDCONN_*

	Spacing	Netname	
I25	GENERIC_ISO	SDCONN_DETECT	41
I26	GENERIC_ISO	SDCONN_ILIM	41
I27	GENERIC_ISO	SDCONN_OC_L	41

SD_*

	Spacing	Netname	
I28	GENERIC_ISO	SD_DETECT_LVL	41

SLG_*

	Spacing	Netname	
I29	PM	SLG_ENET_RESET_L	41
I30	PM	SLG_ENET_RESET_R_L	41

SMBUS_*

	Physical	Spacing	Netname	
I31	SMB_PHY	SMB	SMBUS_PCH_CLK	18 50
I32	SMB_PHY	SMB	SMBUS_PCH_CLK_R	50
I33	SMB_PHY	SMB	SMBUS_PCH_DATA	18 50
I34	SMB_PHY	SMB	SMBUS_PCH_DATA_R	50
I35	SMB_PHY	SMB	SMBUS_SMC_0_S0_SCL	47 50
I36	SMB_PHY	SMB	SMBUS_SMC_0_S0_SDA	47 50
I37	SMB_PHY	SMB	SMBUS_SMC_1_S0_SCL	47 50
I38	SMB_PHY	SMB	SMBUS_SMC_1_S0_SDA	47 50
I39	SMB_PHY	SMB	SMBUS_SMC_2_S4_SCL	47 50
I40	SMB_PHY	SMB	SMBUS_SMC_2_S4_SDA	47 50
I41	SMB_PHY	SMB	SMBUS_SMC_3_SCL	47 50
I42	SMB_PHY	SMB	SMBUS_SMC_3_SDA	47 50
I43	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SCL	47 48 50
I44	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SDA	47 48 50
I45	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SCL	47 48 50
I46	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SDA	47 48 50

SMCISNS_*

	Physical	Spacing	Netname	
I47	SNS_PHY	SENSE	SMCISNS_CPUAXG	48 51
I48	SNS_PHY	SENSE	SMCISNS_CPUCORE	48 51
I49	SNS_PHY	SENSE	SMCISNS_GPUCORE	48 51
I50	SNS_PHY	SENSE	SMCISNS_P12VG3H	48 51
I51	SNS_PHY	SENSE	SMCISNS_P12VS0_CPU_P1V05	48 55
I52	SNS_PHY	SENSE	SMCISNS_P12VS0_CPU_VCCSA	48 55
I53	SNS_PHY	SENSE	SMCISNS_P12VS0_FBVDDQ	48 51
I54	SNS_PHY	SENSE	SMCISNS_P12VS0_HDD	48 51
I55	SNS_PHY	SENSE	SMCISNS_P12VS0_P1V05	48 55
I56	SNS_PHY	SENSE	SMCISNS_P1V05S0_PCH	48 51
I57	SNS_PHY	SENSE	SMCISNS_P1V5S0_CPU_MEM	48 51
I58	SNS_PHY	SENSE	SMCISNS_P3V3S0_SSD	48 51
I59	SNS_PHY	SENSE	SMCISNS_P3V3S4_AP	48 55
I60	SNS_PHY	SENSE	SMCISNS_P5VS0_HDD	48 51
I61	SNS_PHY	SENSE	SMCISNS_PVDDQS3_DDR	48 51

SMCVSNS_*

	Physical	Spacing	Netname	
I62	SNS_PHY	SENSE	SMCVSNS_CPUAXG	48 51
I63	SNS_PHY	SENSE	SMCVSNS_CPUCORE	48 51
I64	SNS_PHY	SENSE	SMCVSNS_GPUCORE	48 51
I65	SNS_PHY	SENSE	SMCVSNS_P12VG3H	48 51
I66	SNS_PHY	SENSE	SMCVSNS_P1V05S0_PCH	48 51
I67	SNS_PHY	SENSE	SMCVSNS_P1V5S0_CPU_MEM	48 51
I68	SNS_PHY	SENSE	SMCVSNS_P3V3S0	48 51
I69	SNS_PHY	SENSE	SMCVSNS_P5VS0_HDD	48 51
I70	SNS_PHY	SENSE	SMCVSNS_PVDDQS3_DDR	48 51

SMC_*

	Physical	Spacing	Netname	
I71		GENERIC_ISO	SMC_ACDC_ID	6 48
I72		GENERIC_ISO	SMC_ACDC_ID_R	6
I73		PM	SMC_ASSERT_RTCRST	48
I74		GENERIC_ISO	SMC_BLC_MUX_RX_L	48 89
I75		GENERIC_ISO	SMC_BLC_MUX_TX_L	48 89
I76		GENERIC_ISO	SMC_CPU_CATERR_L	47 48
I77	CPU_PHY	CPU	SMC_CPU_PECI	48
I78		PM	SMC_DELAYED_PWRGD	38 47 48 65
I79		GENERIC_ISO	SMC_DP_HPD_L	47 48
I80	CLK_XTAL	XTAL	SMC_EXTAL	47 48
I81		GENERIC_ISO	SMC_FAN_0_CTL	47 54
I82		GENERIC_ISO	SMC_FAN_0_TACH	47 54
I83		GENERIC_ISO	SMC_GFX_OVERTEMP	47 48 82
I84		GENERIC_ISO	SMC_GFX_OVERTEMP_Q	82
I85		GENERIC_ISO	SMC_GFX_OVERTEMP_R_L	82
I86		PM	SMC_GFX_THROTTLE_L	47 82
I87		PM	SMC_GFX_THROTTLE_R_L	82
I88		PM	SMC_LRESET_L	26 47
I89		PM	SMC_MANUAL_RST_L	48
I90		PM	SMC_ONOFF_L	47 48
I91		SENSE	SMC_OOB1_RX_CN	44
I92		SENSE	SMC_OOB1_RX_FILT	44 52
I93		SENSE	SMC_OOB1_RX_L	47 52
I94		SENSE	SMC_OOB1_RX_R	52
I95		SENSE	SMC_OOB1_TX_L	47 48
I96		SENSE	SMC_OOB2_RX_L	44 48
I97		SENSE	SMC_OOB2_TX_L	44 48
I98	CPU_PHY	CPU	SMC_PECI_L	47 48
I99	CPU_PHY	CPU	SMC_PECI_L_R	48
I100		PM	SMC_PME_S4_WAKE_L	35 47 48
I101		GENERIC_ISO	SMC_PM_G2_EN	47 48 74
I102		PM	SMC_PM_PCH_SYS_PWROK	47 48
I103		CPU	SMC_PROCHOT	47 48
I104		PM	SMC_RESET_L	47 48 49

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SMC_*

	Physical	Spacing	Netname	
I1		GENERIC_ISO	SMC_ROMBOOT	48 49
I2		GENERIC_ISO	SMC_RUNTIME_SCI_L	15 21 47
I3		GENERIC_ISO	SMC_RX_L	47 48 49
I4		PM	SMC_S4_WAKESRC_EN	35 47 48
I5		GENERIC_ISO	SMC_TCK	47 48 49
I6		GENERIC_ISO	SMC_TDI	47 48 49
I7		GENERIC_ISO	SMC_TDO	47 48 49
I8		PM	SMC_THRMTRIP	47 48
I9		GENERIC_ISO	SMC_TMS	47 48 49
I10		GENERIC_ISO	SMC_TO_BLC_RX_L	48
I11		GENERIC_ISO	SMC_TO_BLC_TX_L	48
I12		GENERIC_ISO	SMC_TX_L	47 48 49
I13		PM	SMC_WAKE_L	47
I14		GENERIC_ISO	SMC_WAKE_SCI_L	15 21 47
I15	CLK_XTAL	XTAL	SMC_XTAL	47 48
I16	CLK_XTAL	XTAL	SMC_XTAL_R	48

TBT_*

	Physical	Spacing	Voltage	Netname	
I24		GENERIC_ISO	3.3V	TBT_A_BIAS	86
I25	TBT_GEN_55S	TBT_GEN		TBT_A_CONFIG1_RC	86
I26		PM		TBT_A_HV_EN	36 86
I27		GENERIC_ISO	3.3V	TBT_B_BIAS	88
I28	TBT_GEN_55S	TBT_GEN		TBT_B_CONFIG1_RC	88
I29		PM		TBT_B_HV_EN	36 88
I30		GENERIC_ISO		TBT_CLKREQ_ISOL_L	38
I31		GENERIC_ISO		TBT_CLKREQ_L	15 38
I32		GENERIC_ISO		TBT_DDC_XBAR_EN_L	36 85
I33		GENERIC_ISO		TBT_EN_CIO_PWR	38
I34		PM		TBT_EN_CIO_PWR_L	36 38
I35		GENERIC_ISO		TBT_EN_LC_ISOL	38
I36		GENERIC_ISO		TBT_EN_LC_PWR	36 38
I37		GENERIC_ISO		TBT_PCH_CLKREQ_L	15 21
I38		PM		TBT_PWR_EN	15 26 36
I39		PM		TBT_PWR_EN_PCH	18 26
I40		PM		TBT_PWR_ON_POC_RST_L	36 38
I41		PM		TBT_PWR_REQ_L	15 20 36
I42		PM		TBT_S0_EN	64
I43		PM		TBT_SW_RESET_L	21 38

VREFMRGN_*

	Spacing	Netname	
I74	GENERIC_ISO	VREFMRGN_CA_SODIMMA_EN	34
I75	GENERIC_ISO	VREFMRGN_CA_SODIMMB_EN	34

VTTCLAMP_*

	Spacing	Netname	
I76	GENERIC_ISO	VTTCLAMP_EN	28
I77	GENERIC_ISO	VTTCLAMP_L	28

WOL_*

	Spacing	Netname	
I78	GENERIC_ISO	WOL_EN	15 21 40

SML_*

	Physical	Spacing	Netname	
I17	SMB_PHY	SMB	SML_PCH_0_CLK	18 50
I18	SMB_PHY	SMB	SML_PCH_0_DATA	18 50
I19	SMB_PHY	SMB	SML_PCH_1_CLK	18 50
I20	SMB_PHY	SMB	SML_PCH_1_DATA	18 50

TSNS_*

	Electrical	Physical	Spacing	Netname	
I44	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_1_N	53
I45	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_1_P	53
I46	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_2_N	53
I47	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_2_P	53
I48	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_3_N	53
I49	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_3_P	53
I50	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_1_N	53
I51	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_1_P	53
I52	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_2_N	53
I53	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_2_P	53
I54	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_3_N	53
I55	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_3_P	53
I56	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_4_N	53
I57	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_4_P	53
I58	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_5_N	53
I59	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_5_P	53
I60	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_6_N	53
I61	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_6_P	53
I62	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_7_N	53
I63	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_7_P	53
I64	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_ACDC_N	6 53
I65	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_ACDC_P	6 53
I66	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_SKIN_N	53
I67	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_SKIN_P	53

SPI_*

	Spacing	Netname	
I21	GENERIC_ISO	SPI_DESCRIPTOR_OVERRIDE_L	15 47

SSD_*

	Spacing	Netname	
I22	GENERIC_ISO	SSD_CLKREQ_L	15 18

SYS_*

	Spacing	Netname	
I23	PM	SYS_PWROK_R	65

UVP_*

	Spacing	Netname	
I68	PM	UVP_IN_1	91
I69	PM	UVP_IN_1_REF	91
I70	PM	UVP_IN_2	91
I71	PM	UVP_IN_3	91
I72	PM	UVP_IN_4	91
I73	PM	UVP_REF	91

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